Discrete Data Analysis and Lab

Lecture notes, Chapters 1–11

Based on A. Agresti, *Categorical Data Analysis*, 2nd ed. (Wiley, 2002)

Course map

The course develops methods for categorical responses, mainly from the biomedical and social sciences, and shows how to carry them out with statistical software in the weekly lab. A single generalized linear model framework links logistic regression and loglinear models for discrete data with normal regression for continuous data. The syllabus topics correspond to the chapters of these notes as follows.

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1.	Introduction	1
2.	Contingency table analysis: chi-squared tests, measures of association	2, 3
3.	Generalized linear models	4
4.	Logistic model	5, 6, 7
5.	Loglinear model: three-way and four-way contingency tables	8, 9
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Chapter 1. Distributions and Inference for Categorical Data

1.1 Categorical response data

Motivation. Before choosing a method we must know what kind of variable we are analysing: the measurement scale of a categorical response decides which models and summaries are meaningful.

A *categorical variable* has a measurement scale consisting of a set of categories (political philosophy: liberal / moderate / conservative; a mammogram diagnosis: normal, benign, probably benign, suspicious, malignant). Such scales are pervasive in the social and biomedical sciences, but also occur in genetics, zoology, education, marketing and industrial quality control.

1.1.1 Response–explanatory variable distinction

As in regression, we distinguish a *response* (dependent) variable from *explanatory* (independent) variables. These notes concern methods for *categorical responses*; explanatory variables may be of any type.

1.1.2 Nominal–ordinal scale distinction

Definition (Nominal, ordinal, interval)

(a) A categorical variable is *nominal* if its categories have no natural ordering (religious affiliation, mode of transport, favourite music).
(b) It is *ordinal* if the categories are ordered but the distances between them are unknown (social class, patient condition good/fair/serious/critical).
(c) An *interval* variable has numerical distances between any two values (blood pressure, income); it is a *ratio* variable if ratios are also meaningful.

The classification depends on how a variable is measured: education is nominal (public/private school), ordinal (highest degree) or interval (years of schooling). In the hierarchy interval > ordinal > nominal, methods for a lower level can be used at a higher level (ignoring information) but not conversely. It is usually best to use methods appropriate to the actual scale. The book treats nominal and ordinal variables, discrete interval variables with few values, and continuous variables grouped into a few categories.

1.1.3 Continuous–discrete variable distinction

All measurement is in practice discrete; the useful distinction is between variables taking many values (often treated as continuous) and variables taking few values, which are the subject here.

1.1.4 Quantitative–qualitative variable distinction

Nominal variables are *qualitative* (categories differ in quality) and interval variables are *quantitative*. Ordinal variables sit in between: each category has more or less of the characteristic, and often an underlying continuous variable exists. Analysts exploit this by assigning *scores* to categories or by assuming a latent continuous distribution.

1.1.5 Organization of this book

The models resemble regression but replace normality by binomial, multinomial or Poisson distributions; the two central families are *logistic regression* (binary and multicategory responses) and *loglinear models* (counts). Chapters 1–3 cover distributions, inference and association measures for one- and two-way data (largely pre-1960, non-model-based methods); Chapters 4–9 the generalized linear model family; Chapters 10–13 clustered and repeated responses; Chapters 14–16 theory and history.

1.2 Distributions for categorical data

Motivation. Inference requires a model for the random mechanism generating the counts. Three distributions play the role that the normal distribution plays for continuous responses: binomial, multinomial and Poisson.

1.2.1 Binomial distribution

Let $Y_1, \dots, Y_n$ be independent and identical binary trials with $P(Y_i = 1) = \pi$, $P(Y_i = 0) = 1 - \pi$ (*Bernoulli trials*). The total $Y = \sum_i Y_i$ has the binomial distribution $\text{bin}(n, \pi)$:

$$p(y) = \binom{n}{y} \pi^y (1-\pi)^{n-y}, \qquad y = 0, 1, \dots, n. \tag{1.1}$$

Proposition 1.1 (Binomial moments)

If $Y \sim \text{bin}(n, \pi)$ then $\mu = E(Y) = n\pi$, $\sigma^2 = \text{Var}(Y) = n\pi(1-\pi)$, and

$$\frac{E(Y-\mu)^3}{\sigma^3} = \frac{1-2\pi}{\sqrt{n\pi(1-\pi)}}.$$

In particular the distribution becomes symmetric (and, by the CLT, normal) as $n \to \infty$ for fixed π.

Proof.

Step 1. For one trial $E(Y_i) = 1 \cdot \pi + 0 \cdot (1-\pi) = \pi$ and $E(Y_i^2) = \pi$, so $\text{Var}(Y_i) = \pi - \pi^2 = \pi(1-\pi)$.

Step 2. By linearity and independence, $E(Y) = n\pi$ and $\text{Var}(Y) = n\pi(1-\pi)$.

Step 3. Third central moments of independent variables add (cumulants are additive). For one trial, $E(Y_i - \pi)^3 = (1-\pi)^3\pi + (-\pi)^3(1-\pi) = \pi(1-\pi)[(1-\pi)^2 - \pi^2] = \pi(1-\pi)(1-2\pi)$. Hence $E(Y-\mu)^3 = n\pi(1-\pi)(1-2\pi)$ and dividing by $\sigma^3 = [n\pi(1-\pi)]^{3/2}$ gives the formula, which tends to 0. ■

If trials are not independent and identical the binomial fails; e.g. sampling without replacement from a finite population gives the hypergeometric distribution (Section 3.5).

1.2.2 Multinomial distribution

Suppose each of n independent, identical trials falls in one of c categories. Let $y_{ij} = 1$ if trial i falls in category j, so $\mathbf{y}_i = (y_{i1}, \dots, y_{ic})$ has exactly one 1. Let $n_j = \sum_i y_{ij}$ and $\pi_j = P(y_{ij} = 1)$. Then $(n_1, \dots, n_c)$ is *multinomial*:

$$p(n_1, \dots, n_{c-1}) = \frac{n!}{n_1!\, n_2! \cdots n_c!}\, \pi_1^{n_1} \pi_2^{n_2} \cdots \pi_c^{n_c}. \tag{1.2}$$

Because $\sum_j n_j = n$, the distribution is $(c-1)$-dimensional; $c = 2$ gives the binomial.

Proposition 1.2 (Multinomial moments)

For the multinomial $(n, \boldsymbol{\pi})$ distribution,

$$E(n_j) = n\pi_j, \qquad \mathrm{Var}(n_j) = n\pi_j(1-\pi_j), \qquad \mathrm{Cov}(n_j, n_k) = -n\pi_j\pi_k \ (j \neq k),$$

and each n_j is marginally $\mathrm{bin}(n, \pi_j)$.

Proof. [Supplement]

Step 1. The indicator y_{ij} is Bernoulli(π_j), and n_j is a sum of n independent such indicators, so $n_j \sim \mathrm{bin}(n, \pi_j)$; the mean and variance follow from Proposition 1.1.

Step 2. For $j \neq k$, a single trial cannot fall in two categories, so $y_{ij}y_{ik} = 0$ and $\mathrm{Cov}(y_{ij}, y_{ik}) = 0 - \pi_j\pi_k$.

Step 3. Different trials are independent, so $\mathrm{Cov}(n_j, n_k) = \sum_i \mathrm{Cov}(y_{ij}, y_{ik}) = -n\pi_j\pi_k$. ∎

In matrix form, $\mathrm{Cov}(\mathbf{n}) = n[\mathrm{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top]$; this matrix is used repeatedly below.

1.2.3 Poisson distribution

Counts without a fixed upper limit n (e.g. weekly motorway deaths in Italy) are often modelled by the *Poisson* distribution with mean μ:

$$p(y) = \frac{e^{-\mu}\mu^y}{y!}, \qquad y = 0, 1, 2, \ldots \tag{1.3}$$

It is unimodal with mode $\lfloor \mu \rfloor$, its skewness is $E(Y-\mu)^3/\sigma^3 = 1/\sqrt{\mu}$, and it approaches normality as $\mu \to \infty$. It describes counts of events occurring randomly in time or space with independent counts in disjoint regions.

Proposition 1.3 (Poisson moments and binomial limit)

(a) If $Y \sim \mathrm{Poisson}(\mu)$ then $E(Y) = \mathrm{Var}(Y) = \mu$.
(b) If $n \to \infty$ and $\pi \to 0$ with $n\pi \to \mu$, then $\mathrm{bin}(n, \pi)$ probabilities converge to $\mathrm{Poisson}(\mu)$ probabilities.

Proof. [Supplement]

Step (a). $E(Y) = \sum_{y \geq 1} e^{-\mu}\mu^y/(y-1)! = \mu$. Similarly $E[Y(Y-1)] = \sum_{y \geq 2} e^{-\mu}\mu^y/(y-2)! = \mu^2$, so $\mathrm{Var}(Y) = \mu^2 + \mu - \mu^2 = \mu$.

Step (b). Write

$$\binom{n}{y}\pi^y(1-\pi)^{n-y} = \frac{(n\pi)^y}{y!} \cdot \frac{n(n-1)\cdots(n-y+1)}{n^y} \cdot (1-\pi)^n(1-\pi)^{-y}.$$

For fixed y the second factor tends to 1, $(1-\pi)^{-y} \to 1$, and $(1-\pi)^n = \exp\{n\log(1-\pi)\} = \exp\{-n\pi + O(n\pi^2)\} \to e^{-\mu}$. Hence the probability tends to $e^{-\mu}\mu^y/y!$. ∎

For instance, if 50 million drivers each die in a given week independently with probability 0.000002, the death count is $\mathrm{bin}(5 \times 10^7, 2 \times 10^{-6})$, approximately Poisson with $\mu = 100$. Since Var $= \mu$, counts vary more when their mean is larger.

1.2.4 Overdispersion

Real counts often vary more than the binomial or Poisson allows (*overdispersion*), because the success probability or rate itself varies with unmeasured factors (driving time, seat-belt use, litter effects in toxicology).

Proposition 1.4 (Mixing creates overdispersion)

Let Y given λ have mean $E(Y \mid \lambda)$ and variance $\mathrm{Var}(Y \mid \lambda)$, where λ is random. Then

$$E(Y) = E[E(Y \mid \lambda)], \qquad \mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)].$$

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\mathrm{Var}(Y) = \mu + \mathrm{Var}(\lambda) > \mu$ whenever λ is nondegenerate.

Proof. [Supplement]

Step 1. The first identity is the tower property. For the second,

$$\begin{aligned}\mathrm{Var}(Y) &= E\big[E(Y^2 \mid \lambda)\big] - \big(E[E(Y \mid \lambda)]\big)^2 \\ &= E\big[\mathrm{Var}(Y \mid \lambda) + E(Y \mid \lambda)^2\big] - \big(E[E(Y \mid \lambda)]\big)^2 \\ &= E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)].\end{aligned}$$

Step 2. In the Poisson case $E(Y \mid \lambda) = \mathrm{Var}(Y \mid \lambda) = \lambda$, which gives $E(Y) = \mu$ and $\mathrm{Var}(Y) = \mu + \mathrm{Var}(\lambda)$. ■

The same mechanism (a mixture of binomials with varying π) produces extra-binomial variation; the negative binomial (Section 4.3.4) and random-effects models (Chapters 12–13) address it.

1.2.5 Connection between Poisson and multinomial distributions

Theorem 1.5 (Poisson conditioned on the total is multinomial)

Let $Y_1, \dots, Y_c$ be independent Poisson with means $\mu_1, \dots, \mu_c$. Then $n = \sum_i Y_i \sim \text{Poisson}(\sum_i \mu_i)$, and conditional on n,

$$(Y_1, \dots, Y_c) \mid \textstyle\sum_i Y_i = n \;\sim\; \text{multinomial}(n, \{\pi_i\}), \qquad \pi_i = \mu_i / \textstyle\sum_j \mu_j.$$

Proof.

Step 1. The pgf of Y_i is $E(s^{Y_i}) = e^{\mu_i(s-1)}$; the product $e^{(\sum \mu_i)(s-1)}$ is the pgf of Poisson$(\sum \mu_i)$, so n is Poisson.

Step 2. For counts $\{n_i\}$ with $\sum n_i = n$,

$$\begin{aligned}P\Big(Y_1 = n_1, \dots, Y_c = n_c \,\Big|\, \sum Y_j = n\Big) &= \frac{\prod_i e^{-\mu_i} \mu_i^{n_i}/n_i!}{e^{-\sum_j \mu_j} (\sum_j \mu_j)^n / n!} \\ &= \frac{n!}{\prod_i n_i!} \prod_i \Big(\frac{\mu_i}{\sum_j \mu_j}\Big)^{n_i},\end{aligned}$$

using $\sum_i n_i = n$ to distribute $(\sum_j \mu_j)^n$. This is (1.2). ■

Given n the Y_i are no longer Poisson (each is at most n) nor independent. Because the Poisson and multinomial likelihoods have the same kernel in $\{\pi_i\}$, analyses under the two models usually give the same estimates.

1.3 Statistical inference for categorical data

Motivation. Once a distribution is chosen its parameters are unknown. Maximum likelihood gives estimators with known large-sample behaviour, and three likelihood-based tests (Wald, likelihood ratio, score) together with the confidence intervals obtained by inverting them form the toolkit for the whole course.

1.3.1 Likelihood functions and maximum likelihood estimation

Definition (Likelihood, ML estimate, information)

For data with a chosen distribution, the *likelihood* $\ell(\beta)$ is the probability of the observed data as a function of the parameter β, and $L(\beta) = \log \ell(\beta)$ is the *log likelihood*. The *maximum likelihood (ML) estimate* $\hat\beta$ maximizes ℓ; typically it solves the *likelihood equation* $\partial L/\partial \beta = 0$. For a parameter vector $\boldsymbol\beta$ the *information matrix* has (j,k) element

$$-E\Big(\frac{\partial^2 L(\boldsymbol\beta)}{\partial\beta_j\,\partial\beta_k}\Big). \tag{1.4}$$

Under regularity conditions the asymptotic covariance matrix of $\hat{\boldsymbol\beta}$ is the inverse information matrix; standard errors (SE) are square roots of its diagonal. Greater curvature of L means smaller SEs: the log likelihood drops quickly away from $\hat\beta$.

Lemma 1.6 (Score identities)

Let $f(y;\beta)$ be a density or pmf such that differentiation under $\int dy$ (or $\sum_y$) is allowed twice. For one observation let $u(\beta) = \partial \log f(Y;\beta)/\partial\beta$. Then

$$E_\beta[u(\beta)] = 0, \qquad \operatorname{Var}_\beta[u(\beta)] = -E_\beta\Big[\frac{\partial^2 \log f(Y;\beta)}{\partial\beta^2}\Big] =: i(\beta).$$

Proof. [Supplement]

Step 1. Differentiate $\int f(y;\beta)\,dy = 1$: $0 = \int \partial_\beta f\,dy = \int (\partial_\beta \log f)\, f\,dy = E[u]$.

Step 2. Differentiate again: $0 = \int [(\partial_\beta^2 \log f) f + (\partial_\beta \log f)^2 f]\,dy$, so $E[u^2] = -E[\partial_\beta^2 \log f]$; since $E[u] = 0$ this is $\operatorname{Var}(u)$. ∎

Theorem 1.7 (Asymptotic normality of the ML estimator)

Let $Y_1,\dots,Y_n$ be i.i.d. $f(y;\beta_0)$ with β_0 interior to the parameter space, satisfying Lemma 1.6, $0 < i(\beta_0) < \infty$, and $|\partial_\beta^3 \log f(y;\beta)| \le M(y)$ near β_0 with $E\,M(Y) < \infty$. If $\hat\beta$ is a consistent root of the likelihood equation, then

$$\sqrt{n}(\hat\beta - \beta_0) \xrightarrow{d} N\big(0,\, 1/i(\beta_0)\big).$$

For a vector $\boldsymbol\beta$ the same holds with $i(\boldsymbol\beta_0)$ the per-observation information matrix: $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) \xrightarrow{d} N(\mathbf{0}, i(\boldsymbol\beta_0)^{-1})$, and in fact $\sqrt{n}(\hat{\boldsymbol\beta} - \boldsymbol\beta_0) = i(\boldsymbol\beta_0)^{-1}\, n^{-1/2} \nabla L(\boldsymbol\beta_0) + o_p(1)$.

Proof. [Supplement]

Step 1. Taylor-expand the score $L'(\beta) = \sum_i u_i(\beta)$ about β_0:

$$0 = L'(\hat\beta) = L'(\beta_0) + (\hat\beta - \beta_0)L''(\beta_0) + \tfrac{1}{2}(\hat\beta - \beta_0)^2 L'''(\beta^*),$$

with β^* between $\hat\beta$ and β_0. Rearranging,

$$\sqrt{n}(\hat\beta-\beta_0)=\frac{n^{-1/2}L'(\beta_0)}{-n^{-1}L''(\beta_0)-\tfrac{1}{2}(\hat\beta-\beta_0)\,n^{-1}L'''(\beta^*)}.$$

Step 2. Numerator: $L'(\beta_0)$ is a sum of i.i.d. terms with mean 0 and variance $i(\beta_0)$ (Lemma 1.6), so by the CLT $n^{-1/2}L'(\beta_0)\xrightarrow{d}N(0,i(\beta_0))$.

Step 3. Denominator: by the WLLN $-n^{-1}L''(\beta_0)\xrightarrow{p}i(\beta_0)$. With probability tending to one β^* lies in the neighbourhood, where $|n^{-1}L'''(\beta^*)|\le n^{-1}\sum_i M(Y_i)=O_p(1)$; multiplied by $\hat\beta-\beta_0=o_p(1)$ it vanishes. So the denominator $\xrightarrow{p}i(\beta_0)$.

Step 4. By Slutsky's theorem $\sqrt{n}(\hat\beta-\beta_0)\xrightarrow{d}N(0,i(\beta_0))/i(\beta_0)=N(0,1/i(\beta_0))$, and the representation $\sqrt{n}(\hat\beta-\beta_0)=i(\beta_0)^{-1}n^{-1/2}L'(\beta_0)+o_p(1)$ follows from the same display. The vector case is identical with the Hessian matrix in place of L'' and matrix inversion in place of division. ■

1.3.2 Likelihood function and ML estimate for binomial parameter

The part of the likelihood involving the parameter is its *kernel*; constants such as $\binom{n}{y}$ do not affect the maximizer.

Proposition 1.8 (Binomial ML estimate and information)

For y successes in n trials, the log likelihood, score and information are

$$L(\pi)=y\log\pi+(n-y)\log(1-\pi), \tag{1.5}$$

$$\frac{\partial L}{\partial\pi}=\frac{y}{\pi}-\frac{n-y}{1-\pi}=\frac{y-n\pi}{\pi(1-\pi)}, \tag{1.6}$$

$$-E\Big(\frac{\partial^2 L}{\partial\pi^2}\Big)=\frac{n}{\pi(1-\pi)}. \tag{1.7}$$

The ML estimate is $\hat\pi=y/n$, with $E(\hat\pi)=\pi$ and $\sigma(\hat\pi)=\sqrt{\pi(1-\pi)/n}$, the inverse square root of the information.

Proof.

Step 1. (1.5) is the log of the kernel $\pi^y(1-\pi)^{n-y}$; differentiation gives (1.6), which vanishes iff $\pi=y/n$. Since $\partial^2L/\partial\pi^2=-y/\pi^2-(n-y)/(1-\pi)^2<0$, L is strictly concave and y/n is the global maximizer (for $y=0$ or n, L is monotone and the maximum is at the boundary 0 or 1, again y/n).

Step 2. Taking expectations with $E(Y)=n\pi$: $E[Y/\pi^2+(n-Y)/(1-\pi)^2]=n/\pi+n/(1-\pi)=n/[\pi(1-\pi)]$.

Step 3. $E(\hat\pi)=E(Y)/n=\pi$ and $\mathrm{Var}(\hat\pi)=n\pi(1-\pi)/n^2$, matching the inverse information. ■

1.3.3 Wald–likelihood ratio–score test triad

Consider $H_0\colon\beta=\beta_0$. Write $L_0=L(\beta_0)$ and $L_1=L(\hat\beta)$ (maximized log likelihoods under H_0 and in general).

Definition (The three likelihood-based statistics)

(a) *Wald*: $z_W=(\hat\beta-\beta_0)/\mathrm{SE}$ with SE evaluated at $\hat\beta$; chi-squared form z_W^2. For a vector, $W=(\hat{\boldsymbol\beta}-\boldsymbol\beta_0)^\top[\widehat{\mathrm{Cov}}(\hat{\boldsymbol\beta})]^{-1}(\hat{\boldsymbol\beta}-\boldsymbol\beta_0)$.

(b) *Likelihood ratio (LR)*: $\Lambda = \ell_0/\ell_1 \le 1$ and $-2\log\Lambda = -2(L_0 - L_1)$.
(c) *Score* (Rao): with $u(\beta_0) = \partial L/\partial\beta|_{\beta_0}$ and $\iota(\beta_0) = -E[\partial^2 L/\partial\beta^2]|_{\beta_0}$, $z_S = u(\beta_0)/\iota(\beta_0)^{1/2}$, chi-squared form $u(\beta_0)^2/\iota(\beta_0)$; in the vector case $\mathbf{u}^\top \iota^{-1}\mathbf{u}$ at $\boldsymbol{\beta}_0$.

The Wald test uses the behaviour of L at $\hat\beta$, the score test its slope and curvature at β_0, and the LR test both values (twice the vertical distance in Figure 1).

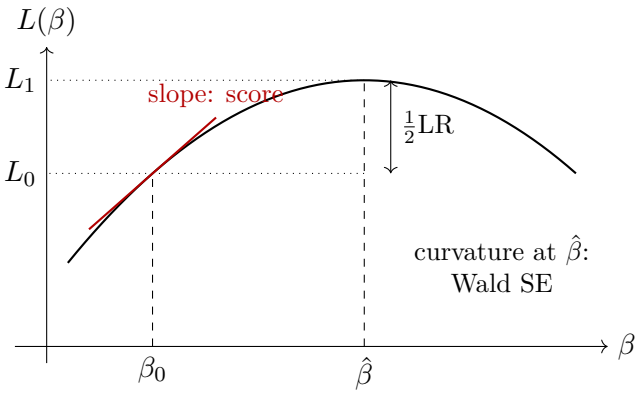


Figure 1: Information used by the three tests of $H_0\colon \beta = \beta_0$.

Lemma 1.9 (Normal quadratic forms)

(a) If $\mathbf{z} \sim N_q(\mathbf{0}, \Sigma)$ with Σ nonsingular, then $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} \sim \chi^2_q$.
(b) If $\mathbf{z} \sim N_c(\mathbf{0}, P)$ with P symmetric idempotent of rank r, then $\mathbf{z}^\top\mathbf{z} \sim \chi^2_r$.

Proof. [Supplement]

Step (a). $\mathbf{w} = \Sigma^{-1/2}\mathbf{z} \sim N_q(\mathbf{0}, I)$, so $\mathbf{z}^\top\Sigma^{-1}\mathbf{z} = \sum_{k=1}^q w_k^2$ is a sum of q squared independent standard normals.

Step (b). Write $P = Q\,\mathrm{diag}(1,\dots,1,0,\dots,0)\,Q^\top$ with Q orthogonal and r ones (eigenvalues of an idempotent matrix are 0 or 1, and their number of ones is $\operatorname{rank} P$). Then $\mathbf{w} = Q^\top\mathbf{z} \sim N(\mathbf{0}, \mathrm{diag}(1,\dots,1,0,\dots,0))$: $w_1,\dots,w_r$ i.i.d. $N(0,1)$ and $w_{r+1} = \cdots = 0$ a.s. Hence $\mathbf{z}^\top\mathbf{z} = \mathbf{w}^\top\mathbf{w} = \sum_{k\le r} w_k^2 \sim \chi^2_r$. ∎

Theorem 1.10 (Asymptotic null distributions of the triad)

Under the conditions of Theorem 1.7 and $H_0\colon \boldsymbol{\beta} = \boldsymbol{\beta}_0$ ($\boldsymbol{\beta} \in \mathbb{R}^q$), the Wald, LR and score statistics each converge in distribution to χ^2_q, and the difference between any two of them converges in probability to 0. For $q = 1$, z_W and z_S are asymptotically $N(0,1)$.

Proof. [Supplement] Write $\iota_1 = i(\boldsymbol{\beta}_0)$ for the per-observation information and $\mathbf{s}_n = n^{-1/2}\nabla L(\boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \iota_1)$ (CLT plus Lemma 1.6).

Step 1 (score). The score statistic is $\nabla L^\top (n\iota_1)^{-1}\nabla L = \mathbf{s}_n^\top \iota_1^{-1}\mathbf{s}_n \xrightarrow{d} \chi^2_q$ by the continuous mapping theorem and Lemma 1.9(a).

Step 2 (Wald). By Theorem 1.7, $\mathbf{z}_n = \sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \iota_1^{-1}\mathbf{s}_n + o_p(1)$. The Wald statistic is $\mathbf{z}_n^\top \hat\iota_1 \mathbf{z}_n$ with $\hat\iota_1 \xrightarrow{p} \iota_1$ (consistency and continuity). Hence it equals $\mathbf{s}_n^\top\iota_1^{-1}\mathbf{s}_n + o_p(1)$, the score statistic plus $o_p(1)$.

Step 3 (LR). Expand L about $\hat{\boldsymbol{\beta}}$, where $\nabla L(\hat{\boldsymbol{\beta}}) = \mathbf{0}$:

$$L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \tfrac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top \nabla^2 L(\tilde{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}}),$$

with $\tilde{\boldsymbol{\beta}}$ on the segment. Hence $-2(L_0 - L_1) = \mathbf{z}_n^\top[-n^{-1}\nabla^2 L(\tilde{\boldsymbol{\beta}})]\mathbf{z}_n$, and $-n^{-1}\nabla^2 L(\tilde{\boldsymbol{\beta}}) \xrightarrow{p} \iota_1$ (WLLN plus the third-derivative bound). So $-2(L_0 - L_1) = \mathbf{z}_n^\top \iota_1 \mathbf{z}_n + o_p(1)$, again the Wald statistic plus $o_p(1)$.

Step 4. All three are $\mathbf{s}_n^\top \iota_1^{-1} \mathbf{s}_n + o_p(1) \xrightarrow{d} \chi^2_q$. For $q = 1$, $z_S = \mathbf{s}_n/\sqrt{\iota_1} \xrightarrow{d} N(0,1)$ and $z_W = z_S + o_p(1)$. ■

More generally, for a composite H_0 Wilks' theorem gives $-2\log\Lambda \xrightarrow{d} \chi^2_{\mathrm{df}}$ with df equal to the difference in dimension of the parameter spaces under $H_0 \cup H_a$ and under H_0; Theorem 1.18 below proves this in the multinomial setting. For small to moderate n the LR test is usually more reliable than the Wald test.

1.3.4 Constructing confidence intervals

A confidence interval results from *inverting* a test: the 95% interval is the set of β_0 for which the test of $H_0\colon \beta = \beta_0$ has P-value exceeding 0.05. Let z_a be the standard normal upper-a point and $\chi^2_{\mathrm{df}}(a)$ the chi-squared upper-a point.

(a) *Wald interval*: $\{\beta_0 : |\hat\beta - \beta_0|/\mathrm{SE} < z_{\alpha/2}\}$, i.e. $\hat\beta \pm z_{\alpha/2}\mathrm{SE}$.

(b) *LR interval*: $\{\beta_0 : -2[L(\beta_0) - L(\hat\beta)] < \chi^2_1(\alpha)\}$, where $\chi^2_1(\alpha) = z^2_{\alpha/2}$.

(c) *Score interval*: $\{\beta_0 : |z_S(\beta_0)| < z_{\alpha/2}\}$.

When $\hat\beta$ is normal the log likelihood is exactly parabolic and the three coincide (as in normal linear regression). With small categorical samples L can be far from parabolic and the methods disagree markedly; such disagreement signals non-normality of $\hat\beta$. The LR interval is preferable for small to moderate n; the Wald interval is the most common because it needs only $\hat\beta$ and its SE.

1.4 Statistical inference for binomial parameters

Motivation. The binomial is the simplest categorical model, yet the three large-sample methods already behave very differently near the boundary; it is the ideal test case for comparing Wald, score, LR and exact inference.

1.4.1 Tests about a binomial parameter

Proposition 1.11 (Binomial test statistics)

For $H_0\colon \pi = \pi_0$ with $\hat\pi = y/n$:

$$z_W = \frac{\hat\pi - \pi_0}{\sqrt{\hat\pi(1-\hat\pi)/n}}, \tag{1.8}$$

$$z_S = \frac{\hat\pi - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}, \tag{1.9}$$

$$-2(L_0 - L_1) = 2\sum \text{observed} \log\frac{\text{observed}}{\text{fitted}} = 2\Big[y\log\frac{y}{n\pi_0} + (n-y)\log\frac{n-y}{n-n\pi_0}\Big]. \tag{1.10}$$

Each is asymptotically $N(0,1)$ or χ^2_1 under H_0.

Proof.

Step 1. The Wald statistic uses $\mathrm{SE} = \sqrt{\hat\pi(1-\hat\pi)/n}$ (Proposition 1.8 with $\pi = \hat\pi$).

Step 2. Score: by (1.6)–(1.7),

$$\frac{u(\pi_0)}{\iota(\pi_0)^{1/2}} = \frac{y - n\pi_0}{\pi_0(1-\pi_0)} \cdot \sqrt{\frac{\pi_0(1-\pi_0)}{n}} = \frac{y - n\pi_0}{\sqrt{n\pi_0(1-\pi_0)}} = \frac{\hat\pi - \pi_0}{\sqrt{\pi_0(1-\pi_0)/n}}.$$

Step 3. LR: $L_0 = y\log\pi_0 + (n-y)\log(1-\pi_0)$ and $L_1 = y\log\hat\pi + (n-y)\log(1-\hat\pi)$, so $-2(L_0 - L_1) = 2[y\log(\hat\pi/\pi_0) + (n-y)\log((1-\hat\pi)/(1-\pi_0))]$, which is (1.10) after multiplying numerator and denominator by n. The asymptotics are Theorem 1.10 with $q = 1$. ■

The score statistic uses the actual null SE rather than an estimate, and its null distribution is closer to $N(0,1)$. Form (1.10) (observed vs. fitted counts) recurs for Poisson and multinomial tests.

1.4.2 Confidence intervals for a binomial parameter

Inverting the Wald test gives the Laplace interval

$$\hat\pi \pm z_{\alpha/2}\sqrt{\hat\pi(1-\hat\pi)/n}, \tag{1.11}$$

which performs poorly unless n is very large: its actual coverage is usually below nominal, much below for π near 0 or 1. (Adding $z_{\alpha/2}^2/2 \approx 2$ observations of each type before applying (1.11) improves it greatly.)

Proposition 1.12 (Wilson score interval)

The score interval $\{\pi_0 : |\hat\pi - \pi_0| < z\sqrt{\pi_0(1-\pi_0)/n}\}$, $z = z_{\alpha/2}$, has endpoints

$$\hat\pi\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big) \;\pm\; z\sqrt{\frac{1}{n+z^2}\Big[\hat\pi(1-\hat\pi)\Big(\frac{n}{n+z^2}\Big) + \frac{1}{2}\cdot\frac{1}{2}\Big(\frac{z^2}{n+z^2}\Big)\Big]}.$$

Its midpoint is $(y + z^2/2)/(n+z^2)$, the sample proportion after adding $z^2/2$ successes and $z^2/2$ failures.

Proof. [Supplement]

Step 1. The endpoints solve $(\hat\pi - \pi_0)^2 = z^2\pi_0(1-\pi_0)/n$, i.e.

$$\Big(1 + \frac{z^2}{n}\Big)\pi_0^2 - \Big(2\hat\pi + \frac{z^2}{n}\Big)\pi_0 + \hat\pi^2 = 0.$$

The quadratic is negative between its roots, which is exactly the set where $|z_S| < z$.

Step 2. Multiplying by n and using the quadratic formula, the roots are

$$\pi_0 = \frac{n\hat\pi + z^2/2 \pm z\sqrt{n\hat\pi(1-\hat\pi) + z^2/4}}{n+z^2}.$$

Step 3. The centre is $(n\hat\pi + z^2/2)/(n+z^2) = \hat\pi\,\frac{n}{n+z^2} + \frac{1}{2}\,\frac{z^2}{n+z^2}$. For the half-width, $z\sqrt{n\hat\pi(1-\hat\pi) + z^2/4}/(n+z^2) = z\sqrt{[\hat\pi(1-\hat\pi)\frac{n}{n+z^2} + \frac{1}{4}\frac{z^2}{n+z^2}]/(n+z^2)}$, which is the stated form. ■

The squared coefficient of z is a weighted average of the variance of a sample proportion at $\pi = \hat\pi$ and at $\pi = 1/2$, with sample size $n + z^2$. The Wilson interval performs much better than the Wald interval. The *LR interval* is the set of π_0 for which $-2[L(\pi_0) - L(\hat\pi)] < \chi_1^2(\alpha)$.

1.4.3 Proportion of vegetarians example

Example 1.13 (Proportion of vegetarians)

Of $n = 25$ students in a statistics class, $y = 0$ said they were vegetarian, so $\hat\pi = 0$.
(a) Wald: $0 \pm 1.96\sqrt{0 \cdot 1/25} = (0, 0)$, a useless answer at the boundary.
(b) Score: $(0.0,\ 0.133)$. E.g. for $\pi_0 = 0.5$, $z_S = (0 - 0.5)/\sqrt{0.5 \cdot 0.5/25} = -5.0$ (rejected), while for $\pi_0 = 0.10$, $z_S = (0 - 0.10)/\sqrt{0.10 \cdot 0.90/25} = -1.67$ (inside the interval).
(c) LR: $L(\pi) = 25 \log(1 - \pi)$ and $L(\hat\pi) = 0$, so the interval is $\{\pi_0 : -50 \log(1 - \pi_0) \le 3.84\}$; the upper bound is $1 - \exp(-3.84/50) = 0.074$, giving $(0.0,\ 0.074)$.
The three large-sample methods differ greatly: for π near 0 and small n, $\hat\pi$ is highly skewed.

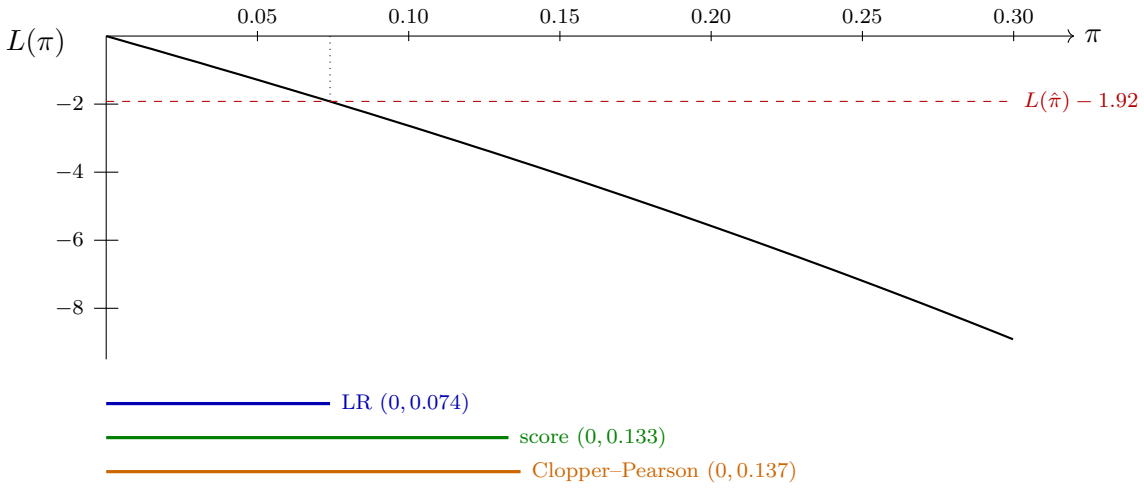


Figure 2: Binomial log likelihood for $y = 0$, $n = 25$; the LR interval is where L is within 3.84/2 of its maximum.

1.5 Statistical inference for multinomial parameters

Motivation. With $c > 2$ categories the natural questions are goodness of fit to a specified distribution, or to a distribution with a few unknown parameters; this leads to Pearson's X^2 and the likelihood-ratio G^2, the two workhorse statistics of categorical data analysis.

1.5.1 Estimation of multinomial parameters

Proposition 1.14 (Multinomial ML estimates)

The multinomial kernel

$$\prod_j \pi_j^{n_j}, \qquad \pi_j \ge 0, \quad \sum_j \pi_j = 1, \tag{1.12}$$

is maximized uniquely by the sample proportions $\hat\pi_j = p_j = n_j/n$.

Proof.

Step 1 (likelihood equations). Treat $L(\boldsymbol\pi) = \sum_j n_j \log \pi_j$ as a function of $\pi_1, \dots, \pi_{c-1}$ with $\pi_c = 1 - \sum_{j<c} \pi_j$, so $\partial \pi_c / \partial \pi_j = -1$. Then

$$\frac{\partial L}{\partial \pi_j} = \frac{n_j}{\pi_j} - \frac{n_c}{\pi_c} = 0, \qquad j = 1, \dots, c-1,$$

so $\hat\pi_j / \hat\pi_c = n_j / n_c$ for all j. Summing, $1/\hat\pi_c = n/n_c$, hence $\hat\pi_c = n_c/n$ and $\hat\pi_j = n_j/n$.

Step 2 (global maximum). [Supplement] For any admissible $\boldsymbol{\pi}$, with $p_j = n_j/n$ and the sum over j with $n_j > 0$, Jensen's inequality for the concave log gives

$$L(\boldsymbol{\pi}) - L(\mathbf{p}) = n \sum_j p_j \log \frac{\pi_j}{p_j} \le n \log \sum_j p_j \frac{\pi_j}{p_j} \le n \log 1 = 0,$$

with equality iff $\pi_j = p_j$ for all j. ■

1.5.2 Pearson statistic for testing a specified multinomial

Karl Pearson's 1900 test (originally for a Monte Carlo roulette wheel) was among the first inferential methods. For $H_0\colon \pi_j = \pi_{j0}$, $j = 1, \dots, c$, the *expected frequencies* are $\mu_j = n\pi_{j0}$ and

$$X^2 = \sum_j \frac{(n_j - \mu_j)^2}{\mu_j}. \tag{1.13}$$

The exact P-value is the sum of null multinomial probabilities of all count vectors with $X^2 \ge X_o^2$ (observed value); for large n, $X^2 \approx \chi^2_{c-1}$ (Theorem 1.16).

1.5.3 Example: testing Mendel's theories

Example 1.15 (Mendel's peas)

Crossing pure yellow with pure green pea strains, Mendel predicted 75% yellow and 25% green second-generation seeds. One experiment gave $n = 8023$ seeds: $n_1 = 6022$ yellow, $n_2 = 2001$ green. Under $H_0\colon \pi_1 = 0.75$, $\mu_1 = 6017.25$ and $\mu_2 = 2005.75$, and $X^2 = 0.015$ (df $= 1$), $P = 0.90$: no contradiction.

Fisher (1936) combined Mendel's experiments using the reproductive property of independent chi-squared statistics ($\sum X_i^2 \sim \chi^2_{\sum \nu_i}$) and obtained $X^2 = 42$ with df $= 84$. Since χ^2_{84} has mean 84 and standard deviation $\sqrt{2 \cdot 84} = 13.0$, $P(\chi^2_{84} \le 42)$ is tiny: the right-tail probability is 0.99996. The fit was *too good*, suggesting the data had been adjusted to agree with expectations.

Goodness-of-fit tests can thus reveal fits better than chance would allow.

1.5.4 Chi-squared theoretical justification

Theorem 1.16 (Asymptotic null distribution of Pearson's X^2)

Under $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$ with all $\pi_{j0} > 0$, $X^2 \xrightarrow{d} \chi^2_{c-1}$ as $n \to \infty$. Moreover X^2 is the score statistic for H_0.

Proof. [Supplement] (Tools: multivariate CLT, Lemma 1.9.) Let $\hat{\boldsymbol{\pi}} = (p_1, \dots, p_{c-1})^\top$, $\boldsymbol{\pi}_0 = (\pi_{10}, \dots, \pi_{c-1,0})^\top$ and $\Sigma_0 = \operatorname{diag}(\boldsymbol{\pi}_0) - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top$.

Step 1 (CLT). $\hat{\boldsymbol{\pi}}$ is the mean of the i.i.d. vectors $(y_{i1}, \dots, y_{i,c-1})$, whose covariance is Σ_0 (Proposition 1.2). Hence $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}_0) \xrightarrow{d} N_{c-1}(\mathbf{0}, \Sigma_0)$.

Step 2 (inverse). $\Sigma_0^{-1} = \operatorname{diag}(1/\pi_{j0}) + \mathbf{1}\mathbf{1}^\top/\pi_{c0}$. Indeed, using $\boldsymbol{\pi}_0^\top \mathbf{1} = 1 - \pi_{c0}$,

$$\Sigma_0 \Big[\operatorname{diag}(1/\pi_{j0}) + \tfrac{\mathbf{1}\mathbf{1}^\top}{\pi_{c0}}\Big] = I + \frac{\boldsymbol{\pi}_0 \mathbf{1}^\top}{\pi_{c0}} - \boldsymbol{\pi}_0 \mathbf{1}^\top - \frac{(1 - \pi_{c0})\boldsymbol{\pi}_0 \mathbf{1}^\top}{\pi_{c0}}$$

$$= I + \boldsymbol{\pi}_0 \mathbf{1}^\top \frac{1 - \pi_{c0} - (1 - \pi_{c0})}{\pi_{c0}} = I.$$

Step 3 (algebra). Put $d_j = p_j - \pi_{j0}$, $j = 1, \dots, c$; then $d_c = -\sum_{j<c} d_j$. So

$$n\,\mathbf{d}^\top \Sigma_0^{-1} \mathbf{d} = n\Big[\sum_{j<c} \frac{d_j^2}{\pi_{j0}} + \frac{\big(\sum_{j<c} d_j\big)^2}{\pi_{c0}}\Big] = n \sum_{j=1}^{c} \frac{d_j^2}{\pi_{j0}} = \sum_{j=1}^{c} \frac{(n_j - n\pi_{j0})^2}{n\pi_{j0}} = X^2. \tag{1.14}$$

Step 4 (limit). By the continuous mapping theorem and Lemma 1.9(a), $X^2 \xrightarrow{d} \mathbf{z}^\top \Sigma_0^{-1} \mathbf{z} \sim \chi^2_{c-1}$ with $\mathbf{z} \sim N(\mathbf{0}, \Sigma_0)$.

Step 5 (score test). From Step 1 of Proposition 1.14, the score has components $u_j = n_j/\pi_j - n_c/\pi_c = n(d_j/\pi_j - d_c/\pi_c)$ at $\boldsymbol{\pi}_0$, i.e. $\mathbf{u} = n\Sigma_0^{-1}\mathbf{d}$ (by Step 2, $(\Sigma_0^{-1}\mathbf{d})_j = d_j/\pi_{j0} + \sum_{k<c} d_k/\pi_{c0} = d_j/\pi_{j0} - d_c/\pi_{c0}$). The information is $n\Sigma_0^{-1}$ (the inverse of $\mathrm{Cov}(\hat{\boldsymbol{\pi}}) = \Sigma_0/n$, or directly by differentiating $\mathbf{u}$). So the score statistic is $\mathbf{u}^\top (n\Sigma_0^{-1})^{-1} \mathbf{u} = n\mathbf{d}^\top \Sigma_0^{-1} \mathbf{d} = X^2$. ∎

Fisher's heuristic

If $n_1, \dots, n_c$ were independent Poisson with large means μ_j, the $z_j = (n_j - \mu_j)/\sqrt{\mu_j}$ would be approximately $N(0,1)$ and $\sum z_j^2 = X^2 \approx \chi^2_c$. Conditioning on $\sum n_j = n$ (Theorem 1.5) imposes one linear constraint $\sum(n_j - \mu_j) = 0$ and loses one degree of freedom. For $c = 2$, $X^2 = z_S^2$ with z_S from (1.9): for Mendel's data $\hat{\pi}_1 = 6022/8023$, $z_S = 0.123$ and $X^2 = 0.123^2 = 0.015$.

1.5.5 Likelihood-ratio chi-squared

Under H_0 the kernel (1.12) is maximized at $\boldsymbol{\pi}_0$, in general at $p_j = n_j/n$. Hence $\Lambda = \prod_j (\pi_{j0}/p_j)^{n_j}$ and

$$G^2 = -2 \log \Lambda = 2 \sum_j n_j \log \frac{n_j}{n\pi_{j0}}, \tag{1.15}$$

of the form (1.10). The general parameter space has dimension $c - 1$, the null space dimension 0, so $G^2 \xrightarrow{d} \chi^2_{c-1}$ under H_0 (Theorem 1.10 with $q = c - 1$).

Theorem 1.17 (Asymptotic equivalence of X^2 and G^2)

Let μ_j be (possibly estimated) expected frequencies with $\sum_j \mu_j = n$ and $(n_j - \mu_j)/\mu_j = O_p(n^{-1/2})$ for each j, as holds under H_0. Then $G^2 - X^2 \xrightarrow{p} 0$; in particular $G^2 \xrightarrow{d} \chi^2_{c-1}$ under $H_0 \colon \boldsymbol{\pi} = \boldsymbol{\pi}_0$.

Proof. [Supplement]

Step 1. Write $n_j = \mu_j(1 + x_j)$ with $x_j = O_p(n^{-1/2})$. Since $(1 + x)\log(1 + x) = x + x^2/2 + R(x)$ with $|R(x)| \le C|x|^3$ for $|x| \le 1/2$,

$$G^2 = 2 \sum_j \mu_j (1 + x_j) \log(1 + x_j) = 2 \sum_j \mu_j x_j + \sum_j \mu_j x_j^2 + 2 \sum_j \mu_j R(x_j).$$

Step 2. $\sum_j \mu_j x_j = \sum_j (n_j - \mu_j) = 0$, and $\sum_j \mu_j x_j^2 = \sum_j (n_j - \mu_j)^2/\mu_j = X^2$.

Step 3. $\mu_j = O(n)$ and $R(x_j) = O_p(n^{-3/2})$, so the remainder is $O_p(n^{-1/2}) \xrightarrow{p} 0$. Hence $G^2 = X^2 + o_p(1)$, and Slutsky with Theorem 1.16 gives the limit. ∎

When H_0 is false both statistics grow proportionally to n but need not be close. For fixed c, X^2 usually approaches chi-squared faster than G^2; the approximation is often poor for G^2 when

$n/c < 5$, while for X^2 it can be adequate with n/c as small as 1 if the table does not mix very small and moderately large expected frequencies. Exact distributions can also be computed from the multinomial.

1.5.6 Testing with estimated expected frequencies

Often $\pi_{j0} = \pi_j(\boldsymbol{\theta})$ depends on unknown $\boldsymbol{\theta}$ with $\dim \boldsymbol{\theta} = p$; the ML estimate $\hat{\boldsymbol{\theta}}$ gives $\hat{\mu}_j = n\pi_j(\hat{\boldsymbol{\theta}})$, and $X^2 = \sum (n_j - \hat{\mu}_j)^2/\hat{\mu}_j$. Pearson did not realize that estimating $\boldsymbol{\theta}$ changes the degrees of freedom.

Theorem 1.18 (Degrees of freedom with estimated parameters)

Assume $\boldsymbol{\pi}(\boldsymbol{\theta})$ has continuous second derivatives, $\pi_j(\boldsymbol{\theta}_0) > 0$, the $c \times p$ Jacobian $\partial \boldsymbol{\pi}/\partial \boldsymbol{\theta}^\top$ has rank p at $\boldsymbol{\theta}_0$, and $\hat{\boldsymbol{\theta}}$ is a consistent ML estimate. If $H_0\colon \boldsymbol{\pi} = \boldsymbol{\pi}(\boldsymbol{\theta})$ holds, then

$$X^2 = \sum_j \frac{(n_j - \hat{\mu}_j)^2}{\hat{\mu}_j} \xrightarrow{d} \chi^2_{c-1-p}, \qquad \text{and } G^2 - X^2 \xrightarrow{p} 0.$$

Proof. [Supplement] (Tools: multivariate CLT, Theorem 1.7, Slutsky, Lemma 1.9(b).) Let $\boldsymbol{\pi}_0 = \boldsymbol{\pi}(\boldsymbol{\theta}_0)$ (all c components), $D = \operatorname{diag}(\boldsymbol{\pi}_0)$, $\mathbf{s} = (\sqrt{\pi_{10}}, \dots, \sqrt{\pi_{c0}})^\top$ (a unit vector), and $A = D^{-1/2}\, \partial \boldsymbol{\pi}/\partial \boldsymbol{\theta}^\top$ at $\boldsymbol{\theta}_0$ ($c \times p$, rank p).

Step 1. Since $\sum_j \pi_j(\boldsymbol{\theta}) = 1$, $\mathbf{s}^\top A = \sum_j \partial \pi_j/\partial \boldsymbol{\theta}^\top = \mathbf{0}$.

Step 2. With $\mathbf{p} = (p_1, \dots, p_c)^\top$, $\operatorname{Cov}(\sqrt{n}\,\mathbf{p}) = D - \boldsymbol{\pi}_0 \boldsymbol{\pi}_0^\top$, so by the CLT

$$\mathbf{z}_n := \sqrt{n}\, D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0},\, I - \mathbf{s}\mathbf{s}^\top).$$

Step 3. The log likelihood is $L(\boldsymbol{\theta}) = n \sum_j p_j \log \pi_j(\boldsymbol{\theta})$, with score at $\boldsymbol{\theta}_0$

$$\nabla L = n \sum_j p_j \frac{\partial \pi_j/\partial \boldsymbol{\theta}}{\pi_j} = n\, A^\top D^{-1/2} \mathbf{p} = n\, A^\top D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}_0) = \sqrt{n}\, A^\top \mathbf{z}_n,$$

using $A^\top D^{-1/2} \boldsymbol{\pi}_0 = A^\top \mathbf{s} = \mathbf{0}$. The per-observation information is $\sum_j (\partial \pi_j/\partial \boldsymbol{\theta})(\partial \pi_j/\partial \boldsymbol{\theta})^\top/\pi_j = A^\top A$ (nonsingular). By Theorem 1.7, $\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) = (A^\top A)^{-1} A^\top \mathbf{z}_n + o_p(1)$.

Step 4. By the delta method (a first-order Taylor expansion of $\boldsymbol{\pi}$), $\sqrt{n}\, D^{-1/2}(\boldsymbol{\pi}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\pi}_0) = A\sqrt{n}(\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0) + o_p(1)$. Therefore, with $H = A(A^\top A)^{-1} A^\top$,

$$\mathbf{e}_n := \sqrt{n}\, D^{-1/2}(\mathbf{p} - \boldsymbol{\pi}(\hat{\boldsymbol{\theta}})) = (I - H)\mathbf{z}_n + o_p(1) \xrightarrow{d} (I - H)\mathbf{z}.$$

Step 5. $X^2 = \sum_j n(p_j - \pi_j(\hat{\boldsymbol{\theta}}))^2/\pi_j(\hat{\boldsymbol{\theta}})$ differs from $\mathbf{e}_n^\top \mathbf{e}_n$ only by replacing $\pi_j(\hat{\boldsymbol{\theta}})$ by π_{j0} in the denominators, a factor $\xrightarrow{p} 1$; so $X^2 = \mathbf{e}_n^\top \mathbf{e}_n + o_p(1)$.

Step 6. The limit $(I - H)\mathbf{z}$ is normal with covariance $(I - H)(I - \mathbf{s}\mathbf{s}^\top)(I - H) = I - H - \mathbf{s}\mathbf{s}^\top$, because $H\mathbf{s} = A(A^\top A)^{-1} A^\top \mathbf{s} = \mathbf{0}$. Here H and $\mathbf{s}\mathbf{s}^\top$ are orthogonal projections onto orthogonal subspaces of dimensions p and 1, so $P = I - H - \mathbf{s}\mathbf{s}^\top$ is a symmetric idempotent matrix of rank $c - p - 1$. By Lemma 1.9(b), $X^2 \xrightarrow{d} \chi^2_{c-1-p}$.

Step 7. Step 4 shows $(n_j - \hat{\mu}_j)/\hat{\mu}_j = O_p(n^{-1/2})$, and $\sum_j \hat{\mu}_j = n$; so Theorem 1.17 gives $G^2 - X^2 \xrightarrow{p} 0$. ■

Example 1.19 (Pneumonia infections in calves)

156 dairy calves were classified by primary pneumonia infection within 60 days of birth and, for infected calves, secondary infection within two weeks after recovery. "No primary, yes secondary" is impossible (a *structural zero*). Estimated expected frequencies are in parentheses.

	Secondary infection	
Primary infection	Yes	No
Yes	30 (38.1)	63 (39.0)
No	–	63 (78.9)

The hypothesis that $P(\text{primary}) = \pi$ equals $P(\text{secondary} \mid \text{primary})$ gives trinomial probabilities $(\pi^2,\ \pi(1-\pi),\ 1-\pi)$ for (yes,yes), (yes,no), (no,no). The kernel is

$$(\pi^2)^{n_{11}}[\pi(1-\pi)]^{n_{12}}(1-\pi)^{n_{22}} = \pi^{2n_{11}+n_{12}}(1-\pi)^{n_{12}+n_{22}},$$

so the likelihood equation $\frac{2n_{11}+n_{12}}{\pi} - \frac{n_{12}+n_{22}}{1-\pi} = 0$ gives

$$\hat{\pi} = \frac{2n_{11}+n_{12}}{2n_{11}+2n_{12}+n_{22}} = \frac{123}{249} = 0.494.$$

Then $\hat{\mu}_{11} = n\hat{\pi}^2 = 38.1$, $\hat{\mu}_{12} = n(\hat{\pi} - \hat{\pi}^2) = 39.0$, $\hat{\mu}_{22} = n(1-\hat{\pi}) = 78.9$, and $X^2 = 19.7$ with $\text{df} = 3-1-1 = 1$ ($P < 0.00001$). Far more calves had a primary but no secondary infection than predicted: the primary infection seems to have an immunizing effect.

Chapter 2. Describing Contingency Tables

2.1 Probability structure for contingency tables

Motivation. The relationship between two categorical variables is entirely contained in their joint distribution, from which the marginal and conditional distributions follow. How the data were sampled determines which of these distributions the observed table can estimate.

2.1.1 Contingency tables and their distributions

Let X have I categories and Y have J categories. A rectangular table with I rows and J columns whose IJ cells hold frequency counts is a *contingency table* (Karl Pearson, 1904), or $I \times J$ cross-classification table.

Definition (Joint, marginal and conditional distributions)

(a) The *joint distribution* is $\{\pi_{ij}\}$, $\pi_{ij} = P(X = i, Y = j)$.
(b) The *marginal distributions* are $\pi_{i+} = \sum_j \pi_{ij}$ and $\pi_{+j} = \sum_i \pi_{ij}$, with $\sum_i \pi_{i+} = \sum_j \pi_{+j} = \sum_i \sum_j \pi_{ij} = 1$ (a "+" denotes summation over that index).
(c) When Y is a response, the *conditional distribution* of Y in row i is $\pi_{j|i} = P(Y = j \mid X = i) = \pi_{ij}/\pi_{i+}$, with $\sum_j \pi_{j|i} = 1$.

If X is fixed by design (not random), only the conditional distributions are meaningful. A principal aim of many studies is to compare the conditional distributions of Y across levels of X.

Example 2.1 (Aspirin and heart attacks)

The Physicians' Health Study was a five-year randomized, blind trial of whether regular aspirin intake reduces cardiovascular mortality. Every other day physicians took aspirin or a placebo.

	Myocardial infarction			
Group	Fatal	Nonfatal	No attack	Total
Placebo	18	171	10,845	11,034
Aspirin	5	99	10,933	11,037

Of 11,034 placebo takers 18 had fatal attacks, against 5 of 11,037 aspirin takers.

2.1.2 Sensitivity and specificity

For a diagnostic test let X = true disease status and Y = diagnosis (positive/negative).

Definition (Sensitivity, specificity)

Sensitivity $= P(Y = \text{positive} \mid X = \text{diseased}) = \pi_{1|1}$; *specificity* $= P(Y = \text{negative} \mid X = \text{not diseased}) = \pi_{2|2}$.

For mammography combined with clinical breast examination, a literature survey estimated sensitivity 0.82 and specificity 0.99: 82% of women with breast cancer, and 99% of those without, are diagnosed correctly.

2.1.3 Independence of categorical variables

Definition (Independence)

X and Y are *independent* if $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i, j.

Proposition 2.2 (Independence as homogeneity)

If all $\pi_{i+} > 0$, then X and Y are independent if and only if $\pi_{j|1} = \pi_{j|2} = \cdots = \pi_{j|I}$ for every j; the common value is then π_{+j}.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \pi_{+j}$ for every i.

Step 2. Conversely, if $\pi_{j|i} = c_j$ for all i, then $\pi_{+j} = \sum_i \pi_{i+}\pi_{j|i} = c_j \sum_i \pi_{i+} = c_j$, so $\pi_{ij} = \pi_{i+}\pi_{j|i} = \pi_{i+}\pi_{+j}$. ■

When Y is a response this *homogeneity of conditional distributions* is the more natural definition. Sample analogues: $p_{ij} = n_{ij}/n$, $p_{j|i} = n_{ij}/n_{i+}$, $n = \sum_i \sum_j n_{ij}$; estimates may also be written $\hat{\pi}_{ij}$.

2.1.4 Poisson, binomial, and multinomial sampling

(a) *Poisson sampling*: the IJ counts Y_{ij} are independent Poisson(μ_{ij}), with joint pmf $\prod_i \prod_j e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!$; nothing is fixed.

(b) *Multinomial sampling*: n is fixed and $\{n_{ij}\}$ is multinomial over the IJ cells, $\frac{n!}{\prod\prod n_{ij}!} \prod\prod \pi_{ij}^{n_{ij}}$.

(c) *Independent (product) multinomial sampling*: the row totals $n_i = n_{i+}$ are fixed and each row is an independent multinomial sample with probabilities $\{\pi_{j|i}\}$:

$$\prod_i \Big[\frac{n_i!}{\prod_j n_{ij}!} \prod_j \pi_{j|i}^{n_{ij}} \Big]. \tag{2.1}$$

(d) When both margins are fixed the distribution is hypergeometric (Section 3.5).

Theorem 2.3 (Conditioning on row totals gives product multinomial)

If $\{n_{ij}\}$ arise from independent Poisson sampling with means $\{\mu_{ij}\}$, or from multinomial sampling with probabilities $\pi_{ij} = \mu_{ij}/n$, then conditional on the row totals $\{n_{i+}\}$ the rows are independent multinomial samples with probabilities $\pi_{j|i} = \mu_{ij}/\mu_{i+}$.

Proof. [Supplement]

Step 1 (Poisson). The rows are independent sets of independent Poisson counts. By Theorem 1.5 applied to row i, given n_{i+} the counts $(n_{i1}, \ldots, n_{iJ})$ are multinomial$(n_{i+}, \{\mu_{ij}/\mu_{i+}\})$. Since row i's counts and total depend only on row i's Poisson variables, conditioning on all row totals keeps the rows independent.

Step 2 (multinomial). The row totals $(n_{1+}, \ldots, n_{I+})$ are multinomial$(n, \{\pi_{i+}\})$ (collapsing categories of a multinomial). Hence for tables with these row totals,

$$P(\{n_{ij}\} \mid \{n_{i+}\}) = \frac{\frac{n!}{\prod_i \prod_j n_{ij}!} \prod_i \prod_j \pi_{ij}^{n_{ij}}}{\frac{n!}{\prod_i n_{i+}!} \prod_i \pi_{i+}^{n_{i+}}} = \prod_i \Big[\frac{n_{i+}!}{\prod_j n_{ij}!} \prod_j \Big(\frac{\pi_{ij}}{\pi_{i+}} \Big)^{n_{ij}} \Big],$$

using $\pi_{i+}^{n_{i+}} = \prod_j \pi_{i+}^{n_{ij}}$. This is (2.1) with $\pi_{j|i} = \pi_{ij}/\pi_{i+} = \mu_{ij}/\mu_{i+}$. ■

So when X is explanatory it is sensible to analyse conditional on the row totals, as if they formed separate independent samples, even when the design did not fix them.

2.1.5 Seat belt example

Example 2.4 (Seat belts and crash outcome)

A study of seat-belt use (yes/no) by crash outcome (fatal/nonfatal) on the Massachusetts Turnpike illustrates the sampling schemes:
(a) recording all accidents in the coming year: n random, the four counts independent Poisson;
(b) sampling 200 police records from last year: $n = 200$ fixed, multinomial over four cells;
(c) sampling 100 fatal and 100 nonfatal records: column totals fixed, each column an independent binomial;
(d) the (unethical) experiment randomizing 100 of 200 subjects to wear belts and forcing crashes: row totals fixed, independent binomial rows.

2.1.6 Types of studies

Example 2.5 (Smoking and lung cancer: a case–control study)

Doll and Hill queried 709 lung cancer patients in 20 London hospitals about smoking, and for each a matched noncancer control (same hospital, gender, 5-year age group).

	Lung cancer	
Smoker	Cases	Controls
Yes	688	650
No	21	59
Total	709	709

The design fixes the lung cancer margin; we can estimate $P(\text{smoker} \mid \text{case}) = 688/709 = 0.970$ and $P(\text{smoker} \mid \text{control}) = 650/709 = 0.917$, but not $P(\text{cancer} \mid \text{smoking})$ without knowing the population prevalence (Bayes' theorem).

Definition (Study designs)

(a) *Retrospective* (case–control): samples are taken at fixed levels of the outcome Y and the explanatory X is looked up in the past; each column is a multinomial sample on X.
(b) *Prospective*: groups defined by X are followed into the future; *clinical trials* randomize subjects to X, *cohort studies* let subjects choose. Each row is a multinomial sample on Y.
(c) *Cross-sectional*: subjects are sampled and classified on both variables at once; the IJ counts are one multinomial sample.

Case–control, cohort and cross-sectional studies are *observational*; a clinical trial is *experimental*, and randomization balances the groups on other variables. Observational studies are common but more prone to bias.

2.2 Comparing two proportions

Motivation. With two groups and a binary response the association is a comparison of two success probabilities π_1, π_2. Three summaries are used—difference, ratio, and odds ratio—and they behave very differently under change of sampling design.

In a 2×2 table write $\pi_i = \pi_{1|i}$ for the success probability in row i, so $\pi_{2|i} = 1 - \pi_i$.

2.2.1 Difference of proportions

The *difference of proportions* $\pi_1 - \pi_2 \in [-1, 1]$ is the same whether we compare successes or failures, since $(1-\pi_1) - (1-\pi_2) = -(\pi_1 - \pi_2)$. It is 0 iff the rows have identical conditional distributions, i.e. iff Y is independent of the row classification. The difference comparing columns is in general different.

2.2.2 Relative risk

A difference of fixed size may matter more near 0 or 1: $0.010 - 0.001$ and $0.410 - 0.401$ are both 0.009, but the ratios are 10.0 and 1.02.

Definition (Relative risk)

The *relative risk* is $\pi_1/\pi_2 \in [0, \infty)$; it equals 1 under independence. Comparing the second response category gives a different value $(1-\pi_1)/(1-\pi_2)$.

2.2.3 Odds ratio

Definition (Odds, odds ratio)

For success probability π the *odds* are $\Omega = \pi/(1-\pi)$, inverted by $\pi = \Omega/(\Omega+1)$. With $\Omega_i = \pi_i/(1-\pi_i) = \pi_{i1}/\pi_{i2}$, the *odds ratio* is

$$\theta = \frac{\Omega_1}{\Omega_2} = \frac{\pi_1/(1-\pi_1)}{\pi_2/(1-\pi_2)} = \frac{\pi_{11}\pi_{22}}{\pi_{12}\pi_{21}}, \tag{2.2}$$

also called the *cross-product ratio* (Yule). The sample odds ratio is $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$.

E.g. $\pi = 0.75$ gives $\Omega = 3$: a success is three times as likely as a failure; $\Omega = 1/3$ gives $\pi = 0.25$.

2.2.4 Properties of the odds ratio

Proposition 2.6 (Properties of the odds ratio)

Assume all $\pi_{ij} > 0$.
(a) $\theta = 1$ iff $\Omega_1 = \Omega_2$ iff $\pi_1 = \pi_2$ (independence); $\theta > 1$ iff $\pi_1 > \pi_2$.
(b) Reversing the order of the rows (or of the columns) replaces θ by $1/\theta$, i.e. $\log\theta$ by $-\log\theta$; transposing the table leaves θ unchanged.
(c) θ is the same whichever conditional direction is used:

$$\theta = \frac{P(Y=1 \mid X=1)/P(Y=2 \mid X=1)}{P(Y=1 \mid X=2)/P(Y=2 \mid X=2)} = \frac{P(X=1 \mid Y=1)/P(X=2 \mid Y=1)}{P(X=1 \mid Y=2)/P(X=2 \mid Y=2)}. \tag{2.3}$$

(d) $\hat{\theta}$ is unchanged when both counts in a row, or both counts in a column, are multiplied by a positive constant.

Proof.

Step (a). $x \mapsto x/(1-x)$ is strictly increasing on $(0,1)$, so $\Omega_1 \gtreqless \Omega_2$ iff $\pi_1 \gtreqless \pi_2$.

Step (b). Swapping rows maps $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$ to $\pi_{21}\pi_{12}/(\pi_{22}\pi_{11}) = 1/\theta$; similarly for columns. Transposing swaps π_{12} and π_{21}, which leaves the cross-product ratio unchanged.

Step (c). The first ratio is $(\pi_{11}/\pi_{12})/(\pi_{21}/\pi_{22})$ after cancelling π_{1+} and π_{2+}; the second is $(\pi_{11}/\pi_{21})/(\pi_{12}/\pi_{22})$ after cancelling π_{+1}, π_{+2}. Both equal $\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})$.

Step (d). Multiplying row 1 by c multiplies numerator $n_{11}n_{22}$ and denominator $n_{12}n_{21}$ each by c; columns likewise. ■

Thus $\theta = 4$ means the odds (not the probability) in row 1 are four times those in row 2, and $\log 4 = 1.39$ and $\log 0.25 = -1.39$ represent equally strong associations. By (c) and (d), θ is equally valid for prospective, retrospective and cross-sectional designs: $\hat{\theta}$ estimates the same parameter whether a flu–vaccination case–control study samples $100 + 100$ or $40 + 160$ subjects. The sample difference and relative risk are invariant to scaling within rows, but not within columns or under transposition.

2.2.5 Aspirin and heart attacks revisited

Example 2.7 (Three measures for the aspirin study)

Combining fatal and nonfatal attacks in Example 2.1: placebo 189/11,034 = 0.0171, aspirin 104/11,037 = 0.0094.
(a) Difference: $0.0171 - 0.0094 = 0.0077$.
(b) Relative risk: $0.0171/0.0094 = 1.82$.
(c) Odds ratio: $\hat{\theta} = \dfrac{189 \times 10{,}933}{10{,}845 \times 104} = 1.83$; the odds of heart attack were 83% higher with placebo.

2.2.6 Case–control studies and the odds ratio

In a case–control study we can estimate $P(X = i \mid Y = j)$ but usually not $P(Y = j \mid X = i)$, hence neither the difference nor the relative risk of the outcome. By (2.3), however, the odds ratio is determined by conditional probabilities in either direction.

Example 2.8 (Odds ratio from Doll and Hill's data)

From Example 2.5,

$$\hat{\theta} = \frac{(688/709)/(21/709)}{(650/709)/(59/709)} = \frac{688 \times 59}{650 \times 21} = 3.0.$$

Using the direction of interest: the estimated odds of lung cancer for smokers were 3.0 times those for nonsmokers.

2.2.7 Relationship between odds ratio and relative risk

Proposition 2.9 (Odds ratio versus relative risk)

$$\theta = \frac{\pi_1}{\pi_2} \times \frac{1-\pi_2}{1-\pi_1}.$$

Hence $\theta \approx$ relative risk when π_1 and π_2 are both close to 0, and θ is farther from 1 than the relative risk (on the same side).

Proof.

Step 1. Directly from (2.2): $\theta = \frac{\pi_1}{1-\pi_1} \cdot \frac{1-\pi_2}{\pi_2}$.

Step 2. If $\pi_1 > \pi_2$ then $(1-\pi_2)/(1-\pi_1) > 1$ so $\theta > \pi_1/\pi_2 > 1$; if $\pi_1 < \pi_2$ the factor is < 1 and $\theta < \pi_1/\pi_2 < 1$. As $\pi_1, \pi_2 \to 0$ the factor tends to 1. ■

In the aspirin study ($\pi_i < 0.02$) the relative risk 1.82 and odds ratio 1.83 almost agree. Hence in case–control studies of rare diseases $\hat{\theta}$ roughly estimates the unestimable relative risk (Cornfield, 1951): smokers had roughly 3 times the lung cancer rate of nonsmokers.

2.3 Partial association in stratified 2×2 tables

Motivation. In observational studies an apparent effect of X on Y may reflect a third variable Z that affects both (confounding). Controlling for Z by holding it fixed can change the association dramatically, even reversing its direction.

To study a passive-smoking effect on lung cancer, for example, one must control age and socioeconomic status: spouses of nonsmokers may be younger, and younger people get lung cancer less often.

2.3.1 Partial tables

Definition (Partial and marginal tables)

Fixing the levels of Z, the two-way cross-sections of an $X \times Y \times Z$ table are the *partial tables*; their associations are *conditional associations*. Summing the partial tables gives the XY *marginal table*, which ignores Z rather than controlling it.

2.3.2 Death penalty example

Example 2.10 (Death penalty verdict by defendant's and victims' race)

674 defendants in Florida multiple-murder indictments, 1976–1987. Y = death penalty, X = defendant's race, Z = victims' race.

Victims' race	Defendant's race	Yes	No	Percent yes
White	White	53	414	11.3
	Black	11	37	22.9
Black	White	0	16	0.0
	Black	4	139	2.8
Total	White	53	430	11.0
	Black	15	176	7.9

(a) Controlling for victims' race, black defendants received the death penalty more often: by $22.9 - 11.3 = 11.6$ points with white victims and 2.8 points with black victims.
(b) Ignoring victims' race, white defendants received it more often (11.0% vs. 7.9%): the association reverses.
(c) Explanation: victims' race is strongly associated with defendant's race (marginal odds ratio $467 \times 143/(48 \times 16) = 87.0$), and the death penalty is much more likely when the victims are white. Whites mostly kill whites, and killing whites more often brings the death penalty.

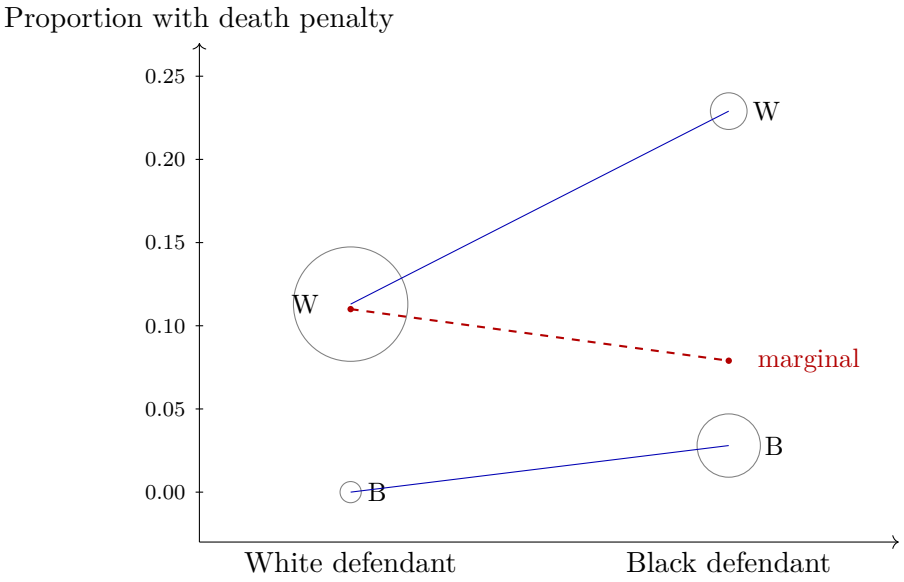


Figure 3: Simpson's paradox in the death penalty data. Letters give victims' race; circle area is proportional to the number of cases. Both conditional lines rise, the marginal line falls, since the summary proportions lie near the large circles.

Definition (Simpson's paradox)

A marginal association whose direction differs from that of each conditional association is an instance of *Simpson's paradox* (Yule 1903, Simpson 1951).

It warns against causal interpretation of an X–Y association. Fisher argued that some factor (e.g. genetic) might explain away the smoking–lung cancer association; Cornfield showed that for a very strong association, the confounder would itself need a very strong association with both X and Y.

2.3.3 Conditional and marginal odds ratios

For a $2 \times 2 \times K$ table with expected frequencies μ_{ijk}:

$$\theta_{XY(k)} = \frac{\mu_{11k}\mu_{22k}}{\mu_{12k}\mu_{21k}} \quad \text{(conditional)}, \qquad \theta_{XY} = \frac{\mu_{11+}\mu_{22+}}{\mu_{12+}\mu_{21+}} \quad \text{(marginal)},$$

with $\mu_{ij+} = \sum_k \mu_{ijk}$; sample versions use counts.

Example 2.11 (Odds ratios for the death penalty data)

(a) White victims: $\hat{\theta}_{XY(1)} = \dfrac{53 \times 37}{414 \times 11} = 0.43$; the odds of the death penalty for white defendants were 43% of those for black defendants.
(b) Black victims: $\hat{\theta}_{XY(2)} = \dfrac{0 \times 139}{16 \times 4} = 0.0$ (no white defendant with black victims got the death penalty).
(c) Marginal: $\hat{\theta}_{XY} = \dfrac{53 \times 176}{430 \times 15} = 1.45$; the odds were 45% higher for white defendants.

2.3.4 Marginal versus conditional independence

Definition (Conditional independence)

X and Y are *conditionally independent at level* k of Z if they are independent in partial table k, i.e. $P(Y = j \mid X = i, Z = k) = P(Y = j \mid Z = k)$ for all i, j. They are *conditionally independent given* Z if this holds for every k.

Proposition 2.12 (Factorization under conditional independence)

For a single multinomial over the three-way table with $\pi_{ijk} = P(X = i, Y = j, Z = k)$ and all $\pi_{++k} > 0$, X and Y are conditionally independent given Z iff

$$\pi_{ijk} = \frac{\pi_{i+k}\,\pi_{+jk}}{\pi_{++k}} \qquad \text{for all } i, j, k.$$

Conditional independence does not imply marginal independence: in general $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k} \neq \pi_{i++}\pi_{+j+}$.

Proof.

Step 1. Always $\pi_{ijk} = P(X = i, Z = k)\,P(Y = j \mid X = i, Z = k) = \pi_{i+k}\,P(Y = j \mid X = i, Z = k)$.

Step 2. Under conditional independence the last factor equals $P(Y = j \mid Z = k) = \pi_{+jk}/\pi_{++k}$, giving the factorization. Conversely, the factorization gives $P(Y = j \mid X = i, Z = k) = \pi_{ijk}/\pi_{i+k} = \pi_{+jk}/\pi_{++k}$ whenever $\pi_{i+k} > 0$.

Step 3. Summing over k gives $\pi_{ij+} = \sum_k \pi_{i+k}\pi_{+jk}/\pi_{++k}$, which involves k in every factor and does not reduce to $\pi_{i++}\pi_{+j+}$; Example 2.13 is an explicit counterexample. ■

For $2 \times 2 \times K$ tables, conditional independence means $\theta_{XY(k)} = 1$ for all k.

Example 2.13 (Conditional but not marginal independence)

Expected frequencies for Y = response, X = treatment, Z = clinic:

Clinic	Treatment	Success	Failure
1	A	18	12
	B	12	8
2	A	2	8
	B	8	32
Total	A	20	20
	B	20	40

(a) $\theta_{XY(1)} = \dfrac{18 \times 8}{12 \times 12} = 1.0$ and $\theta_{XY(2)} = \dfrac{2 \times 32}{8 \times 8} = 1.0$: conditional independence.

(b) Marginal: $\theta_{XY} = \dfrac{20 \times 40}{20 \times 20} = 2.0$.

(c) The XZ and YZ conditional odds ratios all equal 6.0, e.g. $\dfrac{18 \times 8}{12 \times 2} = 6.0$: clinic 1 uses treatment A more and also has more successes (perhaps its patients are healthier). Concluding from the marginal table that A is better would be misleading.

2.3.5 Homogeneous association

Definition (Homogeneous association)

A $2 \times 2 \times K$ table has *homogeneous XY association* if $\theta_{XY(1)} = \theta_{XY(2)} = \cdots = \theta_{XY(K)}$. Conditional independence is the special case where the common value is 1. If the conditional odds ratios vary with k there is *interaction*, and Z is an *effect modifier*.

Proposition 2.14 (Homogeneous association is symmetric)

For a $2 \times 2 \times 2$ table with positive probabilities,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\theta_{YZ(1)}}{\theta_{YZ(2)}},$$

where $\theta_{XZ(j)}$ is the XZ odds ratio at level j of Y and $\theta_{YZ(i)}$ the YZ odds ratio at level i of X. Hence, in a $2 \times 2 \times K$ table, homogeneous XY association holds iff for every pair of levels k, k' of Z the XZ (and the YZ) odds ratio formed with k, k' is the same at both levels of Y (of X).

Proof. [Supplement]

Step 1. Directly,

$$\frac{\theta_{XY(1)}}{\theta_{XY(2)}} = \frac{\pi_{111}\pi_{221}}{\pi_{121}\pi_{211}} \cdot \frac{\pi_{122}\pi_{212}}{\pi_{112}\pi_{222}}, \qquad \frac{\theta_{XZ(1)}}{\theta_{XZ(2)}} = \frac{\pi_{111}\pi_{212}}{\pi_{112}\pi_{211}} \cdot \frac{\pi_{122}\pi_{221}}{\pi_{121}\pi_{222}}.$$

Both equal $\dfrac{\pi_{111}\pi_{122}\pi_{212}\pi_{221}}{\pi_{112}\pi_{121}\pi_{211}\pi_{222}}$ (numerator: the cells whose index sum is odd; denominator: those whose index sum is even). The same expression arises for $\theta_{YZ(1)}/\theta_{YZ(2)}$ by the symmetry of this formula in the three indices.

Step 2. For $2 \times 2 \times K$, apply Step 1 to the $2 \times 2 \times 2$ subtable formed by levels k, k' of Z: $\theta_{XY(k)} = \theta_{XY(k')}$ iff the XZ odds ratio for (k, k') is the same at $Y = 1$ and $Y = 2$, and likewise for YZ. ■

So homogeneous association is a property of the three variables jointly ("no three-factor interaction"). For instance, if for smoking and lung cancer the odds ratios were 1.2, 3.9, 8.8 in age groups

< 45, 45–65, > 65, age would be an effect modifier. For the death penalty data $\hat{\theta}_{XY(1)} = 0.43$ and $\hat{\theta}_{XY(2)} = 0.0$; the latter is unstable (zero count), and adding 1/2 to each cell gives 0.94, so the data need not contradict homogeneity (tests in Section 6.3).

2.4 Extensions for $I \times J$ tables

Motivation. Beyond 2×2 tables a single number rarely captures the association. A minimal set of odds ratios describes it completely, while summary indices (entropy-based for nominal, concordance-based for ordinal variables) trade completeness for interpretability.

2.4.1 Odds ratios in $I \times J$ tables

Any pair of rows a, b and columns c, d gives an odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$; there are $\binom{I}{2}\binom{J}{2}$ of them, highly redundant.

Definition (Local and baseline odds ratios)

The *local odds ratios* are

$$\theta_{ij} = \frac{\pi_{ij}\,\pi_{i+1,j+1}}{\pi_{i,j+1}\,\pi_{i+1,j}}, \qquad i = 1, \dots, I-1,\ j = 1, \dots, J-1. \tag{2.4}$$

An alternative basic set uses the last row and column as baseline:

$$\alpha_{ij} = \frac{\pi_{ij}\,\pi_{IJ}}{\pi_{Ij}\,\pi_{iJ}}, \qquad i = 1, \dots, I-1,\ j = 1, \dots, J-1. \tag{2.5}$$

Proposition 2.15 ($(I-1)(J-1)$ odds ratios describe the association)

Assume all $\pi_{ij} > 0$.
(a) Every odds ratio $\pi_{ac}\pi_{bd}/(\pi_{bc}\pi_{ad})$ with $a < b$, $c < d$ is the product $\prod_{i=a}^{b-1}\prod_{j=c}^{d-1}\theta_{ij}$.
(b) Given the marginal distributions $\{\pi_{i+}\}, \{\pi_{+j}\}$, the local odds ratios determine the cell probabilities uniquely.
(c) Independence holds iff all $(I-1)(J-1)$ local odds ratios equal 1.

Proof. [Supplement]

Step (a). Let $\lambda_{ij} = \log \pi_{ij}$ and $\log \theta_{ij} = \lambda_{ij} - \lambda_{i,j+1} - \lambda_{i+1,j} + \lambda_{i+1,j+1}$. Summing over $j = c, \dots, d-1$ telescopes to $\lambda_{ic} - \lambda_{id} - \lambda_{i+1,c} + \lambda_{i+1,d}$; summing that over $i = a, \dots, b-1$ telescopes to $\lambda_{ac} - \lambda_{ad} - \lambda_{bc} + \lambda_{bd}$, the log of the stated odds ratio.

Step (b). Suppose $\{\pi_{ij}\}$ and $\{\pi'_{ij}\}$ have the same margins and the same local odds ratios. Let $\delta_{ij} = \log(\pi'_{ij}/\pi_{ij})$. Its local contrasts $\delta_{ij} - \delta_{i,j+1} - \delta_{i+1,j} + \delta_{i+1,j+1}$ vanish, so by the telescoping in (a) so does $\delta_{ij} - \delta_{iJ} - \delta_{Ij} + \delta_{IJ}$. With $r_i = \delta_{iJ} - \delta_{IJ}$ and $c_j = \delta_{Ij}$ this gives $\delta_{ij} = r_i + c_j$. Hence

$$\sum_i \sum_j (\pi'_{ij} - \pi_{ij}) \log \frac{\pi'_{ij}}{\pi_{ij}} = \sum_i r_i(\pi'_{i+} - \pi_{i+}) + \sum_j c_j(\pi'_{+j} - \pi_{+j}) = 0.$$

Each term $(\pi' - \pi)\log(\pi'/\pi)$ is ≥ 0, with equality iff $\pi' = \pi$ (log is increasing). Hence $\pi'_{ij} = \pi_{ij}$ for all i, j.

Step (c). If $\pi_{ij} = \pi_{i+}\pi_{+j}$ then every $\theta_{ij} = 1$. Conversely, the independence table $\{\pi_{i+}\pi_{+j}\}$ has the same margins and all local odds ratios 1; by (b) it is the only such table. ∎

In this sense $(I-1)(J-1)$ parameters describe any association in an $I \times J$ table. For Example 2.1 the sample local odds ratios are $\frac{18\times 99}{171\times 5} = 2.08$ (fatal vs. nonfatal) and $\frac{171\times 10{,}933}{99\times 10{,}845} = 1.74$ (nonfatal vs. none); their product 3.63 is the odds ratio for the first and third columns. In each case the more serious outcome was more prevalent under placebo. For $I \times J \times K$ tables, homogeneous XY association means every conditional XY odds ratio is the same at each level of Z.

2.4.2 Summary measures of association

A *proportional reduction in variation* measure, with $V(Y)$ a variation measure for the marginal distribution $\{\pi_{+j}\}$ and $V(Y \mid i)$ that for $\{\pi_{j|i}\}$, is

$$\frac{V(Y) - E[V(Y \mid X)]}{V(Y)}, \qquad E[V(Y \mid X)] = \sum_i \pi_{i+} V(Y \mid i), \tag{2.6}$$

generalizing R^2. For a nominal response Theil used the *entropy* $V(Y) = -\sum_j \pi_{+j} \log \pi_{+j}$, giving the *uncertainty coefficient*

$$U = -\frac{\sum_i \sum_j \pi_{ij} \log(\pi_{ij}/(\pi_{i+}\pi_{+j}))}{\sum_j \pi_{+j} \log \pi_{+j}}. \tag{2.7}$$

Proposition 2.16 (Range of the uncertainty coefficient)

If more than one $\pi_{+j} > 0$, then $0 \le U \le 1$; $U = 0$ iff X and Y are independent, and $U = 1$ iff for each i some $\pi_{j|i} = 1$.

Proof. [Supplement] Write

$$H(Y) = -\sum_j \pi_{+j} \log \pi_{+j} > 0, \qquad H(Y \mid X) = -\sum_i \pi_{i+} \sum_j \pi_{j|i} \log \pi_{j|i} \ge 0.$$

Step 1. (2.7) is (2.6) with entropy: since $\log \pi_{j|i} = \log \pi_{ij} - \log \pi_{i+}$, $H(Y) - H(Y \mid X) = \sum_i \sum_j \pi_{ij} \log \frac{\pi_{ij}}{\pi_{i+}\pi_{+j}}$, so $U = [H(Y) - H(Y \mid X)]/H(Y)$.

Step 2. By Jensen, $-\sum\sum \pi_{ij} \log \frac{\pi_{i+}\pi_{+j}}{\pi_{ij}} \ge -\log \sum\sum \pi_{i+}\pi_{+j} \ge 0$, with equality iff $\pi_{ij} = \pi_{i+}\pi_{+j}$ on all cells (then the sum is 1). So $U \ge 0$, with equality iff independence.

Step 3. $U \le 1$ iff $H(Y \mid X) \ge 0$, true since each $-\pi \log \pi \ge 0$; equality iff every conditional distribution is degenerate. ■

It is hard to develop intuition for such measures (what is a "30% reduction in entropy"?); summaries are easier to interpret when both variables are ordinal.

2.4.3 Ordinal trends: concordant and discordant pairs

For ordinal X and Y a monotone trend is common. A pair of subjects is *concordant* if the subject ranking higher on X also ranks higher on Y, *discordant* if it ranks lower on Y, and *tied* if they agree on X and/or Y.

Example 2.17 (Job satisfaction and income)

Black males in the 1996 General Social Survey ($n = 96$); VD = very dissatisfied, LD = a little dissatisfied, MS = moderately satisfied, VS = very satisfied.

Income ($)	VD	LD	MS	VS
$<$ 15,000	1	3	10	6
15,000–25,000	2	3	10	7
25,000–40,000	1	6	14	12
$>$ 40,000	0	1	9	11

Each cell's count times the total count below and to the right gives concordant pairs:

$$\begin{aligned} C = {} & 1(3+10+7+6+14+12+1+9+11) + 3(10+7+14+12+9+11) \\ & + 10(7+12+11) + 2(6+14+12+1+9+11) + 3(14+12+9+11) + 10(12+11) \\ & + 1(1+9+11) + 6(9+11) + 14(11) = 1331, \end{aligned}$$

and similarly (below and to the left) $D = 3(2{+}1{+}0){+}10(2{+}3{+}1{+}6{+}0{+}1){+}\cdots{+}12(0{+}1{+}9) = 849$. Since $C > D$, low income tends to occur with low satisfaction.

For two independent observations from $\{\pi_{ij}\}$ the probabilities of concordance and discordance are

$$\Pi_c = 2\sum_i\sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k>j}\pi_{hk}\Big), \qquad \Pi_d = 2\sum_i\sum_j \pi_{ij}\Big(\sum_{h>i}\sum_{k<j}\pi_{hk}\Big), \tag{2.8}$$

the factor 2 because either observation may be the one in cell (i, j).

2.4.4 Ordinal measure of association: gamma

Definition (Goodman–Kruskal gamma)

Given that a pair is untied on both variables, $\Pi_c/(\Pi_c+\Pi_d)$ is the probability of concordance. *Gamma* is

$$\gamma = \frac{\Pi_c - \Pi_d}{\Pi_c + \Pi_d}, \qquad \text{sample version } \hat\gamma = \frac{C-D}{C+D}. \tag{2.9}$$

Proposition 2.18 (Properties of gamma)

(a) $-1 \le \gamma \le 1$; $\gamma = 1$ iff $\Pi_d = 0$ and $\gamma = -1$ iff $\Pi_c = 0$ (perfect monotonicity, not linearity).
(b) Reversing the category order of one variable changes the sign of γ; γ is symmetric in X and Y.
(c) Independence implies $\gamma = 0$; the converse is false.

Proof. [Supplement]

Step (a). $\Pi_c, \Pi_d \ge 0$, so $|\Pi_c - \Pi_d| \le \Pi_c + \Pi_d$, with the extreme cases as stated.

Step (b). Reversing one ordering turns concordant pairs into discordant ones and vice versa, swapping Π_c and Π_d. Interchanging the roles of X and Y does not change whether a pair is concordant.

Step (c). Let (X_1, Y_1), (X_2, Y_2) be independent draws. Then $\Pi_c = 2P(X_1 > X_2, Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2, Y_1 < Y_2)$. Under independence of X and Y, (X_1, X_2) is independent of (Y_1, Y_2), so $\Pi_c = 2P(X_1 > X_2)P(Y_1 > Y_2)$ and $\Pi_d = 2P(X_1 > X_2)P(Y_1 < Y_2)$; since Y_1, Y_2 are i.i.d., $P(Y_1 > Y_2) = P(Y_1 < Y_2)$, so $\Pi_c = \Pi_d$ and $\gamma = 0$. For the converse, a U-shaped joint distribution can have $\Pi_c = \Pi_d$ without independence, e.g. the 3×3 table with $\pi_{11} = \pi_{31} = \pi_{22} = 1/3$ has $\Pi_c = 2\pi_{11}\pi_{22} = 2/9 = 2\pi_{22}\pi_{31} = \Pi_d$. ∎

2.4.5 Gamma for job satisfaction example

Example 2.19 (Gamma for income and job satisfaction)

From Example 2.17, $\hat{\gamma} = (1331-849)/(1331+849) = 0.221$: of the untied pairs, the proportion concordant exceeds the proportion discordant by 0.221, a weak tendency for satisfaction to increase with income.

Chapter 3. Inference for Contingency Tables

3.1 Confidence intervals for association parameters

Motivation. Point estimates of the odds ratio, relative risk or difference of proportions mean little without a measure of precision. Large-sample standard errors come from one general tool, the delta method, and better intervals come from inverting score or likelihood-ratio tests.

3.1.1 Interval estimation of odds ratios

The sample odds ratio $\hat{\theta} = n_{11}n_{22}/(n_{12}n_{21})$ equals 0 or ∞ if some $n_{ij} = 0$, and is undefined if a whole row or column is zero. These events have positive probability, so $E(\hat{\theta})$ and $\mathrm{Var}(\log\hat{\theta})$ do not exist. The amended estimator

$$\tilde{\theta} = \frac{(n_{11}+0.5)(n_{22}+0.5)}{(n_{12}+0.5)(n_{21}+0.5)}$$

(and $\log\tilde{\theta}$) behaves well in terms of bias and mean squared error; $\hat{\theta}$ and $\tilde{\theta}$ have the same asymptotic normal distribution. Unless n is large the distribution of $\hat{\theta}$ is very skewed (for $\theta = 1$ it cannot be much below 1 but can be much above), whereas $\log\hat{\theta}$, with additive structure, approaches normality faster. Its estimated standard error is (Corollary 3.7)

$$\hat{\sigma}(\log\hat{\theta}) = \Big(\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}\Big)^{1/2}. \tag{3.1}$$

Definition (Woolf interval)

$\log\hat{\theta} \pm z_{\alpha/2}\,\hat{\sigma}(\log\hat{\theta})$ is a Wald interval for $\log\theta$; exponentiating its endpoints gives the Woolf interval for θ.

It works well, usually slightly conservatively. When $\hat{\theta} = 0$ (or ∞) take 0 as the lower (or ∞ as the upper) limit and compute the other bound after an adjustment such as replacing n_{ij} by $n_{ij} + 0.5$; a less ad hoc approach inverts score or LR tests.

3.1.2 Aspirin and myocardial infarction example

Example 3.1 (Swedish aspirin study: odds ratio)

1360 stroke patients were randomized to low-dose aspirin or placebo and followed about three years for death from myocardial infarction (MI).

	MI death		
	Yes	No	Total
Placebo	28	656	684
Aspirin	18	658	676

$\hat{\theta} = \frac{28\times658}{656\times18} = 1.56$ (close to $\tilde{\theta} = 1.55$), $\log\hat{\theta} = 0.445$, and by (3.1) $\hat{\sigma}(\log\hat{\theta}) = (1/28 + 1/656 + 1/18 + 1/658)^{1/2} = 0.307$. The 95% interval for $\log\theta$ is $0.445 \pm 1.96(0.307) = (-0.157,\ 1.047)$, so for θ it is $(e^{-0.157}, e^{1.047}) = (0.85,\ 2.85)$. Since it contains 1, equal odds of MI death for aspirin and placebo are plausible; a small benefit would need a large sample to detect.

3.1.3 Interval estimation of difference of proportions

Treat the rows as independent binomials: $y_i \sim \text{bin}(n_i, \pi_i)$, $\hat{\pi}_i = y_i/n_i$.

Proposition 3.2 (Difference of proportions)

$E(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1 - \pi_2$ and

$$\sigma(\hat{\pi}_1 - \hat{\pi}_2) = \left[\frac{\pi_1(1-\pi_1)}{n_1} + \frac{\pi_2(1-\pi_2)}{n_2}\right]^{1/2}. \tag{3.2}$$

As $n_1, n_2 \to \infty$, $(\hat{\pi}_1 - \hat{\pi}_2 - (\pi_1 - \pi_2))/\hat{\sigma} \xrightarrow{d} N(0,1)$, where $\hat{\sigma}$ replaces π_i by $\hat{\pi}_i$; so $(\hat{\pi}_1 - \hat{\pi}_2) \pm z_{\alpha/2}\hat{\sigma}$ is a Wald interval.

Proof.

Step 1. Linearity and independence give the mean and variance, using $\text{Var}(\hat{\pi}_i) = \pi_i(1-\pi_i)/n_i$ (Proposition 1.8).

Step 2. [Supplement] Each $(\hat{\pi}_i - \pi_i)/\sigma(\hat{\pi}_i) \xrightarrow{d} N(0,1)$ by the CLT, independently, so the standardized difference is asymptotically $N(0,1)$; $\hat{\sigma}/\sigma \xrightarrow{p} 1$ by the WLLN, and Slutsky's theorem completes the proof. ■

Like the single-proportion Wald interval (1.11), its coverage is usually below nominal, especially when π_1, π_2 are near 0 or 1 (adding one success and one failure to each sample helps greatly).

3.1.4 Interval estimation of relative risk

The sample relative risk $r = \hat{\pi}_1/\hat{\pi}_2$ also converges to normality faster on the log scale, with asymptotic standard error

$$\sigma(\log r) = \left[\frac{1-\pi_1}{\pi_1 n_1} + \frac{1-\pi_2}{\pi_2 n_2}\right]^{1/2} \tag{3.3}$$

(Corollary 3.5(b)). The Wald interval exponentiates $\log r \pm z_{\alpha/2}\hat{\sigma}(\log r)$; it works well but can be somewhat conservative.

Example 3.3 (Swedish aspirin study: relative risk and difference)

Sample MI death proportions are $28/684 = 0.0409$ (placebo) and $18/676 = 0.0266$ (aspirin).
(a) $r = 1.54$; $\log 1.54 \pm 1.96(0.297)$ gives (0.86, 2.75) for the relative risk.
(b) Difference: $0.014 \pm 1.96(0.0098)$, i.e. $(-0.005, 0.033)$.
Substantial benefit is plausible, but so are no effect and a slight harm; the larger Physicians' Health Study (Example 2.7) does show a benefit.

3.1.5 Deriving standard errors with the delta method*

Theorem 3.4 (Delta method)

Suppose $\sqrt{n}(T_n - \theta) \xrightarrow{d} N(0, \sigma^2)$ and g is differentiable at θ. Then

$$\sqrt{n}\,[g(T_n) - g(\theta)] \xrightarrow{d} N(0,\ [g'(\theta)]^2\sigma^2).$$

If moreover g' is continuous at θ, $g'(\theta) \neq 0$, and $\hat{\sigma} \xrightarrow{p} \sigma > 0$, then $\sqrt{n}[g(T_n) - g(\theta)]/(|g'(T_n)|\hat{\sigma}) \xrightarrow{d} N(0,1)$.

Proof. [Supplement]

Step 1. Define $h(t) = \frac{g(t)-g(\theta)}{t-\theta} - g'(\theta)$ for $t \neq \theta$ and $h(\theta) = 0$; differentiability means h is continuous at θ.

Step 2. $T_n - \theta = n^{-1/2} \cdot \sqrt{n}(T_n - \theta) = o_p(1)$, so $T_n \xrightarrow{p} \theta$ and $h(T_n) \xrightarrow{p} 0$ (continuous mapping).

Step 3. Then $\sqrt{n}[g(T_n) - g(\theta)] = [g'(\theta) + h(T_n)]\sqrt{n}(T_n - \theta)$. The bracket $\xrightarrow{p} g'(\theta)$, so by Slutsky the product $\xrightarrow{d} g'(\theta)N(0, \sigma^2) = N(0, g'(\theta)^2\sigma^2)$ (recall $cY \sim N(0, c^2\sigma^2)$ when $Y \sim N(0, \sigma^2)$).

Step 4. $|g'(T_n)|\hat{\sigma} \xrightarrow{p} |g'(\theta)|\sigma > 0$, and a further application of Slutsky gives the standardized version. ■

Locally g is approximately linear with slope $g'(\theta)$, so $g(T_n)$ is approximately normal and its spread around $g(\theta)$ is $|g'(\theta)|$ times the spread of T_n around θ (Figure 4). A large-sample 95% Wald interval for $g(\theta)$ is $g(T_n) \pm 1.96|g'(T_n)|\hat{\sigma}/\sqrt{n}$.

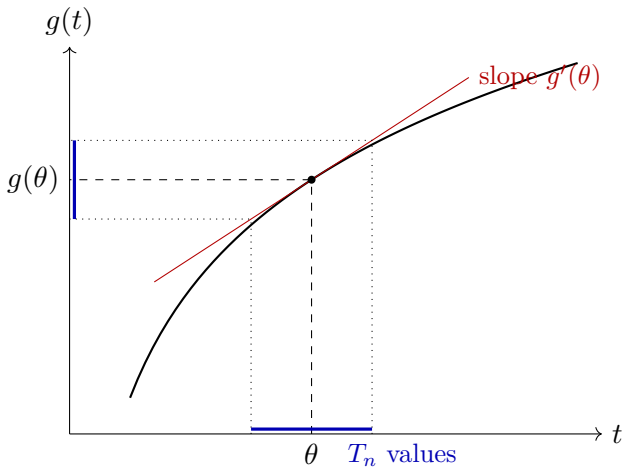


Figure 4: The delta method: near θ, g is almost linear, so a band of T_n values maps to a band of $g(T_n)$ values about $|g'(\theta)|$ times as wide.

3.1.6 Delta method applied to sample logit*

Corollary 3.5 (Sample logit and log proportion)

Let $\hat{\pi} = y/n$ with $y \sim \text{bin}(n, \pi)$, $0 < \pi < 1$.
(a) The sample logit satisfies

$$\sqrt{n}\Big[\log\frac{\hat{\pi}}{1-\hat{\pi}} - \log\frac{\pi}{1-\pi}\Big] \xrightarrow{d} N\Big(0, \frac{1}{\pi(1-\pi)}\Big).$$

(b) $\sqrt{n}(\log\hat{\pi} - \log\pi) \xrightarrow{d} N(0, (1-\pi)/\pi)$; hence for independent samples the log relative risk has asymptotic variance (3.3).

Proof.

Step (a). $\sqrt{n}(\hat{\pi} - \pi) \xrightarrow{d} N(0, \pi(1-\pi))$ by the CLT. With $g(\pi) = \log[\pi/(1-\pi)]$, $g'(\pi) = 1/\pi + 1/(1-\pi) = 1/[\pi(1-\pi)]$; Theorem 3.4 gives variance $\pi(1-\pi) \cdot [\pi(1-\pi)]^{-2}$.

Step (b). With $g = \log$, $g'(\pi) = 1/\pi$ and the variance is $\pi(1-\pi)/\pi^2 = (1-\pi)/\pi$. Dividing by n_i and adding over the two independent samples gives (3.3) for $\log r = \log\hat{\pi}_1 - \log\hat{\pi}_2$. ■

The *asymptotic* variance $[n\pi(1-\pi)]^{-1}$ is the variance of the approximating normal distribution,

not an approximation to the true variance—which does not exist, since $\hat{\pi} = 0$ or 1 (logit $\mp\infty$) with positive probability. That probability vanishes rapidly, so the asymptotic variance is the more useful quantity. For the same reason the bootstrap, which mimics the true sampling variance, is unhelpful for SEs of many discrete measures.

3.1.7 Delta method for log odds ratio*

Theorem 3.6 (Multivariate delta method for multinomial proportions)

Let $(n_1, \ldots, n_c)$ be multinomial$(n, \boldsymbol{\pi})$, $\hat{\boldsymbol{\pi}} = \mathbf{n}/n$, and let g be differentiable at $\boldsymbol{\pi}$ with $\phi_i = \partial g/\partial \pi_i$. Define

$$\sigma^2 = \sum_i \pi_i \phi_i^2 - \Big(\sum_i \pi_i \phi_i\Big)^2. \tag{3.4}$$

If $\sigma > 0$ then $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]/\sigma \xrightarrow{d} N(0,1)$, and the same holds with σ replaced by its ML estimate $\hat{\sigma}$ (substituting $\hat{\pi}_i$ for π_i in (3.4)).

Proof. [Supplement]

Step 1. By the multivariate CLT, $\sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) \xrightarrow{d} \mathbf{z} \sim N_c(\mathbf{0}, \Sigma)$ with $\Sigma = \operatorname{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$ (Proposition 1.2).

Step 2. As in Theorem 3.4, differentiability gives $g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi}) = \boldsymbol{\phi}^\top(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) + o_p(\|\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}\|)$, and $\|\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}\| = O_p(n^{-1/2})$. Hence $\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})] = \boldsymbol{\phi}^\top \sqrt{n}(\hat{\boldsymbol{\pi}} - \boldsymbol{\pi}) + o_p(1) \xrightarrow{d} \boldsymbol{\phi}^\top \mathbf{z}$.

Step 3. $\boldsymbol{\phi}^\top \mathbf{z} \sim N(0, \boldsymbol{\phi}^\top \Sigma \boldsymbol{\phi})$ and $\boldsymbol{\phi}^\top \Sigma \boldsymbol{\phi} = \sum_i \pi_i \phi_i^2 - (\sum_i \pi_i \phi_i)^2 = \sigma^2$.

Step 4. By the WLLN $\hat{\boldsymbol{\pi}} \xrightarrow{p} \boldsymbol{\pi}$; if the ϕ_i are continuous, $\hat{\sigma} \xrightarrow{p} \sigma$ and $\sigma/\hat{\sigma} \xrightarrow{p} 1$. Writing

$$\frac{\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]}{\hat{\sigma}} = \frac{\sqrt{n}[g(\hat{\boldsymbol{\pi}}) - g(\boldsymbol{\pi})]}{\sigma} \cdot \frac{\sigma}{\hat{\sigma}},$$

Slutsky's theorem gives the limit $N(0,1)$ (convergence is slower with $\hat{\sigma}$). ■

Corollary 3.7 (Standard error of the log odds ratio)

For a multinomial 2×2 table, $\log \hat{\theta}$ has asymptotic standard error $\big(\sum_i \sum_j 1/(n\pi_{ij})\big)^{1/2}$, estimated by (3.1).

Proof.

Step 1. $g(\boldsymbol{\pi}) = \log \pi_{11} + \log \pi_{22} - \log \pi_{12} - \log \pi_{21}$ has $\phi_{11} = 1/\pi_{11}$, $\phi_{12} = -1/\pi_{12}$, $\phi_{21} = -1/\pi_{21}$, $\phi_{22} = 1/\pi_{22}$.

Step 2. $\sum\sum \pi_{ij}\phi_{ij} = 1 - 1 - 1 + 1 = 0$ and $\sum\sum \pi_{ij}\phi_{ij}^2 = \sum\sum 1/\pi_{ij}$. By Theorem 3.6, $\sigma(\log \hat{\theta}) = \sigma/\sqrt{n} = (\sum\sum 1/(n\pi_{ij}))^{1/2}$, and substituting $\hat{\pi}_{ij} = n_{ij}/n$ gives $(\sum\sum 1/n_{ij})^{1/2}$. ■

The same result holds for independent binomial rows (Corollary 3.5(a): $\log \hat{\theta}$ is a difference of two independent sample logits, with variance $\frac{1}{n_1\pi_1(1-\pi_1)} + \frac{1}{n_2\pi_2(1-\pi_2)}$, estimated by (3.1)). A Wald interval for θ itself, $\hat{\theta} \pm z_{\alpha/2}\hat{\sigma}(\hat{\theta})$, is not recommended: $\hat{\theta}$ approaches normality slowly, the interval can include negative values, and it is not equivalent to the interval for $1/\theta$.

3.2 Testing independence in two-way contingency tables

Motivation. The null hypothesis of no association, $\pi_{ij} = \pi_{i+}\pi_{+j}$, is the starting point of contingency-table analysis. Pearson's X^2 and the LR statistic G^2 test it, and getting their degrees

of freedom right was the occasion for Fisher's introduction of the concept.

For multinomial sampling H_0: $\pi_{ij} = \pi_{i+}\pi_{+j}$ for all i, j; for independent multinomial rows this is homogeneity of the conditional distributions. The same tests apply in both cases.

3.2.1 Pearson and likelihood-ratio chi-squared tests

Proposition 3.8 (ML estimates under independence)

Under H_0, the ML estimates are $\hat{\pi}_{i+} = n_{i+}/n$, $\hat{\pi}_{+j} = n_{+j}/n$, so the estimated expected frequencies are

$$\hat{\mu}_{ij} = n\hat{\pi}_{i+}\hat{\pi}_{+j} = \frac{n_{i+}n_{+j}}{n}.$$

Proof. Under H_0 the multinomial kernel is $\prod_i \prod_j (\pi_{i+}\pi_{+j})^{n_{ij}} = \prod_i \pi_{i+}^{n_{i+}} \cdot \prod_j \pi_{+j}^{n_{+j}}$, a product of two multinomial kernels in separate parameters. Each is maximized by the sample proportions (Proposition 1.14). ■

$$X^2 = \sum_i \sum_j \frac{(n_{ij} - \hat{\mu}_{ij})^2}{\hat{\mu}_{ij}}, \qquad G^2 = 2\sum_i \sum_j n_{ij} \log \frac{n_{ij}}{\hat{\mu}_{ij}}. \tag{3.5}$$

G^2 is $-2\log\Lambda$ with $\Lambda = \prod\prod(n_{i+}n_{+j})^{n_{ij}}/(n^n \prod\prod n_{ij}^{n_{ij}})$, the ratio of the kernel at $\hat{\pi}_{ij} = n_{i+}n_{+j}/n^2$ to that at $\hat{\pi}_{ij} = n_{ij}/n$.

Theorem 3.9 (Null distribution of X^2 and G^2 for independence)

Under H_0 with all $\pi_{i+}, \pi_{+j} > 0$, X^2 and G^2 both converge in distribution to $\chi^2_{(I-1)(J-1)}$, and $X^2 - G^2 \xrightarrow{p} 0$.

Proof. [Supplement] Apply Theorem 1.18 to the $c = IJ$ cells with $p = (I-1) + (J-1)$ parameters

$$\boldsymbol{\theta} = (\pi_{1+}, \ldots, \pi_{I-1,+}, \pi_{+1}, \ldots, \pi_{+,J-1}).$$

Step 1 (rank). With $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, a tangent direction is $d\pi_{ij} = b_j\, da_i + a_i\, db_j$ with $\sum_i da_i = \sum_j db_j = 0$. If $d\pi_{ij} = 0$ for all i, j, summing over j gives $da_i = 0$, and then $a_i\, db_j = 0$ gives $db_j = 0$. So the Jacobian has full rank p.

Step 2 (df). Hence $X^2 \xrightarrow{d} \chi^2_{IJ-1-p}$ with

$$IJ - 1 - (I-1) - (J-1) = IJ - I - J + 1 = (I-1)(J-1),$$

and $G^2 - X^2 \xrightarrow{p} 0$ by the same theorem. ■

Pearson originally claimed df $= IJ - 1$; Fisher (1922) corrected him, introducing the notion of degrees of freedom. Equivalently, the general parameter space has dimension $IJ - 1$ and the null space $(I-1) + (J-1)$; their difference is the df. The limits also hold for Poisson and product multinomial sampling. Convergence is quicker for X^2 than G^2; the approximation for G^2 is often poor when $n/(IJ) < 5$, while X^2 can be adequate with some $\hat{\mu}_{ij}$ near 1 if most exceed 5. Otherwise use small-sample methods (Section 3.5).

3.2.2 Education and religious fundamentalism example

Example 3.10 (Education and religious beliefs)

1996 General Social Survey. Each cell shows the count, the estimate $\hat{\mu}_{ij}$ under independence, and the standardized Pearson residual (3.6).

Highest degree	Fundamentalist		Moderate		Liberal		Total
< High school	178 (137.8)	4.5	138 (161.5)	−2.6	108 (124.7)	−1.9	424
HS / junior college	570 (539.5)	2.6	648 (632.1)	1.3	442 (488.4)	−4.0	1660
Bachelor / graduate	138 (208.7)	−6.8	252 (244.5)	0.7	252 (188.9)	6.3	642
Total	886		1038		802		2726

E.g. $\hat{\mu}_{11} = 424\times886/2726 = 137.8$. Here $X^2 = 69.2$ and $G^2 = 69.8$ with $\text{df} = (3-1)(3-1) = 4$, $P < 0.0001$: extremely strong evidence of association.

3.3 Following-up chi-squared tests

Motivation. A small P-value says that an association exists, not what it looks like or how strong it is. Residuals, partitions of G^2 and estimated odds ratios reveal its nature.

3.3.1 Pearson and standardized residuals

Raw differences $n_{ij} - \hat{\mu}_{ij}$ tend to be larger where $\hat{\mu}_{ij}$ is larger (under Poisson sampling the standard deviation of n_{ij} is $\sqrt{\mu_{ij}}$).

Definition (Pearson and standardized residuals)

The *Pearson residual* and *standardized Pearson residual* for independence are

$$e_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{\hat{\mu}_{ij}^{1/2}}, \qquad r_{ij} = \frac{n_{ij} - \hat{\mu}_{ij}}{[\hat{\mu}_{ij}(1-p_{i+})(1-p_{+j})]^{1/2}}, \tag{3.6}$$

with $p_{i+} = n_{i+}/n$, $p_{+j} = n_{+j}/n$. Note $\sum_i \sum_j e_{ij}^2 = X^2$.

Theorem 3.11 (Asymptotic variance of Pearson residuals)

Under independence, $e_{ij} \xrightarrow{d} N(0, (1-\pi_{i+})(1-\pi_{+j}))$; hence $r_{ij} \xrightarrow{d} N(0,1)$. The average asymptotic variance of the e_{ij} is $(I-1)(J-1)/(IJ) < 1$.

Proof. [Supplement] Use the notation of the proof of Theorem 1.18, indexing cells by (i,j). Let $a_i = \pi_{i+}$, $b_j = \pi_{+j}$, and the unit vectors $\mathbf{r} = (\sqrt{a_i})_i \in \mathbb{R}^I$, $\mathbf{s}_c = (\sqrt{b_j})_j \in \mathbb{R}^J$; then $\mathbf{s} = (\sqrt{\pi_{ij}}) = \mathbf{r} \otimes \mathbf{s}_c$.

Step 1. By that proof the vector of Pearson residuals is $(I-H)\mathbf{z}_n + o_p(1)$ with limiting covariance $P = I - H - \mathbf{s}\mathbf{s}^\top$, where H is the projection onto the column space of $A = D^{-1/2}\partial\boldsymbol{\pi}/\partial\boldsymbol{\theta}^\top$.

Step 2. A tangent direction $d\pi_{ij} = b_j\, da_i + a_i\, db_j$ (with $\sum da_i = \sum db_j = 0$) gives $D^{-1/2}d\boldsymbol{\pi}$ with entries $\sqrt{b_j}\, x_i + \sqrt{a_i}\, y_j$, where $x_i = da_i/\sqrt{a_i}$, $y_j = db_j/\sqrt{b_j}$; the constraints say $\mathbf{x} \perp \mathbf{r}$, $\mathbf{y} \perp \mathbf{s}_c$. So the column space of A is $(\mathbf{r}^\perp \otimes \mathbf{s}_c) \oplus (\mathbf{r} \otimes \mathbf{s}_c^\perp)$, two orthogonal subspaces, and

$$H = (I_I - \mathbf{r}\mathbf{r}^\top) \otimes \mathbf{s}_c\mathbf{s}_c^\top + \mathbf{r}\mathbf{r}^\top \otimes (I_J - \mathbf{s}_c\mathbf{s}_c^\top).$$

Step 3. The (ij, ij) diagonal entry of H is $(1-a_i)b_j + a_i(1-b_j)$ and that of $\mathbf{s}\mathbf{s}^\top$ is $a_i b_j$. Hence the asymptotic variance of e_{ij} is

$$1 - a_i b_j - (1-a_i)b_j - a_i(1-b_j) = 1 - a_i - b_j + a_i b_j = (1-a_i)(1-b_j).$$

Step 4. Since $p_{i+} \xrightarrow{p} a_i$ and $p_{+j} \xrightarrow{p} b_j$, Slutsky gives $r_{ij} = e_{ij}/[(1-p_{i+})(1-p_{+j})]^{1/2} \xrightarrow{d} N(0,1)$. Finally $\frac{1}{IJ}\sum_i \sum_j (1-a_i)(1-b_j) = \frac{(I-1)(J-1)}{IJ}$. ■

So comparing e_{ij} with $N(0,1)$ is conservative, while $|r_{ij}|$ above about 2 or 3 indicates lack of fit of H_0 in that cell (larger thresholds when df is large, since some large values then occur by chance).

3.3.2 Education and religious fundamentalism revisited

Example 3.12 (Residuals for education and religion)

For the first cell $p_{1+} = 424/2726 = 0.156$, $p_{+1} = 886/2726 = 0.325$ and

$$r_{11} = \frac{178 - 137.8}{[137.8(1-0.156)(1-0.325)]^{1/2}} = 4.5.$$

Large positive residuals occur for (less than high school, fundamentalist) and (bachelor or graduate, liberal), large negative ones for (bachelor or graduate, fundamentalist) and (less than high school, liberal). The corner 2×2 table has odds ratio $\frac{178 \times 252}{108 \times 138} = 3.0$: for the most educated, the odds of liberal rather than fundamentalist were 3 times those for the least educated.

3.3.3 Partitioning chi-squared

If $Z_1, \ldots, Z_\nu$ are i.i.d. $N(0,1)$ then $Z_1^2 + \cdots + Z_\nu^2 \sim \chi^2_\nu$; so a χ^2_ν variable splits into independent components, and conversely independent $\chi^2_{\nu_1}$ and $\chi^2_{\nu_2}$ variables sum to $\chi^2_{\nu_1+\nu_2}$. A partition of a test statistic can show which categories drive an association.

Theorem 3.13 (Exact partition of G^2 by successive columns)

For an $I \times J$ table, for $j = 1, \ldots, J-1$ let G_j^2 be the G^2 statistic for independence in the $I \times 2$ table whose first column merges columns $1, \ldots, j$ and whose second column is column $j+1$. Then

$$G^2 = \sum_{j=1}^{J-1} G_j^2,$$

and each G_j^2 has df $= I - 1$, so the df also add up to $(I-1)(J-1)$.

Proof. [Supplement]

Step 1. For any table with counts m_{ij}, row totals m_{i+}, column totals m_{+j} and total m, (3.5) gives

$$G^2 = 2\Big[\sum_i \sum_j m_{ij} \log m_{ij} - \sum_i m_{i+} \log m_{i+} - \sum_j m_{+j} \log m_{+j} + m \log m\Big],$$

since $\log(m_{ij}/\hat{\mu}_{ij}) = \log m_{ij} + \log m - \log m_{i+} - \log m_{+j}$. Write $\Phi(x) = x \log x$.

Step 2. Let $S_{ij} = \sum_{l \le j} n_{il}$ (so $S_{i1} = n_{i1}$, $S_{iJ} = n_{i+}$) and $S_{+j} = \sum_i S_{ij}$. The jth subtable has cells S_{ij}, $n_{i,j+1}$, row totals $S_{i,j+1}$, column totals S_{+j}, $n_{+,j+1}$ and total $S_{+,j+1}$. Hence $G_j^2/2$ equals

$$\sum_i [\Phi(S_{ij}) + \Phi(n_{i,j+1}) - \Phi(S_{i,j+1})] - \Phi(S_{+j}) - \Phi(n_{+,j+1}) + \Phi(S_{+,j+1}).$$

Step 3. Sum over $j = 1, \ldots, J-1$. The terms $\Phi(S_{ij}) - \Phi(S_{i,j+1})$ telescope to $\Phi(n_{i1}) - \Phi(n_{i+})$, and adding $\sum_{j \geq 2} \Phi(n_{ij})$ gives $\sum_j \Phi(n_{ij}) - \Phi(n_{i+})$. Likewise $\sum_j [\Phi(S_{+,j+1}) - \Phi(S_{+j})] - \sum_{j \geq 2} \Phi(n_{+j}) = \Phi(n) - \sum_j \Phi(n_{+j})$.

Step 4. The total is exactly $\sum\sum \Phi(n_{ij}) - \sum_i \Phi(n_{i+}) - \sum_j \Phi(n_{+j}) + \Phi(n) = G^2/2$ by Step 1. ■

Applying the theorem again to each $I \times 2$ component, with rows merged successively, gives a finer partition into $(I-1)(J-1)$ components with df $= 1$ (Lancaster), based on the 2×2 tables

$$\begin{matrix} \sum_{a<i} \sum_{b<j} n_{ab} & \sum_{a<i} n_{aj} \\ \sum_{b<j} n_{ib} & n_{ij} \end{matrix} \qquad i = 2, \ldots, I, \;\; j = 2, \ldots, J. \tag{3.7}$$

Comparing each column with the last one instead does *not* give components that are independent or sum to G^2 (the corresponding log-odds-ratio contrasts are not orthogonal). In Section 9.2.4 the partition is justified as a sequence of nested models.

3.3.4 Origin of schizophrenia example

Example 3.14 (School of psychiatric thought and origin of schizophrenia)

School	Biogenic	Environmental	Combination
Eclectic	90	12	78
Medical	13	1	6
Psychoanalytic	19	13	50

$G^2 = 23.04$, df $= 4$. Partitioning by (3.7):

Rows compared	Columns compared	G^2 (df $= 1$)
Ecl vs. Med	Bio vs. Env	0.29
Ecl vs. Med	(Bio+Env) vs. Com	1.36
(Ecl+Med) vs. Psy	Bio vs. Env	12.95
(Ecl+Med) vs. Psy	(Bio+Env) vs. Com	8.43
Sum		23.04

The eclectic and medical schools hardly differ. The psychoanalytic school more often ascribes a combined origin and, among biogenic/environmental choices, more often the environmental one.

3.3.5 Rules for partitioning

Necessary conditions for subtables to give independent components (Goodman; Lancaster):

(a) the subtable df sum to the df of the full table;

(b) each cell count of the full table is a cell count in exactly one subtable;

(c) each marginal total of the full table is a marginal total of exactly one subtable.

If the df sum correctly but the G^2 values do not, the components are not independent. Exact algebraic partitions hold for G^2 but not for X^2; the subtable X^2 values are still valid tests (asymptotically equivalent to G^2 under the null), and are safer for small counts.

3.3.6 Limitations of chi-squared tests

Chi-squared tests only measure evidence of association; one should also study residuals, partitions and estimated odds ratios. They need large samples, and since $\hat{\mu}_{ij}$ depend only on the margins, X^2 and G^2 are invariant to reordering rows or columns: they treat both variables as nominal. With ordinal variables, tests using the ordering are more appropriate (Section 3.4).

3.3.7 Why consider independence?

Independence rarely holds exactly, and with large n small P-values are expected. Yet if independence approximately holds, the model-based estimates $\hat{\pi}_{ij} = p_{i+}p_{+j}$ can beat the sample proportions p_{ij} in mean squared error: they may be biased but estimate fewer parameters, so have smaller variance.

Proposition 3.15 (MSE of the two estimators)

For multinomial sampling,

$$\sum_i \sum_j E(p_{ij} - \pi_{ij})^2 = \frac{1 - \sum_i \sum_j \pi_{ij}^2}{n}, \qquad \sum_i \sum_j E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j (\pi_{i+}\pi_{+j} - \pi_{ij})^2.$$

Under independence, $n \sum \sum E(p_{i+}p_{+j} - \pi_{ij})^2 \to \sum_i \sum_j [\pi_{+j}^2 \pi_{i+}(1 - \pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1 - \pi_{+j})]$.

Proof. [Supplement]

Step 1. p_{ij} is unbiased with $\mathrm{Var}(p_{ij}) = \pi_{ij}(1 - \pi_{ij})/n$; summing gives $(1 - \sum \sum \pi_{ij}^2)/n$.

Step 2. Write $a = p_{i+} - \pi_{i+}$, $b = p_{+j} - \pi_{+j}$; then

$$p_{i+}p_{+j} - \pi_{ij} = (\pi_{i+}\pi_{+j} - \pi_{ij}) + \pi_{+j}a + \pi_{i+}b + ab.$$

Since $E(a^2), E(b^2) = O(n^{-1})$ and $E(a^2b^2) \le [E(a^4)E(b^4)]^{1/2} = O(n^{-2})$ (fourth central moments of a binomial proportion are $O(n^{-2})$), all terms but the first vanish in mean square, giving the second limit.

Step 3. Under independence the first term is 0, and $nE(\pi_{+j}a + \pi_{i+}b)^2 = \pi_{+j}^2 \pi_{i+}(1 - \pi_{i+}) + \pi_{i+}^2 \pi_{+j}(1 - \pi_{+j}) + 2\pi_{i+}\pi_{+j}\, n\,\mathrm{Cov}(a, b)$. Here $n\,\mathrm{Cov}(p_{i+}, p_{+j}) = \pi_{ij} - \pi_{i+}\pi_{+j} = 0$ (indicator covariance), and the remaining terms are $o(n^{-1})$ by Cauchy–Schwarz. ∎

Example 3.16 (A 3×3 illustration)

Let $\pi_{i+} = \pi_{+j} = 1/3$ and $\pi_{ij} = \frac{1}{9}[1 + \delta(i - 2)(j - 2)]$, $-1 < \delta < 1$: the corner cells are $(1 \pm \delta)/9$, the others $1/9$; $\delta = 0$ is independence.
(a) At $\delta = 0$: $\sum \sum \pi_{ij}^2 = 1/9$, so the sample proportions have total MSE $8/(9n)$, while the independence estimator has total MSE $\approx 9 \cdot 2 \cdot \frac{1}{9} \cdot \frac{1}{3} \cdot \frac{2}{3}/n = 4/(9n)$: about half.
(b) For $\delta \neq 0$ the independence estimator is inconsistent: its total MSE tends to $4(\delta/9)^2 = 4\delta^2/81$, while that of p_{ij} tends to 0. Still, near independence it wins for moderately large n (e.g. $n = 500$ when $\delta = 0.1$).

3.4 Two-way tables with ordered classifications

Motivation. X^2 and G^2 ignore category order. When both variables are ordinal, a single-parameter description of the trend (a correlation or gamma) yields tests with one degree of freedom and much greater power against monotone alternatives.

3.4.1 Linear trend alternative to independence

Assign row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$ (treating the scales as interval). The covariation $\sum_i \sum_j u_i v_j n_{ij}$, standardized, gives the sample correlation r between the scores over the n subjects:

$$r = \frac{\sum\sum u_i v_j n_{ij} - (\sum_i u_i n_{i+})(\sum_j v_j n_{+j})/n}{\left[\sum_i u_i^2 n_{i+} - \frac{(\sum_i u_i n_{i+})^2}{n}\right]^{1/2}\left[\sum_j v_j^2 n_{+j} - \frac{(\sum_j v_j n_{+j})^2}{n}\right]^{1/2}}.$$

Definition (Linear trend statistic)

$M^2 = (n-1)r^2$ (Mantel, 1963). Large values contradict independence; the P-value is the right-tail χ^2_1 probability. For one-sided alternatives use $M = \sqrt{n-1}\, r$ with the normal tail.

Theorem 3.17 (Null distribution of M^2)

Under independence of X and Y (multinomial sampling, each score variable nondegenerate), $\sqrt{n}\, r \xrightarrow{d} N(0,1)$; hence $M \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$.

Proof. [Supplement] Let (U_k, V_k), $k = 1, \ldots, n$, be the score pairs of the n subjects: i.i.d., with U independent of V, means μ, ν and variances $\sigma^2, \tau^2 > 0$.

Step 1. The sample covariance is $S_{UV} = \frac{1}{n}\sum_k (U_k - \mu)(V_k - \nu) - (\bar{U} - \mu)(\bar{V} - \nu)$. The summands $W_k = (U_k - \mu)(V_k - \nu)$ are i.i.d. with $E(W_k) = E(U-\mu)E(V-\nu) = 0$ and $\mathrm{Var}(W_k) = E(U-\mu)^2 E(V-\nu)^2 = \sigma^2\tau^2$, by independence. So $\sqrt{n} \cdot \frac{1}{n}\sum_k W_k \xrightarrow{d} N(0, \sigma^2\tau^2)$ (CLT).

Step 2. $\sqrt{n}(\bar{U} - \mu)(\bar{V} - \nu) = \sqrt{n}(\bar{U} - \mu) \cdot (\bar{V} - \nu) = O_p(1) o_p(1) \xrightarrow{p} 0$. Hence $\sqrt{n} S_{UV} \xrightarrow{d} N(0, \sigma^2\tau^2)$.

Step 3. $S_{UU} \xrightarrow{p} \sigma^2$ and $S_{VV} \xrightarrow{p} \tau^2$ (WLLN). By Slutsky, $\sqrt{n}\, r = \sqrt{n} S_{UV} / (S_{UU} S_{VV})^{1/2} \xrightarrow{d} N(0,1)$.

Step 4. $\sqrt{(n-1)/n} \to 1$, so $M = \sqrt{n-1}\, r \xrightarrow{d} N(0,1)$ and $M^2 \xrightarrow{d} \chi^2_1$. ∎

A small P-value does not mean the association is linear, only that looking for a linear component helped power. The test is symmetric in X and Y.

3.4.2 Job satisfaction example revisited

Example 3.18 (Linear trend test for income and job satisfaction)

For Example 2.17 ($n = 96$), $X^2 = 6.0$ and $G^2 = 6.8$ with df $= 9$ ($P = 0.74$, 0.66): little evidence. With job satisfaction scores $1, 2, 3, 4$ and income scores $7.5, 20, 32.5, 60$ (approximate category midpoints, thousands of dollars), $r = 0.200$ and

$$M^2 = (96-1)(0.200)^2 = 3.81 \quad (P = 0.051),$$
$$M = \sqrt{95}\,(0.200) = 1.95 \quad (\text{one-sided } P = 0.026).$$

Exploiting the trend gives far stronger evidence than X^2 or G^2.

3.4.3 Monotone trend alternatives to independence

A strictly ordinal analysis uses an ordinal measure such as gamma (Section 2.4.4), whose sample version is approximately normal for large samples with SE from the delta method. Then $z = \hat{\gamma}/\mathrm{SE}$

tests independence against monotone association. For Example 2.17, $\hat{\gamma} = 0.221$ with SE $= 0.117$: $z = 1.89$ (one-sided $P = 0.03$) and the 95% interval $0.221 \pm 1.96(0.117) = (-0.01, 0.45)$. The association is at best moderately positive.

3.4.4 Extra power with ordinal tests

X^2 and G^2 have df $= (I-1)(J-1)$ because the alternative has that many more parameters (the nonredundant odds ratios), and they are designed to detect any pattern in them. The ordinal statistics (M^2, $(\hat{\gamma}/\text{SE})^2$) concentrate on one parameter and have df $= 1$. Since the chi-squared mean equals its df, a value with df $= 1$ lies much farther in the tail than a comparable value with df $= (I-1)(J-1)$, giving a smaller P-value; the power advantage grows with I and J (theory in Section 6.4).

3.4.5 Choice of scores

Cochran: any scores fixed before seeing the data give a valid test, but poor scores give an insensitive one. Linear transformations of scores (e.g. $1, 2, 3, 4$ vs. $0, 2, 4, 6$) leave $|r|$ and M^2 unchanged. Results can depend on the scores when the data are very unbalanced.

Example 3.19 (Maternal drinking and congenital malformations)

	Alcohol consumption (average drinks per day)				
Malformation	0	< 1	1–2	3–5	≥ 6
Absent	17,066	14,464	788	126	37
Present	48	38	5	1	1

With two rows, any distinct row scores give the same M^2.
(a) Column scores 0, 0.5, 1.5, 4.0, 7.0: $M^2 = 6.57$, $P = 0.010$.
(b) Equally spaced $1, 2, 3, 4, 5$: $M^2 = 1.83$, $P = 0.18$.
(c) *Midranks* (average rank of a category's subjects in the full ranking: (1+17,114)/2 = 8557.5, then 24,365.5, 32,013, 32,473, 32,555.5): $M^2 = 0.35$, $P = 0.55$.
Midranks treat the sparse last three categories as nearly equal, putting "1–2 drinks" closer to "≥ 6" than to "0", which is inappropriate.

Prefer scores reflecting distances between categories, and do a sensitivity analysis with two or three sensible choices; equal spacing is a reasonable compromise when labels suggest nothing better. With midrank scores for both variables, r is *Spearman's rho*.

3.4.6 Trend tests for $I \times 2$ and $2 \times J$ tables

For a $2 \times J$ table with row scores $u_1 = 0$, $u_2 = 1$, the covariation $\sum\sum u_i v_j n_{ij} = \sum_j v_j n_{2j}$ is the total of the Y scores in row 2, so M^2 detects a difference between the two row means of the Y scores. With midrank scores this is the large-sample *Wilcoxon–Mann–Whitney* test (with ties): its z^2 equals M^2, and it is asymptotically equivalent to statistics based on concordant and discordant pairs. For an $I \times 2$ table M^2 tests a linear trend in the probability of a response category across rows: the *Cochran–Armitage trend test* (Section 5.3.5).

3.4.7 Nominal–ordinal tables

When one variable is nominal with more than two categories, correlation and gamma are inappropriate; statistics summarizing the variation among the mean scores of the ordinal variable across the nominal categories are used instead (Section 7.5.3).

3.5 Small-sample tests of independence

Motivation. For small n the chi-squared approximations fail. Conditioning on the margins eliminates the nuisance parameters and gives an exact null distribution, the hypergeometric, which underlies Fisher's exact test.

3.5.1 Fisher's exact test for 2×2 tables

Theorem 3.20 (Hypergeometric null distribution)

Under independence, for Poisson, multinomial, or independent binomial sampling, conditional on both sets of marginal totals,

$$p(t) = P(n_{11} = t) = \frac{\binom{n_{1+}}{t}\binom{n_{2+}}{n_{+1}-t}}{\binom{n}{n_{+1}}}, \qquad m_- \le t \le m_+, \tag{3.8}$$

where $m_- = \max(0,\, n_{1+} + n_{+1} - n)$ and $m_+ = \min(n_{1+},\, n_{+1})$.

Proof.

Step 1. By Theorem 2.3, conditioning a Poisson or multinomial table on its row totals gives independent binomials $n_{11} \sim \text{bin}(n_{1+}, \pi_1)$, $n_{21} \sim \text{bin}(n_{2+}, \pi_2)$; under independence $\pi_1 = \pi_2 = \pi$.

Step 2. Given the row totals, the table is determined by (n_{11}, n_{21}), and $n_{+1} = n_{11} + n_{21} \sim \text{bin}(n, \pi)$. So

$$P(n_{11} = t \mid n_{+1}) = \frac{\binom{n_{1+}}{t}\pi^t(1-\pi)^{n_{1+}-t}\binom{n_{2+}}{n_{+1}-t}\pi^{n_{+1}-t}(1-\pi)^{n_{2+}-n_{+1}+t}}{\binom{n}{n_{+1}}\pi^{n_{+1}}(1-\pi)^{n-n_{+1}}},$$

and the powers of π and $1-\pi$ cancel, leaving (3.8).

Step 3. The range: $0 \le t \le n_{1+}$, $0 \le n_{+1} - t \le n_{2+}$ give $m_- \le t \le m_+$. ∎

Given the margins, n_{11} determines the table. For $H_a\colon \theta > 1$, the P-value is $P(n_{11} \ge t_o)$ for observed t_o (Fisher, Irwin, Yates, 1934–35).

Proposition 3.21 (Orderings of 2×2 tables with fixed margins)

With all margins fixed, the sample odds ratio and the difference of proportions $n_{11}/n_{1+} - n_{21}/n_{2+}$ are increasing in n_{11}, and

$$X^2 = \frac{n^3\big(n_{11} - E(n_{11})\big)^2}{n_{1+}n_{2+}n_{+1}n_{+2}}, \qquad E(n_{11}) = \frac{n_{1+}n_{+1}}{n}.$$

Hence one-sided P-values agree for all three orderings, and $P(X^2 \ge X_o^2) = P(|n_{11} - E(n_{11})| \ge |t_o - E(n_{11})|)$.

Proof. [Supplement]

Step 1. With $n_{12} = n_{1+} - n_{11}$, $n_{21} = n_{+1} - n_{11}$, $n_{22} = n - n_{1+} - n_{+1} + n_{11}$, the numerator of $\hat{\theta}$ increases and its denominator decreases in n_{11}; the difference $n_{11}/n_{1+} - (n_{+1} - n_{11})/n_{2+}$ is increasing too.

Step 2. $n_{11}n_{22} - n_{12}n_{21} = n_{11}(n - n_{1+} - n_{+1} + n_{11}) - (n_{1+} - n_{11})(n_{+1} - n_{11}) = n\,n_{11} - n_{1+}n_{+1} = n[n_{11} - E(n_{11})]$.

Step 3. The familiar 2×2 form $X^2 = n(n_{11}n_{22} - n_{12}n_{21})^2/(n_{1+}n_{2+}n_{+1}n_{+2})$ follows from (3.5): all four $n_{ij} - \hat{\mu}_{ij}$ equal $\pm(n_{11}n_{22} - n_{12}n_{21})/n$, and $\sum_{i,j} 1/\hat{\mu}_{ij} = n^3/(n_{1+}n_{2+}n_{+1}n_{+2})$. Substituting Step 2 gives the display. The mean $E(n_{11}) = n_{1+}n_{+1}/n$ is the hypergeometric mean. ■

3.5.2 Fisher's tea drinker

Example 3.22 (Fisher's tea-tasting experiment)

Muriel Bristol claimed she could tell whether milk or tea was poured first. She tasted 8 cups, 4 of each kind, knowing this, and had to name the 4 milk-first cups (order randomized). A possible result:

	Guess poured first		
Poured first	Milk	Tea	Total
Milk	3	1	4
Tea	1	3	4
Total	4	4	8

Both margins are fixed by design, so (3.8) is the natural null distribution. For $H_a\colon \theta > 1$,

$$P(n_{11} = 3) = \frac{\binom{4}{3}\binom{4}{1}}{\binom{8}{4}} = \frac{16}{70} = 0.229, \qquad P(n_{11} = 4) = \frac{1}{70} = 0.014,$$

so $P = P(n_{11} \geq 3) = 0.243$. Such a small experiment cannot establish her ability (though she reportedly did convince Fisher).

3.5.3 Two-sided P-values for Fisher's exact test

For a two-sided alternative, different criteria give different P-values:

(a) sum $p(t)$ over t with $p(t) \leq p(t_o)$: $P = P[p(n_{11}) \leq p(t_o)]$ (the most popular);

(b) sum over tables at least as far from H_0: $P = P(|n_{11} - E(n_{11})| \geq |t_o - E(n_{11})|)$, which equals $P(X^2 \geq X_o^2)$ by Proposition 3.21;

(c) $2\min[P(n_{11} \geq t_o), P(n_{11} \leq t_o)]$, which can exceed 1;

(d) the smaller tail probability plus the largest attainable probability in the other tail not exceeding it (Blaker).

Ordering by a distance such as X^2 extends to $I \times J$ tables. Two-sided tests dominate practice; regulatory guidance (ICH E9) recommends one-sided tests at half the usual level, consistent with two-sided intervals.

3.5.4 Discreteness and conservatism issues

In the tea experiment $n_{11} \in \{0, \ldots, 4\}$, so one-sided P-values can only be 0.014, 0.243, 0.757, 0.986, 1.0; rejecting when $P \leq 0.05$ has actual size 0.014. The test is *conservative*.

Example 3.23 (Randomized test of exact size 0.05)

Reject when $n_{11} = 4$; when $n_{11} = 3$ reject with probability 0.157 (if an independent $U \sim \text{Uniform}(0, 1)$ is below 0.157); otherwise accept. Then

$$P(\text{reject } H_0) = E[P(\text{reject} \mid n_{11})] = 1.0(0.014) + 0.157(0.229) + 0.0 \cdot P(n_{11} \leq 2) = 0.05.$$

With randomization Fisher's test is uniformly most powerful unbiased (Tocher, 1950), but data-unrelated randomization is unacceptable in practice. Report the P-value, or the *mid-P-value* (half the probability of the observed outcome plus the probability of more extreme ones): for the tea data, mid-$P = \frac{1}{2}P(n_{11} = 3) + P(n_{11} > 3) = 0.129$. It no longer guarantees type I error $\leq$ nominal, but is rarely much above it.

3.6 Extensions for multiway tables and nontabulated responses

Motivation. The tests of this chapter are special cases of model-based inference; models extend them to conditional independence in multiway tables and to continuous explanatory variables.

Tests of independence in two-way tables extend to tests of conditional independence in three-way tables (Chapter 6); with models, the methods also apply when some explanatory variables are continuous.

3.6.1 Categorical data need not be contingency tables

Data files usually have one line per subject (e.g. subject, gender, race, education coded 1–3, opinion coded 1–2), and software forms contingency tables when needed. The generalized linear models of the next chapter contain all methods of this chapter as inferences about parameters of simple special cases.

Chapter 4. Introduction to Generalized Linear Models

Chapters 2–3 treated two-way tables. Real studies have several explanatory variables, some continuous, and we want to describe their effects on a response. This chapter introduces generalized linear models (GLMs), a single framework that contains logistic regression, Poisson loglinear models and ordinary regression. Sections 4.1–4.3 present the models; Sections 4.4–4.6 give the general likelihood theory (likelihood equations, information, deviance, fitting algorithms).

4.1 Generalized linear model

Motivation. Ordinary regression assumes a normal response whose mean is linear in the predictors. For counts and proportions neither assumption is natural; a GLM keeps the linear predictor but lets the response distribution and the scale on which linearity holds be chosen separately.

4.1.1 Components of generalized linear models

Definition (Natural exponential family)

A family of densities or mass functions of the form

$$f(y_i;\theta_i) = a(\theta_i)\, b(y_i) \exp\{y_i\, Q(\theta_i)\} \tag{4.1}$$

is a *natural exponential family*; $Q(\theta)$ is the *natural parameter*. The parameter θ_i may vary with i through explanatory variables.

Definition (Generalized linear model)

A GLM has three components.
(i) *Random component*: independent observations $y_1, \ldots, y_N$ of Y, each from a natural exponential family (4.1) (more generally, the exponential dispersion family of Section 4.4).
(ii) *Systematic component*: a *linear predictor* $\eta_i = \sum_{j=1}^{p} \beta_j x_{ij}$, $i = 1, \ldots, N$, where x_{ij} is the value of predictor j for subject i (usually $x_{i1} \equiv 1$ gives an intercept, often written α).
(iii) *Link function*: a monotone differentiable g with

$$g(\mu_i) = \eta_i = \sum_j \beta_j x_{ij}, \qquad \mu_i = E(Y_i). \tag{4.2}$$

The *identity link* $g(\mu) = \mu$ gives a linear model for the mean. The link with $g(\mu_i) = Q(\theta_i)$, transforming the mean to the natural parameter, is the *canonical link*.

In short, a GLM is a linear model for a transformed mean of a response whose distribution lies in the natural exponential family.

4.1.2 Binomial logit models for binary data

Code a binary outcome as 1 (success) and 0 (failure), with $P(Y = 1) = \pi = E(Y)$. The Bernoulli mass function is

$$f(y;\pi) = \pi^y (1-\pi)^{1-y} = (1-\pi) \exp\Big[y \log \frac{\pi}{1-\pi}\Big], \qquad y = 0, 1,$$

which is (4.1) with $\theta = \pi$, $a(\pi) = 1-\pi$, $b(y) = 1$ and $Q(\pi) = \log[\pi/(1-\pi)]$. The natural parameter is the log odds, the *logit* of π; the logit is the canonical link, and GLMs with this link are *logit models*.

4.1.3 Poisson loglinear models for count data

For a count Y with mean μ,

$$f(y;\mu) = \frac{e^{-\mu}\mu^{y}}{y!} = \exp(-\mu)\Big(\frac{1}{y!}\Big)\exp(y\log\mu), \qquad y = 0, 1, 2, \ldots,$$

so $\theta = \mu$, $a(\mu) = e^{-\mu}$, $b(y) = 1/y!$, $Q(\mu) = \log\mu$. The canonical link is the log link, and

$$\log\mu_i = \sum_j \beta_j x_{ij}, \qquad i = 1, \ldots, N, \tag{4.3}$$

is the *Poisson loglinear model*.

4.1.4 Generalized linear models for continuous responses

The normal distribution belongs to an exponential family with a dispersion parameter, and its natural parameter is the mean; so ordinary regression, ANOVA and ANCOVA are GLMs with normal random component and identity link.

Random component	Link	Systematic component	Model
Normal	identity	continuous	regression
Normal	identity	categorical	analysis of variance
Normal	identity	mixed	analysis of covariance
Binomial	logit	mixed	logistic regression (Ch. 5–6)
Poisson	log	mixed	loglinear (Ch. 8–9)
Multinomial	generalized logit	mixed	multinomial response (Ch. 7)

The traditional alternative transforms Y to approximate normality with constant variance and applies least squares. In a GLM the choice of link is separate from the choice of random component: a link need only make the linear predictor plausible; it need not stabilize variance or produce normality, because fitting maximizes the likelihood of the chosen distribution.

4.1.5 Deviance

Let $L(\boldsymbol{\mu};\mathbf{y})$ be the log likelihood expressed through the means $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_N)$ and $L(\hat{\boldsymbol{\mu}};\mathbf{y})$ its maximum under a model. Over all models the largest achievable value is $L(\mathbf{y};\mathbf{y})$, attained by the *saturated model*, which has one parameter per observation and fits perfectly ($\hat{\mu}_i = y_i$). The saturated model gives no data reduction but is a baseline for comparison.

Definition (Deviance)

For a Poisson or binomial GLM the *deviance* is

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = -2\big[L(\hat{\boldsymbol{\mu}};\mathbf{y}) - L(\mathbf{y};\mathbf{y})\big],$$

the likelihood-ratio statistic for testing that the model holds against the saturated alternative.

When N stays fixed while the individual counts grow, the deviance is asymptotically χ^2 with $\mathrm{df} = N - p$, the difference in the numbers of parameters of the saturated and the fitted model (Theorem 4.19).

Proposition 4.1 (Deviance of the homogeneity model)

Let $y_i \sim \text{bin}(n_i, \pi_i)$ independently, $i = 1, \ldots, N$, and consider the model $\pi_i = \pi$ for all i ($p = 1$). Its deviance equals the statistic G^2 for testing independence in the $N \times 2$ table of successes and failures, and has df $= N - 1$.

Proof.

Step 1. Write $n = \sum n_i$ and $s = \sum y_i$ (here y_i is a count). Up to a constant, $L = \sum_i [y_i \log \pi_i + (n_i - y_i) \log(1 - \pi_i)]$. Under homogeneity the MLE is $\hat{\pi} = s/n$; in the saturated model $\tilde{\pi}_i = y_i/n_i$.

Step 2. Hence

$$D = 2 \sum_i \Big[y_i \log \frac{y_i/n_i}{\hat{\pi}} + (n_i - y_i) \log \frac{1 - y_i/n_i}{1 - \hat{\pi}} \Big]$$
$$= 2 \sum_i \Big[y_i \log \frac{y_i}{n_i s/n} + (n_i - y_i) \log \frac{n_i - y_i}{n_i(n - s)/n} \Big].$$

Step 3. In the $N \times 2$ table with row totals n_i and column totals s, $n - s$, the fitted values under independence are $n_i s/n$ and $n_i(n - s)/n$. So $D = 2 \sum \text{observed} \log(\text{observed}/\text{fitted}) = G^2$. The saturated model has N parameters and the model one, giving df $= N - 1$. ■

Components of the deviance serve as residuals, and comparing deviances generalizes the analysis of variance (Section 4.5).

4.1.6 Advantages of the GLM formulation

GLMs give a unified theory for the most important continuous and discrete models. A single algorithm, iteratively reweighted least squares (Section 4.6), computes the ML estimates for every member of the exponential family and every link; this is the reason for restricting the random component to that family.

4.2 Generalized linear models for binary data

Motivation. For a binary response the mean is a probability $\pi(\mathbf{x}) = P(Y = 1)$ and $\text{Var}(Y) = \pi(\mathbf{x})[1 - \pi(\mathbf{x})]$. Which function of $\pi(\mathbf{x})$ should be linear in $\mathbf{x}$? The answer (identity, logit, probit) determines how effects are interpreted.

We use a single predictor x for simplicity.

4.2.1 Linear probability model

The *linear probability model* $\pi(x) = \alpha + \beta x$ is the binomial GLM with identity link. Its structural defect: probabilities lie in $[0, 1]$ while $\alpha + \beta x$ eventually leaves that range; fitting can fail when $\hat{\pi}(x)$ falls outside $[0, 1]$ for some observed x. Over a restricted range it can be adequate, and then β has the simple meaning "change in probability per unit of x".

It should be fitted by ML with the binomial distribution, not by least squares: since $\text{Var}(Y) = \pi(x)[1 - \pi(x)]$ depends on x, the constant-variance condition making least squares optimal fails, the binomial ML estimator is more efficient, and Y is far from normal. ML and least squares agree closely only when the fitted probabilities stay in about 0.3–0.7, where the variance is nearly constant.

4.2.2 Snoring and heart disease example

Example 4.2 (Snoring and heart disease)

An epidemiological survey of 2484 subjects classified snoring level (reported by the spouse) and heart disease. With scores $x = (0, 2, 4, 5)$ for the snoring levels (the last two treated as closer), the rows are independent binomial samples.

	Heart disease		Proportion	Linear	Logit
Snoring	Yes	No	yes	fit	fit
Never ($x = 0$)	24	1355	0.017	0.017	0.021
Occasionally ($x = 2$)	35	603	0.055	0.057	0.044
Nearly every night ($x = 4$)	21	192	0.099	0.096	0.093
Every night ($x = 5$)	30	224	0.118	0.116	0.132

The ML fit of the linear probability model is $\hat{\pi}(x) = 0.0172 + 0.0198x$ ($\widehat{\text{se}}(\hat{\beta}) = 0.0028$). The estimated risk is about 0.02 for nonsnorers and rises by $2(0.0198) \approx 0.04$ for occasional snorers, another 0.04 for those snoring nearly every night and another 0.02 for those snoring every night. The ML estimates are the same whether the data are entered as 2484 binary observations or as the four binomial totals. The logistic fit (Section 4.2.3) is $\text{logit}\,\hat{\pi}(x) = -3.87 + 0.40x$; over this narrow range of probabilities the two fits are similar (Figure 5).

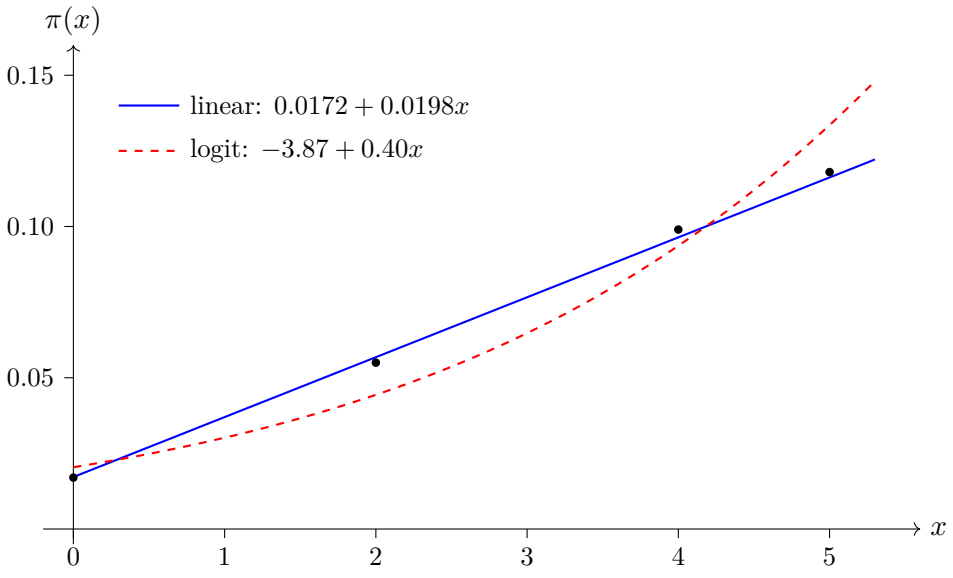


Figure 5: Snoring data: sample proportions (dots) with the linear probability and logistic regression fits.

4.2.3 Logistic regression model

Usually $\pi(x)$ is nonlinear in x: a fixed change in x matters less when $\pi(x)$ is near 0 or 1 than near 0.5. Monotone S-shaped curves are typical, the most important being

$$\pi(x) = \frac{\exp(\alpha + \beta x)}{1 + \exp(\alpha + \beta x)}. \tag{4.4}$$

As $x \to \infty$, $\pi(x) \downarrow 0$ if $\beta < 0$ and $\pi(x) \uparrow 1$ if $\beta > 0$.

Proposition 4.3 (Logistic regression is the binomial GLM with logit link)

Model (4.4) holds if and only if

$$\log \frac{\pi(x)}{1-\pi(x)} = \alpha + \beta x.$$

Hence logistic regression is the binomial GLM with the canonical (logit) link.

Proof. From (4.4), $1-\pi(x) = 1/[1+\exp(\alpha+\beta x)]$, so the odds are $\pi(x)/[1-\pi(x)] = \exp(\alpha+\beta x)$; take logs. Conversely, solving $\log[\pi/(1-\pi)] = \eta$ for π gives $\pi = e^{\eta}/(1+e^{\eta})$. The logit is the binomial natural parameter (Section 4.1.2). ■

The logit ranges over $\mathbb{R}$, matching the range of $\alpha+\beta x$, so logistic regression does not have the structural problem of the linear probability model. For the snoring data $\hat{\beta} = 0.40 > 0$ reflects increasing risk with snoring.

4.2.4 Binomial GLM for 2×2 contingency tables

Proposition 4.4 (Effect parameters for a binary predictor)

Let $x \in \{0,1\}$ and $\text{link}[\pi(x)] = \alpha + \beta x$. Then $\beta = \text{link}[\pi(1)] - \text{link}[\pi(0)]$; in particular
(a) identity link: $\beta = \pi(1) - \pi(0)$, the difference of proportions;
(b) log link: $\beta = \log[\pi(1)/\pi(0)]$, the log relative risk;
(c) logit link: $\beta = \log \dfrac{\pi(1)/[1-\pi(1)]}{\pi(0)/[1-\pi(0)]}$, the log odds ratio.

Proof. Subtract $\text{link}[\pi(0)] = \alpha$ from $\text{link}[\pi(1)] = \alpha + \beta$ and specialize the link; for (c), $\text{logit}\,\pi(1) - \text{logit}\,\pi(0)$ is a difference of log odds, i.e. a log odds ratio. ■

Thus the association measures of Chapter 2 are effect parameters of binary-response GLMs.

4.2.5 Probit and inverse cdf link functions

A monotone increasing curve has the shape of a cdf, suggesting $\pi(x) = F(x)$ for a continuous cdf F. Taking a standard cdf Φ of a location–scale class and writing

$$\pi(x) = \Phi(\alpha+\beta x) \quad \Longleftrightarrow \quad \Phi^{-1}[\pi(x)] = \alpha + \beta x \tag{4.5}$$

gives all members of the class: β sets the rate of increase (or decrease when $\beta < 0$) and α shifts the curve. The link is Φ^{-1}, mapping $(0,1)$ onto $(-\infty,\infty)$. With Φ the standard normal cdf, (4.5) is the *probit model* (Section 6.6).

The logistic distribution with location μ and scale $\tau > 0$ has cdf

$$F(x) = \frac{\exp[(x-\mu)/\tau]}{1+\exp[(x-\mu)/\tau]}, \qquad -\infty < x < \infty;$$

its density is symmetric and bell-shaped with standard deviation $\tau\pi/\sqrt{3}$, slightly thicker-tailed than the normal (kurtosis like a t_9). For the standard logistic cdf $\Phi(x) = e^x/(1+e^x)$ we have $\Phi^{-1}(\pi) = \log[\pi/(1-\pi)]$, so the logit link is the inverse cdf link of the logistic distribution, and (4.4) with $\beta > 0$ is a logistic cdf.

4.3 Generalized linear models for counts

Motivation. Counts are nonnegative and their variance typically grows with the mean. The Poisson GLM with log link is the basic model; overdispersion leads to the negative binomial and, later, to quasi-likelihood.

4.3.1 Poisson loglinear models

The log mean is the Poisson natural parameter, so the log link is canonical. With one predictor,

$$\log \mu = \alpha + \beta x, \qquad \mu = \exp(\alpha + \beta x) = e^{\alpha}(e^{\beta})^{x}. \tag{4.6}$$

A one-unit increase in x multiplies the mean by e^{β}.

4.3.2 Horseshoe crab mating example

Example 4.5 (Horseshoe crab satellites)

For 173 nesting female horseshoe crabs the response is her number of *satellites* (additional males residing nearby); predictors are color (1 light medium, 2 medium, 3 dark medium, 4 dark), spine condition, weight (kg) and carapace width (cm). Mean width is 26.3 cm (sd 2.1). Grouping width into eight categories gives:

Width (cm)	Cases	Satellites	Sample mean	Sample variance
< 23.25	14	14	1.00	2.77
23.25–24.25	14	20	1.43	8.88
24.25–25.25	28	67	2.39	6.54
25.25–26.25	39	105	2.69	11.38
26.25–27.25	22	63	2.86	6.88
27.25–28.25	24	93	3.87	8.81
28.25–29.25	18	71	3.94	16.88
> 29.25	14	72	5.14	8.29

The means increase roughly linearly with width. Using the ungrouped data and $x =$ width:
(a) Log link: $\log \hat{\mu} = -3.305 + 0.164x$ ($\widehat{\text{se}} = 0.020$). At $x = 26.3$, $\hat{\mu} = \exp(-3.305 + 0.164 \cdot 26.3) = 2.74$; $e^{0.164} = 1.18$, so a 1-cm increase raises the estimated mean by 18% (at $x = 27.3$, $\hat{\mu} = 3.23 = 1.18 \times 2.74$).
(b) Identity link: $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\text{se}} = 0.06$). On average about 2 cm more width brings one more satellite.
The two fits diverge at extreme widths but agree where most data lie.

4.3.3 Overdispersion for Poisson GLMs

In the table above the sample variances far exceed the means, whereas a Poisson has equal mean and variance. Greater variability than the random component predicts is *overdispersion*. A common cause is unobserved heterogeneity.

Proposition 4.6 (Mixtures of Poissons are overdispersed)

If $Y \mid \lambda \sim \text{Poisson}(\lambda)$ and λ is random with $E(\lambda) = \mu$, then $E(Y) = \mu$ and $\text{Var}(Y) = \mu + \text{Var}(\lambda) \geq \mu$, with equality only when λ is degenerate.

Proof. [Supplement] By iterated expectations $E(Y) = E[E(Y \mid \lambda)] = E(\lambda) = \mu$, and by the law of total variance

$$\mathrm{Var}(Y) = E[\mathrm{Var}(Y \mid \lambda)] + \mathrm{Var}[E(Y \mid \lambda)] = E(\lambda) + \mathrm{Var}(\lambda) = \mu + \mathrm{Var}(\lambda).$$

■

If, say, Y is Poisson given width, weight, color and spine, then crabs of a given width form a mixture over the other variables, and Y given width alone is overdispersed. When the mean model is right but the distribution is not Poisson, the ML estimates remain consistent but the standard errors are wrong. Overdispersion is not an issue for normal regression, which has a separate variance parameter.

4.3.4 Negative binomial GLMs

The negative binomial mass function is

$$f(y; k, \mu) = \frac{\Gamma(y+k)}{\Gamma(k)\Gamma(y+1)}\Big(\frac{k}{\mu+k}\Big)^k\Big(1-\frac{k}{\mu+k}\Big)^y, \qquad y = 0, 1, 2, \ldots \tag{4.7}$$

Proposition 4.7 (Negative binomial as a gamma mixture)

If $Y \mid \lambda \sim \mathrm{Poisson}(\lambda)$ and λ has the gamma density with shape k and mean μ, then Y has the mass function (4.7), with

$$E(Y) = \mu, \qquad \mathrm{Var}(Y) = \mu + \mu^2/k.$$

For fixed k, (4.7) is a natural exponential family with natural parameter $\log[\mu/(\mu+k)]$, so negative binomial regression is a GLM. As $k^{-1} \to 0$ it converges to the Poisson.

Proof. [Supplement]

Step 1. The gamma density is $h(\lambda) = (k/\mu)^k \lambda^{k-1} e^{-k\lambda/\mu}/\Gamma(k)$, $\lambda > 0$. Then

$$\begin{aligned} P(Y=y) &= \int_0^\infty \frac{e^{-\lambda}\lambda^y}{y!}\, h(\lambda)\, d\lambda = \frac{(k/\mu)^k}{\Gamma(k)\, y!}\int_0^\infty \lambda^{y+k-1} e^{-\lambda(1+k/\mu)} d\lambda \\ &= \frac{\Gamma(y+k)}{\Gamma(k)\, y!}\Big(\frac{k}{\mu}\Big)^k\Big(\frac{\mu}{\mu+k}\Big)^{y+k} = \frac{\Gamma(y+k)}{\Gamma(k)\, y!}\Big(\frac{k}{\mu+k}\Big)^k\Big(\frac{\mu}{\mu+k}\Big)^y, \end{aligned}$$

which is (4.7) since $1 - k/(\mu+k) = \mu/(\mu+k)$.

Step 2. The gamma law has mean μ and variance μ^2/k, so Proposition 4.6 gives $E(Y) = \mu$ and $\mathrm{Var}(Y) = \mu + \mu^2/k$.

Step 3. With $Q = \log[\mu/(\mu+k)]$, (4.7) reads $a\, b(y) \exp(yQ)$ with $a = (1-e^Q)^k$ and $b(y) = \Gamma(y+k)/[\Gamma(k)y!]$, the form (4.1).

Step 4. As $k \to \infty$, $\Gamma(y+k)/[\Gamma(k)(\mu+k)^y] \to 1$ and $(1+\mu/k)^{-k} \to e^{-\mu}$, so $f(y) \to e^{-\mu}\mu^y/y!$. ■

The index k^{-1} is a *dispersion parameter*, usually unknown and common to all observations; log and identity links are both used (Section 13.4).

Example 4.8 (Crab satellites: negative binomial fit)

With identity link and width predictor, the Poisson GLM gives $\hat{\mu} = -11.53 + 0.55x$ ($\widehat{\mathrm{se}} = 0.06$) and the negative binomial GLM gives $\hat{\mu} = -11.15 + 0.53x$ ($\widehat{\mathrm{se}} = 0.11$) with $\hat{k}^{-1} = 0.98$, so the estimated variance is about $\hat{\mu} + \hat{\mu}^2$ rather than $\hat{\mu}$. Fitted values are similar, but the larger

standard error reflects the overdispersion the Poisson model ignores.

4.3.5 Poisson regression for rates

When events occur over time, space or population of varying size t_i, the rate μ_i/t_i is more relevant than the count (e.g. homicides per city population). The loglinear model $\log(\mu_i/t_i) = \sum_j \beta_j x_{ij}$ is $\log \mu_i = \log t_i + \sum_j \beta_j x_{ij}$, a Poisson GLM with the known *offset* $\log t_i$ (Section 9.7).

4.3.6 Poisson GLM of independence in $I \times J$ contingency tables

Proposition 4.9 (Poisson form of independence)

Let Y_{ij} be independent Poisson with means $\mu_{ij} = \mu\alpha_i\beta_j$, where $\alpha_i, \beta_j > 0$ and $\sum_i \alpha_i = \sum_j \beta_j = 1$; equivalently

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \qquad \lambda = \log \mu,\ \lambda_i^X = \log \alpha_i,\ \lambda_j^Y = \log \beta_j. \tag{4.8}$$

Conditional on $n = \sum_{i,j} Y_{ij}$, the cell counts are multinomial with probabilities $\pi_{ij} = \alpha_i\beta_j$, which satisfy $\pi_{i+} = \alpha_i$, $\pi_{+j} = \beta_j$ and $\pi_{ij} = \pi_{i+}\pi_{+j}$.

Proof.

Step 1. A sum of independent Poissons is Poisson, so $n \sim \text{Poisson}(\sum \mu_{ij}) = \text{Poisson}(\mu)$.

Step 2. For counts $\{n_{ij}\}$ with total n,

$$P(\{Y_{ij} = n_{ij}\} \mid n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}}/n_{ij}!}{e^{-\mu}\mu^n/n!} = \frac{n!}{\prod_{ij} n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu}\Big)^{n_{ij}},$$

a multinomial with $\pi_{ij} = \mu_{ij}/\mu = \alpha_i\beta_j$.

Step 3. Summing, $\pi_{i+} = \alpha_i \sum_j \beta_j = \alpha_i$ and $\pi_{+j} = \beta_j$, so $\pi_{ij} = \pi_{i+}\pi_{+j}$. ■

So the Chapter 3 inferences about independence are inferences for the loglinear model (4.8) or the corresponding multinomial model; Chapters 8–9 develop richer loglinear models.

4.4 Moments and likelihood for generalized linear models

Motivation. All GLM computations rest on a few facts about the exponential dispersion family: the mean and variance come from derivatives of one function $b(\theta)$, and the likelihood equations and the information matrix depend on the distribution only through the mean and variance.

Definition (Exponential dispersion family)

The observations $y_1, \ldots, y_N$ are independent with density or mass function

$$f(y_i; \theta_i, \phi) = \exp\Big\{\frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi)\Big\}. \tag{4.9}$$

θ_i is the *natural parameter* and ϕ the *dispersion parameter*. Usually $a(\phi) = \phi/\omega_i$ with a known weight ω_i (e.g. $\omega_i = n_i$ when y_i is a mean of n_i readings).

For known ϕ, (4.9) is the natural exponential family (4.1) with $Q(\theta) = \theta/a(\phi)$, $a(\theta) = \exp[-b(\theta)/a(\phi)]$ and $b(y) = \exp[c(y, \phi)]$.

4.4.1 Mean and variance functions for the random component

Let $L_i = \log f(y_i; \theta_i, \phi)$, so the log likelihood is $L = \sum_i L_i$ and

$$L_i = \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + c(y_i, \phi). \tag{4.10}$$

Theorem 4.10 (Moments of the exponential dispersion family)

For (4.9), with θ_i interior to the natural parameter space,

$$\mu_i = E(Y_i) = b'(\theta_i), \qquad \mathrm{Var}(Y_i) = b''(\theta_i)\, a(\phi). \tag{4.11}$$

Proof. [Supplement]

Step 1. (Bartlett identities.) Write $f = f(y;\theta)$ and $\ell = \log f$. Because $\int f\, dy = 1$ for all θ and, for an exponential family with θ in the interior of the parameter space, differentiation under the integral sign is permitted (the support does not depend on θ and the integrand has locally dominated derivatives),

$$0 = \frac{d}{d\theta}\int f\, dy = \int \frac{\partial \ell}{\partial \theta} f\, dy = E\Big(\frac{\partial \ell}{\partial \theta}\Big).$$

Differentiating once more, using $\partial f/\partial\theta = f\,\partial\ell/\partial\theta$,

$$0 = \int \Big[\frac{\partial^2 \ell}{\partial \theta^2} + \Big(\frac{\partial \ell}{\partial \theta}\Big)^2\Big] f\, dy, \quad \text{i.e.} \quad -E\Big(\frac{\partial^2 \ell}{\partial \theta^2}\Big) = E\Big(\frac{\partial \ell}{\partial \theta}\Big)^2.$$

(For discrete Y replace integrals by sums.)

Step 2. From (4.10),

$$\frac{\partial L_i}{\partial \theta_i} = \frac{y_i - b'(\theta_i)}{a(\phi)}, \qquad \frac{\partial^2 L_i}{\partial \theta_i^2} = -\frac{b''(\theta_i)}{a(\phi)}.$$

Step 3. The first identity gives $E[Y_i - b'(\theta_i)]/a(\phi) = 0$, i.e. $\mu_i = b'(\theta_i)$.

Step 4. The second identity gives

$$\frac{b''(\theta_i)}{a(\phi)} = E\Big[\frac{(Y_i - b'(\theta_i))^2}{a(\phi)^2}\Big] = \frac{\mathrm{Var}(Y_i)}{a(\phi)^2},$$

hence $\mathrm{Var}(Y_i) = b''(\theta_i)a(\phi)$. ∎

Since $\mathrm{Var}(Y_i) > 0$, $b'' > 0$: b' is strictly increasing, so θ_i and μ_i determine each other. The function b is the *cumulant function*.

4.4.2 Mean and variance functions for Poisson and binomial

Proposition 4.11 (Poisson and binomial in dispersion form)

(a) Poisson: $\theta_i = \log \mu_i$, $b(\theta_i) = e^{\theta_i}$, $a(\phi) = 1$, $c = -\log y_i!$; then $E(Y_i) = \mathrm{Var}(Y_i) = \mu_i$.
(b) Binomial proportion: if $n_i Y_i \sim \mathrm{bin}(n_i, \pi_i)$, then $\theta_i = \log[\pi_i/(1-\pi_i)]$, $b(\theta_i) = \log(1 + e^{\theta_i})$, $a(\phi) = 1/n_i$, $c = \log \binom{n_i}{n_i y_i}$; then $E(Y_i) = \pi_i$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$.

Proof.

(a) $f(y_i;\mu_i) = e^{-\mu_i}\mu_i^{y_i}/y_i! = \exp[y_i\theta_i - e^{\theta_i} - \log y_i!]$. By Theorem 4.10, $E(Y_i) = b'(\theta_i) = e^{\theta_i} = \mu_i$ and $\mathrm{Var}(Y_i) = b''(\theta_i) = \mu_i$.

(b) With $\pi_i = e^{\theta_i}/(1+e^{\theta_i})$ we have $\log(1-\pi_i) = -\log(1+e^{\theta_i})$, so

$$f(y_i;\pi_i,n_i) = \binom{n_i}{n_i y_i}\pi_i^{n_i y_i}(1-\pi_i)^{n_i - n_i y_i} = \binom{n_i}{n_i y_i}\exp\{n_i[y_i\theta_i + \log(1-\pi_i)]\}$$
$$= \exp\Big[\frac{y_i\theta_i - \log(1+e^{\theta_i})}{1/n_i} + \log\binom{n_i}{n_i y_i}\Big].$$

Then $b'(\theta_i) = e^{\theta_i}/(1+e^{\theta_i}) = \pi_i$ and $b''(\theta_i)a(\phi) = \dfrac{e^{\theta_i}}{(1+e^{\theta_i})^2}\cdot\dfrac{1}{n_i} = \dfrac{\pi_i(1-\pi_i)}{n_i}$. ■

Normal case

For $N(\mu_i,\sigma^2)$: $\theta_i = \mu_i$, $b(\theta) = \theta^2/2$, $a(\phi) = \phi = \sigma^2$, giving $E(Y_i) = \mu_i$ and $\mathrm{Var}(Y_i) = \sigma^2$.

4.4.3 Systematic component and link function

In matrix form the linear predictor is $\boldsymbol{\eta} = X\boldsymbol{\beta}$, with $\boldsymbol{\eta} = (\eta_1,\ldots,\eta_N)^\top$, $\boldsymbol{\beta} = (\beta_1,\ldots,\beta_p)^\top$ and X the $N \times p$ *model matrix*; the link gives $\eta_i = g(\mu_i)$.

Proposition 4.12 (Canonical link)

The canonical link (the g with $g(\mu_i) = \theta_i$) is $g = (b')^{-1}$. For the Poisson it is the log link; for the binomial it is the logit link.

Proof. By Theorem 4.10, $\mu_i = b'(\theta_i)$ with b' strictly increasing, so $\theta_i = (b')^{-1}(\mu_i)$. Poisson: $b'(\theta) = e^\theta$, inverse log. Binomial: $b'(\theta) = e^\theta/(1+e^\theta)$, inverse logit. ■

With the canonical link, $\theta_i = \sum_j \beta_j x_{ij}$: the natural parameter itself is linear in the predictors.

4.4.4 Likelihood equations for a GLM

The log likelihood is

$$L(\boldsymbol{\beta}) = \sum_i L_i = \sum_i \frac{y_i\theta_i - b(\theta_i)}{a(\phi)} + \sum_i c(y_i,\phi), \tag{4.12}$$

which depends on $\boldsymbol{\beta}$ through $\theta_i = \theta_i(\mu_i)$, $\mu_i = g^{-1}(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$.

Theorem 4.13 (GLM likelihood equations)

For a GLM with exponential dispersion random component,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{(y_i-\mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}, \tag{4.13}$$

and the likelihood equations are

$$\sum_{i=1}^N \frac{(y_i-\mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i} = 0, \qquad j = 1,\ldots,p. \tag{4.14}$$

Proof.

Step 1. By the chain rule,

$$\frac{\partial L_i}{\partial \beta_j} = \frac{\partial L_i}{\partial \theta_i}\frac{\partial \theta_i}{\partial \mu_i}\frac{\partial \mu_i}{\partial \eta_i}\frac{\partial \eta_i}{\partial \beta_j}.$$

Step 2. From the proof of Theorem 4.10, $\partial L_i/\partial\theta_i = [y_i - b'(\theta_i)]/a(\phi) = (y_i-\mu_i)/a(\phi)$.

Step 3. Since $\mu_i = b'(\theta_i)$, $\partial\mu_i/\partial\theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$, so $\partial\theta_i/\partial\mu_i = a(\phi)/\mathrm{Var}(Y_i)$.

Step 4. $\partial\eta_i/\partial\beta_j = x_{ij}$, and $\partial\mu_i/\partial\eta_i = 1/g'(\mu_i)$ depends on the link.

Step 5. Multiplying, $a(\phi)$ cancels and (4.13) follows; summing over i and setting to zero gives (4.14). ∎

Although $\boldsymbol{\beta}$ does not appear explicitly, it enters through $\mu_i = g^{-1}(\sum_j \beta_j x_{ij})$. Crucially, (4.14) depends on the distribution only through μ_i and the *variance function* $\mathrm{Var}(Y_i) = v(\mu_i)$, e.g. $v(\mu)=\mu$ (Poisson), $v(\mu)=\mu(1-\mu)$ (Bernoulli), $v(\mu)=\sigma^2$ (normal). Within the natural exponential family, the mean–variance relation characterizes the distribution (Jorgensen 1987): $v(\mu)=\mu$ forces the Poisson.

4.4.5 Likelihood equations for binomial GLMs

Corollary 4.14 (Binomial likelihood equations)

Let $n_iY_i \sim \mathrm{bin}(n_i,\pi_i)$ and $\pi_i = \Phi(\eta_i)$, $\eta_i = \sum_j \beta_j x_{ij}$, for a continuous cdf Φ with density ϕ. The likelihood equations are

$$\sum_i \frac{n_i(y_i-\pi_i)\,x_{ij}}{\pi_i(1-\pi_i)}\,\phi(\eta_i) = 0, \qquad j = 1,\dots,p. \tag{4.15}$$

For the logit link they reduce to

$$\sum_i n_i(y_i-\pi_i)x_{ij} = 0, \qquad j = 1,\dots,p. \tag{4.16}$$

Proof. Here $\partial\pi_i/\partial\eta_i = \phi(\eta_i)$ and $\mathrm{Var}(Y_i) = \pi_i(1-\pi_i)/n_i$; substitute in (4.14). For the logit, $\eta_i = \log[\pi_i/(1-\pi_i)]$ gives $\partial\eta_i/\partial\pi_i = 1/[\pi_i(1-\pi_i)]$, so $\phi(\eta_i) = \pi_i(1-\pi_i)$ cancels the denominator. ∎

4.4.6 Asymptotic covariance matrix of model parameter estimators

Theorem 4.15 (Information matrix of a GLM)

The expected information matrix $\mathcal{I}$ with entries $E[-\partial^2 L(\boldsymbol{\beta})/\partial\beta_h\partial\beta_j]$ is

$$\mathcal{I} = X^\top W X, \qquad W = \mathrm{diag}(w_1,\dots,w_N), \quad w_i = \frac{(\partial\mu_i/\partial\eta_i)^2}{\mathrm{Var}(Y_i)}. \tag{4.17}$$

Proof.

Step 1. By the second Bartlett identity (proof of Theorem 4.10, applied to $\boldsymbol{\beta}$; valid under the same regularity),

$$E\Big(\frac{\partial^2 L_i}{\partial\beta_h\partial\beta_j}\Big) = -E\Big(\frac{\partial L_i}{\partial\beta_h}\,\frac{\partial L_i}{\partial\beta_j}\Big).$$

Step 2. By (4.13),

$$E\Big(\frac{\partial L_i}{\partial\beta_h}\frac{\partial L_i}{\partial\beta_j}\Big) = \frac{E(Y_i-\mu_i)^2\,x_{ih}x_{ij}}{\mathrm{Var}(Y_i)^2}\Big(\frac{\partial\mu_i}{\partial\eta_i}\Big)^2 = \frac{x_{ih}x_{ij}}{\mathrm{Var}(Y_i)}\Big(\frac{\partial\mu_i}{\partial\eta_i}\Big)^2.$$

Step 3. Summing over the independent observations, $E[-\partial^2 L/\partial\beta_h\partial\beta_j] = \sum_i x_{ih}w_i x_{ij}$, the (h,j) entry of $X^\top W X$. ∎

The estimated asymptotic covariance matrix of $\hat{\boldsymbol{\beta}}$ is therefore

$$\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = \hat{\mathcal{I}}^{-1} = (X^\top \hat{W} X)^{-1}, \tag{4.18}$$

with $\hat{W}$ evaluated at $\hat{\boldsymbol{\beta}}$; W depends on the link. The justification is the following theorem.

Theorem 4.16 (Asymptotic normality of the ML estimator)

Let $\boldsymbol{\beta}_0$ be the true value, interior to the parameter space, and write $\boldsymbol{u}(\boldsymbol{\beta}) = \partial L/\partial \boldsymbol{\beta}$, $H(\boldsymbol{\beta}) = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$, $\mathcal{I}_N(\boldsymbol{\beta}) = X^\top W(\boldsymbol{\beta}) X$. Assume:
(R1) $N^{-1}\mathcal{I}_N(\boldsymbol{\beta}_0) \to \Sigma$, positive definite;
(R2) Lindeberg condition: for each $\boldsymbol{c} \in \mathbb{R}^p$, the terms $\boldsymbol{c}^\top \partial L_i/\partial\boldsymbol{\beta}$ $(i \le N)$, normalized by $(N\boldsymbol{c}^\top\Sigma\boldsymbol{c})^{1/2}$, satisfy Lindeberg's condition;
(R3) $\sup_{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le\delta} \|N^{-1}H(\boldsymbol{\beta}) + \Sigma\| \xrightarrow{p} 0$ as $N \to \infty$ and then $\delta \to 0$.
Then with probability tending to one the likelihood equations have a root $\hat{\boldsymbol{\beta}}$ with $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$, and

$$\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma^{-1}), \qquad \text{i.e.} \qquad \hat{\boldsymbol{\beta}} \approx N_p(\boldsymbol{\beta}_0, (X^\top W X)^{-1}).$$

Proof. [Supplement]

Step 1. (Score CLT.) The summands $\partial L_i/\partial\boldsymbol{\beta}$ are independent with mean $\mathbf{0}$ (first Bartlett identity) and covariance summing to $\mathcal{I}_N(\boldsymbol{\beta}_0)$ (Theorem 4.15). By (R1), (R2), the Lindeberg–Feller CLT and the Cramér–Wold device, $N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} N_p(\mathbf{0}, \Sigma)$. In particular $N^{-1}\boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{p} \mathbf{0}$.

Step 2. (Existence of a consistent root.) Let $\lambda > 0$ be the smallest eigenvalue of Σ. Fix small $\delta > 0$ so that, by (R3), with probability $\to 1$, $-N^{-1}H(\boldsymbol{\beta})$ has all eigenvalues $\ge \lambda/2$ on the ball $B_\delta = \{\|\boldsymbol{\beta} - \boldsymbol{\beta}_0\| \le \delta\}$. For $\boldsymbol{\beta}$ on the boundary of B_δ, Taylor's theorem with some $\bar{\boldsymbol{\beta}}$ on the segment gives

$$L(\boldsymbol{\beta}) - L(\boldsymbol{\beta}_0) = \boldsymbol{u}(\boldsymbol{\beta}_0)^\top(\boldsymbol{\beta} - \boldsymbol{\beta}_0) + \tfrac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}_0)^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta} - \boldsymbol{\beta}_0) \le N\delta\Big(\frac{\|\boldsymbol{u}(\boldsymbol{\beta}_0)\|}{N} - \frac{\lambda\delta}{4}\Big).$$

By Step 1 the right side is negative with probability $\to 1$. Then the maximum of the continuous L over the compact B_δ is attained in the interior, where $\boldsymbol{u} = \mathbf{0}$. Hence a root $\hat{\boldsymbol{\beta}} \in B_\delta$ exists with probability $\to 1$; as δ is arbitrary, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$.

Step 3. (Expansion.) Applying the mean value theorem to each coordinate of $\boldsymbol{u}$ along the segment from $\boldsymbol{\beta}_0$ to $\hat{\boldsymbol{\beta}}$,

$$\mathbf{0} = \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \boldsymbol{u}(\boldsymbol{\beta}_0) + \tilde{H}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0),$$

where row j of $\tilde{H}$ is row j of H at an intermediate point. By consistency and (R3), $-N^{-1}\tilde{H} \xrightarrow{p} \Sigma$.

Step 4. Therefore $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = (-N^{-1}\tilde{H})^{-1}N^{-1/2}\boldsymbol{u}(\boldsymbol{\beta}_0)$, and Slutsky's theorem with Step 1 gives the limit $N_p(\mathbf{0}, \Sigma^{-1}\Sigma\Sigma^{-1}) = N_p(\mathbf{0}, \Sigma^{-1})$. Finally $N^{-1}X^\top\hat{W}X \xrightarrow{p} \Sigma$ by continuity, which justifies (4.18). ∎

For grouped binomial or Poisson data the same argument applies with N fixed and the counts $n_i \to \infty$ (normalize by $n = \sum n_i$ instead of N).

4.4.7 Likelihood equations and covariance matrix for Poisson loglinear model

Corollary 4.17 (Poisson loglinear model)

For $\log \boldsymbol{\mu} = X\boldsymbol{\beta}$ with Poisson responses, the likelihood equations are

$$\sum_i (y_i - \mu_i) x_{ij} = 0, \qquad j = 1, \dots, p, \tag{4.19}$$

and $\widehat{\mathrm{Cov}}(\hat{\boldsymbol{\beta}}) = (X^\top \hat{W} X)^{-1}$ with $\hat{W} = \mathrm{diag}(\hat{\mu}_1, \dots, \hat{\mu}_N)$.

Proof. For the log link $\mu_i = e^{\eta_i}$, so $\partial \mu_i / \partial \eta_i = \mu_i = \mathrm{Var}(Y_i)$. Then (4.14) becomes (4.19), and $w_i = \mu_i^2/\mu_i = \mu_i$ in (4.17). ■

Equations (4.19) equate the sufficient statistics $\sum_i y_i x_{ij}$ to their expected values (see Theorem 4.27).

4.5 Inference for generalized linear models

Motivation. Once a GLM is fitted, the Wald, score and likelihood-ratio methods apply to any parameter. Likelihood-ratio inference is conveniently organized through the deviance, which plays the role of the residual sum of squares.

Theorem 4.18 (Wald, likelihood-ratio and score tests)

Under the conditions of Theorem 4.16, for $H_0 : \boldsymbol{\beta} = \boldsymbol{\beta}_0$ each of

$$\begin{aligned} W &= (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)^\top \hat{\mathcal{I}}\, (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) && \text{(Wald)}, \\ \Lambda &= -2\big[L(\boldsymbol{\beta}_0) - L(\hat{\boldsymbol{\beta}})\big] && \text{(likelihood ratio)}, \\ S &= \boldsymbol{u}(\boldsymbol{\beta}_0)^\top \mathcal{I}(\boldsymbol{\beta}_0)^{-1} \boldsymbol{u}(\boldsymbol{\beta}_0) && \text{(score)} \end{aligned}$$

converges in distribution to χ^2_p, and the three differ by $o_p(1)$. For nested models $M_0 \subset M_1$, where M_0 imposes q linear restrictions on $\boldsymbol{\beta}$, $\Lambda = -2[L(\hat{\boldsymbol{\beta}}_0) - L(\hat{\boldsymbol{\beta}}_1)] \xrightarrow{d} \chi^2_q$.

Proof. [Supplement] Let $\boldsymbol{Z}_N = N^{-1/2} \boldsymbol{u}(\boldsymbol{\beta}_0) \xrightarrow{d} \boldsymbol{Z} \sim N_p(\mathbf{0}, \Sigma)$ and, from Steps 3–4 of Theorem 4.16, $\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 1. (Score.) $S = \boldsymbol{Z}_N^\top (N^{-1}\mathcal{I}_N)^{-1} \boldsymbol{Z}_N \xrightarrow{d} \boldsymbol{Z}^\top \Sigma^{-1} \boldsymbol{Z} \sim \chi^2_p$, since $\Sigma^{-1/2}\boldsymbol{Z} \sim N_p(\mathbf{0}, I)$.

Step 2. (Wald.) $W = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (N^{-1}\hat{\mathcal{I}}) [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$.

Step 3. (LR.) Expand L about $\hat{\boldsymbol{\beta}}$, where $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$:

$$L(\boldsymbol{\beta}_0) = L(\hat{\boldsymbol{\beta}}) + \tfrac{1}{2}(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}})^\top H(\bar{\boldsymbol{\beta}})(\boldsymbol{\beta}_0 - \hat{\boldsymbol{\beta}}),$$

so $\Lambda = [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)]^\top (-N^{-1} H(\bar{\boldsymbol{\beta}})) [\sqrt{N}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0)] = \boldsymbol{Z}_N^\top \Sigma^{-1} \boldsymbol{Z}_N + o_p(1)$ by (R3).

Step 4. (Nested models.) Write M_0 as $\boldsymbol{\beta} = \boldsymbol{\beta}_0 + A\boldsymbol{\gamma}$ with A a $p \times (p-q)$ matrix of full rank. Applying Step 3 to M_1 and to the reparametrized model M_0 (whose score is $A^\top \boldsymbol{u}$ and information $A^\top \mathcal{I} A$),

$$\Lambda = \boldsymbol{Z}_N^\top \big[\Sigma^{-1} - A(A^\top \Sigma A)^{-1} A^\top\big] \boldsymbol{Z}_N + o_p(1).$$

With $\boldsymbol{Z} = \Sigma^{1/2}\boldsymbol{\varepsilon}$, $\boldsymbol{\varepsilon} \sim N_p(\mathbf{0}, I)$, the limit is $\boldsymbol{\varepsilon}^\top (I - P)\boldsymbol{\varepsilon}$, where $P = \Sigma^{1/2} A (A^\top \Sigma A)^{-1} A^\top \Sigma^{1/2}$ is a symmetric idempotent matrix of rank $p-q$. Thus $I-P$ is a projection of rank q, and $\boldsymbol{\varepsilon}^\top (I-P)\boldsymbol{\varepsilon} \sim \chi^2_q$. ■

Confidence regions follow by inverting these tests; e.g. the Wald interval $\hat{\beta}_j \pm z_{\alpha/2} \widehat{\mathrm{se}}(\hat{\beta}_j)$ and the profile-likelihood interval $\{\beta_j : \Lambda(\beta_j) \le \chi^2_1(\alpha)\}$.

4.5.1 Deviance and goodness of fit

Let $\tilde{\theta}_i$ be the natural parameter of the saturated model ($\tilde{\mu}_i = y_i$) and $\hat{\theta}_i$ that of the fitted model. By (4.12) with $a(\phi) = \phi/\omega_i$,

$$-2\big[L(\hat{\boldsymbol{\mu}}; \mathbf{y}) - L(\mathbf{y}; \mathbf{y})\big] = 2\sum_i \frac{\omega_i\big[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\big]}{\phi}.$$

Definition (Scaled deviance and deviance)

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2\sum_i \omega_i\big[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)\big] \tag{4.20}$$

is the *deviance*, and $D(\mathbf{y}; \hat{\boldsymbol{\mu}})/\phi$ the *scaled deviance*. The larger it is, the poorer the fit. For Poisson and binomial models $\phi = 1$, and the two coincide.

Theorem 4.19 (Asymptotic distribution of the deviance)

For a binomial or Poisson GLM with a fixed number N of observations whose counts grow ($n_i \to \infty$, or $\mu_i \to \infty$), if the model with p parameters holds then $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) \xrightarrow{d} \chi^2_{N-p}$.

Proof. [Supplement] The saturated model is a GLM with N parameters (one per observation, θ_i free), and the working model is the nested submodel obtained by $N-p$ linear restrictions $\boldsymbol{\theta} \in \mathrm{col}(X)$ (canonical scale; for other links apply a smooth reparametrization). The deviance is $-2[L(\hat{\boldsymbol{\mu}}) - L(\tilde{\boldsymbol{\mu}})]$, the likelihood-ratio statistic for these nested models; Theorem 4.18 (with the growing counts playing the role of sample size) gives the χ^2_{N-p} limit. ■

Normal deviance

For the normal linear model ($b(\theta) = \theta^2/2$, $\omega_i = 1$), $D = \sum_i (y_i - \hat{\mu}_i)^2$, the residual sum of squares, and D/σ^2 is the scaled deviance.

4.5.2 Deviance for Poisson models

Proposition 4.20 (Poisson deviance)

For a Poisson GLM,

$$D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2\sum_i \big[y_i \log(y_i/\hat{\mu}_i) - y_i + \hat{\mu}_i\big]. \tag{4.21}$$

If the model has log link and an intercept, $\sum_i y_i = \sum_i \hat{\mu}_i$ and $D(\mathbf{y}; \hat{\boldsymbol{\mu}}) = 2\sum_i y_i \log(y_i/\hat{\mu}_i)$.

Proof. Here $\hat{\theta}_i = \log \hat{\mu}_i$, $b(\hat{\theta}_i) = \hat{\mu}_i$, $\tilde{\theta}_i = \log y_i$, $b(\tilde{\theta}_i) = y_i$, $\omega_i = 1$; (4.20) gives (4.21). With an intercept ($x_{i1} \equiv 1$), the first equation of (4.19) is $\sum_i (y_i - \hat{\mu}_i) = 0$, so the linear terms cancel. (Terms with $y_i = 0$ are 0, using $0 \log 0 = 0$.) ■

For a two-way table and the independence fit $\hat{\mu}_{ij}$, this is G^2 of Chapter 3. For a fixed number of cells and large means, the deviance is approximately chi-squared (Section 14.3).

4.5.3 Deviance for binomial models: grouped and ungrouped data

Proposition 4.21 (Binomial deviance)

For binomial proportions y_i based on n_i trials,

$$D(\mathbf{y};\hat{\boldsymbol{\mu}}) = 2\sum_i n_i y_i \log\frac{n_i y_i}{n_i\hat{\pi}_i} + 2\sum_i (n_i - n_i y_i)\log\frac{n_i - n_i y_i}{n_i - n_i\hat{\pi}_i}, \tag{4.22}$$

that is, $D = 2\sum \text{observed} \times \log(\text{observed/fitted})$ over the $2N$ cells of successes and failures.

Proof. Here $\hat{\theta}_i = \operatorname{logit}\hat{\pi}_i$, $b(\hat{\theta}_i) = -\log(1-\hat{\pi}_i)$, $\tilde{\theta}_i = \operatorname{logit} y_i$, $b(\tilde{\theta}_i) = -\log(1-y_i)$, $\omega_i = n_i$. By (4.20),

$$\begin{aligned} D &= 2\sum_i n_i\Big[y_i\log\frac{y_i}{1-y_i} - y_i\log\frac{\hat{\pi}_i}{1-\hat{\pi}_i} + \log(1-y_i) - \log(1-\hat{\pi}_i)\Big] \\ &= 2\sum_i n_i\Big[y_i\log\frac{y_i}{\hat{\pi}_i} + (1-y_i)\log\frac{1-y_i}{1-\hat{\pi}_i}\Big], \end{aligned}$$

which is (4.22) after multiplying numerator and denominator by n_i. ■

Grouped versus ungrouped data

Binomial data may be entered as success/failure counts at each predictor setting (*grouped*) or as individual 0/1 observations (*ungrouped*). The ML estimates agree, but the saturated model and hence the deviance differ. The chi-squared approximation for the deviance holds for grouped data with the number of settings fixed and $n_i \to \infty$, not for ungrouped data, where the number of saturated-model parameters grows with n.

4.5.4 Likelihood-ratio model comparison using the deviance

Proposition 4.22 (Deviance difference)

Let $M_0 \subset M_1$ be nested Poisson or binomial GLMs with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$. Then $D(\mathbf{y};\hat{\boldsymbol{\mu}}_1) \le D(\mathbf{y};\hat{\boldsymbol{\mu}}_0)$ and the likelihood-ratio statistic for $H_0 : M_0$ given M_1 is

$$-2\big[L(\hat{\boldsymbol{\mu}}_0;\mathbf{y}) - L(\hat{\boldsymbol{\mu}}_1;\mathbf{y})\big] = D(\mathbf{y};\hat{\boldsymbol{\mu}}_0) - D(\mathbf{y};\hat{\boldsymbol{\mu}}_1),$$

asymptotically χ^2 with df equal to the difference in numbers of parameters. With log link and intercept (Poisson) or for binomial data, it equals $2\sum \text{observed} \times \log(\text{fitted}_1/\text{fitted}_0)$.

Proof. Maximizing over the smaller parameter space cannot give a larger maximum, so $L(\hat{\boldsymbol{\mu}}_0) \le L(\hat{\boldsymbol{\mu}}_1)$. The identity follows by adding and subtracting $L(\mathbf{y};\mathbf{y})$, which is the same for both models. The distribution is Theorem 4.18. For the last form, subtract (4.22) (or (4.21) without linear terms) for the two models: the log observed terms cancel. ■

Since the saturated part cancels, the comparison does not depend on whether binomial data are grouped or ungrouped, and it can be valid even when each deviance alone is not approximately chi-squared.

4.5.5 Residuals for GLMs

Definition (Deviance and Pearson residuals)

Write $D(\mathbf{y};\hat{\boldsymbol{\mu}}) = \sum_i d_i$ with $d_i = 2\omega_i[y_i(\tilde{\theta}_i - \hat{\theta}_i) - b(\tilde{\theta}_i) + b(\hat{\theta}_i)]$. The *deviance residual* is $\sqrt{d_i}\,\operatorname{sgn}(y_i - \hat{\mu}_i)$, and the *Pearson residual* is

$$e_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)]^{1/2}} \qquad (\text{Poisson: } e_i = (y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i}).$$

For the independence model in a two-way table, $\sum e_{ij}^2 = X^2$ and $\sum d_{ij} = G^2$. These residuals are less variable than $N(0,1)$ because they compare y_i with $\hat{\mu}_i$ rather than μ_i. Standardizing uses the *hat matrix*

$$\mathrm{HatM} = W^{1/2}X(X^\top WX)^{-1}X^\top W^{1/2}. \tag{4.23}$$

Proposition 4.23 (Leverages and residual covariance)

(a) HatM is symmetric and idempotent with trace p; its diagonal elements (*leverages*) satisfy $0 \le h_i \le 1$ and $\sum_i h_i = p$.
(b) To first order, with $V = \operatorname{Cov}(\boldsymbol{Y}) = \operatorname{diag}\{\operatorname{Var}(Y_i)\}$,

$$\operatorname{Cov}(\boldsymbol{Y} - \hat{\boldsymbol{\mu}}) \approx V^{1/2}(I - \mathrm{HatM})V^{1/2}, \qquad \operatorname{Var}(Y_i - \hat{\mu}_i) \approx \operatorname{Var}(Y_i)(1 - h_i).$$

Hence the *standardized Pearson residual* is

$$r_i = \frac{y_i - \hat{\mu}_i}{[\widehat{\operatorname{Var}}(Y_i)(1-\hat{h}_i)]^{1/2}} = \frac{e_i}{\sqrt{1-\hat{h}_i}} \qquad (\text{Poisson: } \frac{y_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i(1-\hat{h}_i)}}). \tag{4.24}$$

Proof. [Supplement] (a) Let $A = W^{1/2}X$. Then $\mathrm{HatM} = A(A^\top A)^{-1}A^\top$ is symmetric with $\mathrm{HatM}^2 = \mathrm{HatM}$ and $\operatorname{tr}\mathrm{HatM} = \operatorname{tr}[(A^\top A)^{-1}A^\top A] = \operatorname{tr} I_p = p$. By symmetry and idempotence $h_i = \sum_j h_{ij}^2 \ge h_i^2$, so $0 \le h_i \le 1$.

(b)

Step 1. Let $G = \operatorname{diag}(\partial\eta_i/\partial\mu_i)$. Since $\partial\mu_i/\partial\eta_i = 1/G_{ii}$, (4.13) gives $\boldsymbol{u} = X^\top WG(\boldsymbol{y} - \boldsymbol{\mu})$.

Step 2. From Theorem 4.16, $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \approx (X^\top WX)^{-1}X^\top WG(\boldsymbol{y} - \boldsymbol{\mu})$, and $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx G^{-1}X(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$. Hence

$$\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx [I - G^{-1}X(X^\top WX)^{-1}X^\top WG](\boldsymbol{y} - \boldsymbol{\mu}).$$

Step 3. Because $w_i = 1/(G_{ii}^2 \operatorname{Var}(Y_i))$, we have $V = G^{-2}W^{-1}$, i.e. $V^{1/2} = G^{-1}W^{-1/2} =: M$. Then $G^{-1}X(X^\top WX)^{-1}X^\top WG = M\,\mathrm{HatM}\,M^{-1}$, so $\boldsymbol{y} - \hat{\boldsymbol{\mu}} \approx M(I - \mathrm{HatM})M^{-1}(\boldsymbol{y} - \boldsymbol{\mu})$.

Step 4. With $\operatorname{Cov}(\boldsymbol{y} - \boldsymbol{\mu}) = V = M^2$ and HatM symmetric idempotent,

$$\operatorname{Cov}(\boldsymbol{y} - \hat{\boldsymbol{\mu}}) \approx M(I - \mathrm{HatM})M^{-1}M^2M^{-1}(I - \mathrm{HatM})M = M(I - \mathrm{HatM})M.$$

Its ith diagonal entry is $\operatorname{Var}(Y_i)(1 - h_i)$. ∎

As in linear models, larger leverage means greater potential influence. Unlike ordinary regression, the hat values depend on the fit through W, so extreme predictor values need not have high leverage. Applied to a linearized $g(\mathbf{y})$, HatM yields (after rescaling) the fitted linear predictor $\hat{\boldsymbol{\eta}}$.

4.6 Fitting generalized linear models

Motivation. The likelihood equations (4.14) are nonlinear in $\boldsymbol{\beta}$. Newton–Raphson and Fisher scoring solve them iteratively; Fisher scoring turns out to be a sequence of weighted least squares fits, which is how all GLM software works.

4.6.1 Newton–Raphson method

Let $\boldsymbol{u} = \partial L/\partial \boldsymbol{\beta}$ and $H = \partial^2 L/\partial\boldsymbol{\beta}\partial\boldsymbol{\beta}^\top$ (the Hessian), and let $\boldsymbol{u}^{(t)}, H^{(t)}$ be their values at the guess $\boldsymbol{\beta}^{(t)}$. Near $\boldsymbol{\beta}^{(t)}$,

$$L(\boldsymbol{\beta}) \approx L(\boldsymbol{\beta}^{(t)}) + \boldsymbol{u}^{(t)\top}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)}) + \tfrac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)})^\top H^{(t)}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)}).$$

Setting the gradient of this quadratic, $\boldsymbol{u}^{(t)} + H^{(t)}(\boldsymbol{\beta} - \boldsymbol{\beta}^{(t)})$, to $\mathbf{0}$ gives the *Newton–Raphson* step

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} - (H^{(t)})^{-1}\boldsymbol{u}^{(t)}, \tag{4.25}$$

assuming $H^{(t)}$ nonsingular (in practice one solves the linear system rather than inverting). Iterate until L changes negligibly. With several local maxima a good starting value is crucial. Figure 6 shows one cycle.

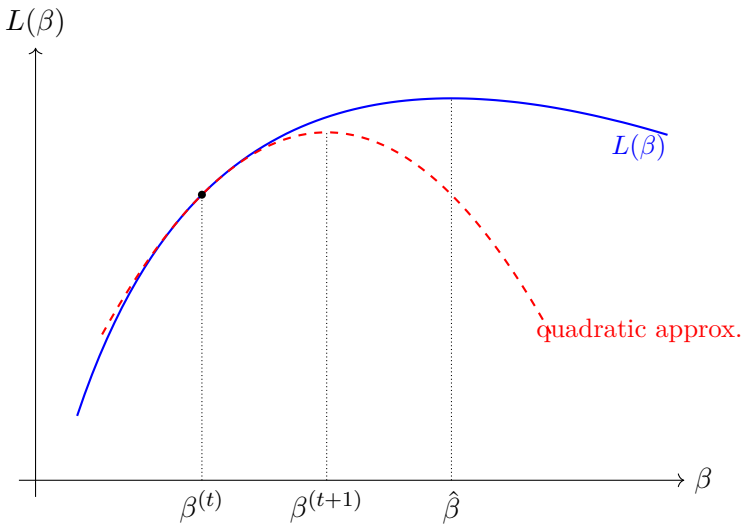


Figure 6: One Newton–Raphson cycle: the next guess maximizes the second-order Taylor approximation at $\beta^{(t)}$.

Example 4.24 (Newton–Raphson for a binomial parameter)

For one $\mathrm{bin}(n, \pi)$ observation y, $L(\pi) = y\log\pi + (n-y)\log(1-\pi)$, so

$$u = \frac{y - n\pi}{\pi(1-\pi)}, \qquad H = -\Big[\frac{y}{\pi^2} + \frac{n-y}{(1-\pi)^2}\Big],$$

$$\pi^{(t+1)} = \pi^{(t)} + \Big[\frac{y}{\pi^{(t)2}} + \frac{n-y}{(1-\pi^{(t)})^2}\Big]^{-1} \frac{y - n\pi^{(t)}}{\pi^{(t)}(1-\pi^{(t)})}.$$

The step moves $\pi^{(t)}$ up if $y/n > \pi^{(t)}$ and down otherwise; at $\pi^{(t)} = y/n$ no adjustment occurs. Starting from $\pi^{(0)} = \frac{1}{2}$ one gets $\pi^{(1)} = y/n$ at once; from other starting values four or five iterations usually suffice.

Theorem 4.25 (Quadratic convergence of Newton–Raphson)

Suppose $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$, H is Lipschitz on a ball B around $\hat{\boldsymbol{\beta}}$ with constant γ ($\|H(\boldsymbol{a}) - H(\boldsymbol{b})\| \le \gamma\|\boldsymbol{a} - \boldsymbol{b}\|$), and $\|H(\boldsymbol{\beta})^{-1}\| \le M$ on B. If $\boldsymbol{\beta}^{(t)} \in B$, then

$$\|\boldsymbol{\beta}^{(t+1)} - \hat{\boldsymbol{\beta}}\| \le \frac{M\gamma}{2}\,\|\boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}\|^2.$$

In particular, if $\boldsymbol{\beta}^{(0)} \in B$ and $\|\boldsymbol{\beta}^{(0)} - \hat{\boldsymbol{\beta}}\| < 2/(M\gamma)$, the iterates converge to $\hat{\boldsymbol{\beta}}$ and the number of correct digits roughly doubles each step.

Proof. [Supplement]

Step 1. Let $\boldsymbol{e}_t = \boldsymbol{\beta}^{(t)} - \hat{\boldsymbol{\beta}}$. Since $\boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$,

$$\boldsymbol{u}^{(t)} = \boldsymbol{u}^{(t)} - \boldsymbol{u}(\hat{\boldsymbol{\beta}}) = \int_0^1 H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)\,ds\;\boldsymbol{e}_t.$$

Step 2. By (4.25),

$$\boldsymbol{e}_{t+1} = \boldsymbol{e}_t - (H^{(t)})^{-1}\boldsymbol{u}^{(t)} = (H^{(t)})^{-1}\int_0^1 [H(\hat{\boldsymbol{\beta}} + \boldsymbol{e}_t) - H(\hat{\boldsymbol{\beta}} + s\boldsymbol{e}_t)]ds\;\boldsymbol{e}_t.$$

Step 3. The integrand has norm at most $\gamma(1-s)\|\boldsymbol{e}_t\|$, and $\int_0^1 (1-s)\,ds = \frac{1}{2}$. Hence $\|\boldsymbol{e}_{t+1}\| \le M \cdot \frac{\gamma}{2}\|\boldsymbol{e}_t\|^2$.

Step 4. If $\rho = \frac{M\gamma}{2}\|\boldsymbol{e}_0\| < 1$, then by induction $\|\boldsymbol{e}_{t+1}\| \le \rho\|\boldsymbol{e}_t\|$, the iterates stay in B, and $\|\boldsymbol{e}_t\| \to 0$; the quadratic bound then shows the error is roughly squared at each step. ∎

4.6.2 Fisher scoring method

Fisher scoring replaces the Hessian (minus the *observed information*) by its expectation, minus the *expected information* $\mathcal{I}$. With $\mathcal{I}^{(t)}$ evaluated at $\boldsymbol{\beta}^{(t)}$,

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + (\mathcal{I}^{(t)})^{-1}\boldsymbol{u}^{(t)} \quad\Longleftrightarrow\quad \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t+1)} = \mathcal{I}^{(t)}\boldsymbol{\beta}^{(t)} + \boldsymbol{u}^{(t)}. \tag{4.26}$$

For the binomial parameter, $\mathcal{I} = n/[\pi(1-\pi)]$ and

$$\pi^{(t+1)} = \pi^{(t)} + \frac{\pi^{(t)}(1-\pi^{(t)})}{n}\cdot\frac{y - n\pi^{(t)}}{\pi^{(t)}(1-\pi^{(t)})} = \frac{y}{n},$$

the answer after one step from any start. For a GLM, $\mathcal{I}^{(t)} = X^\top W^{(t)} X$ by Theorem 4.15, and $\boldsymbol{u}^{(t)}$ has elements (4.14):

$$u_j = \sum_{i=1}^{N} \frac{(y_i - \mu_i)x_{ij}}{\mathrm{Var}(Y_i)}\frac{\partial \mu_i}{\partial \eta_i}. \tag{4.27}$$

At convergence $(\mathcal{I}^{(t)})^{-1}$ is the estimated covariance (4.18), a free by-product.

Observed or expected information?

For canonical links the two coincide (Theorem 4.27). Otherwise Fisher scoring has the advantages that $\mathcal{I}$ is automatically nonnegative definite, it yields the covariance matrix, and it is weighted least squares (next subsection); but it need not converge quadratically, and for complex models the observed information may be easier to compute. Efron and Hinkley (1978) argued that observed-information variances better approximate the relevant conditional vari-

ance and agree more closely with Bayesian analyses.

4.6.3 ML as iterative reweighted least squares

For the linear model $\boldsymbol{z} = X\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathrm{Cov}(\boldsymbol{\varepsilon}) = V$, the weighted least squares (WLS) estimator is $(X^\top V^{-1} X)^{-1} X^\top V^{-1} \boldsymbol{z}$.

Theorem 4.26 (Fisher scoring is IRLS)

For a GLM, the Fisher scoring step (4.26) is

$$\boldsymbol{\beta}^{(t+1)} = (X^\top W^{(t)} X)^{-1} X^\top W^{(t)} \boldsymbol{z}^{(t)},$$

the WLS fit of the *working response*

$$z_i^{(t)} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) \frac{\partial \eta_i^{(t)}}{\partial \mu_i^{(t)}} \tag{4.28}$$

on X with weight matrix (inverse covariance) $W^{(t)}$.

Proof.

Step 1. By Theorem 4.15, $\mathcal{I}^{(t)} \boldsymbol{\beta}^{(t)} = X^\top W^{(t)} X \boldsymbol{\beta}^{(t)} = X^\top W^{(t)} \boldsymbol{\eta}^{(t)}$.

Step 2. Since $w_i = (\partial \mu_i / \partial \eta_i)^2 / \operatorname{Var}(Y_i)$, each term of (4.27) is

$$\frac{(y_i - \mu_i) x_{ij}}{\operatorname{Var}(Y_i)} \frac{\partial \mu_i}{\partial \eta_i} = x_{ij}\, w_i\, (y_i - \mu_i) \frac{\partial \eta_i}{\partial \mu_i},$$

so $\boldsymbol{u}^{(t)} = X^\top W^{(t)} \boldsymbol{d}^{(t)}$ with $d_i^{(t)} = (y_i - \mu_i^{(t)})\, \partial \eta_i^{(t)} / \partial \mu_i^{(t)}$.

Step 3. Adding, the right side of (4.26) is $X^\top W^{(t)} (\boldsymbol{\eta}^{(t)} + \boldsymbol{d}^{(t)}) = X^\top W^{(t)} \boldsymbol{z}^{(t)}$. Thus $X^\top W^{(t)} X \boldsymbol{\beta}^{(t+1)} = X^\top W^{(t)} \boldsymbol{z}^{(t)}$, the WLS normal equations. ∎

The working response is the linearization of the link at the data,

$$g(y_i) \approx g(\mu_i) + (y_i - \mu_i) g'(\mu_i) = \eta_i + (y_i - \mu_i) \frac{\partial \eta_i}{\partial \mu_i} = z_i,$$

and W^{-1} is approximately its covariance matrix. Each cycle regresses $\boldsymbol{z}^{(t)}$ on X with weights $W^{(t)}$, giving $\boldsymbol{\beta}^{(t+1)}$, then $\boldsymbol{\eta}^{(t+1)} = X\boldsymbol{\beta}^{(t+1)}$ and new $\boldsymbol{z}^{(t+1)}$, $W^{(t+1)}$: *iteratively reweighted least squares* (IRLS). A simple start uses $\hat{\boldsymbol{\mu}}^{(0)} = \mathbf{y}$, adjusting values where $g(y_i)$ is infinite (e.g. $y_i = 0$ with log link: use $y_i = \frac{1}{2}$ in the first cycle only).

4.6.4 Simplifications for canonical links

Theorem 4.27 (Canonical link properties)

Consider a GLM with the canonical link, $\theta_i = \eta_i = \sum_j \beta_j x_{ij}$, and a common $a(\phi)$ for all i (e.g. Poisson; binomial with all $n_i = 1$). Then:
(a) $\sum_i y_i x_{ij}$, $j = 1, \ldots, p$, are sufficient statistics for $\boldsymbol{\beta}$;
(b) $\partial L_i / \partial \beta_j = (y_i - \mu_i) x_{ij} / a(\phi)$, so the likelihood equations are

$$\sum_i x_{ij} y_i = \sum_i x_{ij} \mu_i, \qquad j = 1, \ldots, p, \qquad \text{i.e.} \quad X^\top \mathbf{y} = X^\top \boldsymbol{\mu}; \tag{4.29}$$

(c) $\partial^2 L/\partial\beta_h\partial\beta_j = -\sum_i x_{ih}x_{ij}b''(\theta_i)/a(\phi)$ does not involve $\mathbf{y}$; hence observed and expected information coincide, $H = -\mathcal{I}$, and Newton–Raphson equals Fisher scoring;
(d) L is concave in $\boldsymbol{\beta}$, strictly if X has full column rank; so a solution of (4.29), if one exists, is the unique ML estimate.
With weights $a(\phi) = \phi/\omega_i$ the same holds with y_i, μ_i replaced by $\omega_i y_i, \omega_i \mu_i$.

Proof.

Step 1. By (4.12) the part of L involving both data and parameters is

$$\frac{1}{a(\phi)}\sum_i y_i\theta_i = \frac{1}{a(\phi)}\sum_i y_i \sum_j \beta_j x_{ij} = \frac{1}{a(\phi)}\sum_j \beta_j\Big(\sum_i y_i x_{ij}\Big),$$

and the rest depends on $\boldsymbol{\beta}$ only or on $\mathbf{y}$ only; the factorization theorem gives (a).

Step 2. For the canonical link $\partial\mu_i/\partial\eta_i = \partial\mu_i/\partial\theta_i = b''(\theta_i) = \mathrm{Var}(Y_i)/a(\phi)$; inserting into (4.13) gives $\partial L_i/\partial\beta_j = (y_i - \mu_i)x_{ij}/a(\phi)$, and (b) follows.

Step 3. Differentiating again, $\partial\mu_i/\partial\beta_h = b''(\theta_i)x_{ih}$, so $\partial^2 L_i/\partial\beta_h\partial\beta_j = -x_{ij}x_{ih}b''(\theta_i)/a(\phi)$, free of y_i; its expectation is itself, giving (c).

Step 4. For any $\boldsymbol{c} \in \mathbb{R}^p$, $\boldsymbol{c}^\top H\boldsymbol{c} = -\sum_i b''(\theta_i)(\boldsymbol{x}_i^\top \boldsymbol{c})^2/a(\phi) \le 0$ since $b'' > 0$; equality forces $X\boldsymbol{c} = \mathbf{0}$, impossible for $\boldsymbol{c} \neq \mathbf{0}$ under full rank. So L is (strictly) concave; a stationary point of a strictly concave function is its unique global maximum, giving (d). ■

Equations (4.29) equate sufficient statistics to their expected values (Nelder and Wedderburn 1972): for the normal linear model they are the normal equations; for Poisson loglinear models they are (4.19); for binary logistic regression they are (4.16) with $n_i = 1$.

Chapter 5. Logistic Regression

Logistic regression is the most important model for categorical responses, used in biomedicine, social science, marketing, credit scoring and genetics. Section 5.1 interprets its parameters, Section 5.2 presents inference, Sections 5.3–5.4 extend to several and to qualitative predictors, and Section 5.5 applies the GLM theory of Chapter 4 to the likelihood equations and their solution.

5.1 Interpreting parameters in logistic regression

Motivation. The logistic model is linear on the logit scale, but people think in probabilities. We need several readings of β: an odds ratio, a local slope $\beta\pi(1-\pi)$ for the probability, and changes in π over the range of x.

For binary Y and predictor X let $\pi(x) = P(Y = 1 \mid X = x)$. The logistic regression model is

$$\pi(x) = \frac{\exp(\alpha+\beta x)}{1+\exp(\alpha+\beta x)}, \qquad \text{equivalently} \qquad \operatorname{logit}\pi(x) = \log\frac{\pi(x)}{1-\pi(x)} = \alpha+\beta x. \tag{5.1}$$

5.1.1 Interpreting β: odds, probabilities, and linear approximations

The sign of β says whether $\pi(x)$ increases or decreases; the curve gets steeper as $|\beta|$ grows and flattens to a constant as $\beta \to 0$, where Y and X are independent. For $\beta > 0$ the curve is a logistic cdf; the symmetry of the logistic density means $\pi(x)$ approaches 1 at the same rate as it approaches 0.

Proposition 5.1 (Odds-ratio interpretation)

Under (5.1) the odds are $\pi(x)/[1-\pi(x)] = e^{\alpha}(e^{\beta})^{x}$; hence for every x,

$$\frac{\pi(x+1)/[1-\pi(x+1)]}{\pi(x)/[1-\pi(x)]} = e^{\beta}:$$

the odds multiply by e^{β} for each unit increase in x, and e^{β} is an odds ratio.

Proof. Exponentiate the logit form: odds $= \exp(\alpha+\beta x)$. The ratio at $x+1$ and x is $\exp(\alpha+\beta x+\beta)/\exp(\alpha+\beta x) = e^{\beta}$. ∎

Proposition 5.2 (Linear approximation (Berkson 1951))

Under (5.1),

$$\frac{d\pi(x)}{dx} = \beta\,\pi(x)[1-\pi(x)].$$

(a) The slope is greatest, $\beta/4$, where $\pi(x) = \frac{1}{2}$, i.e. at $x = -\alpha/\beta$ (the median effective level EL_{50}; in toxicology the LD_{50}). At $\pi(x) = 0.9$ or 0.1 it is 0.09β, and it tends to 0 as $\pi(x) \to 0$ or 1.
(b) $\pi(-\alpha/\beta \pm 1/\beta) = e^{\pm 1}/(1+e^{\pm 1})$, i.e. 0.73 and 0.27: a change of $1/\beta$ in x from the EL_{50} changes π by about 0.23, close to the value $\frac{1}{4}$ suggested by the tangent line.

Proof. [Supplement]

Step 1. With $\eta = \alpha+\beta x$ and $\pi = e^{\eta}/(1+e^{\eta})$,

$$\frac{d\pi}{d\eta} = \frac{e^{\eta}(1+e^{\eta}) - e^{\eta}e^{\eta}}{(1+e^{\eta})^2} = \frac{e^{\eta}}{(1+e^{\eta})^2} = \pi(1-\pi),$$

and $d\eta/dx = \beta$, so $d\pi/dx = \beta\pi(1-\pi)$.

Step 2. The function $\pi(1-\pi)$ on $(0,1)$ is maximized at $\pi = \frac{1}{2}$ with value $\frac{1}{4}$; $\pi(x) = \frac{1}{2}$ iff $\alpha+\beta x = 0$ iff $x = -\alpha/\beta$. Also $0.9 \times 0.1 = 0.09$.

Step 3. At $x = -\alpha/\beta \pm 1/\beta$, $\eta = \pm 1$, so $\pi = e^{\pm 1}/(1+e^{\pm 1}) = 0.731$ or 0.269. ■

So near a point with probability π, a unit change in x changes the probability by roughly $\beta\pi(1-\pi)$ (Figure 7); the approximation is better for smaller changes. Another summary reports $\pi(x)$ at the quartiles of x; the change over the middle half of the data can be compared across predictors with different units. The intercept is rarely of interest, but after centering ($x \mapsto x - \bar{x}$) it is the logit at $\bar{x}$, and centering reduces correlations among estimates in models with quadratic or interaction terms.

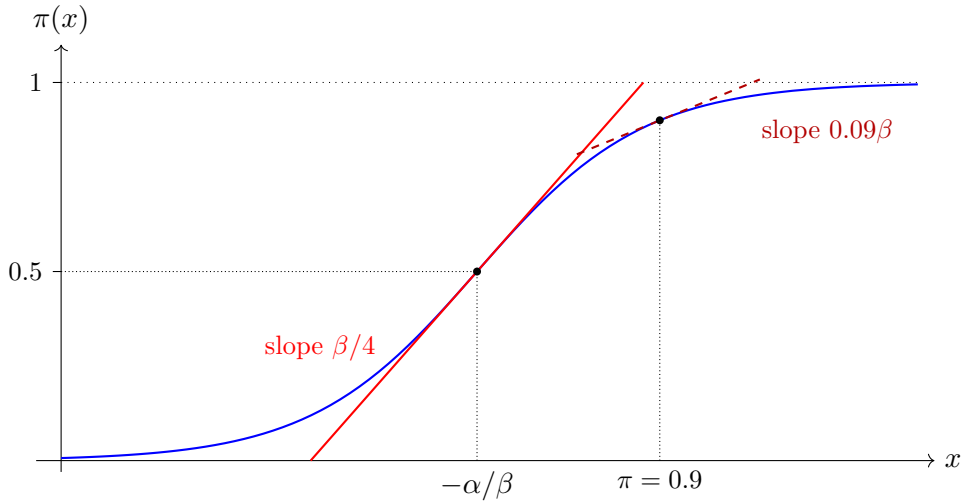


Figure 7: Logistic curve ($\alpha = -3$, $\beta = 1$) with tangent lines: slope $\beta\pi(1-\pi)$ equals $\beta/4$ at $\pi = \frac{1}{2}$ and 0.09β at $\pi = 0.9$.

5.1.2 Looking at the data

Before fitting, check that a logistic curve is plausible. Plotting 0/1 responses against x is uninformative; instead plot sample proportions or sample logits. With y_i successes out of n_i at setting i, the sample logit $\log[y_i/(n_i - y_i)]$ is infinite when $y_i = 0$ or n_i; the adjusted version

$$\log \frac{y_i + \frac{1}{2}}{n_i - y_i + \frac{1}{2}}$$

is the least-biased estimator of this form (bias of order n_i^{-2}, Haldane 1956). The plot should be roughly linear. With continuous x and small n_i, group nearby x values or, better, smooth the data and look for marked departures from an S-shape.

5.1.3 Horseshoe crabs revisited

Example 5.3 (Horseshoe crabs: width and presence of satellites)

For the 173 crabs of Example 4.5, let $Y = 1$ if a female has at least one satellite. The GAM fit and the sample proportions in the eight width categories suggest an increasing S-shaped

trend. The ML fit with $x =$ width (cm) is

$$\hat{\pi}(x) = \frac{\exp(-12.351 + 0.497x)}{1 + \exp(-12.351 + 0.497x)}.$$

Parameter	Estimate	SE	LR 95% limits	Wald χ^2
Intercept	-12.3508	2.6287	$(-17.8097, -7.4573)$	22.07
width	0.4972	0.1017	(0.3084, 0.7090)	23.89

Deviance 194.45 and Pearson $X^2 = 165.14$ on 171 df; log likelihood -97.23.

(a) At the mean width 26.3, $\hat{\pi} = 0.674$; $\hat{\pi} = \frac{1}{2}$ at $x = 12.351/0.497 = 24.8$.
(b) The odds multiply by $e^{0.497} = 1.64$ per cm (a 64% increase).
(c) At $\bar{x}$ the probability rises by about $0.497(0.674)(0.326) = 0.11$ per cm.
(d) At the width quartiles 24.9, 26.1, 27.7, $\hat{\pi} = 0.51,\ 0.65,\ 0.81$: an increase of 0.30 over the middle half.
(e) With weight (kg) instead, $\operatorname{logit}\hat{\pi} = -3.695 + 1.815x$; $\hat{\beta}$'s are not comparable across units, but at the weight quartiles 2.00, 2.35, 2.85 we get $\hat{\pi} = 0.48,\ 0.64,\ 0.81$, an increase of 0.33, similar to width.

5.1.4 Logistic regression with retrospective studies

In case–control studies X is observed for samples of cases ($Y = 1$) and controls ($Y = 0$), so Y is fixed by design. Effects in logistic regression are odds ratios, and odds ratios can be estimated retrospectively.

Theorem 5.4 (Logistic regression under case–control sampling)

Let $Z = 1$ indicate that a subject is sampled, and suppose the sampling probabilities depend only on y: $P(Z = 1 \mid y, x) = \rho_y$ for $y = 0, 1$. If $\operatorname{logit} P(Y = 1 \mid x) = \alpha + \beta x$, then

$$\operatorname{logit} P(Y = 1 \mid Z = 1, x) = \alpha^* + \beta x, \qquad \alpha^* = \alpha + \log(\rho_1/\rho_0).$$

So the slope β is the same as in the prospective model; only the intercept changes.

Proof.

Step 1. By Bayes' theorem,

$$P(Y = 1 \mid Z = 1, x) = \frac{P(Z = 1 \mid Y = 1, x)P(Y = 1 \mid x)}{\sum_{j=0}^{1} P(Z = 1 \mid Y = j, x)P(Y = j \mid x)} = \frac{\rho_1 P(Y = 1 \mid x)}{\rho_0 P(Y = 0 \mid x) + \rho_1 P(Y = 1 \mid x)}.$$

Step 2. Divide numerator and denominator by $P(Y = 0 \mid x)$ and use $P(Y = 1 \mid x)/P(Y = 0 \mid x) = \exp(\alpha + \beta x)$:

$$P(Y = 1 \mid Z = 1, x) = \frac{\rho_1 \exp(\alpha + \beta x)}{\rho_0 + \rho_1 \exp(\alpha + \beta x)} = \frac{\exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}{1 + \exp[\alpha + \log(\rho_1/\rho_0) + \beta x]}.$$

■

If cases are sampled at 10 times the rate of controls, the estimated intercept is $\log 10 = 2.3$ larger than prospectively. For other binary links the effect for Y given X is not recoverable from case–control data; this is a major reason for the popularity of logit models in biomedicine. Matched case–control studies need conditional methods (Section 10.2.5).

Proposition 5.5 (Normal discriminant model implies logistic regression (Cornfield 1962))

Suppose $X \mid Y = i \sim N(\mu_i, \sigma^2)$, $i = 0, 1$, and $P(Y = 1) = \rho$. Then $P(Y = 1 \mid X = x)$ satisfies (5.1) with

$$\beta = \frac{\mu_1 - \mu_0}{\sigma^2}, \qquad \alpha = \log\frac{\rho}{1-\rho} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2}.$$

If the two variances differ, the logit is quadratic in x.

Proof. [Supplement] Let φ_i be the $N(\mu_i, \sigma_i^2)$ density. By Bayes' theorem,

$$\operatorname{logit} P(Y = 1 \mid x) = \log\frac{\rho\varphi_1(x)}{(1-\rho)\varphi_0(x)} = \log\frac{\rho}{1-\rho} + \log\frac{\sigma_0}{\sigma_1} - \frac{(x-\mu_1)^2}{2\sigma_1^2} + \frac{(x-\mu_0)^2}{2\sigma_0^2}.$$

With $\sigma_0 = \sigma_1 = \sigma$ the x^2 terms cancel and

$$-\frac{(x-\mu_1)^2 - (x-\mu_0)^2}{2\sigma^2} = \frac{(\mu_1 - \mu_0)x}{\sigma^2} - \frac{\mu_1^2 - \mu_0^2}{2\sigma^2},$$

giving the stated α, β. If $\sigma_0 \neq \sigma_1$ the coefficient of x^2 is $\frac{1}{2}(\sigma_0^{-2} - \sigma_1^{-2}) \neq 0$. ■

So when a population mixes two roughly normal groups with similar spread, the logistic curve fits well; with unequal spreads $\pi(x)$ is nonmonotone (rises then falls, or the reverse).

5.2 Inference for logistic regression

Motivation. ML estimators in logistic regression are asymptotically normal, so the Wald, likelihood-ratio and score triad of Theorem 4.18 applies. The likelihood-ratio versions are generally the most reliable, and goodness of fit needs care when predictors are continuous.

5.2.1 Types of inference

For $\operatorname{logit}\pi(x) = \alpha + \beta x$, the test of independence $H_0 : \beta = 0$ can use

(i) Wald: $z = \hat\beta/\widehat{\mathrm{se}}$, or $z^2 \sim \chi_1^2$;

(ii) likelihood ratio: $-2(L_0 - L_1)$, twice the difference of maximized log likelihoods under H_0 and the full model, χ_1^2;

(iii) score: based on the derivative of the log likelihood at $\beta = 0$, comparing the sufficient statistic for β with its null expectation (Section 5.3.5).

In large samples they agree. The LR test uses the log likelihood at both H_0 and $\hat\beta$ and is preferred; for large $|\beta|$ the Wald test loses power and can behave aberrantly (Hauck and Donner 1977). Inverting a test gives a confidence interval: Wald gives $\hat\beta \pm z_{\alpha/2}\widehat{\mathrm{se}}$; the LR inversion is the profile-likelihood interval.

Proposition 5.6 (Interval for $\pi(x_0)$)

The estimated variance of the fitted logit at x_0 is

$$\widehat{\operatorname{Var}}(\hat\alpha + \hat\beta x_0) = \widehat{\operatorname{Var}}(\hat\alpha) + x_0^2\,\widehat{\operatorname{Var}}(\hat\beta) + 2x_0\,\widehat{\operatorname{Cov}}(\hat\alpha, \hat\beta),$$

and if (l, u) is a $1-\alpha$ interval for $\operatorname{logit}\pi(x_0)$, then $(e^l/(1+e^l),\ e^u/(1+e^u))$ is a $1-\alpha$ interval for $\pi(x_0)$.

Proof. The first formula is the variance of the linear combination $(1, x_0)(\hat{\alpha}, \hat{\beta})^\top$, namely $(1, x_0)\widehat{\mathrm{Cov}}(\hat{\alpha}, \hat{\beta})(1, x_0)^\top$. Since $t \mapsto e^t/(1+e^t)$ is strictly increasing, $l \le \operatorname{logit}\pi(x_0) \le u$ iff $e^l/(1+e^l) \le \pi(x_0) \le e^u/(1+e^u)$, so the coverage is preserved. ■

Equivalently, refit with $x^* = x - x_0$: the intercept and its SE are the fitted logit at x_0 and its SE.

5.2.2 Inference for horseshoe crab data

Example 5.7 (Horseshoe crabs: tests and intervals)

(a) Wald: $z = 0.497/0.102 = 4.9$, $z^2 = 23.9$ ($P < 0.0001$). The maximized log likelihoods are -112.88 under $\beta = 0$ and -97.23 for the model, so the LR statistic is $-2(-112.88 + 97.23) = 31.3$ (df $= 1$), even stronger evidence.
(b) The Wald interval for β is $0.497 \pm 1.96(0.102) = (0.298, 0.697)$; the profile-likelihood interval is (0.308, 0.709). For the odds ratio per cm: $(e^{0.308}, e^{0.709}) = (1.36, 2.03)$: at least a 36% increase and at most a doubling of the odds.
(c) At $x = 26.5$: logit $= -12.351 + 0.497(26.5) = 0.825$, $\hat{\pi} = 0.695$. With $\widehat{\mathrm{Var}}(\hat{\alpha}) = 6.910$, $\widehat{\mathrm{Var}}(\hat{\beta}) = 0.01035$, $\widehat{\mathrm{Cov}} = -0.2668$,

$$\widehat{\mathrm{Var}} = 6.910 + (26.5)^2(0.01035) + 2(26.5)(-0.2668) = 0.038,$$

so the logit interval is $0.825 \pm 1.96\sqrt{0.038} = (0.44, 1.21)$, i.e. (0.61, 0.77) for $\pi(26.5)$.
(d) Only 6 crabs had $x = 26.5$, 4 with satellites: the sample proportion 0.67 has SE $= \sqrt{0.67(0.33)/6} = 0.19$ and score interval (0.30, 0.90), versus model SE $= 0.04$. The model borrows strength from all 173 observations to estimate two parameters; when it approximates the truth decently, its estimates are much more precise (Section 6.4.5).

5.2.3 Checking goodness of fit: ungrouped and grouped data

One way to detect lack of fit compares the model with a more complex one (quadratic term, interaction) by an LR test; if the extra terms do not help, the model gains support. Global tests compare observed counts with fitted values $n_i\hat{\pi}_i$, $n_i(1-\hat{\pi}_i)$ through X^2 or G^2; for a fixed number of predictor settings, as fitted counts grow, they are χ^2 with residual df = (number of settings) − (number of parameters). This requires categorical predictors: as noted in Section 4.5.3, the saturated model differs for grouped and ungrouped data, and the chi-squared limit needs a fixed number of settings.

5.2.4 Goodness of fit of model for horseshoe crabs

Example 5.8 (Horseshoe crabs: goodness of fit)

(a) Adding a quadratic term with centered width gives logit $\hat{\pi} = 0.618 + 0.533x + 0.040x^2$, with $\widehat{\mathrm{se}} = 0.046$ for the quadratic coefficient; the LR statistic is 0.83 (df $= 1$): no need for it.
(b) Width takes 66 distinct values; treating the data as a 66×2 table violates the chi-squared theory (tiny fitted counts, and more data would add cells). Grouping into the eight width categories of Example 4.5, with fitted values the sums of $\hat{\pi}(x)$ and $1 - \hat{\pi}(x)$ over the crabs in each category:

Width (cm)	Yes	No	Fitted yes	Fitted no
< 23.25	5	9	3.64	10.36
23.25–24.25	4	10	5.31	8.69
24.25–25.25	17	11	13.78	14.22
25.25–26.25	21	18	24.23	14.77
26.25–27.25	15	7	15.94	6.06
27.25–28.25	20	4	19.38	4.62
28.25–29.25	15	3	15.65	2.35
> 29.25	14	0	13.08	0.92

Then $X^2 = 5.3$ and $G^2 = 6.2$ with df $= 8 - 2 = 6$ ($P > 0.4$): no evidence of lack of fit (the theory is still approximate since $\pi(x)$ varies within a category).

5.2.5 Checking goodness of fit with ungrouped data by grouping

With many predictors, grouping on each variable gives a sparse table. Instead, partition the observations by their fitted probabilities.

Definition (Hosmer–Lemeshow statistic)

Order the observations by $\hat{\pi}$ from the ungrouped fit and form g groups of roughly equal size (usually $g = 10$). With y_{ij} the binary outcome and $\hat{\pi}_{ij}$ the fitted probability of observation j in group i ($j = 1, \ldots, n_i$),

$$\mathrm{HL} = \sum_{i=1}^{g} \frac{\left(\sum_j y_{ij} - \sum_j \hat{\pi}_{ij}\right)^2}{\left(\sum_j \hat{\pi}_{ij}\right)\left[1 - \left(\sum_j \hat{\pi}_{ij}\right)/n_i\right]}.$$

Proposition 5.9 (HL as a Pearson statistic)

HL equals the Pearson statistic $\sum(\text{observed} - \text{fitted})^2/\text{fitted}$ for the $g \times 2$ table of observed and fitted counts of successes and failures in the groups.

Proof. [Supplement] In group i let $O_i = \sum_j y_{ij}$, $E_i = \sum_j \hat{\pi}_{ij}$. The failure cell has observed $n_i - O_i$ and fitted $n_i - E_i$, so its deviation is $-(O_i - E_i)$. The two Pearson terms of row i add to

$$(O_i - E_i)^2\Big(\frac{1}{E_i} + \frac{1}{n_i - E_i}\Big) = \frac{(O_i - E_i)^2\, n_i}{E_i(n_i - E_i)} = \frac{(O_i - E_i)^2}{E_i(1 - E_i/n_i)}.$$

Summing over i gives HL. ■

Reference distribution and limitations

HL is not asymptotically chi-squared in the usual sense, since observations within a group do not share a success probability; Hosmer and Lemeshow found by simulation that when the number of distinct covariate patterns equals n, the null distribution is approximately χ^2_{g-2}. Ties in $\hat{\pi}$ make the grouping somewhat arbitrary, so software values differ slightly. For the crab width model with $g = 10$, HL $= 3.5$ on 8 df: a decent fit. Like other global statistics, HL has low power against specific alternatives and does not say what is wrong; comparing with a more complex model is more informative, and diagnostics (Section 6.2) locate problems.

5.3 Logit models with categorical predictors

Motivation. Qualitative predictors enter through dummy variables. Different codings change the individual parameters but not the fit or the odds ratios; for ordered factors, scores lead to the linear logit model and the Cochran–Armitage trend test.

5.3.1 ANOVA-type representation of factors

For a single factor X with I categories, let $y_i \sim \text{bin}(n_i, \pi_i)$ in row i of an $I \times 2$ table. The model

$$\log \frac{\pi_i}{1-\pi_i} = \alpha + \beta_i, \qquad i = 1, \ldots, I, \tag{5.2}$$

resembles one-way ANOVA and has one redundant parameter, so a constraint is imposed, e.g. $\beta_I = 0$.

Proposition 5.10 (Coding invariance and saturation)

(a) Model (5.2) is saturated: for any $\{\pi_i\} \subset (0,1)$ there are parameters satisfying it, and its fitted values are the sample proportions.
(b) Under any constraint, $\alpha + \beta_i = \operatorname{logit} \pi_i$ and $\beta_a - \beta_b$ is the log odds ratio for rows a, b; these and the fit are the same for every coding. With $\beta_I = 0$, $\alpha = \operatorname{logit} \pi_I$ and β_i is the log odds ratio of row i versus row I.
(c) No effect, $\beta_1 = \cdots = \beta_I$, is statistical independence of X and Y.

Proof.

(a) Take $\alpha = \operatorname{logit} \pi_I$ and $\beta_i = \operatorname{logit} \pi_i - \operatorname{logit} \pi_I$. The model has I free parameters for I binomials, so it is saturated and $\hat{\pi}_i = y_i/n_i$.

(b) If $(\alpha, \boldsymbol{\beta})$ satisfies the model, so does $(\alpha - c, \beta_i + c)$ for any c, and every constraint selects one such c; $\alpha + \beta_i$ and $\beta_a - \beta_b$ do not depend on c. In particular $\beta_a - \beta_b = \operatorname{logit} \pi_a - \operatorname{logit} \pi_b$, a log odds ratio.

(c) Equal β_i means equal π_i across rows. ■

5.3.2 Dummy variables in logit models

Equivalently, with dummy variables $x_i = 1$ for row i and 0 otherwise ($i = 1, \ldots, I-1$),

$$\operatorname{logit} \pi = \alpha + \beta_1 x_1 + \cdots + \beta_{I-1} x_{I-1};$$

no dummy for category I corresponds to $\beta_I = 0$ (some software instead sets $\beta_1 = 0$). *Effect coding* imposes $\sum_i \beta_i = 0$; for $I = 2$ it uses $x = 1$ in category 1 and $x = -1$ in category 2, and then the log odds ratio is $\beta_1 - \beta_2 = 2\beta_1$, whereas with a 0/1 dummy it is β_1. A single $\hat{\beta}_i$ is meaningful only in comparison with another category.

5.3.3 Alcohol and infant malformation example revisited

Example 5.11 (Maternal alcohol consumption and malformation)

Alcohol (drinks/day)	Present	Absent	Logit	Observed prop.	Fitted (linear logit)
0	48	17066	−5.87	0.0028	0.0026
< 1	38	14464	−5.94	0.0026	0.0030
1–2	5	788	−5.06	0.0063	0.0041
3–5	1	126	−4.84	0.0079	0.0091
≥ 6	1	37	−3.61	0.0263	0.0231

For (5.2) with malformation as response, $\hat{\alpha} + \hat{\beta}_i$ are the sample logits, e.g. $\log(48/17066) = -5.87$. With $\beta_5 = 0$: $\hat{\alpha} = -3.61$, $\hat{\beta}_1 = -2.26$; with $\beta_1 = 0$: $\hat{\alpha} = -5.87$. Apart from a slight reversal in the first two rows, the logits increase with alcohol. Independence has $\hat{\alpha} = \log(93/32481) = -5.86$; testing it (df $= 4$) gives $X^2 = 12.1$ ($P = 0.02$) but $G^2 = 6.2$ ($P = 0.19$): mixed signals, since the counts are very unbalanced (exact conditional P-values 0.03 and 0.13). Both ignore the ordering of alcohol, and one case more or less in the last rows changes those proportions substantially.

5.3.4 Linear logit model for $I \times 2$ tables

For ordered categories with scores $x_1, \ldots, x_I$ the *linear logit model*

$$\operatorname{logit} \pi_i = \alpha + \beta x_i \tag{5.3}$$

lies between independence ($\beta = 0$) and the saturated model (5.2).

Example 5.12 (Alcohol: linear logit fit)

With scores (0, 0.5, 1.5, 4.0, 7.0) (the last somewhat arbitrary): $\hat{\alpha} = -5.9605$ (0.1154) and $\hat{\beta} = 0.3166$ (0.1254), profile-likelihood interval (0.0187, 0.5236). The estimated odds of malformation multiply by $e^{0.317} = 1.37$ per drink per day. Fit: $G^2 = 1.95$, $X^2 = 2.05$, df $= 3$; log likelihood -635.60. The fitted proportions are in the table of Example 5.11.

5.3.5 Cochran–Armitage trend test

For an $I \times 2$ table with ordered rows and independent $y_i \sim \text{bin}(n_i, \pi_i)$, Cochran (1954) and Armitage (1955) partitioned the Pearson statistic using the linear probability model $\pi_i = \alpha + \beta x_i$ fitted by (weighted) least squares. Let $n = \sum n_i$, $p_i = y_i/n_i$, $p = \sum y_i/n$ and $\bar{x} = \sum n_i x_i/n$. The prediction equation is $\hat{\pi}_i = p + b(x_i - \bar{x})$ with

$$b = \frac{\sum_i n_i (p_i - p)(x_i - \bar{x})}{\sum_i n_i (x_i - \bar{x})^2}.$$

Theorem 5.13 (Partition of the Pearson statistic)

Let $X^2(I) = \sum_i n_i (p_i - p)^2 / [p(1-p)]$ be the Pearson statistic for independence in the $I \times 2$ table. Then

$$X^2(I) = z^2 + X^2(L),$$

where

$$X^2(L) = \frac{\sum_i n_i(p_i - \hat{\pi}_i)^2}{p(1-p)}, \qquad z^2 = \frac{b^2 \sum_i n_i(x_i - \bar{x})^2}{p(1-p)} = \frac{\left[\sum_i (x_i - \bar{x}) y_i\right]^2}{p(1-p) \sum_i n_i (x_i - \bar{x})^2}.$$

Under independence z^2 is asymptotically χ^2_1 (the *Cochran–Armitage trend test* of $\beta = 0$); when the linear probability model holds, $X^2(L)$ is asymptotically χ^2_{I-2} (a test of linearity).

Proof. [Supplement]

Step 1. (The Pearson form.) The cells of row i have observed y_i, $n_i - y_i$ and fitted $n_i p$, $n_i(1-p)$, with deviations $\pm(y_i - n_i p)$. As in Proposition 5.9 the two terms add to $(y_i - n_i p)^2/[n_i p(1-p)] = n_i(p_i - p)^2/[p(1-p)]$.

Step 2. (Decomposition.) Write $p_i - p = (p_i - \hat{\pi}_i) + (\hat{\pi}_i - p)$. Then

$$\sum n_i (p_i - p)^2 = \sum n_i (p_i - \hat{\pi}_i)^2 + \sum n_i (\hat{\pi}_i - p)^2 + 2 \sum n_i (p_i - \hat{\pi}_i)(\hat{\pi}_i - p).$$

Step 3. (Cross term vanishes.) Since $\hat{\pi}_i - p = b(x_i - \bar{x})$,

$$\sum n_i (p_i - \hat{\pi}_i)(x_i - \bar{x}) = \sum n_i (p_i - p)(x_i - \bar{x}) - b \sum n_i (x_i - \bar{x})^2 = 0$$

by the definition of b.

Step 4. (Middle term.) $\sum n_i(\hat{\pi}_i - p)^2 = b^2 \sum n_i (x_i - \bar{x})^2$. Since $\sum n_i (x_i - \bar{x}) = 0$, $\sum n_i (p_i - p)(x_i - \bar{x}) = \sum (y_i - n_i p)(x_i - \bar{x}) = \sum (x_i - \bar{x}) y_i$, which gives the second form of z^2. Dividing Step 2 by $p(1-p)$ yields the partition.

Step 5. (Null distribution of z^2.) Under independence $E \sum (x_i - \bar{x}) Y_i = \pi \sum n_i (x_i - \bar{x}) = 0$ and $\mathrm{Var} \sum (x_i - \bar{x}) Y_i = \pi(1-\pi) \sum n_i (x_i - \bar{x})^2$. By the CLT and $p \xrightarrow{p} \pi$ (Slutsky), $z \xrightarrow{d} N(0,1)$. The limit of $X^2(L)$ follows from the chi-squared theory for the fit of a 2-parameter model to I binomials. ■

Proposition 5.14 (Cochran–Armitage = score test)

The score statistic for $H_0 : \beta = 0$ in the linear logit model (5.3) equals z^2 of Theorem 5.13.

Proof. [Supplement]

Step 1. Under H_0 the MLE is $\tilde{\alpha} = \mathrm{logit}\, p$, so $\tilde{\pi}_i = p$. By (4.16) in count form the score components at $(\tilde{\alpha}, 0)$ are

$$u_\alpha = \sum_i (y_i - n_i p) = 0, \qquad u_\beta = \sum_i x_i (y_i - n_i p) = \sum_i (x_i - \bar{x}) y_i.$$

Step 2. The information at $(\tilde{\alpha}, 0)$ is $X^\top \mathrm{diag}\{n_i p(1-p)\} X$:

$$\mathcal{I} = p(1-p) \begin{pmatrix} n & \sum n_i x_i \\ \sum n_i x_i & \sum n_i x_i^2 \end{pmatrix},$$

and by the formula for the inverse of a 2×2 matrix,

$$[\mathcal{I}^{-1}]_{\beta\beta} = \frac{n}{p(1-p)\left[n \sum n_i x_i^2 - (\sum n_i x_i)^2\right]} = \frac{1}{p(1-p) \sum n_i (x_i - \bar{x})^2}.$$

Step 3. The score statistic $\boldsymbol{u}^\top \mathcal{I}^{-1} \boldsymbol{u}$ with $u_\alpha = 0$ is $u_\beta^2 [\mathcal{I}^{-1}]_{\beta\beta} = z^2$. ■

The statistic also equals $M^2 = (n-1) r^2$ of Section 3.4 for $J = 2$, with $n - 1$ replaced by n; for $I = 2$, $X^2(L) = 0$ and $z^2 = X^2(I)$.

Example 5.15 (Alcohol: trend test)

With the scores above, $z^2 = 6.6$ ($P = 0.010$), and $X^2(I) = 12.1 = 6.6 + 5.5$, where $X^2(L) = 5.5$ (df $= 3$) shows only slight departure from linearity. The correlation is only $r = 0.014$ ($n = 32{,}574$), but r is a poor descriptive measure for such unbalanced tables. The Wald statistic is $(0.317/0.125)^2 = 6.4$ ($P = 0.012$) and the LR statistic 4.25 ($P = 0.039$); given the highly unbalanced counts the LR approach is safest, and the profile interval (0.02, 0.52) is preferable to the Wald interval (0.07, 0.56). Exact methods (Section 6.7.4) are relevant even though n is huge.

5.4 Multiple logistic regression

Motivation. With several predictors each coefficient is a conditional effect, controlling for the others. Comparing nested models by deviance differences is the main tool for deciding which predictors and interactions are needed.

For predictors $\mathbf{x} = (x_1, \ldots, x_p)$,

$$\operatorname{logit} \pi(\mathbf{x}) = \alpha + \beta_1 x_1 + \cdots + \beta_p x_p, \qquad \pi(\mathbf{x}) = \frac{\exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}{1 + \exp(\alpha + \beta_1 x_1 + \cdots + \beta_p x_p)}. \tag{5.4}$$

e^{β_j} is the multiplicative effect on the odds of a unit increase in x_j, at fixed levels of the other predictors.

5.4.1 Logit models for multiway contingency tables

With binary X and Z coded by dummies ($x_1 = z_1 = 1$, $x_2 = z_2 = 0$) and independent binomials at each (i, k), consider

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_1 x_i + \beta_2 z_k. \tag{5.5}$$

Proposition 5.16 (Main-effects model and homogeneous association)

In (5.5), at each level z_k of Z,

$$\operatorname{logit} P(Y = 1 \mid x = 1, z_k) - \operatorname{logit} P(Y = 1 \mid x = 0, z_k) = \beta_1,$$

so e^{β_1} is the conditional XY odds ratio, the same at each level of Z (homogeneous association). $\beta_1 = 0$ is conditional independence of X and Y given Z.

Proof. $(\alpha + \beta_1 + \beta_2 z_k) - (\alpha + \beta_2 z_k) = \beta_1$, free of z_k; a difference of logits is a log odds ratio. If $\beta_1 = 0$ the conditional odds ratios all equal 1. ■

No interaction thus means additivity on the logit scale; on another scale (probit, identity) interaction may appear or disappear, and in some applications another scale is natural. An ANOVA-type form handles factors with any numbers of categories:

$$\operatorname{logit} P(Y = 1) = \alpha + \beta_i^X + \beta_k^Z, \tag{5.6}$$

with one redundant parameter per factor (e.g. $\beta_I^X = \beta_K^Z = 0$; superscripts are labels). Conditional independence of X and Y is $\beta_1^X = \cdots = \beta_I^X$. For two categories each, (5.6) with $\beta_2^X = \beta_2^Z = 0$ is (5.5) with $\beta_1^X = \beta_1$, $\beta_1^Z = \beta_2$.

5.4.2 AIDS and AZT example

Example 5.17 (AIDS symptoms, AZT use and race)

338 veterans with faltering immune systems were randomized to immediate AZT or to AZT only after severe T-cell weakness; the response is development of AIDS symptoms over 3 years.

		Symptoms	
Race	AZT use	Yes	No
White	Yes	14	93
	No	32	81
Black	Yes	11	52
	No	12	43

In (5.5) let $x = 1$ for immediate AZT and $z = 1$ for whites. Fit:

Parameter	Estimate	SE	Wald χ^2
Intercept	-1.0736	0.2629	16.67
AZT	-0.7195	0.2790	6.65
race	0.0555	0.2886	0.04

(a) The estimated conditional odds ratio for AZT is $e^{-0.7195} = 0.487$: for each race the odds of symptoms are about halved by immediate AZT. Wald interval $\exp[-0.720 \pm 1.96(0.279)] = (0.28, 0.84)$; profile interval (0.279, 0.835). Race: 1.057, interval (0.60, 1.86).
(b) $H_0 : \beta_1 = 0$ (conditional independence of AZT and symptoms given race): LR statistic 6.9 (df $= 1$, $P = 0.01$); Wald $(-0.720/0.279)^2 = 6.65$.
(c) Codings: for (last $= 0$, first $= 0$, sum $= 0$) the intercept is $(-1.074, -1.738, -1.406)$, AZT-yes $(-0.720, 0, -0.360)$, white $(0.055, 0, 0.028)$. In each, intercept + AZT-yes + white $= -1.738$, so $\hat{\pi} = e^{-1.738}/(1+e^{-1.738}) = 0.15$ for whites with immediate AZT, and $\hat{\beta}_1^X - \hat{\beta}_2^X = -0.720$ always.
(d) Fitted probabilities (95% limits): white/AZT 0.150 (0.099, 0.220); white/no 0.265 (0.197, 0.348); black/AZT 0.143 (0.087, 0.225); black/no 0.255 (0.170, 0.364).

5.4.3 Goodness of fit as a likelihood-ratio test

Write $G^2(M_0 \mid M_1) = -2(L_0 - L_1)$ for the LR statistic testing M_0 within M_1. The goodness-of-fit statistic $G^2(M)$ is the case $M_0 = M$, $M_1 =$ saturated, with df = (number of binomials) − (number of parameters). By Proposition 4.22, with L_S the saturated log likelihood,

$$G^2(M_0 \mid M_1) = -2(L_0 - L_S) - [-2(L_1 - L_S)] = G^2(M_0) - G^2(M_1).$$

Example 5.18 (AIDS: goodness of fit)

Whites with immediate AZT have $\hat{\pi} = 0.150$, so the fitted counts are $107(0.150) = 16.0$ and $107(0.850) = 91.0$; similarly for all 8 cells. Then $G^2 = 1.38$, $X^2 = 1.39$, with df $= 4 - 3 = 1$ ($P > 0.2$). Since the model has no interaction, this is also a test of homogeneous odds ratios; adding the interaction gives the saturated model. For H_0: no race effect, $G^2(M_0) - G^2(M_1) = 1.42 - 1.38 = 0.04$.

The comparison statistic is often nearly chi-squared even when the separate $G^2(M_i)$ are not (continuous predictors, small fitted values), provided the models differ by few parameters.

5.4.4 Horseshoe crab example revisited

Example 5.19 (Horseshoe crabs: width and color)

Color has four observed categories (medium light, medium, medium dark, dark), a surrogate for age. With dummies c_1, c_2, c_3 for the first three (dark is the reference),

$$\operatorname{logit}\pi = \alpha + \beta_1 c_1 + \beta_2 c_2 + \beta_3 c_3 + \beta_4 x. \tag{5.7}$$

Parameter	Estimate	SE	LR 95% limits	χ^2
intercept	−12.7151	2.7618	(−18.4564, −7.5788)	21.20
c_1	1.3299	0.8525	(−0.2738, 3.1354)	2.43
c_2	1.4023	0.5484	(0.3527, 2.5260)	6.54
c_3	1.1061	0.5921	(−0.0279, 2.3138)	3.49
width	0.4680	0.1055	(0.2713, 0.6870)	19.66

Deviance 187.46, Pearson 168.66 on 168 df; log likelihood −93.73.

For dark crabs $\operatorname{logit}\hat{\pi} = -12.715 + 0.468x$; for medium-light, $-11.385 + 0.468x$. At $x = 26.3$, $\hat{\pi} = 0.399$ (dark) and 0.715 (medium light). Without interaction the curves for the colors are horizontal shifts of each other and never cross; a 1-cm increase multiplies the odds by $e^{0.468} = 1.60$ for every color. Medium-light versus dark: $e^{1.330} = 3.8$ at any width (at 26.3: $2.51/0.66 = 3.8$).

5.4.5 Model comparison

Example 5.20 (Horseshoe crabs: testing color and interaction)

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0$ (no color effect given width): $-2(L_0 - L_1) = 7.0$, df = 3, $P = 0.07$: slight evidence of a color effect. Adding the three color×width products (equivalent to separate logistic fits per color, so color comparisons depend on width) gives LR = 4.4, df = 3, $P = 0.22$: weak evidence of interaction.

5.4.6 Quantitative treatment of ordinal predictor

Example 5.21 (Horseshoe crabs: color scores)

With scores $c = 1, 2, 3, 4$,

$$\operatorname{logit}\pi = \alpha + \beta_1 c + \beta_2 x, \qquad \hat{\beta}_1 = -0.509\ (0.224), \quad \hat{\beta}_2 = 0.458\ (0.104).$$

Each step darker multiplies the odds by $e^{-0.509} = 0.60$ at fixed width. Against (5.7), LR = 1.7 (df = 2, $P = 0.44$): the color parameters are consistent with a linear trend in the scores. The qualitative estimates (1.33, 1.40, 1.11, 0) suggest the binary scoring dark (0) versus other (1); against (5.7) it gives LR = 0.5 (df = 2), and the fit is

$$\operatorname{logit}\hat{\pi} = -12.980 + 1.300c + 0.478x \qquad (\widehat{\text{se}}:\ 0.526,\ 0.104). \tag{5.8}$$

At a given width, non-dark crabs have $e^{1.300} = 3.7$ times the odds of dark crabs. All three models say dark crabs are least likely to have satellites; deciding the best scoring needs more data. Treating an ordinal predictor quantitatively, when it fits, is simpler and gives more powerful one-df tests (Section 6.4).

5.4.7 Standardized and probability-based interpretations

To compare quantitative predictors with different units, one can fit with standardized predictors $(x_j - \bar{x}_j)/s_{x_j}$, or equivalently multiply $\hat{\beta}_j$ by s_{x_j}: the effect of a one-standard-deviation change. For a probability scale, Proposition 5.2 extends: at a setting with $\hat{\pi}$, a unit increase in x_j (others fixed) changes the probability by about $\hat{\beta}_j \hat{\pi}(1-\hat{\pi})$. More precisely, set the other predictors at their means and report $\hat{\pi}$ at the quartiles of x_j; for a dummy predictor report $\hat{\pi}$ at its two values.

Example 5.22 (Horseshoe crabs: effects on the probability scale)

For (5.8) at $\hat{\pi} = 0.5$, a 1-cm width increase raises π by about $0.478(0.5)(0.5) = 0.12$, considerable since 1 cm is under half a standard deviation. With $\bar{c} = 0.873$, $\hat{\pi} = 0.51$ at the lower width quartile 24.9 and 0.80 at the upper quartile 27.7. At $x = 26.3$, $\hat{\pi} = 0.40$ for dark and 0.71 for other colors.

Variable	Estimate	SE	Comparison	Change in probability
No interaction model				
Intercept	−12.980	2.727		
Color (0 dark, 1 other)	1.300	0.526	$(1, 0)$ at $\bar{x}$	$0.31 = 0.71 - 0.40$
Width x (cm)	0.478	0.104	(UQ, LQ) at $\bar{c}$	$0.29 = 0.80 - 0.51$
Interaction model				
Intercept	−5.854	6.694		
Color	−6.958	7.318		
Width x	0.200	0.262	(UQ, LQ) at $c = 0$	$0.13 = 0.43 - 0.30$
Width × color	0.322	0.286	(UQ, LQ) at $c = 1$	$0.29 = 0.84 - 0.55$

The width effect appears larger for lighter crabs, but the interaction is not significant.

5.5 Fitting logistic regression models

Motivation. Logistic regression is a canonical-link binomial GLM, so the theory of Sections 4.4–4.6 specializes neatly: the likelihood is concave, Newton–Raphson coincides with Fisher scoring and IRLS, and finite estimates exist exactly when the two outcome groups overlap.

Let $\mathbf{x}_i = (x_{i1}, \ldots, x_{ip})$ be setting i $(i = 1, \ldots, N)$, with $x_{i1} = 1$ for the intercept, and let y_i be the number of successes in n_i trials (with continuous predictors typically $N = n$ and $n_i = 1$). The model is

$$\pi(\mathbf{x}_i) = \frac{\exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}{1 + \exp\left(\sum_{j=1}^{p} \beta_j x_{ij}\right)}. \tag{5.9}$$

5.5.1 Likelihood equations

Theorem 5.23 (Logistic log likelihood and likelihood equations)

For independent $y_i \sim \text{bin}(n_i, \pi(\mathbf{x}_i))$ under (5.9),

$$L(\boldsymbol{\beta}) = \sum_j \Big(\sum_i y_i x_{ij}\Big)\beta_j - \sum_i n_i \log\Big[1 + \exp\Big(\sum_j \beta_j x_{ij}\Big)\Big], \tag{5.10}$$

so $\{\sum_i y_i x_{ij}\}$ are sufficient, and the likelihood equations are

$$\sum_i y_i x_{ij} - \sum_i n_i \hat{\pi}_i x_{ij} = 0, \quad j = 1, \dots, p, \qquad \text{i.e.} \qquad X^\top \mathbf{y} = X^\top \hat{\boldsymbol{\mu}}, \tag{5.11}$$

where $\hat{\mu}_i = n_i \hat{\pi}_i$.

Proof.

Step 1. The joint mass function is proportional to

$$\prod_i \pi_i^{y_i} (1-\pi_i)^{n_i - y_i} = \Big\{ \prod_i (1-\pi_i)^{n_i} \Big\} \exp\Big[\sum_i y_i \log \frac{\pi_i}{1-\pi_i} \Big].$$

Step 2. With $\operatorname{logit} \pi_i = \sum_j \beta_j x_{ij}$, the exponent is $\sum_i y_i \sum_j \beta_j x_{ij} = \sum_j \beta_j \sum_i y_i x_{ij}$, and $1 - \pi_i = [1 + \exp(\sum_j \beta_j x_{ij})]^{-1}$. Taking logs gives (5.10); the factorization theorem gives sufficiency.

Step 3. Differentiating,

$$\frac{\partial L}{\partial \beta_j} = \sum_i y_i x_{ij} - \sum_i n_i x_{ij} \frac{\exp(\sum_k \beta_k x_{ik})}{1 + \exp(\sum_k \beta_k x_{ik})} = \sum_i (y_i - n_i \pi_i) x_{ij}.$$

■

These are the binomial GLM equations (4.16) written with counts, an instance of Theorem 4.27: for canonical links the likelihood equations equate sufficient statistics to their estimated expected values (they are the normal equations of ordinary regression for the identity link). They are nonlinear and need iteration.

5.5.2 Asymptotic covariance matrix of parameter estimators

Theorem 5.24 (Information matrix for logistic regression)

$$-\frac{\partial^2 L}{\partial \beta_a \partial \beta_b} = \sum_i x_{ia} x_{ib}\, n_i \pi_i (1 - \pi_i),$$

which does not involve $\mathbf{y}$; so observed and expected information coincide, and

$$\widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) = \{ X^\top \operatorname{diag}[n_i \hat{\pi}_i (1 - \hat{\pi}_i)] X \}^{-1}. \tag{5.12}$$

Proof. From Step 3 above, $\partial L / \partial \beta_a = \sum_i (y_i - n_i \pi_i) x_{ia}$, and by Proposition 5.2, Step 1, $\partial \pi_i / \partial \beta_b = \pi_i (1 - \pi_i) x_{ib}$. Hence $\partial^2 L / \partial \beta_a \partial \beta_b = -\sum_i n_i \pi_i (1 - \pi_i) x_{ia} x_{ib}$. This is free of y_i, so it equals its expectation, and Theorem 4.15 gives (5.12): the GLM form $(X^\top \hat{W} X)^{-1}$ with $w_i = n_i \pi_i (1 - \pi_i)$. ■

The square roots of the diagonal of (5.12) are the standard errors of $\hat{\boldsymbol{\beta}}$.

5.5.3 Distribution of probability estimators

For a setting $\mathbf{x}$, the fitted logit $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$ has estimated variance $\mathbf{x}^\top \widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{x}$, so

$$\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm z_{\alpha/2} \sqrt{\mathbf{x}^\top \widehat{\operatorname{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{x}}$$

is a large-sample interval for $\operatorname{logit} \pi(\mathbf{x})$; applying $t \mapsto e^t/(1+e^t)$ to the endpoints gives one for $\pi(\mathbf{x})$ (Proposition 5.6).

5.5.4 Newton–Raphson method applied to logistic regression

Proposition 5.25 (Newton–Raphson / IRLS for logistic regression)

With $\pi_i^{(t)} = \exp(\mathbf{x}_i^\top \boldsymbol{\beta}^{(t)})/[1+\exp(\mathbf{x}_i^\top \boldsymbol{\beta}^{(t)})]$ and $\mu_i^{(t)} = n_i \pi_i^{(t)}$, the Newton–Raphson step is

$$\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \{X^\top \mathrm{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X\}^{-1} X^\top (\mathbf{y} - \boldsymbol{\mu}^{(t)}). \tag{5.13}$$

Equivalently $\boldsymbol{\beta}^{(t+1)} = (X^\top V_t^{-1} X)^{-1} X^\top V_t^{-1} \boldsymbol{z}^{(t)}$, with $V_t = \mathrm{diag}\{1/[n_i \pi_i^{(t)}(1-\pi_i^{(t)})]\}$ and

$$z_i^{(t)} = \log \frac{\pi_i^{(t)}}{1-\pi_i^{(t)}} + \frac{y_i - n_i \pi_i^{(t)}}{n_i \pi_i^{(t)}(1-\pi_i^{(t)})}. \tag{5.14}$$

At convergence, $-H^{-1}$ is the estimated covariance (5.12).

Proof. By Theorems 5.23 and 5.24, $\boldsymbol{u}^{(t)} = X^\top(\mathbf{y} - \boldsymbol{\mu}^{(t)})$ and $H^{(t)} = -X^\top \mathrm{diag}[n_i \pi_i^{(t)}(1-\pi_i^{(t)})] X$; substitute in (4.25). Since $H = -\mathcal{I}$ (canonical link), this is also Fisher scoring, hence IRLS by Theorem 4.26. For the logit, $\partial \eta_i / \partial \mu_i = 1/[n_i \pi_i (1-\pi_i)]$ when $\mu_i = n_i \pi_i$, so the working response (4.28) is (5.14), and $W = V_t^{-1}$. ■

The elements of V_t are the estimated asymptotic variances of sample logits (Section 3.1.6), and $\boldsymbol{z}^{(t)}$ is the linearized logit of the sample data. Starting from $\boldsymbol{\beta}^{(0)}$ (e.g. $\mathbf{0}$), the iterates converge to $\hat{\boldsymbol{\beta}}$ (Walker and Duncan 1967).

5.5.5 Convergence and existence of finite estimates

Theorem 5.26 (Concavity and uniqueness)

If X has full column rank, $L(\boldsymbol{\beta})$ in (5.10) is strictly concave; hence the ML estimate, when it exists, is unique, and any root of (5.11) is it.

Proof. By Theorem 5.24, $\boldsymbol{c}^\top H \boldsymbol{c} = -\sum_i n_i \pi_i (1-\pi_i)(\mathbf{x}_i^\top \boldsymbol{c})^2$ with every $n_i \pi_i (1-\pi_i) > 0$. It is < 0 unless $X\boldsymbol{c} = \mathbf{0}$, i.e. $\boldsymbol{c} = \mathbf{0}$. A strictly concave function has at most one maximizer, and a stationary point of a concave function is a global maximum. ■

Theorem 5.27 (Existence of the ML estimate: overlap versus separation)

Write the data as n individual binary observations $(\mathbf{x}_i, y_i)$ and let $s_i = 2y_i - 1 \in \{-1, 1\}$. Assume X has full column rank. Call the data *separated* if some $\boldsymbol{d} \neq \mathbf{0}$ satisfies $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i, with strict inequality for at least one i (a hyperplane with all $y = 1$ on one side and all $y = 0$ on the other, ties on it allowed). Otherwise the data *overlap*.
(a) If the data are separated, no ML estimate exists: L increases indefinitely along $\boldsymbol{d}$ and some $|\hat{\beta}_j|$ is "infinite".
(b) If the data overlap, a finite ML estimate exists, and it is unique.

Proof. [Supplement]

Step 1. (A convenient form.) For one binary observation, $y\eta - \log(1+e^\eta) = -\log(1+e^{-s\eta})$ (check $y = 1$ and $y = 0$). Hence

$$L(\boldsymbol{\beta}) = -\sum_i \log[1 + \exp(-s_i \mathbf{x}_i^\top \boldsymbol{\beta})] < 0.$$

Each term is a strictly increasing function of $s_i \mathbf{x}_i^\top \boldsymbol{\beta}$.

Step 2. (a) Suppose $\boldsymbol{d}$ separates and $\hat{\boldsymbol{\beta}}$ were a maximizer. For $t > 0$, $s_i \mathbf{x}_i^\top (\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) \geq s_i \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$ for all i, with strict inequality for some i. By Step 1, $L(\hat{\boldsymbol{\beta}} + t\boldsymbol{d}) > L(\hat{\boldsymbol{\beta}})$, a contradiction. So the supremum

is not attained; along $\boldsymbol{\beta} + t\boldsymbol{d}$ the likelihood keeps increasing as $t \to \infty$ (under complete separation, $s_i \mathbf{x}_i^\top \boldsymbol{d} > 0$ for all i, it tends to the perfect-fit value $L = 0$).

Step 3. (b) Overlap and full rank imply: for every unit vector $\boldsymbol{d}$, $m(\boldsymbol{d}) = \min_i s_i \mathbf{x}_i^\top \boldsymbol{d} < 0$. Indeed, otherwise $s_i \mathbf{x}_i^\top \boldsymbol{d} \geq 0$ for all i; not separated forces all equalities, so $X\boldsymbol{d} = \mathbf{0}$, contradicting full rank.

Step 4. m is continuous on the compact unit sphere, so $\max_{\|\boldsymbol{d}\|=1} m(\boldsymbol{d}) = -c < 0$. For $\boldsymbol{\beta} = r\boldsymbol{d}$ with $r > 0$, $\|\boldsymbol{d}\| = 1$, choose i with $s_i \mathbf{x}_i^\top \boldsymbol{d} \leq -c$; then by Step 1

$$L(r\boldsymbol{d}) \leq -\log[1 + \exp(rc)] \leq -rc.$$

Step 5. Hence $\{\boldsymbol{\beta} : L(\boldsymbol{\beta}) \geq L(\mathbf{0})\} \subset \{\|\boldsymbol{\beta}\| \leq -L(\mathbf{0})/c\}$ is bounded; it is closed by continuity, hence compact and nonempty. The continuous L attains its maximum on it, which is a global maximum. Uniqueness is Theorem 5.26. ■

Similar results hold for the probit and some other links (Silvapulle 1981). In practice software rarely detects $\hat{\beta} = \pm\infty$: after a few cycles the log likelihood looks flat, convergence criteria are met, and enormous standard errors are reported.

Example 5.28 (Perfect discrimination)

Let $y = 0$ at $x = 10, 20, 30, 40$ and $y = 1$ at $x = 60, 70, 80, 90$. The ideal fit has $\hat{\pi} = 0$ for $x \leq 40$ and $\hat{\pi} = 1$ for $x \geq 60$. Taking $\alpha = -50\beta$ (so $\hat{\pi}(50) = 0.5$) and letting $\beta \to \infty$ gives ever larger likelihood approaching a perfect fit (Figure 8). PROC GENMOD reports $\operatorname{logit} \hat{\pi} = -192.2 + 3.8x$ with standard errors 8.0×10^8 and 1.5×10^7, numerical artifacts of $\hat{\beta} = \infty$.

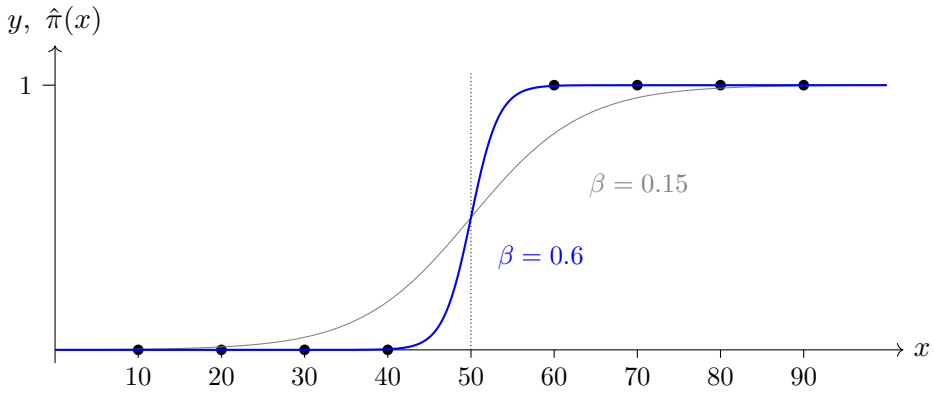


Figure 8: Complete separation: with $\alpha = -50\beta$ the fitted curve sharpens toward a step at $x = 50$ as $\beta \to \infty$, and the likelihood increases without attaining a maximum.

Chapter 6. Building and Applying Logistic Regression Models

6.1 Strategies in model selection

Motivation. With several explanatory variables the number of candidate logistic models explodes. We need a model complex enough to fit well but simple enough to interpret, and principled tools (tests, AIC, subject-matter reasoning) to choose among candidates.

Confirmatory studies compare a few models chosen in advance by the study questions; exploratory studies search more widely and treat the result as a source of hypotheses. In either case, first look at the marginal effect of each predictor (a smoothed plot for a continuous x, a contingency table for a categorical one).

Practical guidelines

(a) *Events per variable.* A rough rule asks for at least 10 outcomes of each type per predictor: if $y = 1$ only 30 times among $n = 1000$, use at most about three x terms.
(b) *Multicollinearity.* Strongly correlated predictors can make each one look unimportant given the others, even when the global test is highly significant. Dropping a redundant predictor reduces the standard errors of the others.

6.1.1 Horseshoe crab example revisited

Example 6.1 (Horseshoe crabs: all main effects)

For 173 female crabs, $y = 1$ if she has at least one satellite. Predictors: color C (4 categories), spine condition S (3), weight, and carapace width W. The main-effects model

$$\text{logit}\, P(Y = 1) = \alpha + \beta_1\, \text{weight} + \beta_2\, \text{width} + \beta_3 c_1 + \beta_4 c_2 + \beta_5 c_3 + \beta_6 s_1 + \beta_7 s_2$$

(dummies for the first three colors and first two spine conditions) gives the LR test of $H_0 : \beta_1 = \cdots = \beta_7 = 0$ equal to 40.6 ($df = 7$, $P < 0.0001$). Yet no single effect is clearly significant:

Parameter	Estimate	SE	Parameter	Estimate	SE
Intercept	−9.273	3.838	color 2	1.506	0.567
weight	0.826	0.704	color 3	1.120	0.593
width	0.263	0.195	spine 1	−0.400	0.503
color 1	1.609	0.936	spine 2	−0.496	0.629

A significant global test with insignificant individual effects is a warning sign of multicollinearity: weight and width have correlation 0.887. We keep width W and use color C and spine condition S as factors. Models are named by their highest-order terms: $C + S + W$ (main effects), $C + S{*}W$ (adds the $S \times W$ interaction), etc. A model with an interaction should also contain the corresponding main effects.

6.1.2 Stepwise procedures

Forward selection adds, at each stage, the term giving the greatest improvement (smallest P-value, since different terms have different df); the *stepwise* variant re-tests earlier terms at each stage.

Backward elimination starts from a complex model and at each stage removes the term whose removal hurts least (largest P-value), stopping when every deletion significantly worsens the fit.

Cautions

(a) Add or drop a factor (or an interaction containing it) as a whole, never single dummy variables; otherwise the result depends on the coding.
(b) Backward elimination is usually preferred: forward selection may stop early because one test in the sequence has low power.
(c) With many tests, unimportant terms can look impressive by chance, and the largest sample effect tends to overestimate its true effect. If forward, backward and best-subset searches (or a search on a bootstrap sample) disagree, the selected model is of dubious value.
(d) A variable central to the study may be kept and reported even if not significant; it can reduce bias in other effects and make results comparable with other studies.

6.1.3 Backward elimination for the horseshoe crab example

The deviance G^2 against the saturated model is not approximately χ^2 with a continuous predictor, but differences of deviances between nested models differing by a few parameters are LR statistics $-2(L_0 - L_1)$ with approximate χ^2 null distributions.

Example 6.2 (Horseshoe crabs: backward elimination)

Selected rows of the book's Table 6.2 (C color, S spine, W width; "vs." gives the comparison and deviance difference):

	Model	G^2	df	AIC	Comparison	Difference
1	$C{*}S{*}W$	170.44	152	212.4	–	–
2	$C{*}S + C{*}W + S{*}W$	173.68	155	209.7	(2)–(1)	3.2 ($df = 3$)
3c	$C{*}S + C{*}W$	173.69	157	205.7	(3c)–(2)	0.0 ($df = 2$)
4b	$W + C{*}S$	177.61	160	203.6	(4b)–(3c)	3.9 ($df = 3$)
5	$C + S + W$	186.61	166	200.6	(5)–(4b)	9.0 ($df = 6$)
6c	$C + W$	187.46	168	197.5	(6c)–(5)	0.8 ($df = 2$)
7a	C	212.06	169	220.1	(7a)–(6c)	24.5 ($df = 1$)
7b	W	194.45	171	198.5	(7b)–(6c)	7.0 ($df = 3$)
8	$C = \text{dark} + W$	187.96	170	194.0	(8)–(6c)	0.5 ($df = 2$)
9	None	225.76	172	227.8	(9)–(8)	37.8 ($df = 2$)

Starting from $C{*}S{*}W$, the three-factor interaction is dropped ($P = 0.36$), then $S \times W$, then the remaining two-factor interactions ($P = 0.27$ and then $P = 0.17$). Among main effects S can go, but removing C from $C + W$ costs 7.0 on $df = 3$ ($P = 0.07$) and removing W costs 24.5. Replacing the four colors by a single dummy (dark vs. not) loses almost nothing. Final model: width plus a "dark" indicator.

6.1.4 AIC, model selection, and the "correct" model

No model is exactly true; a parsimonious model with small bias may estimate quantities of interest (e.g. cell probabilities) better than a complex one. The *Akaike information criterion* judges a model by how close its fitted values are expected to be to the truth.

Definition (AIC)

For a model with p parameters and maximized log likelihood L_M,

$$\text{AIC} = -2\,(L_M - p) = -2L_M + 2p.$$

Smaller is better: the first term rewards fit, the second penalizes complexity.

Proposition 6.3 (AIC and the adjusted deviance)

Let models for a categorical response be fitted to the same data, with N observations in grouped form (so the saturated model has N parameters and the residual degrees of freedom are $df = N - p$). Then

$$\text{AIC} = [G^2 - 2\,df] + \text{const},$$

where the constant $-2L_S + 2N$ is the same for all models. Hence ordering models by AIC is the same as ordering them by $G^2 - 2\,df$.

Proof. By definition $G^2 = -2(L_M - L_S)$, so $-2L_M = G^2 - 2L_S$. Substituting $p = N - df$,

$$\text{AIC} = G^2 - 2L_S + 2(N - df) = [G^2 - 2\,df] + (2N - 2L_S).$$

The last bracket depends only on the data and the saturated model. ■

Theorem 6.4 (Akaike's bias correction)

Let $Y_1, \ldots, Y_n$ be i.i.d. with density $f(\cdot;\theta_0)$ belonging to a regular parametric model with p-dimensional parameter, $\hat\theta$ the MLE and $\ell(\theta) = \sum_i \log f(Y_i;\theta)$. Let $Y_1^*, \ldots, Y_n^*$ be an independent copy of the data and $\ell^*(\theta)$ its log likelihood. Then, to first order,

$$E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p.$$

Thus $-2\ell(\hat\theta) + 2p = \text{AIC}$ is an approximately unbiased estimate of $-2E[\ell^*(\hat\theta)]$, the expected out-of-sample deviance (equivalently, of $2n$ times the Kullback–Leibler discrepancy of $f(\cdot;\hat\theta)$ from the truth, up to a constant).

Proof. [Supplement] Let I be the Fisher information per observation, so $-\ell''(\theta_0)/n \to I$ and $\sqrt{n}(\hat\theta - \theta_0) \xrightarrow{d} N(0, I^{-1})$.

Step 1. Expand ℓ about $\hat\theta$, where $\ell'(\hat\theta) = 0$:

$$\ell(\theta_0) \approx \ell(\hat\theta) - \tfrac{n}{2}(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0).$$

Since $n(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0) \xrightarrow{d} \chi^2_p$ with mean p, $E[\ell(\hat\theta)] \approx E[\ell(\theta_0)] + p/2$.

Step 2. Expand ℓ^* about θ_0. Because $\ell^{*\prime}(\theta_0)$ is independent of $\hat\theta$ with mean zero,

$$E[\ell^*(\hat\theta)] \approx E[\ell^*(\theta_0)] + E[\ell^{*\prime}(\theta_0)^\top(\hat\theta - \theta_0)] - \tfrac{n}{2}E[(\hat\theta - \theta_0)^\top I(\hat\theta - \theta_0)] = E[\ell^*(\theta_0)] - \tfrac{p}{2}.$$

Step 3. $\ell(\theta_0)$ and $\ell^*(\theta_0)$ have the same distribution, so subtracting the two displays gives $E[\ell(\hat\theta)] - E[\ell^*(\hat\theta)] \approx p$. Hence $E[-2\ell(\hat\theta) + 2p] \approx -2E[\ell^*(\hat\theta)]$. Finally $-2E[\ell^*(\theta)]/n = 2\,\mathrm{KL}(f_{\theta_0} \| f_\theta) - 2E\log f(Y;\theta_0)$, and the last term does not depend on the model. ■

Example 6.5 (Horseshoe crabs: AIC)

Among models built from C, S, W, AIC is smallest for $C + W$ (197.5); the dark-indicator model does better still (194.0). Its advantage should be weighed against the fact that it was suggested by the $C + W$ fit.

6.1.5 Using causal hypotheses to guide model building

A time ordering among variables suggests a sequence of logit models, each response being explained by variables earlier in the chain.

Example 6.6 (Divorce and pre/extramarital sex)

A British study cross-classified G = gender, P = premarital sex, E = extramarital sex, M = marital status:

	Women				Men			
PMS	Yes		No		Yes		No	
EMS	Yes	No	Yes	No	Yes	No	Yes	No
Divorced	17	54	36	214	28	60	17	68
Still married	4	25	4	322	11	42	4	130

The ordering $G \to P \to E \to M$ suggests a causal diagram in which G affects P; G, P affect E; and E, P (not G directly) affect M. Fitting each stage:

Response	Explanatory	G^2	*df*
P	none	75.3	1
E	none / P / $G + P$	48.9 / 2.9 / 0.0	3 / 2 / 1
M	$E + P$ / $E{*}P$ / $E{*}P + G$	18.2 / 5.2 / 0.7	5 / 4 / 3

Stage 1 shows a strong G–P association (odds ratio 0.27). At stage 2, adding G to P reduces G^2 by only 2.9, weak evidence of a direct G effect; the EP conditional odds ratio is 4.0. At stage 3, the hypothesized $E + P$ fits poorly; an $E \times P$ interaction is needed. The causal reasoning led close to, but not exactly to, a well-fitting model.

6.1.6 New model-building strategies for data mining

For huge data sets (*data mining*) almost any effect is significant, so tests are of little use; complex models may still be valuable for prediction. Nevertheless, beyond some point new predictors are nearly linear combinations of existing ones and add no predictive power. Summary measures of predictive power (Section 6.2) become more relevant than inference.

6.2 Logistic regression diagnostics

Motivation. A global goodness-of-fit statistic can hide where and why a model fails. Residuals and influence measures examine the fit observation by observation and suggest improvements.

6.2.1 Pearson, deviance, and standardized residuals

Let y_i be binomial with n_i trials at setting i ($i = 1, \dots, N$) of the predictors, $\hat{\pi}_i$ the fitted probability, and $\hat{h}_i$ the leverage, the ith diagonal element of the estimated GLM hat matrix

$$\widehat{\text{Hat}} = \hat{W}^{1/2} X (X^\top \hat{W} X)^{-1} X^\top \hat{W}^{1/2}, \qquad \hat{W} = \text{diag}(n_i \hat{\pi}_i (1 - \hat{\pi}_i)).$$

Definition (Residuals for binomial GLMs)

(a) Pearson residual:

$$e_i = \frac{y_i - n_i \hat{\pi}_i}{\sqrt{n_i \hat{\pi}_i (1 - \hat{\pi}_i)}}. \tag{6.1}$$

(b) Standardized (adjusted) Pearson residual: $r_i = e_i / \sqrt{1 - \hat{h}_i}$.
(c) Deviance residual: $\sqrt{d_i}\ \text{sgn}(y_i - n_i \hat{\pi}_i)$, where

$$d_i = 2\Big[y_i \log \frac{y_i}{n_i \hat{\pi}_i} + (n_i - y_i) \log \frac{n_i - y_i}{n_i - n_i \hat{\pi}_i} \Big].$$

By construction $X^2 = \sum_i e_i^2$ and $G^2 = \sum_i d_i$.

Theorem 6.7 (Approximate distribution of residuals)

Suppose the logistic model holds with p parameters and the n_i are large. Then, approximately,
(a) $\text{Var}(y_i - n_i \hat{\pi}_i) \approx n_i \pi_i (1 - \pi_i)(1 - h_i)$, so $\text{Var}(e_i) \approx 1 - h_i < 1$ while r_i is approximately $N(0, 1)$;
(b) $E(\sum_i e_i^2) \approx N - p = df$, so $E(\sum_i e_i^2)/N \approx df/N < 1$.

Proof. [Supplement] Write $\mu = (n_i \pi_i)$, $W = \text{diag}(n_i \pi_i (1 - \pi_i)) = \text{Cov}(y)$, $A = X^\top W X$.

Step 1. The likelihood equations for the canonical (logit) link are $X^\top (y - \hat{\mu}) = 0$. Since $\partial \mu / \partial \beta^\top = WX$, linearization gives $\hat{\mu} - \mu \approx WX(\hat{\beta} - \beta)$, and expanding the likelihood equations,

$$0 = X^\top (y - \hat{\mu}) \approx X^\top (y - \mu) - A(\hat{\beta} - \beta) \quad \Longrightarrow \quad \hat{\beta} - \beta \approx A^{-1} X^\top (y - \mu).$$

Step 2. Hence $y - \hat{\mu} \approx (I - WXA^{-1}X^\top)(y - \mu)$ and

$$\begin{aligned} \text{Cov}(y - \hat{\mu}) &\approx (I - WXA^{-1}X^\top)\, W\, (I - XA^{-1}X^\top W) \\ &= W - 2WXA^{-1}X^\top W + WXA^{-1}(X^\top W X)A^{-1}X^\top W = W - WXA^{-1}X^\top W. \end{aligned}$$

Step 3. Pre- and post-multiplying by $W^{-1/2}$: $\text{Cov}(W^{-1/2}(y - \hat{\mu})) \approx I - H$ with $H = W^{1/2} X A^{-1} X^\top W^{1/2}$, whose diagonal is (h_i). The ith component of $W^{-1/2}(y - \hat{\mu})$ is e_i (with π_i in place of $\hat{\pi}_i$, a negligible change), proving (a); asymptotic normality of e_i follows from that of y, and dividing by $\sqrt{1 - h_i}$ gives unit variance.

Step 4. H is a projection of rank p, so $E \sum e_i^2 \approx \text{tr}(I - H) = N - p$. ■

When residuals are uninformative

With continuous predictors usually $n_i = 1$; then $y_i \in \{0, 1\}$, e_i takes only two values, residual plots show two parallel bands, and the deviance itself carries no information about fit. Group observations with common predictor values whenever possible. With very small fitted counts, residuals lose relevance just as X^2 and G^2 do.

6.2.2 Heart disease example

Example 6.8 (Blood pressure and heart disease)

Framingham men aged 40–59, classified by blood pressure; response: coronary heart disease within six years. Scores x_i: $111.5, 121.5, \ldots, 161.5, 176.5, 191.5$.

Blood	Sample	Heart	Fitted		Std. residual	
pressure	size	disease	Indep.	Linear	Indep.	Linear
< 117	156	3	10.8	5.2	−2.62	−1.11
117–126	252	17	17.4	10.6	−0.12	2.37
127–136	284	12	19.7	15.1	−2.02	−0.95
137–146	271	16	18.8	18.1	−0.74	−0.57
147–156	139	12	9.6	11.6	0.84	0.13
157–166	85	8	5.9	8.9	0.93	−0.33
167–186	99	16	6.9	14.2	3.76	0.65
> 186	43	8	3.0	8.4	3.07	−0.18

The independence model $\text{logit}\,\pi_i = \alpha$ fits poorly ($G^2 = 30.0$, $X^2 = 33.4$, $df = 7$) and its residuals increase with x, suggesting the linear logit model $\text{logit}\,\pi_i = \alpha + \beta x_i$. For that model the trend disappears and $G^2 = 5.9$, $X^2 = 6.3$ ($df = 6$). Only category 2 has a largish residual ($r_2 = 2.37$; Pearson 2.01, deviance 1.85), unsurprising among eight residuals.

6.2.3 Graduate admissions example

Example 6.9 (Admissions at the University of Florida)

Applications to 23 departments, by gender G and admission A; π_{ik} = admission probability for gender i in department k. The model with no gender effect, $\text{logit}\,\pi_{ik} = \alpha + \beta_k^D$, fits rather poorly ($G^2 = 44.7$, $X^2 = 40.9$, $df = 23$). Admitted/rejected counts and the standardized residual for (female, admitted):

	Female		Male				Female		Male		
Dept	Yes	No	Yes	No	r	Dept	Yes	No	Yes	No	r
anth	32	81	21	41	−0.76	ling	21	10	7	8	1.37
astr	6	0	3	8	2.87	math	25	18	31	37	1.29
chem	12	43	34	110	−0.27	phil	3	0	9	6	1.34
clas	3	1	4	0	−1.07	phys	10	11	25	53	1.32
comm	52	149	5	10	−0.63	poli	25	34	39	49	−0.23
comp	8	7	6	12	1.16	psyc	2	123	4	41	−2.27
engl	35	100	30	112	0.94	reli	3	3	0	2	1.26
geog	9	1	11	11	2.17	roma	29	13	6	3	0.14
geol	6	3	15	6	−0.26	soci	16	33	7	17	0.30
germ	17	0	4	1	1.89	stat	23	9	36	14	−0.01
hist	9	9	21	19	−0.18	zool	4	62	10	54	−1.76
lati	26	7	25	16	1.65						

Astronomy and geography admitted significantly more females than predicted, psychology fewer; without these three departments the model fits well ($G^2 = 24.4$, $df = 20$). Adding a gender effect barely helps ($G^2 = 42.4$, $df = 22$), because the deviations go in different directions. That model's conditional GA odds ratio is 1.19 (odds of admission 19% higher for females given department), whereas the marginal odds ratio is 0.94: Simpson's paradox.

Proposition 6.10 (Residuals under conditional independence)

In a $2 \times 2 \times K$ table with $y_{ik} \sim \text{bin}(n_{ik}, \pi_{ik})$, the model $\text{logit}\,\pi_{ik} = \alpha + \beta_k$ has $\hat{\pi}_{1k} = \hat{\pi}_{2k} = y_{+k}/n_{+k}$, and in each stratum the raw residuals satisfy $y_{1k} - n_{1k}\hat{\pi}_{1k} = -(y_{2k} - n_{2k}\hat{\pi}_{2k})$. Consequently the two standardized residuals in stratum k are equal in magnitude and opposite in sign, whereas the two ordinary Pearson residuals need not be.

Proof.

Step 1. The model is saturated for the column $(y_{1k} + y_{2k})$ in each stratum and has no gender term, so the likelihood equations $\sum_i y_{ik} = \sum_i n_{ik}\hat{\pi}_{ik}$ with $\hat{\pi}_{1k} = \hat{\pi}_{2k}$ give $\hat{\pi}_{ik} = y_{+k}/n_{+k}$.

Step 2. Then

$$y_{1k} - n_{1k}\hat{\pi}_{1k} = y_{1k} - \frac{n_{1k}(y_{1k} + y_{2k})}{n_{+k}} = \frac{n_{2k}y_{1k} - n_{1k}y_{2k}}{n_{+k}} = -(y_{2k} - n_{2k}\hat{\pi}_{2k}).$$

Step 3. Two random variables that are negatives of each other have the same (estimated) standard error, so after standardizing by the true standard error of the raw residual the two residuals are $\pm$ the same number. The ordinary Pearson residuals divide instead by $\sqrt{n_{ik}\hat{\pi}_{ik}(1 - \hat{\pi}_{ik})}$, which differs for $i = 1, 2$ when $n_{1k} \neq n_{2k}$. ■

6.2.4 Influence diagnostics for logistic regression

Deleting a poorly fitted observation and refitting is equivalent to adding a parameter that fits it perfectly. High leverage means high potential influence, but in logistic regression the leverage depends on the fit through $\hat{W}$: an extreme x_i with $\hat{\pi}_i$ near 0 or 1 can have small leverage. Influence measures for observation i (a binary response or a binomial count) include:

(1) *Dfbeta*: the change in each $\hat{\beta}_j$ when i is deleted, divided by its SE;

(2) *confidence interval displacement* c_i: the change in a joint confidence region for β;

(3) ΔX_i^2, ΔG_i^2: the decrease in the fit statistics when i is deleted.

Theorem 6.11 (One-step influence approximations)

Let $A = X^\top \hat{W} X$, $w_i = n_i\hat{\pi}_i(1 - \hat{\pi}_i)$, $h_i = w_i x_i^\top A^{-1} x_i$, and let $\hat{\beta}_{(i)}$ be the one-step Newton approximation (started from $\hat{\beta}$) to the estimate without observation i. Then

$$\hat{\beta} - \hat{\beta}_{(i)} = \frac{A^{-1}x_i\,(y_i - n_i\hat{\pi}_i)}{1 - h_i}, \qquad c_i = (\hat{\beta} - \hat{\beta}_{(i)})^\top A(\hat{\beta} - \hat{\beta}_{(i)}) = \frac{e_i^2 h_i}{(1 - h_i)^2},$$

and the score statistic for the extra parameter that fits observation i perfectly equals $r_i^2 = e_i^2/(1 - h_i)$, the usual approximation to ΔX_i^2.

Proof. [Supplement]

Step 1. At $\hat{\beta}$ the full score is zero: $\sum_j x_j(y_j - n_j\hat{\pi}_j) = 0$. Hence the score of the reduced data is $U_{(i)}(\hat{\beta}) = -x_i(y_i - n_i\hat{\pi}_i)$, and its information is $A - w_i x_i x_i^\top$. One Newton step gives $\hat{\beta}_{(i)} = \hat{\beta} + (A - w_i x_i x_i^\top)^{-1} U_{(i)}(\hat{\beta})$.

Step 2. By the Sherman–Morrison formula,

$$(A - w_i x_i x_i^\top)^{-1} x_i = A^{-1}x_i + \frac{w_i A^{-1} x_i x_i^\top A^{-1} x_i}{1 - w_i x_i^\top A^{-1} x_i} = \frac{A^{-1}x_i}{1 - h_i},$$

which yields the formula for $\hat{\beta} - \hat{\beta}_{(i)}$.

Step 3. Then $c_i = (y_i - n_i\hat{\pi}_i)^2 x_i^\top A^{-1} x_i/(1-h_i)^2 = (y_i - n_i\hat{\pi}_i)^2 h_i/\{w_i(1-h_i)^2\} = e_i^2 h_i/(1-h_i)^2$.

Step 4. Add a parameter γ with covariate $z = \mathbf{e}_i$ (indicator of observation i). At $(\hat{\beta}, \gamma = 0)$ the score for γ is $U_\gamma = y_i - n_i\hat{\pi}_i$, and the information blocks are $I_{\gamma\gamma} = w_i$, $I_{\gamma\beta} = w_i x_i^\top$, $I_{\beta\beta} = A$. The efficient information for γ is $w_i - w_i^2 x_i^\top A^{-1} x_i = w_i(1-h_i)$, so the score statistic is $(y_i - n_i\hat{\pi}_i)^2/\{w_i(1-h_i)\} = r_i^2$. Fitting γ is the same as deleting observation i, so this approximates the resulting drop in X^2 (and G^2). ■

Example 6.12 (Heart disease: influence)

For the linear logit model of Example 6.8, the second category has by far the largest Dfbeta (-1.14), c (2.26), ΔX^2 (5.64) and ΔG^2 (5.04); all others are below 1.4. For the independence model, deleting an extreme category changes X^2 by up to 14.2, but that removes a whole binomial sample, not one subject.

6.2.5 Summarizing predictive power: R and R-squared measures

No analog of R^2 is as universally useful for categorical responses as R^2 is for linear models. Three proposals:

(a) The correlation $r(y, \hat{\mu})$ between observations and fitted values (for binary data, $\hat{\mu}_i = \hat{\pi}_i$). For the crabs, $r = 0.402$ for W, 0.452 for $C + W$, 0.447 for dark+W, 0.526 for $C{*}S{*}W$; the last is inflated by its many parameters (a jackknife adjustment removes most of the difference).

(b) Proportional reduction in squared error $1 - \sum(y_i - \hat{\pi}_i)^2/\sum(y_i - \bar{y})^2$. Neither (a) nor (b) need be monotone in model complexity for logistic regression.

(c) Likelihood-based measures, below.

Proposition 6.13 (Likelihood R^2 measures)

Let $L_0 \le L_M \le L_S$ be the maximized log likelihoods of the null (intercept-only), the given and the saturated models. Then $(L_M - L_0)/(L_S - L_0) \in [0, 1]$, and:
(a) for n ungrouped Bernoulli observations, $L_S = 0$ and the measure reduces to McFadden's $D = (L_0 - L_M)/L_0$, where $L_0 = n[\bar{y}\log\bar{y} + (1-\bar{y})\log(1-\bar{y})]$;
(b) for N grouped binomial counts it equals $D^* = \{G^2(0) - G^2(M)\}/G^2(0)$.

Proof. The parameter spaces are nested (null $\subset M \subset$ saturated), so $L_0 \le L_M \le L_S$, and all are ≤ 0 since probabilities are at most 1; this gives the range.

(a) For Bernoulli data the saturated model sets $\hat{\pi}_i = y_i \in \{0, 1\}$, so each term $y_i \log\hat{\pi}_i + (1 - y_i)\log(1-\hat{\pi}_i) = 0$ and $L_S = 0$. The null model has $\hat{\pi}_i = \bar{y}$, giving L_0.

(b) $G^2(M) = -2(L_M - L_S)$ and $G^2(0) = -2(L_0 - L_S)$, so $(L_M - L_0)/(L_S - L_0) = \{G^2(0) - G^2(M)\}/G^2(0)$. ■

Why D^* is misleading

$D^* = 1$ for any perfectly fitting model, however poorly individual responses can be predicted. If the model holds but the null model does not, as $n \to \infty$ with N fixed $G^2(M)$ stays $O_p(1)$ while $G^2(0) \to \infty$, so $D^* \to 1$. It confounds goodness of fit with predictive power; use D on ungrouped data.

6.2.6 Summarizing predictive power: classification tables and ROC curves

A *classification table* cross-classifies y with the prediction $\hat{y} = 1$ if $\hat{\pi}_i > \pi_0$ (often $\pi_0 = 0.5$), summarized by sensitivity $P(\hat{y} = 1 \mid y = 1)$ and specificity $P(\hat{y} = 0 \mid y = 0)$. It discards information, depends on the arbitrary π_0, and is sensitive to the relative frequencies of the outcomes.

Definition (ROC curve and concordance index)

The *ROC curve* plots sensitivity against 1−specificity as π_0 ranges over $[0, 1]$; it joins $(0, 0)$ to $(1, 1)$. The *concordance index* is

$$c = \frac{1}{n_1 n_0} \sum_{i:\, y_i = 1} \sum_{j:\, y_j = 0} \Big[\mathbb{1}\{\hat{\pi}_i > \hat{\pi}_j\} + \tfrac{1}{2} \mathbb{1}\{\hat{\pi}_i = \hat{\pi}_j\} \Big],$$

the proportion of (case, non-case) pairs in which the case has the larger fitted probability (n_1, n_0 = numbers of 1's and 0's).

Proposition 6.14 (Area under the ROC curve)

The area under the empirical ROC curve (with linear interpolation across ties) equals the concordance index c.

Proof. [Supplement]

Step 1. Lower the cutoff π_0 from 1 to 0. Each time it passes a distinct fitted value v, the curve moves from (a, b) by $(m_0(v)/n_0,\, m_1(v)/n_1)$, where $m_1(v), m_0(v)$ count cases and non-cases with $\hat{\pi} = v$. At that moment $b = \#\{i : y_i = 1, \hat{\pi}_i > v\}/n_1$.

Step 2. The area added by this segment is a trapezoid:

$$\frac{m_0(v)}{n_0} \Big(b + \frac{m_1(v)}{2n_1} \Big) = \frac{1}{n_0 n_1} \sum_{j:\, y_j = 0,\ \hat{\pi}_j = v} \Big[\#\{i : y_i = 1, \hat{\pi}_i > v\} + \tfrac{1}{2} \#\{i : y_i = 1, \hat{\pi}_i = v\} \Big].$$

Step 3. Summing over all distinct v runs over every non-case j exactly once, giving the double sum defining c. ■

A value $c = 0.5$ corresponds to random guessing (the diagonal ROC line of an intercept-only model). For the crabs, $c = 0.639$ (color), 0.742 (width), 0.771 (width and color), 0.772 (width and dark indicator).

6.3 Inference about conditional associations in $2 \times 2 \times K$ tables

Motivation. Comparing two treatments on a binary outcome while controlling for strata (centers, age groups, studies in a meta-analysis) is perhaps the most common application. Logit models, the Cochran–Mantel–Haenszel test and the Mantel–Haenszel estimator are the standard tools, and they turn out to be closely linked.

Example 6.15 (Eight-center clinical trial)

Active drug vs. control cream for curing an infection, at eight centers (μ_{11k}, $\mathrm{Var}(n_{11k})$ are null hypergeometric moments of the number of drug successes):

	Drug		Control				
Center	Succ.	Fail.	Succ.	Fail.	Odds ratio	μ_{11k}	$\mathrm{Var}(n_{11k})$
1	11	25	10	27	1.19	10.36	3.79
2	16	4	22	10	1.82	14.62	2.47
3	14	5	7	12	4.80	10.50	2.41
4	2	14	1	16	2.29	1.45	0.70
5	6	11	0	12	∞	3.52	1.20
6	1	10	0	10	∞	0.52	0.25
7	1	4	1	8	2.0	0.71	0.42
8	4	2	6	1	0.33	4.62	0.62

6.3.1 Using logit models to test conditional independence

For binary Y, binary X ($x_1 = 1$, $x_2 = 0$) and a categorical control Z, let $\pi_{ik} = P(Y = 1 \mid X = i, Z = k)$ and consider

$$\operatorname{logit} \pi_{ik} = \alpha + \beta x_i + \beta_k^Z, \qquad i = 1, 2,\ k = 1, \ldots, K. \tag{6.2}$$

It assumes a common conditional odds ratio $\theta_{XY(k)} = e^{\beta}$. Conditional independence is $H_0 : \beta = 0$, tested by the Wald statistic $(\hat{\beta}/\mathrm{SE})^2$ or the LR statistic comparing (6.2) with

$$\operatorname{logit} \pi_{ik} = \alpha + \beta_k^Z, \tag{6.3}$$

both with $df = 1$. Alternatively, a goodness-of-fit test of (6.3) against the saturated model has $df = K$; it is less powerful when the association is similar across strata (Section 6.4), but can be more powerful when its direction varies.

6.3.2 Cochran–Mantel–Haenszel test of conditional independence

Conditioning on both margins in each partial table, the count n_{11k} determines the table.

Proposition 6.16 (Null hypergeometric moments)

Under conditional independence, given the margins of partial table k, n_{11k} is hypergeometric with

$$\mu_{11k} = E(n_{11k}) = \frac{n_{1+k} n_{+1k}}{n_{++k}}, \qquad \mathrm{Var}(n_{11k}) = \frac{n_{1+k} n_{2+k} n_{+1k} n_{+2k}}{n_{++k}^2 (n_{++k} - 1)}.$$

Proof. [Supplement] Drop the index k and write $n = n_{++}$. Under H_0 and fixed margins, the n_{+1} successes are a random subset of the n subjects. Let $I_s = 1$ if subject s of row 1 is a success ($s = 1, \ldots, n_{1+}$), so $n_{11} = \sum_s I_s$, $P(I_s = 1) = n_{+1}/n$ and, for $s \neq t$, $P(I_s = I_t = 1) = n_{+1}(n_{+1} - 1)/\{n(n-1)\}$.

Step 1. $E(n_{11}) = n_{1+} n_{+1}/n$.

Step 2. With $q = n_{+1}/n$, $\mathrm{Var}(I_s) = q(1-q)$ and $\mathrm{Cov}(I_s, I_t) = \frac{n_{+1}(n_{+1}-1)}{n(n-1)} - q^2 = -\frac{q(1-q)}{n-1}$. Hence

$$\mathrm{Var}(n_{11}) = n_{1+} q(1-q) - n_{1+}(n_{1+} - 1)\frac{q(1-q)}{n-1} = n_{1+} q(1-q)\frac{n - n_{1+}}{n-1} = \frac{n_{1+} n_{2+} n_{+1} n_{+2}}{n^2 (n-1)}.$$

■

Definition (CMH statistic)

$$\text{CMH} = \frac{\left[\sum_k (n_{11k} - \mu_{11k})\right]^2}{\sum_k \text{Var}(n_{11k})}, \tag{6.4}$$

with an approximate χ^2_1 null distribution. Cochran's version uses the binomial-based variance $n_{1+k}n_{2+k}n_{+1k}n_{+2k}/n^3_{++k}$ instead.

When $\theta_{XY(k)} > 1$ in every stratum (or < 1 in every stratum), the deviations $n_{11k} - \mu_{11k}$ tend to share a sign and accumulate; the test works best when the association is similar across strata. The hypergeometric formulation also covers retrospective studies (column margins fixed) and randomized trials (randomization distribution). A continuity correction brings the P-value closer to the exact conditional test but tends to be conservative.

Theorem 6.17 (CMH as a score test for the logit model)

For model (6.2) with independent binomial rows:
(a) the (unconditional) score statistic for $H_0 : \beta = 0$ is Cochran's statistic;
(b) the score statistic for β based on the conditional likelihood given all row and column margins is exactly CMH. In particular CMH and Cochran differ only by the factors $(n_{++k} - 1)/n_{++k}$ in the variance and are asymptotically equivalent to the Wald and LR tests of $\beta = 0$.

Proof. [Supplement] Absorb α into the stratum parameters: $\text{logit}\,\pi_{ik} = \beta x_i + \gamma_k$.

Step 1. (Score.) The log likelihood is $\sum_{ik}[y_{ik}(\beta x_i + \gamma_k) - n_{ik}\log(1 + e^{\beta x_i + \gamma_k})]$, so

$$\frac{\partial L}{\partial \beta} = \sum_k (y_{1k} - n_{1k}\pi_{1k}), \qquad \frac{\partial L}{\partial \gamma_k} = \sum_i (y_{ik} - n_{ik}\pi_{ik}).$$

Under H_0, $\hat\pi_{1k} = \hat\pi_{2k} = \hat\pi_k = n_{+1k}/n_{++k}$ (here $y_{ik} = n_{i1k}$), and the β-score becomes $U = \sum_k (n_{11k} - n_{1+k}n_{+1k}/n_{++k}) = \sum_k (n_{11k} - \mu_{11k})$.

Step 2. (Information.) With $v_k = \pi_k(1 - \pi_k)$ under H_0: $I_{\beta\beta} = \sum_k n_{1+k}v_k$, $I_{\beta\gamma_k} = n_{1+k}v_k$, $I_{\gamma_k\gamma_k} = n_{++k}v_k$, and $I_{\gamma_k\gamma_l} = 0$ for $k \neq l$. The efficient information for β is

$$I_{\beta\beta} - \sum_k \frac{I^2_{\beta\gamma_k}}{I_{\gamma_k\gamma_k}} = \sum_k v_k\Big(n_{1+k} - \frac{n^2_{1+k}}{n_{++k}}\Big) = \sum_k v_k \frac{n_{1+k}n_{2+k}}{n_{++k}}.$$

Plugging in $\hat v_k = n_{+1k}n_{+2k}/n^2_{++k}$ gives $\sum_k n_{1+k}n_{2+k}n_{+1k}n_{+2k}/n^3_{++k}$, so U^2 divided by this is Cochran's statistic, proving (a).

Step 3. (Conditional score.) Given all margins, the conditional distribution of $(n_{111}, \ldots, n_{11K})$ is the product of the stratum-wise noncentral hypergeometric laws:

$$P(n_{11k} = t_k,\ k = 1, \ldots, K) = \prod_k \frac{c_k(t_k)e^{\beta t_k}}{\sum_u c_k(u)e^{\beta u}},$$

a one-parameter exponential family in β with sufficient statistic $T = \sum_k n_{11k}$. Hence $\partial \log P/\partial\beta = T - E_\beta(T)$ and the conditional information is $\text{Var}_\beta(T)$. At $\beta = 0$ these are $\sum_k (n_{11k} - \mu_{11k})$ and $\sum_k \text{Var}(n_{11k})$ with the hypergeometric variance of Proposition 6.16, so the conditional score statistic is CMH, proving (b). The final claim follows from standard asymptotics for score, Wald and LR tests. ■

6.3.3 Multicenter clinical trial example

Example 6.18 (Clinical trial: tests)

In Example 6.15 every center except the last shows a positive association, so combining is sensible: CMH $= 6.38$ ($df = 1$, $P = 0.012$). Model (6.2) gives $\hat{\beta} = 0.777$ (SE 0.307), Wald 6.42 ($P = 0.011$), LR 6.67 ($P = 0.010$).

6.3.4 Estimation of common odds ratio

Under (6.2) the common odds ratio is $\theta = e^{\beta}$, with ML estimate $e^{\hat{\beta}}$. Non-model-based alternatives are Woolf's weighted average of the log odds ratios and the Mantel–Haenszel estimator.

Definition (Mantel–Haenszel estimator)

$$\hat{\theta}_{MH} = \frac{\sum_k n_{11k} n_{22k}/n_{++k}}{\sum_k n_{12k} n_{21k}/n_{++k}} = \frac{\sum_k R_k}{\sum_k S_k} = \frac{R}{S}. \tag{6.5}$$

Theorem 6.19 (Unbiased estimating equation for $\hat{\theta}_{MH}$)

If the rows of each partial table are independent binomials and the odds ratio equals θ in every stratum, then $E(R - \theta S) = 0$. Consequently $\hat{\theta}_{MH}$ is consistent both as $n \to \infty$ with K fixed and as $K \to \infty$ with bounded strata sizes.

Proof. [Supplement]

Step 1. In stratum k, $n_{11k} \sim \text{bin}(n_{1+k}, \pi_{1k})$ and $n_{21k} \sim \text{bin}(n_{2+k}, \pi_{2k})$ independently, so

$$E(n_{11k} n_{22k}) = n_{1+k} n_{2+k} \pi_{1k}(1 - \pi_{2k}), \qquad E(n_{12k} n_{21k}) = n_{1+k} n_{2+k} (1 - \pi_{1k}) \pi_{2k}.$$

Step 2. Since $\theta = \pi_{1k}(1 - \pi_{2k})/\{(1 - \pi_{1k})\pi_{2k}\}$, $E(n_{11k} n_{22k}) = \theta E(n_{12k} n_{21k})$, i.e. $E(R_k - \theta S_k) = 0$. Summing gives $E(R - \theta S) = 0$.

Step 3. Under either asymptotic scheme, R/m and S/m (with $m = n$ or $m = K$) are averages of independent terms with bounded variances relative to m, so by the law of large numbers $(R - \theta S)/m \xrightarrow{p} 0$ and S/m stays bounded away from 0 (assuming the strata are nondegenerate). Hence $\hat{\theta}_{MH} - \theta = (R - \theta S)/S \xrightarrow{p} 0$. ∎

Theorem 6.20 (Unconditional ML breaks down for matched pairs)

In model (6.2) with one matched pair per stratum, let b (c) be the number of pairs in which only the first (only the second) member has $y = 1$. Then the unconditional ML estimator is $\hat{\beta} = 2\log(b/c)$, whereas $\hat{\theta}_{MH} = b/c$. As $K \to \infty$, $\log(b/c) \xrightarrow{p} \beta$, so $\hat{\beta} \xrightarrow{p} 2\beta$.

Proof. [Supplement]

Step 1. A concordant pair's likelihood is maximized by $\gamma_k = \pm\infty$ and then equals 1, contributing nothing about β.

Step 2. Write $u = e^{\gamma_k}$, $\psi = e^{\beta}$. A pair where only member 1 succeeds has likelihood $\frac{u\psi}{1+u\psi} \cdot \frac{1}{1+u}$, and a pair where only member 2 succeeds has $\frac{1}{1+u\psi} \cdot \frac{u}{1+u}$. Both are proportional in u to $u/\{(1+u\psi)(1+u)\}$, maximized at $u = \psi^{-1/2}$ (derivative of its log: $1/u - \psi/(1+u\psi) - 1/(1+u) = 0 \iff u^2\psi = 1$). The maximized values are $\psi/(1+\psi^{1/2})^2$ and $1/(1+\psi^{1/2})^2$.

Step 3. The profile log likelihood is $b\beta - 2(b+c)\log(1+e^{\beta/2})$, with derivative $b - (b+c)e^{\beta/2}/(1+e^{\beta/2})$, zero at $e^{\beta/2} = b/c$. So $\hat{\beta} = 2\log(b/c)$.

Step 4. For pairs, $R_k = \frac{1}{2}$ in a b-type pair, $S_k = \frac{1}{2}$ in a c-type pair, and both vanish for concordant pairs, so $\hat{\theta}_{MH} = b/c$.

Step 5. Given discordance, a pair is of b-type with probability $\pi_1(1-\pi_2)/\{\pi_1(1-\pi_2)+(1-\pi_1)\pi_2\} = \psi/(1+\psi)$ in every stratum, so $b/(b+c) \xrightarrow{p} \psi/(1+\psi)$ and $b/c \xrightarrow{p} \psi$. ■

Variance of $\log \hat{\theta}_{MH}$

Robins, Breslow and Greenland showed $\log \hat{\theta}_{MH} - \log \theta$ is approximately proportional to $R - \theta S$ and derived an estimated variance valid under both asymptotic schemes:

$$\hat{\sigma}^2(\log \hat{\theta}_{MH}) = \frac{\sum_k (n_{11k} + n_{22k}) R_k / n_{++k}}{2R^2} + \frac{\sum_k (n_{12k} + n_{21k}) S_k / n_{++k}}{2S^2} + \frac{\sum_k [(n_{11k} + n_{22k}) S_k + (n_{12k} + n_{21k}) R_k] / n_{++k}}{2RS}.$$

With fixed K, $\log \hat{\theta}_{MH}$ is slightly less efficient than the ML estimator unless $\beta = 0$; with many sparse strata it is preferred.

Example 6.21 (Clinical trial: common odds ratio)

For Example 6.15,

$$\hat{\theta}_{MH} = \frac{(11 \cdot 27)/73 + \cdots + (4 \cdot 1)/13}{(25 \cdot 10)/73 + \cdots + (2 \cdot 6)/13} = 2.13,$$

$\log \hat{\theta}_{MH} = 0.758$ with $\hat{\sigma} = 0.303$, so the 95% CI is $\exp(0.758 \pm 1.96 \cdot 0.303) = (1.18, 3.87)$. The model-based Wald interval for e^{β} is $(1.19, 3.97)$ and the LR interval $(1.20, 4.02)$. The effect is clear but its size is imprecise.

$\hat{\theta}_{MH}$ remains a useful summary, and CMH a powerful test, when the odds ratios vary moderately but mostly in one direction; equality of odds ratios is not required for either.

6.3.5 Testing homogeneity of odds ratios

Homogeneity $\theta_{XY(1)} = \cdots = \theta_{XY(K)}$ is equivalent to model (6.2), so its G^2 and X^2 goodness-of-fit statistics test it with $df = K - 1$ (the $K - 1$ omitted $X \times Z$ interaction terms).

Proposition 6.22 (Fitted counts under a common odds ratio)

Fix the margins of stratum k and a value $\theta > 0$. There is a unique m_k with $\max(0, n_{1+k} + n_{+1k} - n_{++k}) < m_k < \min(n_{1+k}, n_{+1k})$ such that the table with $(1,1)$ entry m_k and the given margins has odds ratio θ:

$$\frac{m_k \, (n_{2+k} - n_{+1k} + m_k)}{(n_{1+k} - m_k)(n_{+1k} - m_k)} = \theta.$$

Proof. [Supplement] The other three cells are $n_{1+k} - m$, $n_{+1k} - m$ and $n_{2+k} - n_{+1k} + m$, all positive exactly on the open interval stated. On that interval the left side is continuous and strictly increasing in m (numerator increasing, denominator decreasing), tends to 0 at the lower end (a numerator factor vanishes) and to $+\infty$ at the upper end (a denominator factor vanishes). By the intermediate value theorem there is exactly one root; it solves a quadratic equation in m. ■

Definition (Breslow–Day statistic)

With m_k from Proposition 6.22 at $\theta = \hat{\theta}_{MH}$ and v_k the variance of n_{11k} under the fitted table (approximately $[1/m_k + 1/(n_{1+k} - m_k) + 1/(n_{+1k} - m_k) + 1/(n_{2+k} - n_{+1k} + m_k)]^{-1}$),

$$\mathrm{BD} = \sum_k \frac{(n_{11k} - m_k)^2}{v_k},$$

approximately χ^2_{K-1} under homogeneity.

Example 6.23 (Clinical trial: homogeneity)

For Example 6.15, $G^2 = 9.7$ and $X^2 = 8.0$ ($df = 7$): no evidence against a common odds ratio, so $\hat{\theta}_{MH} = 2.1$ summarizes all eight centers. A homogeneity test is not a prerequisite for using $\hat{\theta}_{MH}$ or CMH.

6.3.6 Summarizing heterogeneity in odds ratios

Exactly equal odds ratios are unrealistic (unmeasured covariates vary across strata), and the sampled strata (centers) usually represent a larger population. A *random-effects* logit model takes the stratum log odds ratios as a sample from $N(\mu, \sigma^2)$, estimating a mean effect and its variability; inference then extends to the population of strata (Chapter 12).

6.4 Using models to improve inferential power

Motivation. Tests aimed at a narrow, plausible alternative (such as a linear trend) can be far more powerful than omnibus tests. The noncentral chi-squared distribution explains why: power is gained by concentrating the noncentrality on few degrees of freedom.

6.4.1 Directed alternatives

For an $I \times 2$ table with binomial parameters $\{\pi_i\}$, independence is $\operatorname{logit} \pi_i = \alpha$. The ordinary statistics $X^2(I)$ and $G^2(I)$ test it against the saturated model $\operatorname{logit} \pi_i = \alpha + \beta_i$ ($df = I - 1$), treating rows as nominal. With ordered rows, the linear logit model $\operatorname{logit} \pi_i = \alpha + \beta x_i$ gives the directed test of $\beta = 0$ with statistic

$$G^2(I \mid L) = G^2(I) - G^2(L), \qquad df = 1.$$

A large value on $df = 1$ lies farther in the tail than a comparable value on $df = I - 1$.

6.4.2 Noncentral chi-squared distribution

Definition (Noncentral chi-squared)

If $Z_i \sim N(\mu_i, 1)$, $i = 1, \ldots, \nu$, are independent, then $\sum Z_i^2$ has the noncentral chi-squared distribution $\chi^2_{\nu,\lambda}$ with $df = \nu$ and noncentrality $\lambda = \sum \mu_i^2$. The central case is $\lambda = 0$.

Proposition 6.24 (Moments and monotonicity of power)

(a) $E(\chi^2_{\nu,\lambda}) = \nu + \lambda$ and $\mathrm{Var}(\chi^2_{\nu,\lambda}) = 2(\nu + 2\lambda)$; the distribution depends on the μ_i only through λ.
(b) For fixed ν and critical value $c > 0$, $P(\chi^2_{\nu,\lambda} > c)$ is strictly increasing in λ, equal to α at $\lambda = 0$ when $c = \chi^2_\nu(\alpha)$.

Proof. [Supplement]

Step 1. $E(Z_i^2) = 1 + \mu_i^2$. With $Z_i = \mu_i + \varepsilon_i$, $Z_i^2 = \mu_i^2 + 2\mu_i\varepsilon_i + \varepsilon_i^2$ and $\mathrm{Var}(Z_i^2) = 4\mu_i^2 + 2$ (using $E\varepsilon^3 = 0$, $\mathrm{Var}\,\varepsilon^2 = 2$). Summing gives (a)'s moments.

Step 2. An orthogonal matrix Q with first row $\mu^\top/\|\mu\|$ maps $Z \sim N(\mu, I)$ to $QZ \sim N((\sqrt{\lambda}, 0, \ldots, 0)^\top, I)$ with $\|QZ\| = \|Z\|$. Hence $\sum Z_i^2 \overset{d}{=} (\varepsilon_1 + \sqrt{\lambda})^2 + \chi^2_{\nu-1}$, depending only on λ.

Step 3. For $a \geq 0$ and $s > 0$, $g(a) = P(|\varepsilon + a| > s) = 1 - \Phi(s - a) + \Phi(-s - a)$ has $g'(a) = \phi(s - a) - \phi(s + a) > 0$ for $a > 0$. Conditioning on the independent $\chi^2_{\nu-1} = w$ (with $s = \sqrt{c - w}$ when $w < c$; the event is certain when $w \geq c$) and integrating shows that $P(\chi^2_{\nu,\lambda} > c)$ is strictly increasing in $a = \sqrt{\lambda}$. ■

Power decreases with *df*

A fundamental further fact (Das Gupta and Perlman): for fixed λ and α, $P(\chi^2_{\nu,\lambda} > \chi^2_\nu(\alpha))$ decreases as ν increases. So concentrating the noncentrality on a statistic with small *df* increases power.

6.4.3 Increased power for narrower alternatives

Proposition 6.25 (Partitioning G^2)

For the nested models independence $\subset$ linear logit $\subset$ saturated,

$$G^2(I) = G^2(I \mid L) + G^2(L),$$

with *df* $(I - 1) = 1 + (I - 2)$.

Proof. With L_0, L_1, L_S the maximized log likelihoods, $G^2(I) = -2(L_0 - L_S) = -2(L_0 - L_1) - 2(L_1 - L_S) = G^2(I \mid L) + G^2(L)$. The *df* are differences in numbers of parameters: $(I-1) = (2-1) + (I-2)$. ■

When the linear logit model holds, $G^2(L)$ is asymptotically central χ^2_{I-2}, so $G^2(I)$ and $G^2(I \mid L)$ have (approximately) the same noncentrality but *df* $I-1$ versus 1: the directed test is more powerful. If the linear model fails, $G^2(I)$ has larger noncentrality, but as long as the linear model approximates reality reasonably, the *df* $= 1$ test still wins. For instance, for $\alpha = 0.05$ a *df* $= 1$ test beats a *df* $= 7$ test whenever its noncentrality is at least about half as large. As Mantel noted, testing a linear component does not assume linearity: it provides power against any progressive association. The same reasoning favors the *df* $= 1$ test of $\beta = 0$ in (6.2) over the *df* $= K$ goodness-of-fit test of (6.3) when (6.2) holds.

6.4.4 Treatment of leprosy example

Example 6.26 (Leprosy)

Change in clinical condition after 48 weeks of treatment, by degree of infiltration at the start:

Clinical change (score)	High	Low	Prop. high
Worse (−1)	1	11	0.08
Stationary (0)	13	53	0.20
Slight improvement (1)	16	42	0.28
Moderate improvement (2)	15	27	0.36
Marked improvement (3)	7	11	0.39

Comparing mean change between infiltration levels is equivalent to a trend test with infiltration as binary response. $G^2(I) = 7.28$ ($df = 4$, $P = 0.12$) ignores the ordering; the linear logit test gives $G^2(I \mid L) = 6.65$ ($df = 1$, $P = 0.01$), and $G^2(L) = 0.63$ ($df = 3$) shows a good linear fit.

6.4.5 Model smoothing improves precision of estimation

Theorem 6.27 (Model-based estimators are more efficient)

Let p be the sample proportions from a multinomial(n, π) sample with $\pi = \pi(\theta)$ satisfying a smooth model with full-rank $A = \partial\pi/\partial\theta^\top$, and let $\hat\pi = \pi(\hat\theta)$ be the ML fit. For every differentiable g,

$$\text{asymp. Var}\big[\sqrt{n}\, g(\hat\pi)\big] \le \text{asymp. Var}\big[\sqrt{n}\, g(p)\big].$$

Proof. [Supplement] Let $D = \operatorname{diag}(\pi)$ and $c = \nabla g(\pi)$. By the delta method it suffices to show $c^\top \Sigma_{\hat\pi} c \le c^\top \Sigma_p c$ for the asymptotic covariances.

Step 1. $\Sigma_p = D - \pi\pi^\top$. The Fisher information per observation is $A^\top D^{-1} A$, so $\Sigma_{\hat\theta} = (A^\top D^{-1} A)^{-1}$ and $\Sigma_{\hat\pi} = A(A^\top D^{-1} A)^{-1} A^\top$.

Step 2. Put $M = D^{-1/2} A$ and $q = D^{-1/2}\pi = (\sqrt{\pi_i})$, a unit vector. Then $\Sigma_{\hat\pi} = D^{1/2} P_M D^{1/2}$ with $P_M = M(M^\top M)^{-1} M^\top$, and $\Sigma_p = D^{1/2}(I - qq^\top) D^{1/2}$.

Step 3. $M^\top q = A^\top \mathbf{1} = \partial(\sum_i \pi_i)/\partial\theta = 0$, so $q \perp \operatorname{col}(M)$. Then $P_M + qq^\top$ is an orthogonal projection, hence $P_M + qq^\top \le I$, i.e. $P_M \le I - qq^\top$. Sandwiching by $D^{1/2}$ gives $\Sigma_{\hat\pi} \le \Sigma_p$ in the Loewner order. ∎

Even when the model is only approximately true, $\hat\pi_i$ is biased but less variable, and $\operatorname{MSE}(\hat\pi_i) < \operatorname{MSE}(p_i)$ unless n is very large.

Chapter 7. Logit Models for Multinomial Responses

7.1 Nominal responses: baseline-category logit models

Motivation. Many responses have more than two unordered categories (food type, brand, party). Pairing each category with a baseline gives $J-1$ simultaneous logit equations that together describe all pairwise odds, extending binary logistic regression.

Let Y have J categories and $\pi_j(\mathbf{x}) = P(Y = j \mid \mathbf{x})$, $\sum_j \pi_j(\mathbf{x}) = 1$. At a fixed setting $\mathbf{x}$, the counts in the J categories are multinomial with probabilities $\{\pi_1(\mathbf{x}), \ldots, \pi_J(\mathbf{x})\}$. Of the $\binom{J}{2}$ pairwise logits, any suitable $J-1$ determine the rest.

7.1.1 Baseline-category logits

Definition (Baseline-category logit model)

With category J as baseline,

$$\log \frac{\pi_j(\mathbf{x})}{\pi_J(\mathbf{x})} = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.1}$$

The effects $\boldsymbol{\beta}_j$ depend on which category is paired with the baseline.

Proposition 7.1 (All pairwise logits)

Under (7.1), for any categories a, b,

$$\log \frac{\pi_a(\mathbf{x})}{\pi_b(\mathbf{x})} = (\alpha_a - \alpha_b) + (\boldsymbol{\beta}_a - \boldsymbol{\beta}_b)^\top \mathbf{x},$$

with the convention $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. Hence the choice of baseline is arbitrary: changing it reparametrizes but does not change the model or its fit.

Proof. $\log(\pi_a/\pi_b) = \log(\pi_a/\pi_J) - \log(\pi_b/\pi_J)$; substitute (7.1) (the convention makes the formula valid when a or b equals J). The set of probability vectors described is the same for every baseline, since each parametrization is a linear reparametrization of the others. ■

With categorical predictors and nonsparse data, X^2 and G^2 check fit; with continuous predictors or sparse data they remain valid for comparing nested models differing by few terms.

7.1.2 Alligator food choice example

Example 7.2 (Alligator food choice)

For 219 alligators from four Florida lakes, the response is the primary food type by volume (fish, invertebrate, reptile, bird, other); predictors are lake L, gender G and size S (≤ 2.3 m vs. > 2.3 m). The data are sparse (219 observations in 80 cells), so G^2 is more reliable for comparing models than for testing fit. Comparisons give $G^2[(G) \mid (\cdot)] = 2.1$ and $G^2[(L+S) \mid (G+L+S)] = 2.2$, each on $df = 4$, so we collapse over gender. For the collapsed table, $(L+S)$ has $G^2 = 17.1$, $X^2 = 15.0$ ($df = 12$), whereas (L) has 38.2 and (S) 66.2; both L and S matter. Observed counts (fitted values for $L+S$ in parentheses):

Lake	Size	Fish	Invert.	Reptile	Bird	Other
Hancock	≤ 2.3	23 (20.9)	4 (3.6)	2 (1.9)	2 (2.7)	8 (9.9)
	> 2.3	7 (9.1)	0 (0.4)	1 (1.1)	3 (2.3)	5 (3.1)
Oklawaha	≤ 2.3	5 (5.2)	11 (12.0)	1 (1.5)	0 (0.2)	3 (1.1)
	> 2.3	13 (12.8)	8 (7.0)	6 (5.5)	1 (0.8)	0 (1.9)
Trafford	≤ 2.3	5 (4.4)	11 (12.4)	2 (2.1)	1 (0.9)	5 (4.2)
	> 2.3	8 (8.6)	7 (5.6)	6 (5.9)	3 (3.1)	5 (5.8)
George	≤ 2.3	16 (18.5)	19 (16.9)	1 (0.5)	2 (1.2)	3 (3.8)
	> 2.3	17 (14.5)	1 (3.1)	0 (0.5)	1 (1.8)	3 (2.2)

Only two of the 40 standardized residuals exceed 2 in absolute value. With fish as baseline, dummies $s = 1$ for small size and z_H, z_O, z_T for Hancock, Oklawaha, Trafford (George is the reference), the ML estimates (SE) are:

Logit	Intercept	Size ≤ 2.3	Hancock	Oklawaha	Trafford
$\log(\pi_I/\pi_F)$	-1.55	1.46 (0.40)	-1.66 (0.61)	0.94 (0.47)	1.12 (0.49)
$\log(\pi_R/\pi_F)$	-3.31	-0.35 (0.58)	1.24 (1.19)	2.46 (1.12)	2.94 (1.12)
$\log(\pi_B/\pi_F)$	-2.09	-0.63 (0.64)	0.70 (0.78)	-0.65 (1.20)	1.09 (0.84)
$\log(\pi_O/\pi_F)$	-1.90	0.33 (0.45)	0.83 (0.56)	0.01 (0.78)	1.52 (0.62)

Given lake, small alligators have estimated odds of invertebrates rather than fish $e^{1.46} = 4.3$ times those of large alligators (Wald 95% CI $e^{1.46 \pm 1.96(0.396)} = (2.0, 9.3)$). Those odds are relatively higher in Trafford and Oklawaha and lower in Hancock than in George. Other pairs follow by subtraction (Proposition 7.1), e.g.

$$\begin{aligned}\log(\hat{\pi}_I/\hat{\pi}_O) &= (-1.55 + 1.46s - 1.66z_H + 0.94z_O + 1.12z_T) \\ &\quad - (-1.90 + 0.33s + 0.83z_H + 0.01z_O + 1.52z_T) \\ &= 0.35 + 1.13s - 2.48z_H + 0.93z_O - 0.39z_T.\end{aligned}$$

7.1.3 Estimating response probabilities

Proposition 7.3 (Response probabilities)

Model (7.1) is equivalent to

$$\pi_j(\mathbf{x}) = \frac{\exp(\alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x})}{1 + \sum_{h=1}^{J-1} \exp(\alpha_h + \boldsymbol{\beta}_h^\top \mathbf{x})}, \qquad j = 1, \ldots, J, \tag{7.2}$$

with $\alpha_J = 0$, $\boldsymbol{\beta}_J = \mathbf{0}$. For $J = 2$ this is binary logistic regression.

Proof. From (7.1), $\pi_j = \pi_J \exp(\eta_j)$ with $\eta_j = \alpha_j + \boldsymbol{\beta}_j^\top \mathbf{x}$, valid also for $j = J$ with $\eta_J = 0$. Summing over j, $1 = \pi_J \sum_{h=1}^{J} e^{\eta_h} = \pi_J(1 + \sum_{h<J} e^{\eta_h})$. Solve for π_J and substitute. Conversely (7.2) gives $\pi_j/\pi_J = e^{\eta_j}$. ■

The parameters of the baseline category are set to zero for identifiability: adding a common vector to all $(\alpha_j, \boldsymbol{\beta}_j)$, $j = 1, \ldots, J$, leaves (7.2) unchanged.

Example 7.4 (Alligators: fitted probabilities)

For a large alligator in Lake Hancock ($s = 0$, $z_H = 1$),

$$\hat{\pi}_I = \frac{e^{-1.55-1.66}}{1 + e^{-1.55-1.66} + e^{-3.31+1.24} + e^{-2.09+0.70} + e^{-1.90+0.83}} = 0.023,$$

and similarly $\hat{\pi}_R = 0.072$, $\hat{\pi}_B = 0.141$, $\hat{\pi}_O = 0.194$, $\hat{\pi}_F = 0.570$. With a quantitative predictor (actual length), plotting $\hat{\pi}_j(x)$ against x is informative; with $J > 2$ a single $\hat{\pi}_j(x)$ need not be monotone in x.

7.2 Ordinal responses: cumulative logit models

Motivation. When the categories are ordered, a model that uses the ordering is more parsimonious and has more power (Section 6.4). Cumulative logits with a common effect across cutpoints give the most popular such model, motivated by an underlying continuous latent response.

7.2.1 Cumulative logits

Definition (Cumulative logits)

With $P(Y \le j \mid \mathbf{x}) = \pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})$,

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \log \frac{P(Y \le j \mid \mathbf{x})}{1 - P(Y \le j \mid \mathbf{x})} = \log \frac{\pi_1(\mathbf{x}) + \cdots + \pi_j(\mathbf{x})}{\pi_{j+1}(\mathbf{x}) + \cdots + \pi_J(\mathbf{x})}, \quad j = 1, \ldots, J-1.$$

Each cumulative logit uses all J categories; a model for one of them alone is a binary logit model for the collapsing $\{1, \ldots, j\}$ vs. $\{j+1, \ldots, J\}$.

7.2.2 Proportional odds model

Definition (Proportional odds model)

$$\operatorname{logit} P(Y \le j \mid \mathbf{x}) = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1, \tag{7.3}$$

with intercepts $\alpha_1 < \alpha_2 < \cdots < \alpha_{J-1}$ (since $P(Y \le j \mid \mathbf{x})$ increases in j) and a single effect vector $\boldsymbol{\beta}$ common to all cutpoints.

Proposition 7.5 (Parallel curves and cumulative odds ratios)

Under (7.3):
(a) for a single predictor and $\beta \neq 0$, the curve of $P(Y \le k \mid x)$ is the curve of $P(Y \le j \mid x)$ shifted horizontally: $P(Y \le k \mid X = x) = P(Y \le j \mid X = x + (\alpha_k - \alpha_j)/\beta)$;
(b) for any $\mathbf{x}_1, \mathbf{x}_2$ and every j, the cumulative odds ratio is

$$\frac{P(Y \le j \mid \mathbf{x}_1)/P(Y > j \mid \mathbf{x}_1)}{P(Y \le j \mid \mathbf{x}_2)/P(Y > j \mid \mathbf{x}_2)} = \exp[\boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)],$$

the same for all j and proportional (on the log scale) to the distance between $\mathbf{x}_1$ and $\mathbf{x}_2$;
(c) the conditional distributions are stochastically ordered: for each pair $\mathbf{x}_1, \mathbf{x}_2$, either $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ for all j, or $\ge$ for all j.

Proof. Let $F(z) = e^z/(1+e^z)$, so $P(Y \le j \mid \mathbf{x}) = F(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x})$.

(a) $F(\alpha_j + \beta[x + (\alpha_k - \alpha_j)/\beta]) = F(\alpha_k + \beta x)$.

(b) The log of the cumulative odds ratio is the difference of logits, $(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_1) - (\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_2) = \boldsymbol{\beta}^\top(\mathbf{x}_1 - \mathbf{x}_2)$.

(c) F is increasing, so $P(Y \le j \mid \mathbf{x}_1) \le P(Y \le j \mid \mathbf{x}_2)$ iff $\boldsymbol{\beta}^\top \mathbf{x}_1 \le \boldsymbol{\beta}^\top \mathbf{x}_2$, a condition not involving j. ■

With one predictor, e^β is the cumulative odds ratio for $x_1 - x_2 = 1$, the same for every collapsing of Y into $(\le j, > j)$: hence McCullagh's name *proportional odds*. The response curves for $j = 1, 2, 3$ ($J = 4$) have exactly the same shape (Figure 9).

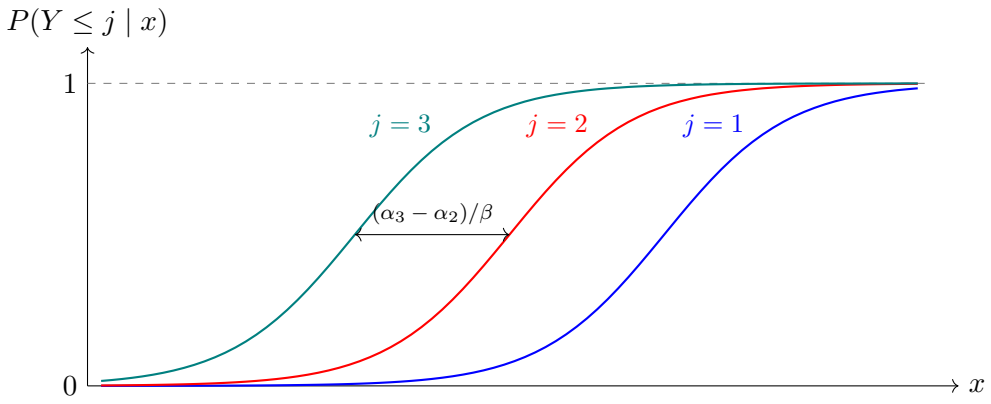


Figure 9: Proportional odds model with $J = 4$ and $\beta > 0$: the three cumulative probability curves are logistic curves with a common slope, horizontally displaced from one another.

Because the $J - 1$ curves are constrained to share a shape, the fit differs from separate binary fits. With y_{ij} indicators of the response of subject i, the likelihood is

$$\prod_{i=1}^{n} \prod_{j=1}^{J} \left[\frac{\exp(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_i)}{1 + \exp(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}_i)} - \frac{\exp(\alpha_{j-1} + \boldsymbol{\beta}^\top \mathbf{x}_i)}{1 + \exp(\alpha_{j-1} + \boldsymbol{\beta}^\top \mathbf{x}_i)} \right]^{y_{ij}} \tag{7.4}$$

with $\alpha_0 = -\infty$, $\alpha_J = +\infty$; it is maximized by Fisher scoring.

7.2.3 Mental impairment example

Example 7.6 (Mental impairment)

A random sample of 40 adults in Alachua County, Florida. Y = mental impairment (1 well, 2 mild symptom formation, 3 moderate symptom formation, 4 impaired); x_1 = life events index (number and severity of important events in the past three years); x_2 = SES (1 high, 0 low). Listed as (x_2, x_1) within each response category:

Well	(1,1) (1,9) (1,4) (1,3) (0,2) (1,0) (0,1) (1,3) (1,3) (1,7) (0,1) (0,2)
Mild	(1,5) (0,6) (1,3) (0,1) (1,8) (1,2) (0,5) (1,5) (1,9) (0,3) (1,3) (1,1)
Moderate	(0,0) (1,4) (0,3) (0,9) (1,6) (0,4) (0,3)
Impaired	(1,8) (1,2) (1,7) (0,5) (0,4) (0,4) (1,8) (0,8) (0,9)

The fit of logit $P(Y \le j \mid \mathbf{x}) = \alpha_j + \beta_1 x_1 + \beta_2 x_2$:

Parameter	Estimate	SE	LR 95% CI
α_1	-0.282	0.642	$(-1.56,\ 0.98)$
α_2	1.213	0.661	$(-0.05,\ 2.57)$
α_3	2.209	0.721	$(0.86,\ 3.71)$
life (β_1)	-0.319	0.121	$(-0.57,\ -0.09)$
SES (β_2)	1.111	0.611	$(-0.06,\ 2.35)$

At the mean life events $\bar{x}_1 = 4.275$ and low SES,

$$\hat{P}(Y=1) = \frac{\exp[-0.282 - 0.319(4.275)]}{1+\exp[-0.282 - 0.319(4.275)]} = 0.16.$$

The cumulative probability from the "well" end decreases with life events and is higher at high SES: given life events, the odds of impairment below any fixed level at high SES are $e^{1.111} = 3.0$ times those at low SES. In probability terms, at $\bar{x}_1$, $\hat{P}(Y=1)$ is 0.37 (high SES) vs. 0.16 (low SES); between the life-events quartiles 2.0 and 6.5 it falls from 0.55 to 0.22 (high SES) and from 0.28 to 0.09 (low SES). The score test of proportional odds (separate β's for each logit, 4 extra parameters) gives 2.33, $df = 4$, $P = 0.68$.

7.2.4 More complex models

Interactions, factors and so on enter as in binary logistic regression; only the intercept is replaced by $J-1$ cutpoint intercepts. For the mental impairment data, adding the interaction gives

$$\text{logit}\,\hat{P}(Y \le j \mid \mathbf{x}) = \hat{\alpha}_j - 0.420x_1 + 0.371x_2 + 0.181x_1x_2,$$

interaction SE 0.238: the life-events effect is -0.420 at low SES and -0.239 at high SES, not significantly different.

When proportional odds fails

Replacing $\boldsymbol{\beta}$ by $\boldsymbol{\beta}_j$ gives nonparallel curves, which necessarily cross for some $\mathbf{x}$, violating the ordering of cumulative probabilities. A small P-value in the proportional odds test may reflect a large n more than important differences among the $\hat{\boldsymbol{\beta}}_j$, and the simpler model may have smaller MSE even if biased. If the fit is poor in practical terms, options include:
(1) a link with a nonsymmetric response curve (complementary log-log);
(2) extra terms such as interactions;
(3) dispersion parameters;
(4) separate effects for some predictors only (partial proportional odds);
(5) a baseline-category logit model, using the ordering informally in interpretation.

7.3 Ordinal responses: cumulative link models

Motivation. The latent-variable argument works for any error distribution, so the logit can be replaced by other inverse cdfs. Probit and complementary log-log links give cumulative models with their own interpretations, and a scale component handles dispersion differences.

Definition (Cumulative link model)

For a continuous cdf G,

$$G^{-1}[P(Y \le j \mid \mathbf{x})] = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}, \qquad j = 1, \ldots, J-1. \tag{7.5}$$

Model (7.5) arises from crude measurement of a latent Y^* following a location family $G(y^* - \boldsymbol{\beta}^\top \mathbf{x})$; the α_j are cutpoints on a standardized latent scale (using $-\boldsymbol{\beta}$ instead of $+\boldsymbol{\beta}$ only flips signs).

7.3.1 Types of cumulative links

(a) *Cumulative probit* ($G = \Phi$): appropriate when Y^* is normal. In $\Phi^{-1}[P(Y \le j)] = \alpha_j - \boldsymbol{\beta}^\top \mathbf{x}$, a unit increase in x_k raises $E(Y^*)$ by β_k standard deviations of ε. Logit and probit fits are similar; logit parameters are easier to interpret.

(b) *Complementary log-log* ($G(z) = 1 - \exp(-e^z)$, from an extreme-value latent distribution): $\log\{-\log[1 - P(Y \le j \mid \mathbf{x})]\} = \alpha_j + \boldsymbol{\beta}^\top \mathbf{x}$. It is the grouped-time version of the proportional hazards model and is sometimes called a *proportional hazards* model. $P(Y \le j)$ approaches 1 faster than it approaches 0. The log-log link suits the case where cloglog holds for the reversed categories (reversing the order of the categories turns G into $1 - G(-z)$).

Proposition 7.7 (Power interpretation of the cumulative cloglog model)

Under the cumulative complementary log-log model, for all j,

$$P(Y > j \mid \mathbf{x}_1) = P(Y > j \mid \mathbf{x}_2)^{\exp[\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)]}.$$

Proof. $P(Y > j \mid \mathbf{x}) = 1 - G(\alpha_j + \boldsymbol{\beta}^\top \mathbf{x}) = \exp\left[-e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}}\right]$. Hence

$$\log P(Y > j \mid \mathbf{x}_1) = -e^{\alpha_j} e^{\boldsymbol{\beta}^\top \mathbf{x}_2} e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} = e^{\boldsymbol{\beta}^\top (\mathbf{x}_1 - \mathbf{x}_2)} \log P(Y > j \mid \mathbf{x}_2);$$

exponentiate. ■

7.3.2 Estimation for cumulative link models

Cumulative link models are multivariate GLMs. McCullagh's Fisher scoring algorithm uses the likelihood written as in (7.4) with G in place of the logistic cdf; for large n the maximum is unique, and the log likelihood is concave for the logit, probit and complementary log-log links, so iterations converge rapidly.

7.3.3 Life table example

Example 7.8 (U.S. life lengths, 1981)

Percent distribution of life length by gender and race (fitted cloglog values in parentheses):

	Males		Females	
Life length	White	Black	White	Black
0–20	2.4 (2.4)	3.6 (4.4)	1.6 (1.2)	2.7 (2.3)
20–40	3.4 (3.5)	7.5 (6.4)	1.4 (1.9)	2.9 (3.4)
40–50	3.8 (4.4)	8.3 (7.7)	2.2 (2.4)	4.4 (4.3)
50–65	17.5 (16.7)	25.0 (26.1)	9.9 (9.6)	16.3 (16.3)
Over 65	72.9 (73.0)	55.6 (55.4)	84.9 (84.9)	73.7 (73.7)

The life-length cdf rises slowly at young ages and sharply at old ages, suggesting the cloglog link (also implied by an exponentially increasing hazard, the Gompertz extreme-value law). For G (1 female) and R (1 black),

$$\log\{-\log[1 - P(Y \le j \mid G = g, R = r)]\} = \alpha_j + \beta_1 g + \beta_2 r, \qquad \hat{\beta}_1 = -0.658,\ \hat{\beta}_2 = 0.626.$$

(Fit statistics are irrelevant: these are population values.) By Proposition 7.7, $P(Y > j \mid G = 0, r) = P(Y > j \mid G = 1, r)^{\exp(0.658)}$: given race, men's survival proportion beyond any age is women's raised to the power 1.93; given gender, blacks' is whites' to the power $e^{0.626} = 1.87$. If white women survive beyond some age with probability π, white men and black women do so with probability about π^2, and black men with about π^4.

Chapter 8. Loglinear Models for Contingency Tables

Loglinear models are Poisson GLMs with log link for the cell counts of a contingency table. They treat all cross-classified variables symmetrically, as joint responses, and their purpose is to describe association and interaction patterns. With one response it is simpler to use logit models; Section 8.5 shows that logit models with categorical predictors are special loglinear models. Throughout, $\{n_{ij}\}$ or $\{n_{ijk}\}$ are observed counts, $\{\mu\}$ expected counts and $\{\pi\}$ cell probabilities; a "+" subscript denotes summation over that index.

8.1 Loglinear models for two-way tables

Motivation. For an $I \times J$ table the familiar independence structure $\pi_{ij} = \pi_{i+}\pi_{+j}$ is multiplicative, so on the log scale it is additive, like a two-way ANOVA without interaction. Adding the "interaction" terms gives a model that can describe any association.

Let $\{\mu_{ij}\}$ be expected counts: either $\mu_{ij} = n\pi_{ij}$ under multinomial sampling, or $\mu_{ij} = E(Y_{ij})$ for $N = IJ$ independent Poisson counts. Formulas in terms of μ_{ij} cover both cases. Denote the row variable by X and the column variable by Y.

8.1.1 Independence model

Definition (Loglinear model of independence)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y, \tag{8.1}$$

with identifiability constraints such as $\lambda_I^X = \lambda_J^Y = 0$ (baseline coding) or $\sum_i \lambda_i^X = \sum_j \lambda_j^Y = 0$.

Theorem 8.1 (Independence as a loglinear model)

For positive $\{\mu_{ij}\}$ with $\mu_{ij} = n\pi_{ij}$, the variables X and Y are independent if and only if (8.1) holds for some parameters. The ML fitted values are $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$.

Proof.

Step 1. If $\pi_{ij} = \pi_{i+}\pi_{+j}$, then $\log \mu_{ij} = \log n + \log \pi_{i+} + \log \pi_{+j}$. Put $\lambda_i^X = \log \pi_{i+} - \log \pi_{I+}$, $\lambda_j^Y = \log \pi_{+j} - \log \pi_{+J}$ and $\lambda = \log n + \log \pi_{I+} + \log \pi_{+J}$; this is (8.1) with baseline constraints.

Step 2. Conversely, if $\mu_{ij} = a_i b_j$ with $a_i = e^{\lambda + \lambda_i^X}$, $b_j = e^{\lambda_j^Y}$, then $\mu_{i+} = a_i b_+$, $\mu_{+j} = a_+ b_j$, $\mu_{++} = a_+ b_+$, so

$$\pi_{ij} = \frac{a_i b_j}{a_+ b_+} = \frac{a_i b_+}{a_+ b_+} \cdot \frac{a_+ b_j}{a_+ b_+} = \pi_{i+}\pi_{+j}.$$

Step 3. The fitted values $n_{i+}n_{+j}/n$ satisfy the model and reproduce the row and column totals; by Birch's result (Theorem 8.19) they are the ML estimates. ∎

Hence the Pearson X^2 and G^2 tests of independence are goodness-of-fit tests of the loglinear model (8.1), with $\text{df} = (I-1)(J-1)$.

8.1.2 Interpretation of parameters

A loglinear GLM takes the N cell counts, not the n subjects, as its observations, and does not distinguish response from explanatory variables. For interpretation it nevertheless helps to form a logit.

Proposition 8.2 (Independence means a constant logit)

Let $J = 2$. Under (8.1),

$$\operatorname{logit} P(Y = 1 \mid X = i) = \lambda_1^Y - \lambda_2^Y \qquad \text{for every } i,$$

so the odds of column 1 equal $\exp(\lambda_1^Y - \lambda_2^Y)$ in every row.

Proof. $P(Y = 1 \mid X = i)/P(Y = 2 \mid X = i) = \mu_{i1}/\mu_{i2}$, hence the logit equals

$$\log \mu_{i1} - \log \mu_{i2} = (\lambda + \lambda_i^X + \lambda_1^Y) - (\lambda + \lambda_i^X + \lambda_2^Y) = \lambda_1^Y - \lambda_2^Y,$$

which does not depend on i. ■

In general, differences of main-effect parameters for one variable are log odds of one category versus another on that variable.

8.1.3 Saturated model

Definition (Saturated two-way model)

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY}. \tag{8.2}$$

The association terms λ_{ij}^{XY} (coefficients of products of dummy variables) measure departures from independence; with baseline constraints $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$.

Theorem 8.3 (Saturated model and odds ratios)

(a) Model (8.2) has IJ nonredundant parameters and describes every positive table $\{\mu_{ij}\}$ exactly.
(b) Every odds ratio is a contrast of the association parameters; for rows a, c and columns b, d,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}.$$

In particular, for a 2×2 table $\log \theta = \lambda_{11}^{XY} + \lambda_{22}^{XY} - \lambda_{12}^{XY} - \lambda_{21}^{XY}$.
(c) Independence holds iff all $\lambda_{ij}^{XY} = 0$; the test has df $= (I - 1)(J - 1)$.

Proof.

Step 1. Counting under baseline constraints: $1 + (I - 1) + (J - 1) + (I - 1)(J - 1) = IJ$.

Step 2. Given positive $\{\mu_{ij}\}$, set $h_{ij} = \log \mu_{ij}$ and define

$$\lambda = h_{IJ}, \quad \lambda_i^X = h_{iJ} - h_{IJ}, \quad \lambda_j^Y = h_{Ij} - h_{IJ}, \quad \lambda_{ij}^{XY} = h_{ij} - h_{iJ} - h_{Ij} + h_{IJ}.$$

These satisfy the baseline constraints and $\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY} = h_{ij}$, so every positive table is represented (uniquely, since the map is invertible).

Step 3. For (b), substitute (8.2): in $h_{ab} + h_{cd} - h_{ad} - h_{cb}$ the terms λ, $\lambda_a^X, \lambda_c^X, \lambda_b^Y, \lambda_d^Y$ each appear once with $+$ and once with $-$, leaving the stated contrast.

Step 4. For (c), by step 2 all $\lambda_{ij}^{XY} = 0$ iff $h_{ij} - h_{iJ} - h_{Ij} + h_{IJ} = 0$ for all i, j, i.e. all odds ratios with the last row and column are 1, which is equivalent to independence. The $(I - 1)(J - 1)$ free association parameters give the df. ■

Hierarchical models

A model is *hierarchical* if whenever it contains a term it also contains all lower-order terms built from the same variables (e.g. λ_{ij}^{XY} requires λ_i^X, λ_j^Y). Nonhierarchical models such as $\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_{ij}^{XY}$ have fits and interpretations that depend on the coding (with baseline coding it forces $\log \mu_{Ij} = \lambda$ in every column) and are rarely sensible. In a model with λ^{XY}, interpretation should focus on the highest-order terms, not on main effects.

8.1.4 Alternative parameter constraints

Instead of $\lambda_{Ij}^{XY} = \lambda_{iJ}^{XY} = 0$ one may use $\sum_i \lambda_{ij}^{XY} = \sum_j \lambda_{ij}^{XY} = 0$. Only contrasts such as the one in Theorem 8.3(b) are invariant. For a 2×2 table with $\log \theta = 2.0$: baseline coding gives $\lambda_{11}^{XY} = 2.0$ and the others 0; sum-to-zero coding gives $\lambda_{11}^{XY} = \lambda_{22}^{XY} = 0.5$, $\lambda_{12}^{XY} = \lambda_{21}^{XY} = -0.5$. Both give $\log \theta = 2.0$. Baseline coding has the advantage that some parameters in a set may have infinite estimates while others stay finite.

8.1.5 Multinomial models for cell probabilities

Proposition 8.4 (Poisson to multinomial)

If Y_{ij} are independent Poisson with $\log \mu_{ij}$ given by (8.2), then conditional on $\sum Y_{ij} = n$ the counts are multinomial$(n, \{\pi_{ij}\})$ with

$$\pi_{ij} = \frac{\exp(\lambda + \lambda_i^X + \lambda_j^Y + \lambda_{ij}^{XY})}{\sum_a \sum_b \exp(\lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}^{XY})}, \tag{8.3}$$

in which λ cancels.

Proof. Let $\mu_{++} = \sum \mu_{ij}$; the total is Poisson(μ_{++}). For counts $\{n_{ij}\}$ summing to n,

$$P(\{Y_{ij} = n_{ij}\} \mid Y_{++} = n) = \frac{\prod_{ij} e^{-\mu_{ij}} \mu_{ij}^{n_{ij}} / n_{ij}!}{e^{-\mu_{++}} \mu_{++}^n / n!} = \frac{n!}{\prod n_{ij}!} \prod_{ij} \Big(\frac{\mu_{ij}}{\mu_{++}} \Big)^{n_{ij}},$$

since $\prod e^{-\mu_{ij}} = e^{-\mu_{++}}$ and $\sum n_{ij} = n$. This is multinomial with $\pi_{ij} = \mu_{ij}/\mu_{++}$, which is (8.3); the common factor e^{λ} cancels. ∎

The intercept λ only describes the total sample size, which is random under Poisson sampling but fixed under multinomial sampling.

8.2 Loglinear models for independence and interaction in three-way tables

Motivation. Three-way tables admit several distinct kinds of independence (mutual, joint, conditional) and of association (homogeneous or not). Each one corresponds to a loglinear model in which certain interaction terms are absent, so the model symbol tells you the independence structure.

The following lemma turns every independence statement into a statement about which terms appear.

Lemma 8.5 (Generating classes and factorization)

Let a hierarchical loglinear model have generating class $\{C_1, \dots, C_m\}$ (the highest-order terms, i.e. the model symbol; e.g. (XZ, YZ) has $C_1 = \{X, Z\}$, $C_2 = \{Y, Z\}$). Then a positive table $\{\mu\}$ satisfies the model if and only if

$$\mu(\mathbf{x}) = \prod_{r=1}^{m} \psi_r(\mathbf{x}_{C_r})$$

for some positive functions ψ_r that depend on the cell $\mathbf{x}$ only through the coordinates in C_r.

Proof. [Supplement]

Step 1. Every term of the model involves a subset of some C_r, hence is a function of $\mathbf{x}_{C_r}$. Grouping terms by a generator containing them, $\log \mu = \sum_r g_r(\mathbf{x}_{C_r})$; exponentiate to get the product form.

Step 2. Conversely, by Theorem 8.3 (step 2, applied to the variables in C_r, and its obvious extension to more than two variables) any function $g(\mathbf{x}_C)$ decomposes as a constant plus main effects plus interactions of every subset of C. Since the model is hierarchical, all these subsets are terms of the model. Summing the decompositions of $\log \psi_r$ gives $\log \mu$ in the model's form. ■

8.2.1 Types of independence

Consider an $I \times J \times K$ table of three responses X, Y, Z with probabilities $\{\pi_{ijk}\}$ (or Poisson means $\{\mu_{ijk}\}$).

Definition (Three kinds of independence)

(i) *Mutual independence*: $\pi_{ijk} = \pi_{i++}\pi_{+j+}\pi_{++k}$ for all i, j, k.
(ii) *Joint independence* of Y from (X, Z): $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$, i.e. two-way independence between Y and the IK combinations of (X, Z).
(iii) *Conditional independence* of X and Y given Z: $\pi_{ij|k} = \pi_{i+|k}\pi_{+j|k}$ for all k, where $\pi_{ij|k} = P(X = i, Y = j \mid Z = k)$; equivalently

$$\pi_{ijk} = \pi_{i+k}\pi_{+jk}/\pi_{++k} \qquad \text{for all } i, j, k. \tag{8.4}$$

Theorem 8.6 (Independence structures as loglinear models)

For positive tables:
(a) mutual independence $\iff$ (X, Y, Z): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$;
(b) Y jointly independent of (X, Z) $\iff$ (XZ, Y): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ}$;
(c) $X \perp Y \mid Z$ $\iff$ (XZ, YZ): $\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}$.
Moreover (a) $\Rightarrow$ (b) $\Rightarrow$ (c).

Proof. By Lemma 8.5 it suffices to match each independence statement with a product form. Write $\mu = n\pi$.

Step 1 (c). If (8.4) holds, $\mu_{ijk} = \psi_1(i,k)\psi_2(j,k)$ with $\psi_1 = n\pi_{i+k}$ and $\psi_2 = \pi_{+jk}/\pi_{++k}$. Conversely, if $\mu_{ijk} = A_{ik}B_{jk}$, then $\mu_{i+k} = A_{ik}B_{+k}$, $\mu_{+jk} = A_{+k}B_{jk}$ and $\mu_{++k} = A_{+k}B_{+k}$, whence

$$\frac{\mu_{i+k}\mu_{+jk}}{\mu_{++k}} = \frac{A_{ik}B_{+k}\, A_{+k}B_{jk}}{A_{+k}B_{+k}} = A_{ik}B_{jk} = \mu_{ijk},$$

which is (8.4) after dividing by n.

Step 2 (b). If $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$ we have a product $\psi_1(i,k)\psi_2(j)$. Conversely if $\mu_{ijk} = A_{ik}B_j$ then $\mu_{i+k} = A_{ik}B_+$, $\mu_{+j+} = A_{++}B_j$, $\mu_{+++} = A_{++}B_+$, and $\mu_{i+k}\mu_{+j+}/\mu_{+++} = A_{ik}B_j$, i.e. $\pi_{ijk} = \pi_{i+k}\pi_{+j+}$.

Step 3 (a). Same computation with $\mu_{ijk} = A_iB_jC_k$: $\mu_{i++}\mu_{+j+}\mu_{++k}/\mu_{+++}^2 = A_iB_jC_k$.

Step 4. The implications follow because the model in (a) is a special case of (b) (set $\lambda^{XZ} = 0$), which is a special case of (c) (set $\lambda^{YZ} = 0$). ■

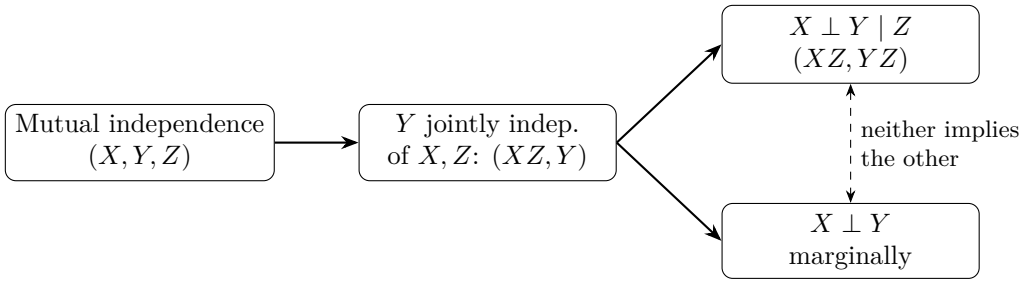


Figure 10: Relationships among types of XY independence (book Figure 8.1).

Conditional independence does not imply marginal independence, and conversely (Simpson's paradox). Both hold when one of the stronger structures applies (Figure 10).

8.2.2 Homogeneous association and three-factor interaction

Definition (Homogeneous association and the saturated model)

The model of *homogeneous association* (no three-factor interaction) is

$$\log \mu_{ijk} = \lambda + \lambda_i^X + \lambda_j^Y + \lambda_k^Z + \lambda_{ij}^{XY} + \lambda_{ik}^{XZ} + \lambda_{jk}^{YZ}, \tag{8.5}$$

symbol (XY, XZ, YZ); equivalently $\pi_{ijk} = \psi_{ij}\phi_{jk}\omega_{ik}$. The general (saturated) model adds λ_{ijk}^{XYZ}, symbol (XYZ).

The saturated model has

$$1+(I-1)+(J-1)+(K-1)+(I-1)(J-1)+(I-1)(K-1)+(J-1)(K-1)+(I-1)(J-1)(K-1) = IJK$$

parameters, one per cell (expand $IJK = [(I-1)+1][(J-1)+1][(K-1)+1]$). Unlike the models of Theorem 8.6, (8.5) has no closed-form expression of ψ, ϕ, ω in terms of margins.

Model symbol	Association terms	Interpretation
(X, Y, Z)	none	mutual independence
(XY, Z)	λ^{XY}	Z jointly independent of X, Y
(XY, YZ)	$\lambda^{XY}, \lambda^{YZ}$	$X \perp Z \mid Y$
(XY, XZ, YZ)	all two-factor	homogeneous association
(XYZ)	all, incl. λ^{XYZ}	saturated

8.2.3 Interpreting model parameters

At level k of Z the XY conditional association is described by the $(I-1)(J-1)$ local odds ratios

$$\theta_{ij(k)} = \frac{\mu_{ijk}\,\mu_{i+1,j+1,k}}{\mu_{i,j+1,k}\,\mu_{i+1,j,k}}, \tag{8.6}$$

and similarly $\theta_{i(j)k}$ for XZ and $\theta_{(i)jk}$ for YZ. Conditional independence of X, Y given Z means all $\theta_{ij(k)} = 1$.

Theorem 8.7 (Homogeneous association)

(a) Under (XY, XZ, YZ),

$$\log \theta_{ij(k)} = \lambda_{ij}^{XY} + \lambda_{i+1,j+1}^{XY} - \lambda_{i,j+1}^{XY} - \lambda_{i+1,j}^{XY} \quad \text{for all } k, \tag{8.7}$$

and similarly for XZ and YZ odds ratios.
(b) Conversely, a positive table satisfies (XY, XZ, YZ) if and only if $\theta_{ij(1)} = \cdots = \theta_{ij(K)}$ for all i, j. Equivalently, iff the XZ odds ratios are the same at every j, or iff the YZ odds ratios are the same at every i.

Proof. [Supplement]

Step 1. For (a), substitute (8.5) into $\log \theta_{ij(k)}$. Each of $\lambda, \lambda^X, \lambda^Y, \lambda_k^Z, \lambda_{ik}^{XZ}, \lambda_{i+1,k}^{XZ}, \lambda_{jk}^{YZ}, \lambda_{j+1,k}^{YZ}$ appears once with each sign and cancels; only the XY contrast remains, and it does not involve k.

Step 2. For (b), suppose $\theta_{ij(k)} = \theta_{ij}$ does not depend on k. Let $h_{ijk} = \log \mu_{ijk}$. Telescoping the local log odds ratios over the rectangle $[1, i] \times [1, j]$,

$$h_{ijk} - h_{1jk} - h_{i1k} + h_{11k} = \sum_{a<i} \sum_{b<j} \log \theta_{ab} =: g(i, j),$$

which is free of k. Therefore

$$h_{ijk} = g(i, j) + [h_{1jk} - h_{11k}] + h_{i1k},$$

a sum of a function of (i, j), a function of (j, k) and a function of (i, k). By Lemma 8.5 the table satisfies (XY, XZ, YZ).

Step 3. The symmetric statements follow by the same argument with the roles of the variables permuted; since each is equivalent to (XY, XZ, YZ), they are equivalent to one another. ■

The three-factor term describes how an odds ratio changes across the third variable. For a $2 \times 2 \times 2$ table, direct substitution of the saturated model gives

$$\log \frac{\theta_{11(1)}}{\theta_{11(2)}} = (\lambda_{111}^{XYZ} + \lambda_{221}^{XYZ} - \lambda_{121}^{XYZ} - \lambda_{211}^{XYZ}) - (\lambda_{112}^{XYZ} + \lambda_{222}^{XYZ} - \lambda_{122}^{XYZ} - \lambda_{212}^{XYZ}),$$

which equals λ_{111}^{XYZ} when second-category parameters are set to 0. When X, Y are binary and second-level parameters are zero, $\log \theta_{11(k)} = \lambda_{11}^{XY}$.

8.2.4 Alcohol, cigarette, and marijuana use example

Example 8.8 (Alcohol, cigarette, marijuana use)

A 1992 survey of 2276 high-school seniors near Dayton, Ohio, recorded whether they had ever used alcohol (A), cigarettes (C) and marijuana (M).

		Observed: M		Fit of (AC, AM, CM)	
A	C	yes	no	yes	no
yes	yes	911	538	910.4	538.6
yes	no	44	456	44.6	455.4
no	yes	3	43	3.6	42.4
no	no	2	279	1.4	279.6

Estimated odds ratios under several models:

	Conditional			Marginal		
Model	AC	AM	CM	AC	AM	CM
(A, C, M)	1.0	1.0	1.0	1.0	1.0	1.0
(AC, M)	17.7	1.0	1.0	17.7	1.0	1.0
(AM, CM)	1.0	61.9	25.1	2.7	61.9	25.1
(AC, AM, CM)	7.8	19.8	17.3	17.7	61.9	25.1
(ACM), M=yes	13.8	24.3	17.5	17.7	61.9	25.1
(ACM), M=no	7.7	13.5	9.7			

A term absent from a model gives a conditional odds ratio 1.0; e.g. under (AM, CM),

$$\frac{909.24 \times 0.24}{45.76 \times 4.76} = \frac{438.84 \times 179.84}{555.16 \times 142.16} = 1.0,$$

yet the fitted marginal AC odds ratio is 2.7: conditional independence does not imply marginal independence. Under (AC, AM, CM) the AC conditional odds ratio is 7.8 at both levels of M, $= \exp(\hat{\lambda}_{11}^{AC} + \hat{\lambda}_{22}^{AC} - \hat{\lambda}_{12}^{AC} - \hat{\lambda}_{21}^{AC})$. Estimated odds ratios depend heavily on the model, so model selection matters.

8.3 Inference for loglinear models

Motivation. Once a loglinear model is fitted, we need to know whether it fits, which association terms are needed, and how large the associations are. The tools are the usual deviance tests and Wald/likelihood intervals for GLMs.

8.3.1 Chi-squared goodness-of-fit tests

$X^2 = \sum(n - \hat{\mu})^2/\hat{\mu}$ and $G^2 = 2\sum n \log(n/\hat{\mu})$ test model fit, with df = number of cells − number of nonredundant parameters. Model selection can also use AIC $= -2$(maximized log likelihood − number of parameters), equivalently minimizing $G^2 - 2\,\text{df}$.

Example 8.9 (Goodness of fit for the student survey)

Model	G^2	X^2	df	P-value
(A, C, M)	1286.0	1411.4	4	< 0.001
(A, CM)	534.2	505.6	3	< 0.001
(C, AM)	939.6	824.2	3	< 0.001
(M, AC)	843.8	704.9	3	< 0.001
(AC, AM)	497.4	443.8	2	< 0.001
(AC, CM)	92.0	80.8	2	< 0.001
(AM, CM)	187.8	177.6	2	< 0.001
(AC, AM, CM)	0.4	0.4	1	0.54
(ACM)	0.0	0.0	0	–

Only the homogeneous association model fits; it also minimizes AIC.

8.3.2 Inference about conditional associations

Tests about a term compare nested models. The likelihood-ratio statistic $-2(L_0 - L_1)$ equals $G^2(M_0) - G^2(M_1)$, the difference of deviances. For XY conditional independence within (XY, XZ, YZ), $H_0 : \lambda_{ij}^{XY} = 0$ for all $(I-1)(J-1)$ parameters, and the statistic is

$$G^2[(XZ, YZ) \mid (XY, XZ, YZ)] = G^2(XZ, YZ) - G^2(XY, XZ, YZ), \qquad \text{df} = (I-1)(J-1).$$

This is a model-based alternative to the generalized CMH test.

Example 8.10 (Conditional associations in the student survey)

For AC conditional independence, $G^2[(AM, CM) \mid (AC, AM, CM)] = 187.8 - 0.4 = 187.4$, df $= 1$, $P < 0.001$. The other two pairwise terms are likewise needed (LR statistics 91.6 for AM, 497.0 for CM). With baseline coding (last category $= 0$), (AC, AM, CM) gives

$$\hat{\lambda}_{11}^{AC} = 2.054 \text{ (SE} = 0.174), \quad \hat{\lambda}_{11}^{AM} = 2.986 \text{ (0.465)}, \quad \hat{\lambda}_{11}^{CM} = 2.848 \text{ (0.164)}.$$

For binary variables these are conditional log odds ratios. The 95% Wald interval for the AC conditional odds ratio is $\exp(2.054 \pm 1.96 \times 0.174) = (5.5, 11.0)$; for AM it is (8.0, 49.2) and for CM (12.5, 23.8). All associations are strong, and the marginal ones are stronger still: controlling for one response moderates the association between the other two.

With large samples, significance alone is not informative; confidence intervals show whether associations are practically important.

8.4 Loglinear models for higher dimensions

Motivation. The three-way ideas extend directly to four or more variables. The difficulties are the large number of possible terms, which makes model selection harder, and the large number of cells, which can make tables sparse.

8.4.1 Four-way contingency tables

For variables W, X, Y, Z, the model with all two-factor terms,

$$\log \mu_{hijk} = \lambda + \lambda_h^W + \lambda_i^X + \lambda_j^Y + \lambda_k^Z$$

$$+\lambda_{hi}^{WX}+\lambda_{hj}^{WY}+\lambda_{hk}^{WZ}+\lambda_{ij}^{XY}+\lambda_{ik}^{XZ}+\lambda_{jk}^{YZ},$$

symbol (WX, WY, WZ, XY, XZ, YZ), has each pair conditionally dependent with the same odds ratios at every combination of levels of the other two variables. Removing a two-factor term means conditional independence of that pair given the other two variables. Under (WXY, WZ, XZ, YZ), the WX, WY, XY associations may vary across levels of the remaining variable of the triple, while every association involving Z is homogeneous. The saturated model adds all three-factor terms and λ^{WXYZ}.

Proposition 8.11 (Odds ratios in models without a variable in higher-order terms)

If a variable V appears in no term of order three or more, then for every other variable U the conditional UV odds ratios are the same at every combination of levels of the remaining variables, and they equal the exponentiated contrast of λ^{UV}.

Proof. [Supplement] Fix levels of the remaining variables and form a UV odds ratio $\mu_{ab\cdot}\mu_{cd\cdot}/(\mu_{ad\cdot}\mu_{cb\cdot})$ (U levels a, c; V levels b, d). In the log, any term not involving V appears with U at level a once with each sign (in μ_{ab} and μ_{ad}), and likewise at level c, so it cancels. Any term not involving U cancels the same way. The only terms involving both U and V are, by hypothesis, the two-factor term λ^{UV}, which does not depend on the other variables. Hence the log odds ratio is $\lambda_{ab}^{UV}+\lambda_{cd}^{UV}-\lambda_{ad}^{UV}-\lambda_{cb}^{UV}$ at every level combination. ■

8.4.2 Automobile accident example

Example 8.12 (Automobile accidents and seat belts)

In 1991, 68,694 passengers in auto accidents in Maine were classified by gender (G), location (L: urban, rural), seat-belt use (S) and injury (I).

Gender	Location	Seat belt	Injury no	Injury yes
Female	Urban	No	7287	996
		Yes	11587	759
	Rural	No	3246	973
		Yes	6134	757
Male	Urban	No	10381	812
		Yes	10969	380
	Rural	No	6123	1084
		Yes	6693	513

Goodness of fit:

Model	G^2	df	P
(G, I, L, S)	2792.8	11	< 0.0001
(GI, GL, GS, IL, IS, LS)	23.4	5	< 0.001
(GIL, GIS, GLS, ILS)	1.3	1	0.25
(GIL, GS, IS, LS)	18.6	4	0.001
(GIS, GL, IL, LS)	22.8	4	< 0.001
(GLS, GI, IL, IS)	7.5	4	0.11
(ILS, GI, GL, GS)	20.6	4	< 0.001

Estimated conditional odds ratios:

Odds ratio	(GI, GL, GS, IL, IS, LS)	(GLS, GI, IL, IS)
GI	0.58	0.58
IL	2.13	2.13
IS	0.44	0.44
GL (S=no / yes)	1.23 / 1.23	1.33 / 1.17
GS (L=urban / rural)	0.63 / 0.63	0.66 / 0.58
LS (G=female / male)	1.09 / 1.09	1.17 / 1.03

Under the pairwise model the IS conditional log odds ratio is -0.814 (SE 0.028), so the 95% interval for the odds ratio is $\exp(-0.814 \pm 1.96 \times 0.028) = (0.42, 0.47)$: wearing a seat belt more than halves the odds of injury at each gender–location combination. Injury is more likely in rural accidents and for females; males have 0.63 times the odds of females of wearing seat belts. In (GLS, GI, IL, IS) the variable I is in no three-factor term, so (Proposition 8.11) its odds ratios with G, L, S are homogeneous; the GL, GS, LS odds ratios vary with the third variable, e.g. $0.66 = (7273.2 \times 10{,}959.2)/(11{,}632.6 \times 10{,}358.9)$ for urban accidents.

8.4.3 Large samples and statistical versus practical significance

(GLS, GI, IL, IS) fits significantly better than the pairwise model: $23.4 - 7.5 = 15.9$, df $= 1$, $P = 0.0001$. Yet the fitted GL, GS, LS odds ratios are similar at both levels of the third variable, so the three-factor interaction is weak; the significance reflects the huge n. With large samples, focus on estimation; the simpler pairwise model is adequate for most purposes.

8.4.4 Dissimilarity index

Definition (Dissimilarity index)

For cell counts $n_i = np_i$ and fitted values $\hat{\mu}_i = n\hat{\pi}_i$,

$$\hat{\Delta} = \frac{\sum_i |n_i - \hat{\mu}_i|}{2n} = \frac{1}{2}\sum_i |p_i - \hat{\pi}_i|.$$

It estimates the population index $\Delta = \frac{1}{2}\sum |\pi_i - \pi_i^{\text{model}}|$.

Proposition 8.13 (Meaning of $\hat{\Delta}$)

If $\sum \hat{\mu}_i = n$ (true for any loglinear model with an intercept), then $0 \le \hat{\Delta} \le 1$, and $n\hat{\Delta}$ is the minimum number of observations that must be moved to other cells to make the observed table equal to the fitted one.

Proof. [Supplement]

Step 1. $\sum |p_i - \hat{\pi}_i| \le \sum p_i + \sum \hat{\pi}_i = 2$, so $\hat{\Delta} \le 1$; clearly $\hat{\Delta} \ge 0$.

Step 2. Let $d_i = n_i - \hat{\mu}_i$. Since $\sum d_i = 0$, $\sum d_i^+ = \sum d_i^- = \frac{1}{2}\sum |d_i| = n\hat{\Delta}$.

Step 3. Moving one observation changes $\sum d_i^+$ by at most 1, so at least $\sum d_i^+$ moves are necessary. Moving d_i^+ observations out of each cell with surplus into cells with deficits (total deficit $\sum d_i^-$, the same amount) is sufficient (allowing fractional "observations" since $\hat{\mu}$ need not be integer). Hence the minimum is $n\hat{\Delta}$. ■

Values $\hat{\Delta} < 0.02$ or 0.03 indicate that the data follow the model closely in practical terms even if

the model is rejected; $\hat{\Delta}$ tends to overestimate Δ when Δ is near 0, especially for small n. For the accident data, $\hat{\Delta} = 0.008$ for the pairwise model and 0.003 for (GLS, GI, IL, IS): moving less than 1% of the data yields a perfect fit.

8.5 Loglinear–logit model connection

Motivation. Loglinear models treat variables symmetrically, logit models single out a response. Forming a logit from a loglinear model helps interpret it, and every logit model with categorical predictors is equivalent to a loglinear model that contains the full interaction among the predictors.

8.5.1 Using logit models to interpret loglinear models

Proposition 8.14 (Logit form of (XY, XZ, YZ))

If Y is binary and (XY, XZ, YZ) holds, then

$$\text{logit}\, P(Y = 1 \mid X = i, Z = k) = \alpha + \beta_i^X + \beta_k^Z, \tag{8.8}$$

with $\alpha = \lambda_1^Y - \lambda_2^Y$, $\beta_i^X = \lambda_{i1}^{XY} - \lambda_{i2}^{XY}$, $\beta_k^Z = \lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}$.

Proof. The logit is $\log(\mu_{i1k}/\mu_{i2k})$. Subtracting (8.5) at $j = 2$ from (8.5) at $j = 1$, the terms $\lambda, \lambda_i^X, \lambda_k^Z, \lambda_{ik}^{XZ}$ cancel, leaving

$$(\lambda_1^Y - \lambda_2^Y) + (\lambda_{i1}^{XY} - \lambda_{i2}^{XY}) + (\lambda_{1k}^{YZ} - \lambda_{2k}^{YZ}).$$

■

The λ^{XZ} terms, which describe association among explanatory variables, cancel: the logit model $(X + Z)$ says nothing about it.

8.5.2 Auto accident example revisited

Example 8.15 (Injury as a response)

For (GLS, GI, IL, IS), forming the logit on injury cancels every term without the index i of I, including λ^{GLS}, giving the logit model $(G + L + S)$:

$$\text{logit}\, P(I = 1 \mid G = g, L = l, S = s) = \alpha + \beta_g^G + \beta_l^L + \beta_s^S, \qquad \beta_s^S = \lambda_{1s}^{IS} - \lambda_{2s}^{IS}.$$

Fitted values, G^2, residual df and standardized residuals are identical for the two models. The seat-belt log odds ratio is $\hat{\beta}_1^S - \hat{\beta}_2^S = -0.817$ in the logit model and $\hat{\lambda}_{11}^{IS} + \hat{\lambda}_{22}^{IS} - \hat{\lambda}_{12}^{IS} - \hat{\lambda}_{21}^{IS} = -0.817$ in the loglinear model. The loglinear model treats the 16 counts as Poisson; the logit model treats the GLS margin as fixed and the 8 counts $\{n_{g1ls}\}$ as binomial.

8.5.3 Correspondence between loglinear and logit models

Forming the logit on Y in (XY, YZ) also gives (8.8), but (XY, YZ) is *not* equivalent to the logit model: it additionally forces $X \perp Z \mid Y$. The equivalent loglinear model must contain the most general interaction among the explanatory variables.

Theorem 8.16 (Loglinear–logit equivalence)

Let Y be the response and $\mathbf{x}$ index the combinations of the explanatory variables. Consider a loglinear model containing the full interaction term among the explanatory variables, $\log \mu_{\mathbf{x}j} = \gamma_{\mathbf{x}} + \eta_{\mathbf{x}j}(\boldsymbol{\beta})$, where $\gamma_{\mathbf{x}}$ is unrestricted and $\eta_{\mathbf{x}j}$ collects the terms involving Y (with $\eta_{\mathbf{x}J} = 0$). Then the ML fitted values under Poisson sampling coincide with those of the baseline-category logit model $\log(\pi_{j|\mathbf{x}}/\pi_{J|\mathbf{x}}) = \eta_{\mathbf{x}j}(\boldsymbol{\beta})$ fitted with the margins $n_{\mathbf{x}+}$ fixed, and the ML estimates of $\boldsymbol{\beta}$ agree.

Proof. [Supplement]

Step 1. Reparametrize: $\mu_{\mathbf{x}+} = e^{\gamma_{\mathbf{x}}} \sum_j e^{\eta_{\mathbf{x}j}}$ and $\pi_{j|\mathbf{x}} = e^{\eta_{\mathbf{x}j}} / \sum_h e^{\eta_{\mathbf{x}h}}$, so $\mu_{\mathbf{x}j} = \mu_{\mathbf{x}+}\pi_{j|\mathbf{x}}$. Since $\gamma_{\mathbf{x}}$ is free, $(\{\mu_{\mathbf{x}+}\}, \boldsymbol{\beta})$ ranges over $(0,\infty)^{\#\mathbf{x}} \times \{\text{all } \boldsymbol{\beta}\}$, a product set (variation independence).

Step 2. The Poisson log likelihood splits:

$$\sum_{\mathbf{x},j}(n_{\mathbf{x}j}\log\mu_{\mathbf{x}j} - \mu_{\mathbf{x}j}) = \sum_{\mathbf{x}}(n_{\mathbf{x}+}\log\mu_{\mathbf{x}+} - \mu_{\mathbf{x}+}) + \sum_{\mathbf{x},j} n_{\mathbf{x}j}\log\pi_{j|\mathbf{x}}(\boldsymbol{\beta}).$$

Step 3. The two parts depend on separate parameters, so they are maximized separately: $\hat{\mu}_{\mathbf{x}+} = n_{\mathbf{x}+}$, and $\hat{\boldsymbol{\beta}}$ maximizes the second part, which is exactly the product-multinomial (binomial if $J = 2$) log likelihood of the logit model. Hence $\hat{\mu}_{\mathbf{x}j} = n_{\mathbf{x}+}\hat{\pi}_{j|\mathbf{x}}$ in both models. ■

Loglinear symbol	Logit model	Logit symbol
(Y, XZ)	α	$(-)$
(XY, XZ)	$\alpha + \beta_i^X$	(X)
(YZ, XZ)	$\alpha + \beta_k^Z$	(Z)
(XY, YZ, XZ)	$\alpha + \beta_i^X + \beta_k^Z$	$(X + Z)$
(XYZ)	$\alpha + \beta_i^X + \beta_k^Z + \beta_{ik}^{XZ}$	$(X*Z)$

With several response variables the loglinear approach is more general; with one response, logit models are more natural.

8.5.4 Generalized loglinear model

Write the counts as a vector $\mathbf{n} = (n_1, \ldots, n_N)'$ with means $\boldsymbol{\mu}$. An ordinary loglinear model is

$$\log \boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}. \tag{8.9}$$

For instance, independence in a 2×2 table with $\lambda_2^X = \lambda_2^Y = 0$ is

$$\begin{pmatrix} \log\mu_{11} \\ \log\mu_{12} \\ \log\mu_{21} \\ \log\mu_{22} \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} \lambda \\ \lambda_1^X \\ \lambda_1^Y \end{pmatrix}.$$

The *generalized loglinear model* is $\mathbf{C}\log(\mathbf{A}\boldsymbol{\mu}) = \mathbf{X}\boldsymbol{\beta}$. With $\mathbf{A} = \mathbf{I}_4$, $\mathbf{C} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix}$, $\mathbf{X} = (1,1)'$ and $\boldsymbol{\beta} = \alpha$, it states that the logits in both rows equal α, i.e. independence again. The class includes logit models and, later, models for marginal distributions.

8.6 Loglinear model fitting: likelihood equations

Motivation. Why are fitted values of loglinear models obtained by matching marginal tables? Because the Poisson model is an exponential family whose sufficient statistics are those margins. This section derives the sufficient statistics, likelihood equations, uniqueness, closed-form fits, degrees of freedom and the asymptotic covariance matrix.

Derivations use Poisson sampling, which needs no constraint on parameters; Theorem 8.21 transfers them to multinomial sampling.

8.6.1 Minimal sufficient statistics

The kernel of the Poisson log likelihood for a three-way table is

$$L(\boldsymbol{\mu}) = \sum_{i,j,k} n_{ijk} \log \mu_{ijk} - \sum_{i,j,k} \mu_{ijk}. \tag{8.10}$$

Substituting the saturated model,

$$\begin{aligned} L = n\lambda + \sum_i n_{i++}\lambda_i^X + \sum_j n_{+j+}\lambda_j^Y + \sum_k n_{++k}\lambda_k^Z + \sum_{i,j} n_{ij+}\lambda_{ij}^{XY} + \sum_{i,k} n_{i+k}\lambda_{ik}^{XZ} \\ + \sum_{j,k} n_{+jk}\lambda_{jk}^{YZ} + \sum_{i,j,k} n_{ijk}\lambda_{ijk}^{XYZ} - \sum_{i,j,k} \exp(\lambda + \lambda_i^X + \cdots + \lambda_{ijk}^{XYZ}). \end{aligned} \tag{8.11}$$

Theorem 8.17 (Sufficient statistics are marginal tables)

For a hierarchical loglinear model, the minimal sufficient statistics are the marginal tables corresponding to the terms in the model symbol: e.g. $\{n_{i++}\}, \{n_{+j+}\}, \{n_{++k}\}$ for (X, Y, Z); $\{n_{ij+}\}, \{n_{++k}\}$ for (XY, Z); $\{n_{ij+}\}, \{n_{+jk}\}$ for (XY, YZ); $\{n_{ij+}\}, \{n_{i+k}\}, \{n_{+jk}\}$ for (XY, XZ, YZ).

Proof.

Step 1. Setting the absent parameters to zero in (8.11), the joint distribution is an exponential family whose canonical statistics are the coefficients of the remaining parameters, i.e. the marginal tables of all terms present. By the factorization theorem these are sufficient; minimality holds because, after removing redundant parameters, the family is full rank.

Step 2. A lower-order margin is a sum of a higher-order one (e.g. $n_{i++} = \sum_j n_{ij+}$), so only the margins of the highest-order terms (the generators) are needed. ■

For the saturated model the full table is sufficient: no data reduction.

8.6.2 Likelihood equations for loglinear models

In the matrix form (8.9), $\log \mu_i = \sum_j x_{ij}\beta_j$ and

$$L(\boldsymbol{\beta}) = \sum_i n_i \Big(\sum_j x_{ij}\beta_j\Big) - \sum_i \exp\Big(\sum_j x_{ij}\beta_j\Big). \tag{8.12}$$

Theorem 8.18 (Likelihood equations match margins)

(a) The likelihood equations are

$$\mathbf{X}'\mathbf{n} = \mathbf{X}'\hat{\boldsymbol{\mu}}\,: \tag{8.13}$$

sufficient statistics equal their fitted (expected) values.
(b) For a hierarchical model, (8.13) is equivalent to equality of the observed and fitted marginal tables in the model symbol. For (XZ, YZ):

$$\hat{\mu}_{i+k} = n_{i+k} \text{ for all } i,k, \qquad \hat{\mu}_{+jk} = n_{+jk} \text{ for all } j,k. \tag{8.14}$$

Proof.

Step 1. Since $\partial \exp(\sum_j x_{ij}\beta_j)/\partial\beta_j = x_{ij}\mu_i$,

$$\frac{\partial L}{\partial \beta_j} = \sum_i n_i x_{ij} - \sum_i \mu_i x_{ij}, \qquad j = 1, \ldots, p.$$

Setting these to 0 gives (8.13).

Step 2. (8.13) says $\mathbf{v}'\mathbf{n} = \mathbf{v}'\hat{\boldsymbol{\mu}}$ for every $\mathbf{v}$ in the column space of $\mathbf{X}$. By Lemma 8.5, this column space is the set of functions $\sum_r g_r(\mathbf{x}_{C_r})$, which is spanned by the indicator vectors $\mathbf{v} = \mathbb{1}\{\mathbf{x}_{C_r} = \mathbf{c}\}$ of the cells of each generator's margin. For such $\mathbf{v}$, $\mathbf{v}'\mathbf{n}$ is the marginal count at $\mathbf{c}$. Hence (8.13) is equivalent to equality of all generator margins. For (XZ, YZ) this is (8.14). ■

Thus fitted values are smoothed versions of the data that agree with them on certain margins but whose associations follow the model.

8.6.3 Birch's results for loglinear models

Theorem 8.19 (Birch)

There is at most one $\hat{\boldsymbol{\mu}}$ that both satisfies the loglinear model and matches the data in the minimal sufficient statistics, and if it exists it is the ML estimate. Conversely, any ML estimate has these two properties.

Proof. [Supplement]

Step 1. Write $\boldsymbol{\eta} = \log \boldsymbol{\mu} \in \mathcal{V} = \text{col}(\mathbf{X})$. As a function of $\boldsymbol{\eta}$, $L = \mathbf{n}'\boldsymbol{\eta} - \sum_i e^{\eta_i}$ is strictly concave on $\mathbb{R}^N$: its Hessian is $-\text{diag}(e^{\eta_i})$, negative definite. Its restriction to the subspace $\mathcal{V}$ is still strictly concave.

Step 2. A point $\boldsymbol{\eta} \in \mathcal{V}$ is stationary for $L|_{\mathcal{V}}$ iff the gradient $\mathbf{n} - \boldsymbol{\mu}$ is orthogonal to $\mathcal{V}$, i.e. $\mathbf{X}'(\mathbf{n} - \boldsymbol{\mu}) = \mathbf{0}$, which is margin matching by Theorem 8.18.

Step 3. For a differentiable concave function, a stationary point is a global maximum; for a strictly concave one, there is at most one maximizer. Hence a $\boldsymbol{\mu}$ in the model that matches the margins is the unique MLE, and conversely the MLE (an interior maximizer) is stationary, so it matches the margins. ■

For example, for independence in a two-way table, $\hat{\mu}_{ij} = n_{i+}n_{+j}/n$ satisfies the model and has $\hat{\mu}_{i+} = n_{i+}$, $\hat{\mu}_{+j} = n_{+j}$; by Birch it is the MLE.

8.6.4 Direct versus iterative calculation of fitted values

Theorem 8.20 (Closed-form (direct) estimates)

The ML fitted values for three-way models other than (XY, XZ, YZ) are:

Model	Probabilistic form	Fitted value $\hat{\mu}_{ijk}$
(X, Y, Z)	$\pi_{i++}\pi_{+j+}\pi_{++k}$	$n_{i++}n_{+j+}n_{++k}/n^2$
(XY, Z)	$\pi_{ij+}\pi_{++k}$	$n_{ij+}n_{++k}/n$
(XY, XZ)	$\pi_{ij+}\pi_{i+k}/\pi_{i++}$	$n_{ij+}n_{i+k}/n_{i++}$
(XY, XZ, YZ)	$\psi_{ij}\phi_{jk}\omega_{ik}$	iterative (IPF)
(XYZ)	none	n_{ijk}

(other models by symmetry, e.g. (XZ, YZ): $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$).

Proof. Take (XZ, YZ); the others are identical in structure.

Step 1. By Theorem 8.6, under the model $\mu_{ijk} = \mu_{i+k}\mu_{+jk}/\mu_{++k}$.

Step 2. Define $\hat{\mu}_{ijk} = n_{i+k}n_{+jk}/n_{++k}$. It has the product form $A_{ik}B_{jk}$, so it satisfies the model.

Step 3. Summing over j: $\hat{\mu}_{i+k} = n_{i+k}n_{++k}/n_{++k} = n_{i+k}$; summing over i: $\hat{\mu}_{+jk} = n_{+jk}$. Thus it matches the sufficient margins (8.14).

Step 4. By Birch's theorem it is the MLE. For (X, Y, Z), (XY, Z), (XY, XZ) the displayed fits likewise have the factorized form and reproduce the sufficient margins (e.g. for (XY, XZ): $\sum_k n_{ij+}n_{i+k}/n_{i++} = n_{ij+}$ and $\sum_j n_{ij+}n_{i+k}/n_{i++} = n_{i+k}$). For (XYZ), $\hat{\mu} = \mathbf{n}$ trivially. ■

Models with closed-form estimates are called *direct* (or *decomposable*: fitted values are products and ratios of fitted margins). Unsaturated models containing all two-factor terms, such as (XY, XZ, YZ), have no direct estimates. Software uses iterative methods for all models anyway.

8.6.5 Connection between multinomial and Poisson loglinear models

Theorem 8.21 (Poisson and multinomial give the same estimates)

Write the Poisson model as $\log \mu_i = \alpha + \mathbf{x}_i'\boldsymbol{\beta}$ (α the intercept). Then the Poisson log likelihood decomposes as

$$L(\alpha, \boldsymbol{\beta}) = \Big[\sum_i n_i \log \pi_i(\boldsymbol{\beta})\Big] + [n \log \tau - \tau], \qquad \pi_i(\boldsymbol{\beta}) = \frac{e^{\mathbf{x}_i'\boldsymbol{\beta}}}{\sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}}, \tag{8.15}$$

where $\tau = \sum_i \mu_i$. The first bracket is the multinomial log likelihood given n, the second the Poisson log likelihood of n. Consequently $\hat{\boldsymbol{\beta}}$ and its asymptotic covariance are the same under Poisson and multinomial sampling.

Proof.

Step 1. $L = \sum n_i \log \mu_i - \sum \mu_i = \sum n_i(\alpha + \mathbf{x}_i'\boldsymbol{\beta}) - \tau = n\alpha + \sum n_i \mathbf{x}_i'\boldsymbol{\beta} - \tau$.

Step 2. Since $\tau = e^{\alpha} \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$, $\alpha = \log \tau - \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}}$. Substituting,

$$L = \sum_i n_i \mathbf{x}_i'\boldsymbol{\beta} - n \log \sum_h e^{\mathbf{x}_h'\boldsymbol{\beta}} + n \log \tau - \tau = \sum_i n_i \log \pi_i(\boldsymbol{\beta}) + (n \log \tau - \tau).$$

Step 3. The map $(\alpha, \boldsymbol{\beta}) \mapsto (\tau, \boldsymbol{\beta})$ is one-to-one onto $(0, \infty) \times \mathbb{R}^p$, and $\boldsymbol{\beta}$ enters only the first bracket. Hence $\hat{\boldsymbol{\beta}}$ maximizes the multinomial log likelihood, $\hat{\tau} = n$, and the information matrix is block diagonal in $(\tau, \boldsymbol{\beta})$, so the $\boldsymbol{\beta}$ block of the inverse information is the same for both sampling schemes. ■

The same argument with several independent multinomial samples works as long as the model contains the term for the margin fixed by the design (e.g. λ^{XZ} when $\{n_{i+k}\}$ are fixed), which

forces $\hat{\mu}_{i+k} = n_{i+k}$; this is Theorem 8.16. For a single multinomial sample, $\widehat{\text{cov}}(\hat{\boldsymbol{\beta}}) = \{\mathbf{X}'[\text{diag}(\hat{\boldsymbol{\mu}}) - \hat{\boldsymbol{\mu}}\hat{\boldsymbol{\mu}}'/n]\mathbf{X}\}^{-1}$, where $\mathbf{X}$ omits the intercept column; it equals the Poisson covariance $\{\mathbf{X}'\,\text{diag}(\hat{\boldsymbol{\mu}})\mathbf{X}\}^{-1}$ with the row and column for α deleted.

Chapter 9. Building and Extending Loglinear–Logit Models

This chapter turns the loglinear models of Chapter 8 into a working toolkit: association graphs that display conditional independence and tell us when a table may be collapsed (Section 9.1), model selection and comparison (Section 9.2), residual diagnostics (Section 9.3), models that use the ordering of categories (Section 9.4), Poisson regression for rates (Section 9.5), and the problems caused by empty cells and sparse tables (Section 9.6).

9.1 Association graphs and collapsibility

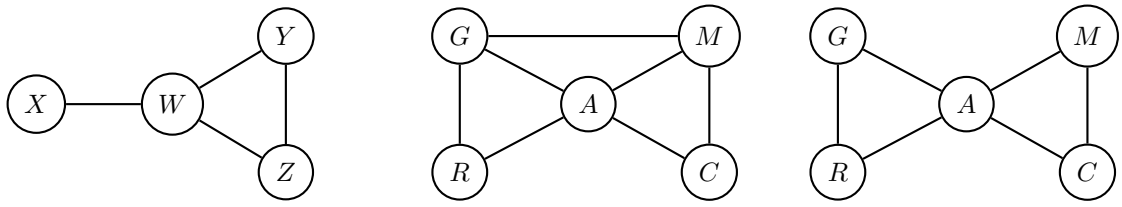
Motivation. With four or five variables it is hard to read conditional independence off a model formula. Drawing the model as a graph turns these questions into questions about paths, and it also tells us when a table can be collapsed over some variables without changing the associations of interest.

9.1.1 Association graphs

Definition (Association graph, path, separation)

The *association graph* of a loglinear model has a vertex for each variable and an edge between two variables if and only if the model contains a term involving both, i.e. if they are conditionally associated. A *path* is a sequence of edges leading from one variable to another. A set S of variables *separates* X and Y if every path between X and Y passes through a vertex of S.

Models with the same pairwise associations have the same graph; e.g. (WX, WY, WZ, YZ) and (WX, WYZ) share Figure 11.



(WX, WY, WZ, YZ) model 6: $(AC, AM, CM, AG, AR, GM, GR)$ model 7: (AC, AM, CM, AG, AR, GR)

Figure 11: Association graphs. Left: book Figure 9.1; W separates X from $\{Y, Z\}$. Middle and right: two models for the five-way student survey (Example 9.5); in model 7, A separates $\{G, R\}$ from $\{C, M\}$.

Theorem 9.1 (Separation implies conditional independence)

In a hierarchical loglinear model, if a set S of variables separates X and Y in the association graph, then X and Y are conditionally independent given S (after marginalizing over all other variables). More generally, if S separates sets A_0 and B_0, then $A_0 \perp B_0 \mid S$.

Proof. [Supplement]

Step 1. Let V be the set of all variables, A the set of vertices reachable from X by paths avoiding S, and $B = V \setminus (A \cup S)$. Since S separates X and Y, $Y \in B$.

Step 2. Every term T of the model lies in $A \cup S$ or in $B \cup S$. Indeed, if T contained $a \in A$ and $b \in B$, then the hierarchical model would contain the two-factor term $\{a, b\} \subseteq T$, so a–b would be an edge and b would be reachable from X avoiding S, i.e. $b \in A$, a contradiction.

Step 3. Grouping terms accordingly, $\mu(\mathbf{x}) = f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)$ (assign terms inside S to f). Hence, for fixed $\mathbf{x}_S$,

$$P(\mathbf{x}_A, \mathbf{x}_B \mid \mathbf{x}_S) = \frac{f(\mathbf{x}_A, \mathbf{x}_S)\, g(\mathbf{x}_B, \mathbf{x}_S)}{\sum_{\mathbf{a}} f(\mathbf{a}, \mathbf{x}_S) \sum_{\mathbf{b}} g(\mathbf{b}, \mathbf{x}_S)},$$

which factors into a function of $\mathbf{x}_A$ times a function of $\mathbf{x}_B$: $\mathbf{x}_A \perp \mathbf{x}_B \mid \mathbf{x}_S$.

Step 4. Summing over the coordinates of A other than X and the coordinates of B other than Y preserves the product form, so $X \perp Y \mid S$. The statement for sets $A_0 \subseteq A$, $B_0 \subseteq B$ is the same computation. ■

In Figure 11 (left), W separates X and Y, so $X \perp Y \mid W$ (and also given $\{W, Z\}$); likewise $X \perp Z \mid W$.

9.1.2 Collapsibility in three-way contingency tables

Theorem 9.2 (Collapsibility conditions)

In a three-way table, the XY marginal odds ratios equal the XY conditional odds ratios (at every level of Z) if Z and X are conditionally independent given Y, or Z and Y are conditionally independent given X; that is, under models (XY, YZ) or (XY, XZ) (or simpler models).

Proof. [Supplement]

Step 1. Under (XY, XZ), by Lemma 8.5, $\mu_{ijk} = A_{ij} C_{ik}$. The conditional odds ratio at level k is

$$\frac{A_{ij} C_{ik}\, A_{i'j'} C_{i'k}}{A_{ij'} C_{ik}\, A_{i'j} C_{i'k}} = \frac{A_{ij} A_{i'j'}}{A_{ij'} A_{i'j}},$$

free of k. The marginal table is $\mu_{ij+} = A_{ij} C_{i+}$, and the row factors $C_{i+}, C_{i'+}$ cancel in the odds ratio, giving the same value.

Step 2. Under (XY, YZ), $\mu_{ijk} = A_{ij} B_{jk}$ and $\mu_{ij+} = A_{ij} B_{j+}$; now column factors cancel, with the same conclusion. ■

Example 9.3 (Collapsibility in the student survey)

In (AM, CM) (graph A–M–C), C is conditionally independent of A, so the AM association is collapsible over C: fitted conditional and marginal AM odds ratios both equal 61.9 (Example 8.8); similarly CM (25.1). The AC association is not collapsible, because M is linked to both A and C: conditional AC odds ratio 1.0, marginal 2.7. In (AC, AM, CM) no collapsibility condition holds, and every pair has quite different conditional and marginal associations.

9.1.3 Collapsibility and logit models

In a multicenter clinical trial with binary treatment x, binary response Y and centers Z, the model $\operatorname{logit} P(Y = 1 \mid X = i, Z = k) = \alpha + \beta x_i + \beta_k^Z$ corresponds to (XY, XZ, YZ), so the common conditional odds ratio e^{β} typically differs from the odds ratio of the marginal 2×2 table. The simpler model without center effects, $\operatorname{logit} = \alpha + \beta x_i$, states $Y \perp Z \mid X$; it is equivalent to (XY, XZ), which is collapsible over Z. So when center effects are negligible the treatment effect is approximately the marginal odds ratio.

9.1.4 Collapsibility and association graphs for multiway tables

Theorem 9.4 (Parametric collapsibility, Bishop et al.)

Partition the variables into disjoint sets A, B, C such that B separates A and C in the association graph. After collapsing the table over the variables in C, the parameters relating variables within A, and those relating variables in A to variables in B, are unchanged.

Proof. [Supplement]

Step 1. No term contains variables from both A and C (that would create an A–C edge, a path avoiding B). Put all terms meeting A into f and all others into g; then each term of f lies in $A \cup B$ and each term of g lies in $B \cup C$:

$$\mu(\mathbf{a}, \mathbf{b}, \mathbf{c}) = f(\mathbf{a}, \mathbf{b})\, g(\mathbf{b}, \mathbf{c}), \qquad \log f = \sum_{T \cap A \neq \emptyset} \lambda^T.$$

Step 2. Collapsing over C: $\mu(\mathbf{a}, \mathbf{b}) = f(\mathbf{a}, \mathbf{b})\, G(\mathbf{b})$ with $G(\mathbf{b}) = \sum_{\mathbf{c}} g(\mathbf{b}, \mathbf{c})$. Thus

$$\log \mu(\mathbf{a}, \mathbf{b}) = \sum_{T \cap A \neq \emptyset} \lambda^T + \log G(\mathbf{b}).$$

Step 3. $\log G(\mathbf{b})$ decomposes into terms involving only variables of B. Since the decomposition of $\log \mu(\mathbf{a}, \mathbf{b})$ into main-effect and interaction terms is unique under fixed (e.g. baseline) constraints, every term involving a variable of A in the collapsed table equals the corresponding λ^T of the full model. ■

For (WX, WY, WZ, YZ): with $A = \{X\}$, $B = \{W\}$, $C = \{Y, Z\}$, the WX association is unchanged after collapsing over Y, Z; with $A = \{Y, Z\}$, $B = \{W\}$, $C = \{X\}$, the associations among W, Y, Z are unchanged after collapsing over X. A variable independent of all others can be collapsed over without affecting other terms. When B has several variables the parameters are unchanged but their ML estimates may differ slightly; estimates also agree ("commutativity of fitting and collapsing") if the model contains the highest-order term relating the variables of B to each other.

9.2 Model selection and comparison

Motivation. Even with five variables there are thousands of hierarchical loglinear models. We need principles (which terms must be included) and tools (likelihood-ratio comparisons of nested models) to find a model that fits well but is simple enough to interpret.

9.2.1 Considerations in model selection

A model should be complex enough to fit well yet simple to interpret, smoothing rather than overfitting. Guidelines:

(i) Distinguish responses from explanatory variables and concentrate on terms linking responses and linking explanatory variables to responses.

(ii) Include the most general interaction among the explanatory variables, and any margin fixed by the sampling design; then the likelihood equations reproduce these totals. For Table 8.8 with responses I, S and predictors G, L, the model should contain GL (so the fit has 20,629 urban women as observed), i.e. be at least as complex as (GL, S, I).

(iii) With a single categorical response, use the equivalent logit model.

(iv) For exploration, fit the model with all single-factor terms, then all two-factor terms, then all three-factor terms, and so on, to locate the needed complexity; automatic searches (backward elimination) should be used with care.

9.2.2 Model building for the Dayton student survey

Example 9.5 (Five-way student survey)

The survey of Example 8.8 also recorded gender (G) and race (R):

		Race = white				Race = other			
		Female		Male		Female		Male	
A	C	M=yes	no	yes	no	yes	no	yes	no
yes	yes	405	268	453	228	23	23	30	19
yes	no	13	218	28	201	2	19	1	18
no	yes	1	17	1	17	0	1	1	8
no	no	1	117	1	133	0	12	0	17

Treating A, C, M as responses and G, R as explanatory, every model contains GR.

Model	G^2	df
1. Mutual independence $+ GR$	1325.1	25
2. Homogeneous association (all two-factor terms)	15.3	16
3. All three-factor terms	5.3	6
4g. (2) $- CR$	15.8	17
5. $(AC, AM, CM, AG, AR, GM, GR, MR)$	16.7	18
6. $(AC, AM, CM, AG, AR, GM, GR)$	19.9	19
7. (AC, AM, CM, AG, AR, GR)	28.8	20

Model 3 improves on model 2 only by $15.3 - 5.3 = 10.0$ on 10 df, so three-factor terms are dropped. Backward elimination from model 2 (removing each of the 9 candidate pairs, models 4a–4i, e.g. removing AC gives 201.2 and removing CM gives 513.5) first removes CR (increase 0.5, df 1), then CG (+0.9), then MR (+3.2); removing AG next would add 5.3 ($P = 0.02$), so we stop at model 6. (These P-values are not literal, since the data chose the tests; many counts are small, but G^2 differences remain useful.)

In model 6 (Figure 11, middle) every path from C to $\{G, R\}$ goes through $\{A, M\}$: given alcohol and marijuana use, cigarette use is independent of gender and race. By Theorem 9.4, the AC and CM conditional associations equal those of (AC, AM, CM) fitted to the ACM table collapsed over G, R. Model 7 drops GM; then A separates $\{G, R\}$ from $\{C, M\}$, so all associations among A, C, M equal those of (AC, AM, CM) in the collapsed table. Model 7 fits reasonably ($G^2 = 28.8$, df $= 20$, $\hat{\Delta} = 0.036$), so one may collapse over gender and race to study A, C, M; the full model additionally estimates the effects of G and R on the responses.

9.2.3 Loglinear model comparison statistics

Let M_0 be a special case of M_1, with fitted values $\hat{\boldsymbol{\mu}}_0, \hat{\boldsymbol{\mu}}_1$.

Theorem 9.6 (Partitioning G^2)

$$G^2(M_0) = G^2(M_1) + G^2(M_0 \mid M_1), \tag{9.1}$$

where the likelihood-ratio statistic for M_0 against M_1 has the form

$$G^2(M_0 \mid M_1) = G^2(M_0) - G^2(M_1) = 2\sum_i \hat{\mu}_{1i} \log \frac{\hat{\mu}_{1i}}{\hat{\mu}_{0i}}, \tag{9.2}$$

which is the G^2 statistic for M_0 with $\hat{\boldsymbol{\mu}}_1$ playing the role of the data.

Proof.

Step 1. $G^2(M) = 2\sum n_i \log(n_i/\hat{\mu}_i)$, so

$$G^2(M_0) - G^2(M_1) = 2\sum_i n_i(\log \hat{\mu}_{1i} - \log \hat{\mu}_{0i}).$$

Step 2. Write $\log \hat{\boldsymbol{\mu}}_1 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ and, since $M_0 \subseteq M_1$, $\log \hat{\boldsymbol{\mu}}_0 = \mathbf{X}_0\hat{\boldsymbol{\beta}}_0 = \mathbf{X}_1\hat{\boldsymbol{\beta}}_0^*$, where $\hat{\boldsymbol{\beta}}_0^*$ is $\hat{\boldsymbol{\beta}}_0$ with zeros appended for the extra parameters. Then

$$2\mathbf{n}'(\log \hat{\boldsymbol{\mu}}_1 - \log \hat{\boldsymbol{\mu}}_0) = 2\,\mathbf{n}'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*) = 2\,\hat{\boldsymbol{\mu}}_1'\mathbf{X}_1(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_0^*),$$

using the likelihood equations $\mathbf{X}_1'\mathbf{n} = \mathbf{X}_1'\hat{\boldsymbol{\mu}}_1$ for M_1 (Theorem 8.18). This is $2\sum \hat{\mu}_{1i}\log(\hat{\mu}_{1i}/\hat{\mu}_{0i})$.

Step 3. $G^2(M_0 \mid M_1)$ has the G^2 form. (Note also $\sum \hat{\mu}_{0i} = \sum \hat{\mu}_{1i} = n$ when both models contain an intercept.) Equation (9.1) is the definition rearranged. ■

$G^2(M_0 \mid M_1)$ depends on the data only through $\hat{\boldsymbol{\mu}}_1$, i.e. through the sufficient statistics of M_1. The Pearson difference $X^2(M_0) - X^2(M_1)$ need not be nonnegative; the natural Pearson analogue is

$$X^2(M_0 \mid M_1) = \sum_i \frac{(\hat{\mu}_{1i} - \hat{\mu}_{0i})^2}{\hat{\mu}_{0i}}. \tag{9.3}$$

When M_0 holds, both are asymptotically χ^2 with $\mathrm{df} = \mathrm{df}(M_0) - \mathrm{df}(M_1)$, and they behave similarly even in fairly sparse tables (Haberman). When M_1 holds but M_0 does not, $G^2(M_1)$ remains χ^2 while the others grow without bound.

9.2.4 Partitioning chi-squared with model comparisons

Applying (9.1) repeatedly to a nested sequence $M_1 \supset M_2 \supset \cdots$ partitions G^2 into components that are asymptotically independent. Example: for a $2 \times J$ table let M_j state that columns $1, \ldots, j$ form an independence table. Then

$$G^2(M_J) = G^2(M_2) + G^2(M_3 \mid M_2) + \cdots + G^2(M_J \mid M_{J-1}),$$

and substituting the fitted values shows that $G^2(M_j \mid M_{j-1})$ is the G^2 for independence in the 2×2 table whose first column combines columns $1, \ldots, j-1$ and whose second column is column j. This justifies the partitioning of Section 3.3.3.

With s planned tests for nested models, using level $1-(1-\alpha)^{1/s}$ for each keeps the overall asymptotic type I error at most α (the tests being asymptotically independent). For $s = 3$ and $\alpha = 0.05$, use $1 - 0.95^{1/3} = 0.017$ per test.

9.2.5 Identical marginal and conditional tests of independence

When both models are direct, they contain the independence linkages that guarantee collapsibility, and the conditional test reduces to a marginal one (Sundberg, 1975): if two direct models differ

only by one pairwise term, $G^2(M_0 \mid M_1)$ equals G^2 for independence in the marginal table of that pair.

Proposition 9.7 (A conditional test that equals a marginal test)

$G^2[(X,Y,Z) \mid (XY,Z)]$ equals $G^2(X,Y)$ for testing independence in the marginal XY table.

Proof. By Theorem 8.20, $\hat{\mu}_{1,ijk} = n_{ij+}n_{++k}/n$ and $\hat{\mu}_{0,ijk} = n_{i++}n_{+j+}n_{++k}/n^2$, so $\hat{\mu}_1/\hat{\mu}_0 = n\,n_{ij+}/(n_{i++}n_{+j+})$ does not depend on k. By (9.2),

$$G^2 = 2\sum_{i,j,k} \frac{n_{ij+}n_{++k}}{n} \log \frac{n\,n_{ij+}}{n_{i++}n_{+j+}} = 2\sum_{i,j} n_{ij+} \log \frac{n_{ij+}}{n_{i++}n_{+j+}/n},$$

after summing n_{++k}/n over k. This is G^2 for independence in the $\{n_{ij+}\}$ table. ■

This is expected: under (XY,Z) the XY association is collapsible over Z.

9.3 Diagnostics for checking models

Motivation. G^2 tells us whether a model fits; residuals tell us where it fails, cell by cell.

9.3.1 Residuals for loglinear models

Definition (Pearson and standardized residuals)

For cell i the *Pearson residual* is

$$e_i = \frac{n_i - \hat{\mu}_i}{\sqrt{\hat{\mu}_i}}, \qquad \sum_i e_i^2 = X^2. \tag{9.4}$$

The *standardized Pearson residual* is $r_i = e_i/\sqrt{1-\hat{h}_i}$, where $\hat{h}_i$ is the leverage (diagonal of the estimated hat matrix).

Pearson residuals have asymptotic variances below 1 (on average residual df/number of cells); r_i is asymptotically $N(0,1)$ and is preferable. Deviance residuals are an alternative.

9.3.2 Student survey example revisited

For the five-way survey with all two-factor terms, the only large standardized residual is 3.2 (fitted 3.1 for a count of 8); for model 6 it is 3.3 (fitted 2.9) in the same cell: nonwhite males who smoked cigarettes but used neither alcohol nor marijuana. Given the many cells and large n, neither model looks problematic.

9.3.3 Correspondence between loglinear and logit residuals

Proposition 9.8 (Logit versus loglinear Pearson residuals)

For equivalent logit and loglinear models with binary response, the logit-model Pearson residual for a success count equals the loglinear Pearson residual for that cell divided by $\sqrt{1-\hat{\pi}}$; hence it is larger in absolute value.

Proof. Let the cell have count y out of N trials in its binomial row. The fitted values agree: $\hat{\mu} = N\hat{\pi}$ (Theorem 8.16). The logit residual is $(y - N\hat{\pi})/\sqrt{N\hat{\pi}(1-\hat{\pi})}$ and the loglinear residual is $(y-\hat{\mu})/\sqrt{\hat{\mu}} = (y - N\hat{\pi})/\sqrt{N\hat{\pi}}$. Their ratio is $1/\sqrt{1-\hat{\pi}} > 1$. ■

After standardization by their own estimated standard errors, the two kinds of residual are identical, another reason to prefer standardized residuals.

9.4 Modeling ordinal associations

Motivation. Nominal loglinear models ignore category order: permuting rows gives the same fit. For two ordinal variables the saturated model is useless and independence often fails mostly in the corner cells; a single "trend" parameter captures this.

Example 9.9 (Premarital sex and teenage birth control)

1991 General Social Survey. Rows: premarital sex is always wrong (1), almost always wrong (2), wrong only sometimes (3), not wrong at all (4). Columns: teenage birth control should be available, strongly disagree (1) to strongly agree (4). Entries: count, (independence fit), [standardized residual], {L×L fit}.

	SD	D	A	SA
1	81 (42.4) [7.6] {80.9}	68 (51.2) [3.1] {67.6}	60 (86.4) [−4.1] {69.4}	38 (67.0) [−4.8] {29.1}
2	24 (16.0) [2.3] {20.8}	26 (19.3) [1.8] {23.1}	29 (32.5) [−0.8] {31.5}	14 (25.2) [−2.8] {17.6}
3	18 (30.1) [−2.7] {24.4}	41 (36.3) [1.0] {36.1}	74 (61.2) [2.2] {65.7}	42 (47.4) [−1.0] {48.8}
4	36 (70.6) [−6.1] {33.0}	57 (85.2) [−4.6] {65.1}	161 (143.8) [2.4] {157.4}	157 (111.4) [6.8] {155.5}

Independence fits poorly, $G^2(I) = 127.6$, df $= 9$; the corner residuals show a positive trend.

9.4.1 Linear-by-linear association in two-way tables

Definition (Linear-by-linear association (L×L) model)

With ordered row scores $u_1 \le \cdots \le u_I$ and column scores $v_1 \le \cdots \le v_J$,

$$\log \mu_{ij} = \lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j, \tag{9.5}$$

the special case $\lambda_{ij}^{XY} = \beta u_i v_j$ of the saturated model: one association parameter instead of $(I-1)(J-1)$. Independence is $\beta = 0$. With $u_i = i$, $v_j = j$ it is the *uniform association* model.

Theorem 9.10 (Odds ratios under L×L)

Under (9.5), for rows $a < c$ and columns $b < d$,

$$\log \frac{\mu_{ab}\mu_{cd}}{\mu_{ad}\mu_{cb}} = \beta(u_c - u_a)(v_d - v_b). \tag{9.6}$$

With unit-spaced scores all local odds ratios equal e^{β}. If $\beta > 0$, then Y tends to increase with X and μ_{ij} exceeds its independence value where X, Y are both high or both low.

Proof.

Step 1. By Theorem 8.3(b) the log odds ratio equals $\lambda_{ab}^{XY} + \lambda_{cd}^{XY} - \lambda_{ad}^{XY} - \lambda_{cb}^{XY}$ with $\lambda_{ij}^{XY} = \beta u_i v_j$:

$$\beta(u_a v_b + u_c v_d - u_a v_d - u_c v_b) = \beta(u_c - u_a)(v_d - v_b).$$

Step 2. For adjacent rows and columns with unit spacing, $(u_{i+1} - u_i)(v_{j+1} - v_j) = 1$.

Step 3. For $\beta > 0$ all odds ratios with $a < c$, $b < d$ exceed 1, which is positive (likelihood-ratio type) dependence. The deviation $\beta u_i v_j$ from independence is largest in absolute value at the corners. ■

The deviation $\beta u_i v_j$ is linear in the column scores within a row (slope βu_i) and linear in the row scores within a column, hence the name. Scores are often midpoints of an underlying continuous scale, or standardized (mean 0, variance 1 under the margins); then β is a log odds ratio for unit standard-deviation changes, and for an underlying bivariate normal with correlation ρ, $\beta \approx \rho/(1-\rho^2)$ ($\approx \rho$ for weak association).

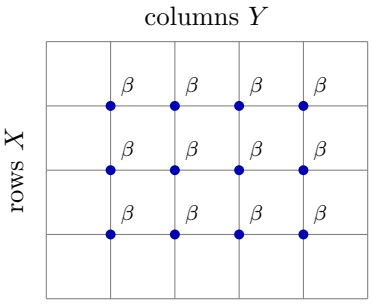


Figure 12: Uniform association: every 2×2 table of adjacent rows and columns has log odds ratio β (book Figure 9.2).

9.4.2 Corresponding logit model for adjacent responses

Proposition 9.11 (Adjacent-categories logit form)

Let $\pi_{j|i} = P(Y = j \mid X = i)$. Under (9.5),

$$\log \frac{\pi_{j+1|i}}{\pi_{j|i}} = (\lambda_{j+1}^{Y} - \lambda_{j}^{Y}) + \beta(v_{j+1} - v_j)u_i,$$

and for unit-spaced $\{v_j\}$ this is $\alpha_j + \beta u_i$: the same linear effect of X on all $J-1$ adjacent-categories logits.

Proof. $\pi_{j+1|i}/\pi_{j|i} = \mu_{i,j+1}/\mu_{ij}$; subtracting (9.5) at j from (9.5) at $j+1$, λ and λ_i^X cancel, leaving the stated expression. ■

9.4.3 Likelihood equations and model fitting

Theorem 9.12 (Likelihood equations for L×L)

The ML fit of (9.5) satisfies

$$\hat{\mu}_{i+} = n_{i+}, \qquad \hat{\mu}_{+j} = n_{+j}, \qquad \sum_{i,j} u_i v_j \hat{\mu}_{ij} = \sum_{i,j} u_i v_j n_{ij}.$$

Hence the fitted and observed distributions have the same marginal means and variances of the scores and the same correlation between them. The residual df are $IJ - I - J$.

Proof.

Step 1. The Poisson log likelihood is

$$L = n\lambda + \sum_i n_{i+}\lambda_i^X + \sum_j n_{+j}\lambda_j^Y + \beta \sum_{i,j} u_i v_j n_{ij} - \sum_{i,j} \exp(\lambda + \lambda_i^X + \lambda_j^Y + \beta u_i v_j).$$

Differentiating with respect to λ_i^X, λ_j^Y and β and equating to zero gives the three sets of equations (the sufficient statistics are the margins and $\sum u_i v_j n_{ij}$).

Step 2. Dividing by n, the fitted proportions $\hat{\pi}_{ij}$ and the sample proportions p_{ij} have equal margins, hence equal means and variances of u and of v, and equal $\sum u_i v_j \hat{\pi}_{ij} = \sum u_i v_j p_{ij}$, hence equal covariance and correlation.

Step 3. The model has $1 + (I-1) + (J-1) + 1$ parameters, so df $= IJ - (I + J)$, unsaturated except for 2×2 tables. ■

The fitted counts therefore reproduce the trend of the data. Fitting uses Newton–Raphson.

9.4.4 Sex opinions example

Example 9.13 (L×L fit to Example 9.9)

With scores $1, 2, 3, 4$ for both variables, $G^2(\text{L}\times\text{L}) = 11.5$, df $= 8$, a dramatic improvement over 127.6 on 9 df, especially in the corners. $\hat{\beta} = 0.286$ (SE 0.028): local odds ratio $e^{0.286} = 1.33$, 95% CI $\exp(0.286 \pm 1.96 \times 0.028) = (1.26, 1.41)$. Nonlocal odds ratios are larger: for the four corner cells, by (9.6),

$$\exp[0.286(4-1)(4-1)] = 13.1 = \frac{80.9 \times 155.5}{29.1 \times 33.0}.$$

Rescaled equally spaced scores (e.g. rows $2, 4, 6, 8$) give the same fit with $\hat{\beta}$ rescaled (0.143). Scores $1, 2, 4, 5$ (treating categories 2 and 3 as farther apart) give $G^2 = 8.8$, $\hat{\beta} = 0.146$. Standardized scores give $\hat{\beta} = 0.374$, and solving $\beta = \rho/(1-\rho^2)$ gives $\hat{\rho} = 0.333$ for an underlying bivariate normal.

9.4.5 Directed ordinal test of independence

Within L×L, independence is $H_0 : \beta = 0$, tested by

$$G^2(I \mid \text{L} \times \text{L}) = G^2(I) - G^2(\text{L} \times \text{L}), \qquad \text{df} = 1.$$

For Example 9.9: $127.6 - 11.5 = 116.1$; the Wald statistic $(\hat{\beta}/\text{SE})^2 = 102.5$; the score statistic is the correlation statistic $M^2 = (n-1)r^2 = 112.6$ of Section 3.4.1.

Why the ordinal test is more powerful

When L×L holds, $G^2(I \mid \mathrm{L} \times \mathrm{L})$ and $G^2(I)$ have (asymptotically) the same noncentrality, but the former has df $= 1$ instead of $(I-1)(J-1)$. For a fixed noncentrality, the power of a χ^2 test decreases as df increase, so the directed test is more powerful, and the advantage grows with I and J.

9.5 Poisson regression for rates

Motivation. When events are counted over different amounts of exposure (time, space, population), the rate, not the raw count, is the natural response. A loglinear model with an offset models the rate while keeping the Poisson count as the observation.

9.5.1 Analyzing rates using loglinear models with offsets

Proposition 9.14 (Rate model with offset)

For counts n_i with exposure (index) t_i, the model

$$\log(\mu_i/t_i) = \alpha + \beta x_i \tag{9.7}$$

is the Poisson loglinear model $\log \mu_i = \log t_i + \alpha + \beta x_i$, in which $\log t_i$ is an offset (a predictor whose coefficient is fixed at 1). Equivalently $\mu_i = t_i \exp(\alpha + \beta x_i)$: the mean is proportional to the exposure, and e^{β} is the multiplicative effect of a unit increase in x on the rate.

Proof. Add $\log t_i$ to both sides of (9.7) and exponentiate. The ratio of rates at $x+1$ and x is $e^{\alpha+\beta(x+1)}/e^{\alpha+\beta x} = e^{\beta}$. ■

The identity link $\mu_i/t_i = \alpha + \beta x_i$, i.e. $\mu_i = \alpha t_i + \beta x_i t_i$, needs no offset: it is a Poisson GLM with identity link, predictors t_i and $x_i t_i$ and no intercept. Effects are additive rather than multiplicative; fitting may fail if negative fitted values arise.

9.5.2 Modeling death rates for heart valve operations

Example 9.15 (Heart valve replacement)

109 patients, classified by valve type and age; time at risk is the total follow-up (months) of all patients in the cell, those who died and those censored.

Age		Aortic	Mitral
< 55	Deaths	4	1
	Time at risk	1259	2082
	Death rate	0.0032	0.0005
55+	Deaths	7	9
	Time at risk	1417	1647
	Death rate	0.0049	0.0055

A binary model for the $2 \times 2 \times 2$ table (died yes/no) would be wrong, since patients had different exposures (3 to 97 months). With dummies a (age 55+) and v (mitral),

$$\log(\mu_{ij}/t_{ij}) = \alpha + \beta_1 a_i + \beta_2 v_j. \tag{9.8}$$

Estimates: $\hat{\beta}_1 = 1.221$ (SE 0.514), $\hat{\beta}_2 = -0.330$ (SE 0.438). Given valve type, the older group's death rate is $e^{1.221} = 3.4$ times the younger group's; 95% Wald CI $\exp(1.221 \pm 1.96 \times 0.514) = (1.2, 9.3)$, likelihood-ratio CI $(1.3, 10.4)$. Fit: $G^2 = 3.2$, $X^2 = 3.1$, df $= 1$; the model without valve type has $G^2 = 3.8$ (df 2); models without age fit poorly. Fitted death rates: < 55: 0.0018 (aortic), 0.0013 (mitral); 55+: 0.0062, 0.0044. The identity-link model fits with $G^2 = 1.1$ (df 1); its $\hat{\beta}_1 = 0.0040$ (SE 0.0014) is an additive difference in death rates. Only 21 of 109 patients died, so estimates are imprecise, but every patient contributes through the times at risk.

9.6 Empty cells and sparseness in modeling contingency tables

Motivation. Multiway tables quickly become sparse: many cells, many zeros. Zeros can make ML estimates infinite or nonexistent, and they spoil the χ^2 approximation for G^2 and X^2. We need to know when estimates exist and which statistics remain trustworthy.

9.6.1 Empty cells: sampling versus structural zeros

Definition (Sampling and structural zeros)

A *sampling zero* is a cell with $n_i = 0$ but $\mu_i > 0$; with larger n it would eventually be positive. A *structural zero* is a cell in which observations are impossible, $\mu_i = 0$ (e.g. prostate cancer among women). Tables with structural zeros are *incomplete*.

Sampling zeros are data and enter the likelihood; structural zeros are not data and are excluded from the model. The rest of this section concerns sampling zeros.

9.6.2 Existence of estimates in loglinear/logit models

Lemma 9.16 (Existence criterion)

Let $\mathcal{V} = \mathrm{col}(\mathbf{X})$ ($\mathbf{X}$ of full rank). Finite ML estimates exist if and only if there is no nonzero $\mathbf{d} \in \mathcal{V}$ with $d_i \le 0$ for all i and $d_i = 0$ whenever $n_i > 0$.

Proof. [Supplement]

Step 1 (necessity). Suppose such $\mathbf{d}$ exists, and let $\boldsymbol{\eta} \in \mathcal{V}$. Since $n_i = 0$ where $d_i < 0$,

$$L(\boldsymbol{\eta} + s\mathbf{d}) = \sum_{i:d_i=0} (n_i\eta_i - e^{\eta_i}) - \sum_{i:d_i<0} e^{\eta_i + sd_i},$$

which is strictly increasing in s (at least one $d_i < 0$). So no $\boldsymbol{\eta}$ is a maximizer.

Step 2 (sufficiency). Suppose no such $\mathbf{d}$ exists; we show $L(\boldsymbol{\eta}) \to -\infty$ as $\|\boldsymbol{\eta}\| \to \infty$ in $\mathcal{V}$. Otherwise there are $\boldsymbol{\eta}_m \in \mathcal{V}$ with $\|\boldsymbol{\eta}_m\| \to \infty$ and $L(\boldsymbol{\eta}_m) \ge c$. Pass to a subsequence with $\boldsymbol{\eta}_m/\|\boldsymbol{\eta}_m\| \to \mathbf{d} \in \mathcal{V}$, $\|\mathbf{d}\| = 1$. Each term $n_i\eta_i - e^{\eta_i}$ is bounded above (by $n_i \log n_i - n_i$, or 0 if $n_i = 0$), so each term is bounded below along the sequence.

Step 3. If $d_i > 0$, then $\eta_{m,i} \to +\infty$ and $n_i\eta_{m,i} - e^{\eta_{m,i}} \to -\infty$, impossible; so $d_i \le 0$. If $n_i > 0$ and $d_i < 0$, then $\eta_{m,i} \to -\infty$ and $n_i\eta_{m,i} \to -\infty$, impossible; so $d_i = 0$ whenever $n_i > 0$. Then $\mathbf{d}$ is a forbidden direction, a contradiction.

Step 4. Hence the superlevel sets of L on $\mathcal{V}$ are bounded and closed, L attains its maximum, and since $\mathbf{X}$ has full rank, $\hat{\boldsymbol{\beta}}$ is finite. ∎

Theorem 9.17 (Haberman's results)

For Poisson sampling:
(1) The log likelihood is a strictly concave function of $\log \boldsymbol{\mu}$.
(2) If an ML estimate of $\boldsymbol{\mu}$ exists, it is unique and satisfies $\mathbf{X}'\mathbf{n} = \mathbf{X}'\hat{\boldsymbol{\mu}}$; conversely, a $\boldsymbol{\mu}$ that satisfies the model and these equations is the ML estimate.
(3) If all $n_i > 0$, ML estimates of the parameters exist.
(4) If ML estimates exist for a model whose likelihood equations match certain marginal tables, those marginal tables have all counts positive.
(5) If ML estimates exist for a model M, they exist for every special case of M.
(6) The ML estimates $\hat{\boldsymbol{\mu}}$ are identical for multinomial and independent Poisson sampling and exist in the same situations.

Proof. [Supplement]

(1) and (2) are Theorem 8.19 (steps 1–3 of its proof).

(3) If all $n_i > 0$, a $\mathbf{d}$ with $d_i = 0$ whenever $n_i > 0$ is $\mathbf{0}$, so Lemma 9.16 gives existence.

(4) If the MLE exists, $\hat{\mu}_i = e^{\hat{\eta}_i} > 0$ for every cell, so all fitted marginal counts are positive, and by (2) they equal the observed marginal counts.

(5) A special case M_0 has $\mathcal{V}_0 \subseteq \mathcal{V}$. A forbidden direction in $\mathcal{V}_0$ would be a forbidden direction in $\mathcal{V}$; since none exists for M, none exists for M_0, and Lemma 9.16 applies.

(6) By Theorem 8.21, $L = [\text{multinomial log likelihood in } \boldsymbol{\beta}] + [n \log \tau - \tau]$, and the second bracket is maximized at the finite value $\tau = n$ for any data. So $\boldsymbol{\beta}$ has a finite maximizer under one scheme iff under the other, the maximizers coincide, and $\hat{\boldsymbol{\mu}} = n\boldsymbol{\pi}(\hat{\boldsymbol{\beta}})$ in both. ■

For the saturated model, $\hat{\boldsymbol{\mu}} = \mathbf{n}$ and the parameters (contrasts of $\log \hat{\mu}_i$) are finite only if all $n_i > 0$. For unsaturated models, estimates exist if all $n_i > 0$ and do not exist if a sufficient marginal count is 0. If some $n_i = 0$ but all sufficient margins are positive, Glonek et al. showed that estimates of a hierarchical model then always exist iff the model is decomposable (direct). For non-decomposable models like (XY, XZ, YZ) estimates exist when only one cell is empty but may fail with two:

Proposition 9.18 (A nonexistence example)

In a $2 \times 2 \times 2$ table with $n_{111} = n_{222} = 0$ and all other counts positive (all two-way margins positive), the ML estimates for (XY, XZ, YZ) do not exist.

Proof. [Supplement]

Step 1. For $2 \times 2 \times 2$ tables, $\mathcal{V}$ for (XY, XZ, YZ) has dimension 7, and the three-factor contrast $c_{ijk} = (-1)^{i+j+k}$ is orthogonal to it: for any $a_{ij} + b_{ik} + e_{jk}$, summing $(-1)^{i+j+k} a_{ij}$ over k gives 0, and similarly for the other terms. Hence $\mathcal{V} = \{\mathbf{d} : \mathbf{c}'\mathbf{d} = 0\}$.

Step 2. Let $d_{111} = d_{222} = -1$ and $d_{ijk} = 0$ otherwise. Then $\mathbf{c}'\mathbf{d} = -(-1)^3 - (-1)^6 = 1 - 1 = 0$, so $\mathbf{d} \in \mathcal{V}$; $\mathbf{d} \leq \mathbf{0}$, $\mathbf{d} \neq \mathbf{0}$, and $d_{ijk} = 0$ wherever $n_{ijk} > 0$.

Step 3. By Lemma 9.16, no finite ML estimate exists. ■

Extended ML estimates and infinite parameters

The supremum of the likelihood is always finite, so *extended* ML estimates of $\boldsymbol{\mu}$ exist, possibly with some $\hat{\mu}_i = 0$ on the boundary (e.g. $\hat{\mu}_i = 0$ when $n_i = 0$ in the saturated model). Then some parameter estimates are $\pm\infty$ and some odds ratios 0 or ∞. A zero XY marginal total gives infinite $\hat{\lambda}_{ij}^{XY}$ in (XY, XZ, YZ), and infinite effects of X in logit models for Y. Software is usually fooled by the nearly flat likelihood: it reports "convergence" with huge estimates

and enormous, unstable standard errors. Infinite estimates are not fatal: the likelihood-ratio confidence interval has one finite endpoint, e.g. $(-\infty, U)$ for $\log\theta$ when $n_{11} = 0$ in a 2×2 table. Zero patterns that force fitted zeros also change the df for testing fit.

9.6.3 Clinical trials example

Example 9.19 (Multicenter trial with zero successes)

Active drug versus placebo for fungal infections at five centers (Y: success/failure, X: treatment, Z: center).

Center	Treatment	Success	Failure	YZ margin	
1	Drug / Placebo	0 / 0	5 / 9	0	14
2	Drug / Placebo	1 / 0	12 / 10	1	22
3	Drug / Placebo	0 / 0	7 / 5	0	12
4	Drug / Placebo	6 / 2	3 / 6	8	9
5	Drug / Placebo	5 / 2	9 / 12	7	21
XY margin	Drug / Placebo	12 / 4	36 / 42		

Centers 1 and 3 have no successes, so the YZ margin has zeros, and for $\text{logit}\, P(Y = 1) = \beta x + \beta_k^Z$ (no intercept), $\hat\beta_1^Z = \hat\beta_3^Z = -\infty$. Software reports about -26 with SE about 200,000. The XY margin is positive and the treatment estimate is finite: $\hat\beta = 1.55$ (SE 0.70), the same as after deleting centers 1 and 3. Combining centers 1–3 gives $\hat\beta = 1.56$ (SE 0.70).

Proposition 9.20 (Uninformative strata)

In the logit model $\text{logit}\,\pi_{ik} = \beta x_i + \beta_k^Z$ with free center parameters, a center in which all responses are failures (or all successes) contributes nothing to the profile likelihood of β; hence $\hat\beta$ (and its likelihood-based inference) equals that obtained by deleting the center.

Proof. [Supplement] Center k with only failures contributes $\prod_i (1-\pi_{ik})^{n_{ik}}$, which for every fixed β increases to 1 as $\beta_k^Z \to -\infty$. Since β_k^Z appears in no other factor, the profile likelihood $\sup_{\{\beta_k^Z\}} \ell(\beta, \{\beta_k^Z\})$ equals the product over the remaining centers of their own profiled contributions, times 1. This is the profile likelihood with center k deleted, so it has the same maximizer and the same curvature. The all-success case is symmetric ($\beta_k^Z \to +\infty$). ■

Such strata also contribute nothing to the CMH test and the exact conditional test, although they do carry information about measures such as the difference of proportions.

9.6.4 Effect of small samples on X^2 and G^2

For fixed N cells, X^2 and G^2 are asymptotically χ^2 as $n \to \infty$; the quality depends on n and N. Cochran's rule for independence tests with df > 1: a minimum expected count of about 1 is acceptable if no more than about 20% of the $\hat\mu_i$ are below 5. X^2 is usable for smaller n and sparser tables than G^2; G^2 is usually poorly approximated when $n/N < 5$. When most $\hat\mu_i < 0.5$, G^2 is highly conservative (reported P-values too large); when most are between 0.5 and 4, it tends to be too liberal. For fixed n, N, tests with smaller df are better approximated: for conditional independence, $G^2[(XZ, YZ) \mid (XY, XZ, YZ)]$ (df $(I-1)(J-1)$) beats $G^2(XZ, YZ)$ (df $K(I-1)(J-1)$), and the ordinal df $= 1$ test of Section 9.4 is better still.

9.6.5 Model-based tests and sparseness

By (9.2)–(9.3), $G^2(M_0 \mid M_1)$ and $X^2(M_0 \mid M_1)$ depend on the data only through the sufficient statistics of M_1, usually marginal tables. Marginal totals are closer to normal than single cells, so these statistics approach χ^2 faster than $G^2(M_0)$ or $X^2(M_0)$. If the sufficient marginal totals of M_1 are mostly at least 5–10, the χ^2 approximation for model comparisons is usually adequate even when cell counts are small.

9.6.6 Alternative asymptotics and alternative statistics

When asymptotics fail, use exact small-sample methods, or Monte Carlo approximations of exact distributions. Alternatively, *sparse asymptotics* let N grow with n (cell means need not grow); then suitably standardized G^2 and X^2 are approximately normal (Koehler and Larntz; Koehler; McCullagh; Zelterman).

9.6.7 Adding constants to cells of a contingency table

Adding a small constant (e.g. $\frac{1}{2}$) to each cell guarantees finite estimates and convergence, and for saturated models it reduces bias in estimating an odds ratio. But for unsaturated models it oversmooths, distorting estimates and test statistics, especially with many cells. Even for a saturated model it replaces an honest infinite estimate (with interval (L, ∞)) by an arbitrary finite one. When in doubt, do a sensitivity analysis: delete or move possibly influential observations and see whether conclusions change. Random effects and Bayesian methods (later chapters) smooth in a less ad hoc way.

Chapter 10. Models for Matched Pairs

Matched pairs arise when each observation in one sample is paired with an observation in the other: the same subject measured twice, a case matched with a control, two raters judging the same slide. The two responses of a pair are dependent, and their cross-classification is a *square* $I \times I$ table with the same categories for rows and columns. This chapter develops the analysis of such tables: comparing the two margins (marginal models), modelling subject-specific effects (conditional models), loglinear models of symmetry type, agreement between raters, and paired preference (Bradley–Terry) models.

10.1 Comparing dependent proportions

Motivation. When the same people answer the same yes/no question twice, the two sample proportions are correlated, so the two-independent-sample formulas are wrong. We need the correct variance of a difference of dependent proportions, and the corresponding test of equal margins, which turns out to use only the discordant pairs.

For each of n matched pairs let π_{ab} be the probability of outcome a for the first observation and b for the second, n_{ab} the number of such pairs and $p_{ab} = n_{ab}/n$. We treat $\{n_{ab}\}$ as multinomial$(n; \{\pi_{ab}\})$. Then p_{a+} is the sample proportion in category a for observation 1 and p_{+a} for observation 2.

Definition (Marginal homogeneity)

The square table satisfies **marginal homogeneity** if $\pi_{a+} = \pi_{+a}$ for all a, i.e. the two observations have the same marginal distribution.

For a 2×2 table, $\pi_{1+} - \pi_{+1} = (\pi_{11} + \pi_{12}) - (\pi_{11} + \pi_{21}) = \pi_{12} - \pi_{21}$, so marginal homogeneity is equivalent to $\pi_{12} = \pi_{21}$: the table is symmetric about the main diagonal.

10.1.1 Inference for dependent proportions

Let $\delta = \pi_{+1} - \pi_{1+}$ and $d = p_{+1} - p_{1+} = p_{21} - p_{12}$.

Theorem 10.1 (Variance of a difference of dependent proportions)

Under multinomial sampling,

$$\begin{aligned} n \operatorname{Var}(d) &= \pi_{1+}(1-\pi_{1+}) + \pi_{+1}(1-\pi_{+1}) - 2(\pi_{11}\pi_{22} - \pi_{12}\pi_{21}) \\ &= (\pi_{12} + \pi_{21}) - (\pi_{12} - \pi_{21})^2. \end{aligned}$$

Hence $d \pm z_{\alpha/2}\,\hat\sigma(d)$ with $\hat\sigma^2(d) = \big[(p_{12} + p_{21}) - (p_{12} - p_{21})^2\big]/n$ is a large-sample confidence interval for δ.

Proof.

Step 1. Multinomial moments: $n \operatorname{Var}(p_{ab}) = \pi_{ab}(1-\pi_{ab})$ and $n \operatorname{Cov}(p_{ab}, p_{cd}) = -\pi_{ab}\pi_{cd}$ for distinct cells.

Step 2. First form. $n \operatorname{Var}(p_{1+}) = \pi_{1+}(1-\pi_{1+})$ and $n \operatorname{Var}(p_{+1}) = \pi_{+1}(1-\pi_{+1})$ because each margin is binomial. For the covariance,

$$\begin{aligned} n \operatorname{Cov}(p_{+1}, p_{1+}) &= n \operatorname{Cov}(p_{11} + p_{21},\, p_{11} + p_{12}) \\ &= \pi_{11}(1-\pi_{11}) - \pi_{11}\pi_{12} - \pi_{21}\pi_{11} - \pi_{21}\pi_{12} \end{aligned}$$

$$= \pi_{11}(1 - \pi_{11} - \pi_{12} - \pi_{21}) - \pi_{12}\pi_{21} = \pi_{11}\pi_{22} - \pi_{12}\pi_{21}.$$

Then $\mathrm{Var}(d) = \mathrm{Var}(p_{+1}) + \mathrm{Var}(p_{1+}) - 2\,\mathrm{Cov}(p_{+1}, p_{1+})$ gives the first line.

Step 3. Second form. Directly, $d = p_{21} - p_{12}$, so

$$n\,\mathrm{Var}(d) = \pi_{21}(1 - \pi_{21}) + \pi_{12}(1 - \pi_{12}) + 2\pi_{12}\pi_{21} = (\pi_{12} + \pi_{21}) - (\pi_{12} - \pi_{21})^2.$$

Step 4. By the multivariate CLT $\sqrt{n}(d - \delta)$ is asymptotically normal, and $\hat{\sigma}(d)$ is consistent, so the interval follows from Slutsky's theorem. ■

The hypothesis of marginal homogeneity is $H_0 : \pi_{1+} = \pi_{+1}$ ($\delta = 0$). The ratio $z = d/\hat{\sigma}(d)$ is a Wald statistic. Under H_0 the variance simplifies to $(\pi_{12} + \pi_{21})/n$, estimated by

$$\hat{\sigma}_0^2(d) = \frac{p_{12} + p_{21}}{n} = \frac{n_{12} + n_{21}}{n^2}.$$

Theorem 10.2 (McNemar's test is the score test)

The score statistic for $H_0 : \pi_{12} = \pi_{21}$ in the multinomial model is z_0^2, where

$$z_0 = \frac{d}{\hat{\sigma}_0(d)} = \frac{n_{21} - n_{12}}{(n_{21} + n_{12})^{1/2}}; \tag{10.1}$$

z_0^2 is asymptotically χ_1^2 under H_0 (**McNemar's test**).

Proof.

Step 1. Reparametrize the four cell probabilities by $(\pi_{11}, \pi_{22}, \sigma, q)$ with $\sigma = \pi_{12} + \pi_{21}$, $q = \pi_{21}/\sigma$, so $\pi_{21} = \sigma q$, $\pi_{12} = \sigma(1 - q)$, subject only to $\pi_{11} + \pi_{22} + \sigma = 1$. Marginal homogeneity is $q = \frac{1}{2}$.

Step 2. With $n^* = n_{12} + n_{21}$ the log-likelihood is

$$L = n_{11}\log\pi_{11} + n_{22}\log\pi_{22} + n^*\log\sigma + n_{21}\log q + n_{12}\log(1 - q).$$

The parameter q does not enter the constraint and separates from $(\pi_{11}, \pi_{22}, \sigma)$, so the information matrix is block diagonal and the score test for q involves only the q-block.

Step 3. The score is $U(q) = n_{21}/q - n_{12}/(1 - q)$ and, since $E(n_{21}) = n\sigma q$, $E(n_{12}) = n\sigma(1 - q)$, the information is $\mathcal{I}(q) = n\sigma/[q(1 - q)]$. At $q = \frac{1}{2}$ with σ replaced by its null ML estimate $\hat{\sigma} = n^*/n$:

$$U = 2(n_{21} - n_{12}), \qquad \mathcal{I} = 4n^*, \qquad \frac{U^2}{\mathcal{I}} = \frac{(n_{21} - n_{12})^2}{n_{21} + n_{12}} = z_0^2.$$

Step 4. Also $d/\hat{\sigma}_0(d) = [(n_{21} - n_{12})/n]/[(n_{12} + n_{21})^{1/2}/n]$, which is (10.1). The χ_1^2 limit is the usual score-test asymptotics. ■

Only discordant pairs matter for the test

McNemar's statistic depends only on the $n_{12} + n_{21}$ discordant pairs; the diagonal counts are irrelevant to *whether* the margins differ. All pairs do contribute, through n, to estimating *how much* they differ (d and its standard error).

10.1.2 Prime Minister approval rating example

Example 10.3 (Prime Minister approval)

A random sample of 1600 British voters rated the Prime Minister's performance twice, six months apart:

	Second survey		
First survey	Approve	Disapprove	Total
Approve	794	150	944
Disapprove	86	570	656
Total	880	720	1600

The sample odds ratio $794 \cdot 570/(150 \cdot 86) = 35.1$ shows strong association between the two surveys. The approval proportions are $p_{1+} = 0.59$ and $p_{+1} = 0.55$. By Theorem 10.1, $\hat{\sigma}(d) = 0.0095$ and a 95% interval for $\pi_{+1} - \pi_{1+}$ is $-0.04 \pm 1.96(0.0095)$, i.e. $(-0.06, -0.02)$. McNemar's statistic is

$$z_0 = \frac{86 - 150}{(86 + 150)^{1/2}} = -4.17 \qquad (z_0^2 = 17.4),$$

strong evidence that approval dropped.

10.1.3 Increased precision with dependent samples

For independent samples of size n each, $n\,\mathrm{Var}(\hat{\pi}_1 - \hat{\pi}_2) = \pi_1(1-\pi_1) + \pi_2(1-\pi_2)$, i.e. the first form in Theorem 10.1 without the covariance term. Matched data usually show *positive* dependence, $\log\theta = \log[\pi_{11}\pi_{22}/(\pi_{12}\pi_{21})] > 0$, and then the covariance term reduces $\mathrm{Var}(d)$. In Example 10.3, $\hat{\sigma}(d) = 0.0095$, while independent samples of size 1600 with the same proportions would give $\left[(0.59 \cdot 0.41 + 0.55 \cdot 0.45)/1600\right]^{1/2} = 0.0175$, almost twice as large. Within-subject designs thus estimate within-subject effects more precisely.

10.1.4 Small-sample test comparing matched proportions

Proposition 10.4 (Exact conditional test)

Conditional on $n^* = n_{12} + n_{21}$, $n_{21} \sim \mathrm{bin}(n^*, \pi_{21}/(\pi_{12} + \pi_{21}))$. Under $H_0 : \pi_{12} = \pi_{21}$ this is $\mathrm{bin}(n^*, \frac{1}{2})$, and the normal approximation to it gives exactly McNemar's z_0.

Proof.

Step 1. Collapse the multinomial to the three categories $\{(1,2), (2,1), \text{diagonal}\}$. Conditioning a multinomial on the total of a subset of cells gives a multinomial on that subset with renormalized probabilities; here the subset has two cells, so $n_{21} \mid n^*$ is binomial with success probability $\pi_{21}/(\pi_{12}+\pi_{21})$, which equals $\frac{1}{2}$ under H_0.

Step 2. The approximating normal has mean $n^*/2$ and variance $n^*/4$, so

$$z = \frac{n_{21} - \frac{1}{2}n^*}{(\frac{1}{4}n^*)^{1/2}} = \frac{n_{21} - n_{12}}{(n_{21} + n_{12})^{1/2}} = z_0.$$

■

For Example 10.3, $n^* = 236$ and the one-sided P-value $P(\mathrm{bin}(236, \frac{1}{2}) \geq 150) = 0.00002$; the two-sided P-value doubles this.

10.1.5 Connection between McNemar and Cochran–Mantel–Haenszel tests

Display each pair as its own 2×2 stratum: rows are the two observations (first, second), columns the two outcomes. Each stratum has row totals $(1, 1)$. For Example 10.3 there are 1600 strata of four types (794 approve–approve, 570 disapprove–disapprove, 86 disapprove–approve, 150 approve–disapprove). We call this $2 \times 2 \times n$ table the **subject-specific table**; the ordinary 2×2 table of pairs is the **population-averaged table**, since its margins estimate population marginal proportions. If, for each subject, the probability of success is the same at both observations, then occasion and response are conditionally independent given subject, which implies marginal homogeneity in the population-averaged table.

Theorem 10.5 (McNemar as a CMH statistic)

The Cochran–Mantel–Haenszel statistic for conditional independence in the $2 \times 2 \times n$ subject-specific table equals $(n_{21} - n_{12})^2/(n_{21} + n_{12})$.

Proof.

Step 1. In stratum k let $c_k \in \{0, 1, 2\}$ be the number of successes (column-1 total). The CMH statistic is $\left[\sum_k (n_{11k} - En_{11k})\right]^2 / \sum_k \mathrm{Var}(n_{11k})$ with hypergeometric moments

$$E(n_{11k}) = \frac{1 \cdot c_k}{2}, \qquad \mathrm{Var}(n_{11k}) = \frac{1 \cdot 1 \cdot c_k(2 - c_k)}{2^2(2 - 1)} = \frac{c_k(2 - c_k)}{4}.$$

Step 2. Concordant strata ($c_k = 0$ or 2) have $\mathrm{Var}(n_{11k}) = 0$ and $n_{11k} = E(n_{11k})$, so they contribute nothing.

Step 3. Discordant strata have $c_k = 1$, variance $\frac{1}{4}$, and $n_{11k} - \frac{1}{2} = +\frac{1}{2}$ for a (success, failure) pair (there are n_{12} of these) and $-\frac{1}{2}$ for a (failure, success) pair (n_{21} of these). Hence

$$\mathrm{CMH} = \frac{\left[\frac{1}{2}(n_{12} - n_{21})\right]^2}{\frac{1}{4}(n_{12} + n_{21})} = \frac{(n_{21} - n_{12})^2}{n_{21} + n_{12}}.$$

■

The connection suggests how to treat more complex matched data: with several categories or several observations per subject, the generalized CMH tests with one stratum per subject test marginal homogeneity (Section 11.1.4 of Chapter 11).

10.2 Conditional logistic regression for binary matched pairs

Motivation. The tests of Section 10.1 correspond to models. Allowing each subject its own success probability gives a logit model with one nuisance parameter per subject; conditioning on their sufficient statistics removes them and yields the simple estimator n_{21}/n_{12} of a subject-specific odds ratio.

10.2.1 Marginal versus conditional models for matched pairs

Let (Y_1, Y_2) be the pair for a randomly selected subject, 1 = success, 0 = failure, and $x_1 = 0$, $x_2 = 1$.

Definition (Marginal and conditional models)

A **marginal** (population-averaged) model describes the marginal distributions, e.g.

$$P(Y_t = 1) = \alpha + \delta x_t, \qquad \text{logit}\, P(Y_t = 1) = \alpha + \beta x_t.$$

A **conditional** (subject-specific) model lets the probabilities depend on the subject i:

$$\text{link}\, P(Y_{it} = 1) = \alpha_i + \beta x_t, \qquad i = 1, \ldots, n.$$

In the marginal logit model, the ML estimate is the log odds ratio of the marginal proportions, $\hat{\beta} = \log[p_{+1}p_{2+}/(p_{+2}p_{1+})]$. In the conditional model β describes the conditional association in the subject-stratified table.

Proposition 10.6 (Subject-specific versus population-averaged effects)

Let α_i be drawn from a distribution F in the conditional model.
(a) With the identity link, $P(Y_2 = 1) - P(Y_1 = 1) = \beta$; the subject-specific and population-averaged effects coincide.
(b) With the logit link, the population-averaged probabilities are $P(Y_t = 1) = E_F[e^{\alpha+\beta x_t}/(1+e^{\alpha+\beta x_t})]$, which in general is not of the form $e^{a+bx_t}/(1+e^{a+bx_t})$ with $b = \beta$.

Proof.

(a) Averaging $P(Y_{i2} = 1) - P(Y_{i1} = 1) = (\alpha_i + \beta) - \alpha_i = \beta$ over F gives β.

(b) The first statement is the law of total probability. For the second, take $\alpha = \pm c$ with probability $\frac{1}{2}$ each and $\beta > 0$. The marginal log odds ratio is $b = \text{logit}\, E[\text{expit}(\alpha + \beta)] - \text{logit}\, E[\text{expit}(\alpha)]$, and $E[\text{expit}(\alpha)] = \frac{1}{2}$ by symmetry, so $b = \text{logit}\,(\frac{1}{2}[\text{expit}(c + \beta) + \text{expit}(\beta - c)])$. As $c \to \infty$ the bracket tends to $\frac{1}{2}(1 + 0)$, so $b \to 0$ while β is fixed. Hence $b \neq \beta$ in general. ■

10.2.2 A logit model with subject-specific probabilities

The model of Cox and Rasch is

$$\text{logit}\, P(Y_{it} = 1) = \alpha_i + \beta x_t, \qquad x_1 = 0,\ x_2 = 1, \tag{10.2}$$

so for subject i, $P(Y_{i1} = 1) = e^{\alpha_i}/(1 + e^{\alpha_i})$ and $P(Y_{i2} = 1) = e^{\alpha_i+\beta}/(1 + e^{\alpha_i+\beta})$; for every subject the odds of success at observation 2 are e^{β} times those at observation 1. Given the parameters, responses are assumed independent, both across subjects and within a pair. Nevertheless the model induces dependence when averaged over subjects.

Proposition 10.7 (Heterogeneity induces positive association)

Suppose $\alpha_1, \ldots, \alpha_n$ are i.i.d. from F and, given α_i, Y_{i1}, Y_{i2} are independent with the probabilities of (10.2). Then $\text{Cov}(Y_{i1}, Y_{i2}) \geq 0$ for every β, with equality if and only if F is degenerate.

Proof. [Supplement]

Step 1. By conditional independence, $E(Y_{i1}Y_{i2} \mid \alpha_i) = f(\alpha_i)g(\alpha_i)$ with $f(a) = \text{expit}(a)$, $g(a) = \text{expit}(a + \beta)$, so $\text{Cov}(Y_{i1}, Y_{i2}) = \text{Cov}\,(f(\alpha), g(\alpha))$, $\alpha \sim F$.

Step 2. Let α' be an independent copy of α. Since f and g are both strictly increasing, $(f(\alpha) - f(\alpha'))(g(\alpha) - g(\alpha')) \geq 0$, and

$$0 \leq E\big[(f(\alpha) - f(\alpha'))(g(\alpha) - g(\alpha'))\big] = 2\big(E[fg] - E[f]E[g]\big) = 2\,\text{Cov}(f(\alpha), g(\alpha)).$$

Step 3. Equality forces the nonnegative product to vanish a.s., i.e. $\alpha = \alpha'$ a.s. (strict monotonicity), which holds for independent copies only if F is degenerate. ■

The more variable the $\{\alpha_i\}$, the stronger the positive association: subjects with large α_i tend to succeed both times, subjects with small α_i to fail both times. So the model accounts for the dependence in matched pairs through its structure. The large number of $\{\alpha_i\}$, however, ruins ordinary ML (Proposition 10.9 below). The remedy is *conditional ML*, which conditions on sufficient statistics for the nuisance parameters. (Terminology: "conditional model" refers to subject-specific effects; "conditional ML" refers to the inference method.)

10.2.3 Conditional ML inference for binary matched pairs

Theorem 10.8 (Conditional ML estimator of the subject-specific odds ratio)

In model (10.2) let $S_i = y_{i1} + y_{i2}$.
(a) S_i is sufficient for α_i. Given $S_i = 0$ or $S_i = 2$ the pair is determined; given $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^{\beta}}{1 + e^{\beta}}, \qquad P(Y_{i1} = 1, Y_{i2} = 0 \mid S_i = 1) = \frac{1}{1 + e^{\beta}}.$$

(b) The conditional likelihood is $\ell_c(\beta) = (e^{\beta})^{n_{21}}/(1 + e^{\beta})^{n^*}$ with $n^* = n_{12} + n_{21}$, i.e. $n_{21} \mid n^* \sim \text{bin}(n^*, e^{\beta}/(1 + e^{\beta}))$.
(c) The conditional ML estimator and its standard error are

$$\hat{\beta} = \log \frac{n_{21}}{n_{12}}, \qquad \text{SE} = \Big(\frac{1}{n_{21}} + \frac{1}{n_{12}}\Big)^{1/2}.$$

Proof.

Step 1. Sufficiency. The joint mass function is

$$\prod_{i=1}^{n} \frac{e^{\alpha_i y_{i1}}}{1 + e^{\alpha_i}} \cdot \frac{e^{(\alpha_i + \beta) y_{i2}}}{1 + e^{\alpha_i + \beta}} \;\propto\; \exp\Big(\sum_i \alpha_i S_i + \beta \sum_i y_{i2}\Big),$$

an exponential family in which S_i is sufficient for α_i; so the conditional distribution given $\{S_i\}$ is free of $\{\alpha_i\}$.

Step 2. Conditional probabilities. Given $S_i = 0$ ($S_i = 2$) the only possible pair is $(0, 0)$ ($(1, 1)$), which carries no information about β. Given $S_i = 1$, write $c_i = (1 + e^{\alpha_i})(1 + e^{\alpha_i + \beta})$. Then

$$P(0, 1) = \frac{e^{\alpha_i + \beta}}{c_i}, \quad P(1, 0) = \frac{e^{\alpha_i}}{c_i}, \quad P(0, 1 \mid S_i = 1) = \frac{e^{\alpha_i + \beta}}{e^{\alpha_i} + e^{\alpha_i + \beta}} = \frac{e^{\beta}}{1 + e^{\beta}}.$$

Step 3. Conditional likelihood. Multiplying over the n^* discordant pairs, n_{21} of which are $(0, 1)$ and n_{12} are $(1, 0)$, gives $\ell_c(\beta) = e^{\beta n_{21}}/(1 + e^{\beta})^{n^*}$; as a sum of n^* independent Bernoulli variables, n_{21} is binomial.

Step 4. Maximization. $\log \ell_c = n_{21}\beta - n^* \log(1 + e^{\beta})$ has derivative $n_{21} - n^* e^{\beta}/(1 + e^{\beta})$, which vanishes when $e^{\beta}/(1 + e^{\beta}) = n_{21}/n^*$, i.e. $e^{\hat{\beta}} = n_{21}/n_{12}$; the log-likelihood is strictly concave, so this is the maximum.

Step 5. Standard error. The observed information is $n^* e^{\beta}/(1 + e^{\beta})^2$; at $\hat{\beta}$ it equals $n^*(n_{21}/n^*)(n_{12}/n^*) = n_{21}n_{12}/n^*$, whose inverse is $n^*/(n_{21}n_{12}) = 1/n_{21} + 1/n_{12}$. ■

Testing $\beta = 0$ conditionally is the binomial test with parameter $\frac{1}{2}$ of Proposition 10.4. When model (10.2) is realistic, this justifies comparing the margins using only the discordant pairs.

Proposition 10.9 (Inconsistency of unconditional ML)

Treating $\alpha_1, \ldots, \alpha_n$ as fixed parameters, the unconditional ML estimator of β in (10.2) is $2\log(n_{21}/n_{12})$, which converges to 2β.

Proof. [Supplement]

Step 1. For a concordant pair the likelihood is maximized by $\alpha_i \to \pm\infty$, with maximal value 1 whatever β is; these pairs drop out of the profile likelihood.

Step 2. For a $(1,0)$ pair the likelihood is $p_1(1-p_2)$ with $p_1 = \text{expit}(\alpha_i)$, $p_2 = \text{expit}(\alpha_i + \beta)$. Its α_i-derivative of the log is $(1-p_1) - p_2$, which is zero when $p_1 + p_2 = 1$, i.e. $\alpha_i = -\beta/2$. The profiled value is $\text{expit}(-\beta/2)^2$. Similarly a $(0,1)$ pair profiles to $\text{expit}(\beta/2)^2$.

Step 3. The profile log-likelihood $2n_{21}\log\text{expit}(\beta/2) + 2n_{12}\log\text{expit}(-\beta/2)$ has the form of Step 4 of Theorem 10.8 in the parameter $\beta/2$, so $\hat\beta/2 = \log(n_{21}/n_{12})$.

Step 4. By the law of large numbers $n_{21}/n^* \to e^\beta/(1+e^\beta)$, so $\log(n_{21}/n_{12}) \to \beta$ and $\hat\beta \to 2\beta$. ∎

10.2.4 Random effects in binary matched-pairs model

Alternatively, treat $\{\alpha_i\}$ as a random sample from $N(\mu, \sigma^2)$ and integrate them out. The resulting marginal likelihood has only the three parameters (β, μ, σ); this is a *generalized linear mixed model* (Chapter 12). When the sample log odds ratio is nonnegative, it also yields $\hat\beta = \log(n_{21}/n_{12})$ (Neuhaus et al. 1994).

Proposition 10.10 (Mantel–Haenszel estimator)

Applied to the n subject-specific 2×2 tables, the Mantel–Haenszel estimator of the common odds ratio comparing observation 2 with observation 1 equals n_{21}/n_{12}.

Proof. [Supplement] Order the rows of stratum k as (observation 2, observation 1) and the columns as (success, failure), so $n_{11k} = y_{k2}$, $n_{12k} = 1 - y_{k2}$, $n_{21k} = y_{k1}$, $n_{22k} = 1 - y_{k1}$, and $n_{++k} = 2$. Then

$$\hat\theta_{MH} = \frac{\sum_k n_{11k}n_{22k}/n_{++k}}{\sum_k n_{12k}n_{21k}/n_{++k}} = \frac{\sum_k y_{k2}(1-y_{k1})}{\sum_k (1-y_{k2})y_{k1}} = \frac{n_{21}}{n_{12}}.$$

∎

Thus the MH estimate, the conditional ML estimate and (with nonnegative log odds ratio) the random-effects ML estimate all give $e^{\hat\beta} = n_{21}/n_{12}$.

	$Y_2 = 1$	$Y_2 = 0$
$Y_1 = 1$	n_{11}	n_{12}
$Y_1 = 0$	n_{21}	n_{22}

shaded diagonal: $S_i \in \{0, 2\}$, no information on β
orange cells: $S_i = 1$, $n_{21} \mid n^* \sim \text{bin}\big(n^*, e^\beta/(1+e^\beta)\big)$

Figure 10.1. Only the discordant cells carry conditional information about the subject-specific effect β.

10.2.5 Logistic regression for matched case–control studies

The two members of a pair need not be the same subject. In a matched case–control study each case $(Y = 1)$ is matched with a control $(Y = 0)$ on characteristics that could affect the response, and both are measured on an explanatory variable X. For subject t in pair i consider

$$\text{logit}\, P(Y_{it} = 1) = \alpha_i + \beta x_{it}. \tag{10.3}$$

Although the design fixes Y and observes X, e^{β} is an XY odds ratio, which is the same for both conditional distributions, so it is estimable from retrospective data.

Example 10.11 (Diabetes and myocardial infarction)

A study of acute myocardial infarction (MI) among Navajo Indians matched 144 MI cases with 144 controls on age and gender, and recorded previous diagnosis of diabetes ($x = 1$ yes, 0 no):

	MI cases		
MI controls	Diabetes	No diabetes	Total
Diabetes	9	16	25
No diabetes	37	82	119
Total	46	98	144

In 37 pairs the case had diabetes and the control did not; in 16 pairs the reverse. The conditional ML estimate is $e^{\hat{\beta}} = 37/16 = 2.3$, so $\hat{\beta} = 0.84$.

10.2.6 Conditional ML for matched pairs with multiple predictors

With p predictors, $x_{it} = (x_{1it}, \ldots, x_{pit})'$, the model is

$$\operatorname{logit} P(Y_{it} = 1) = \alpha_i + \beta_1 x_{1it} + \cdots + \beta_p x_{pit} = \alpha_i + x_{it}'\boldsymbol{\beta}. \tag{10.4}$$

Theorem 10.12 (Conditional likelihood as no-intercept logistic regression)

In model (10.4), for pairs with $S_i = 1$,

$$P(Y_{i1} = 0, Y_{i2} = 1 \mid S_i = 1) = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}} = \frac{e^{x_i^{*\prime}\boldsymbol{\beta}}}{1 + e^{x_i^{*\prime}\boldsymbol{\beta}}}, \qquad x_i^* = x_{i2} - x_{i1}.$$

Hence:
(a) the conditional ML estimate of $\boldsymbol{\beta}$ is obtained by fitting an ordinary logistic regression, with no intercept, to the discordant pairs only, with response $y_i^* = 1$ for $(0,1)$ pairs, $y_i^* = 0$ for $(1,0)$ pairs and predictor x_i^*;
(b) a predictor with $x_{ji1} = x_{ji2}$ for all i (a between-pair effect) cancels from the conditional likelihood, so its coefficient cannot be estimated by conditional ML.

Proof.

Step 1. Exactly as in Step 2 of Theorem 10.8, with $c_i = (1 + e^{\alpha_i + x_{i1}'\boldsymbol{\beta}})(1 + e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})$,

$$P(0,1 \mid S_i = 1) = \frac{e^{\alpha_i + x_{i2}'\boldsymbol{\beta}}/c_i}{(e^{\alpha_i + x_{i1}'\boldsymbol{\beta}} + e^{\alpha_i + x_{i2}'\boldsymbol{\beta}})/c_i} = \frac{e^{x_{i2}'\boldsymbol{\beta}}}{e^{x_{i1}'\boldsymbol{\beta}} + e^{x_{i2}'\boldsymbol{\beta}}};$$

dividing numerator and denominator by $e^{x_{i1}'\boldsymbol{\beta}}$ gives the logistic form in x_i^*. $P(1,0 \mid S_i = 1)$ is the complement.

Step 2. The conditional likelihood is $\prod_{S_i=1} \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})^{y_i^*} [1 - \operatorname{expit}(x_i^{*\prime}\boldsymbol{\beta})]^{1-y_i^*}$, which is the likelihood of a no-intercept logistic regression of y^* on x^*. This proves (a).

Step 3. If $x_{ji1} = x_{ji2}$ for all i, then the jth component of every x_i^* is 0, so β_j does not appear in the conditional likelihood. This proves (b). ∎

For Example 10.11, take $t = 1$ the control and $t = 2$ the case, so $y_i^* = 1$ always, and $x_i^* = -1$ (16 pairs), 0 (91 pairs), +1 (37 pairs). Pairs with $x^* = 0$ contribute a constant, and the no-intercept fit maximizes $37 \log \text{expit}(\beta) + 16 \log \text{expit}(-\beta)$, giving $\hat{\beta} = \log(37/16) = 0.84$, as before.

10.2.7 Marginal models and conditional models: extensions

Both kinds of model extend to multinomial responses and to matched *sets*: in (10.4), i may index a subject with repeated measurements or a cluster (children in a family, fetuses in a litter). By Theorem 10.12(b), conditional ML estimates only *within-cluster* effects, such as those in case–control and crossover studies; information about a between-cluster effect is carried by subject totals, which are fixed by the conditioning. The random-effects approach (Chapter 12) has no such restriction. The rest of this chapter emphasizes marginal models and loglinear models for square tables; Chapter 11 extends marginal models to matched sets with covariates.

10.3 Marginal models for square contingency tables

Motivation. With $I > 2$ categories, comparing the two margins of an $I \times I$ table needs models that act directly on the marginal distributions: cumulative logits for ordinal responses and baseline-category logits (or Wald/score tests) for nominal ones.

For a square table $\{n_{ab}\}$ of outcomes (Y_1, Y_2) with $\pi_{ab} = P(Y_1 = a, Y_2 = b)$, marginal homogeneity is $P(Y_1 = a) = P(Y_2 = a)$, $a = 1, \ldots, I$.

10.3.1 Marginal models for ordinal classifications

With cumulative logits,

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.5}$$

with $x_1 = 0$, $x_2 = 1$. This is a proportional odds structure: the odds of $Y_2 \le j$ are e^β times the odds of $Y_1 \le j$, for every j. The margins are stochastically ordered, $\beta > 0$ meaning that Y_1 tends to be higher than Y_2; marginal homogeneity is $\beta = 0$.

ML fitting must respect the dependence of (Y_1, Y_2): the likelihood is multinomial in the joint $\{\pi_{ab}\}$, but the model constrains only the margins π_{a+}, π_{+b}, so one cannot simply substitute the model into $\sum\sum n_{ab} \log \pi_{ab}$ (specialized constrained-ML algorithms are needed). Model (10.5) describes $2(I-1)$ marginal probabilities with I parameters, so its goodness-of-fit test has df $= I - 2$.

10.3.2 Premarital and extramarital sex example

Example 10.13 (Premarital and extramarital sex)

In the 1989 General Social Survey, 475 subjects rated premarital sex and extramarital sex on the scale 1 = always wrong, 2 = almost always wrong, 3 = wrong only sometimes, 4 = not wrong at all:

	Extramarital sex				
Premarital sex	1	2	3	4	Total
1	144	2	0	0	146
2	33	4	2	0	39
3	84	14	6	1	105
4	126	29	25	5	185
Total	387	49	33	6	475

The cumulative marginal proportions are $(0.307, 0.389, 0.611)$ for premarital and $(0.815, 0.918, 0.987)$ for extramarital sex; with scores $1, 2, 3, 4$ the means are 2.69 and 1.28. Model (10.5) gives $\hat{\beta} = 2.51$ (SE $= 0.13$): responses are much more permissive about premarital sex. Marginal homogeneity fits terribly ($G^2 = 348.1$, df $= 3$); model (10.5) has $G^2 = 35.1$ (df $= 2$), much better but still poor. Section 10.4.7 finds better models.

10.3.3 Marginal models for nominal classifications

For nominal responses, the baseline-category logit model

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \alpha_j + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.6}$$

has $2(I-1)$ parameters for $2(I-1)$ marginal probabilities, so it is saturated; marginal homogeneity is $\beta_1 = \cdots = \beta_{I-1} = 0$. Maximizing the multinomial likelihood under this constraint (iteratively) and comparing fitted with observed counts by G^2 or X^2 tests marginal homogeneity with df $= I-1$.

A non-iterative alternative uses the asymptotic normality of marginal differences. Let $d_a = p_{+a} - p_{a+}$ and $\mathbf{d} = (d_1, \ldots, d_{I-1})'$ (d_I is redundant since $\sum_a d_a = 0$).

Proposition 10.14 (Bhapkar and Stuart statistics)

(a) The covariance matrix $\mathbf{V}$ of $\sqrt{n}\,\mathbf{d}$ has elements

$$v_{ab} = -(\pi_{ab} + \pi_{ba}) - (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+}), \qquad a \neq b,$$
$$v_{aa} = \pi_{+a} + \pi_{a+} - 2\pi_{aa} - (\pi_{+a} - \pi_{a+})^2.$$

(b) Under marginal homogeneity the Wald (Bhapkar) statistic $W = n\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d}$ is asymptotically χ^2_{I-1}. Stuart's statistic $W_0 = n\mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d}$ uses the null version $\hat{v}_{ab0} = -(p_{ab} + p_{ba})$, $\hat{v}_{aa0} = p_{+a} + p_{a+} - 2p_{aa}$, and

$$W = \frac{W_0}{1 - W_0/n}.$$

(c) For $I = 2$, W_0 is McNemar's statistic z_0^2.

Proof. [Supplement]

Step 1. Write $\mathbf{d}$ as the mean of i.i.d. vectors $\mathbf{X}_k$, one per pair, with $X_{ka} = \mathbb{1}\{Y_{k2} = a\} - \mathbb{1}\{Y_{k1} = a\}$. For $a \neq b$, the products $\mathbb{1}\{Y_2 = a\}\mathbb{1}\{Y_2 = b\}$ and $\mathbb{1}\{Y_1 = a\}\mathbb{1}\{Y_1 = b\}$ vanish, so

$$E(X_a X_b) = -P(Y_1 = b, Y_2 = a) - P(Y_1 = a, Y_2 = b) = -(\pi_{ba} + \pi_{ab}).$$

For $a = b$, $E(X_a^2) = P(Y_2 = a) + P(Y_1 = a) - 2P(Y_1 = Y_2 = a) = \pi_{+a} + \pi_{a+} - 2\pi_{aa}$. Subtracting $E(X_a)E(X_b) = (\pi_{+a} - \pi_{a+})(\pi_{+b} - \pi_{b+})$ gives (a).

Step 2. By the CLT, $\sqrt{n}(\mathbf{d} - E\mathbf{d}) \xrightarrow{d} N(\mathbf{0}, \mathbf{V})$, $\hat{\mathbf{V}} \xrightarrow{p} \mathbf{V}$, and under H_0, $E\mathbf{d} = \mathbf{0}$; so $W \xrightarrow{d} \chi^2_{I-1}$ (a quadratic form in an asymptotically $N(\mathbf{0}, \mathbf{V})$ vector with the inverse of a consistent estimate of $\mathbf{V}$).

Step 3. Comparing the sample versions, $\hat{\mathbf{V}} = \hat{\mathbf{V}}_0 - \mathbf{d}\mathbf{d}'$. By the Sherman–Morrison formula, with $q = \mathbf{d}'\hat{\mathbf{V}}_0^{-1}\mathbf{d} = W_0/n$,

$$\hat{\mathbf{V}}^{-1} = \hat{\mathbf{V}}_0^{-1} + \frac{\hat{\mathbf{V}}_0^{-1}\mathbf{d}\mathbf{d}'\hat{\mathbf{V}}_0^{-1}}{1-q}, \qquad \mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} = q + \frac{q^2}{1-q} = \frac{q}{1-q}.$$

Multiplying by n gives $W = W_0/(1 - W_0/n)$.

Step 4. For $I = 2$, $d_1 = p_{21} - p_{12}$ and $\hat{v}_{110} = p_{+1} + p_{1+} - 2p_{11} = p_{12} + p_{21}$, so $W_0 = n(p_{21} - p_{12})^2/(p_{12} + p_{21}) = (n_{21} - n_{12})^2/(n_{12} + n_{21})$. ■

These tests use all $I-1$ degrees of freedom. With ordered categories, large I and strong dependence, a single-df ordinal test is often much more powerful.

10.3.4 Migration example

Example 10.15 (Migration 1980–1985)

Region of residence in 1980 and 1985 for a sample of 55,981 U.S. residents (NE = Northeast, MW = Midwest, S = South, W = West):

	Residence in 1985				
Residence in 1980	NE	MW	S	W	Total
NE	11,607	100	366	124	12,197
MW	87	13,677	515	302	14,581
S	172	225	17,819	270	18,486
W	63	176	286	10,192	10,717
Total	11,929	14,178	18,986	10,888	55,981

(The South–Midwest count is 225, the value consistent with the printed margins and fitted values.) About 95% of observations lie on the diagonal. The ML fit of marginal homogeneity has $G^2 = 240.8$ (df = 3), and Bhapkar's $W = 236.5$. The marginal proportions were $(0.218, 0.260, 0.330, 0.191)$ in 1980 and $(0.213, 0.253, 0.339, 0.194)$ in 1985. Applying Theorem 10.1 to the 2×2 table that collapses the other regions, a 95% interval for $\pi_{+1} - \pi_{1+}$ is $0.2131 - 0.2179 \pm 1.96(0.00054)$, i.e. -0.005 ± 0.001; for the other regions the intervals are -0.007 ± 0.001, 0.009 ± 0.001, 0.003 ± 0.001. The changes are statistically clear only because n is huge; they are small.

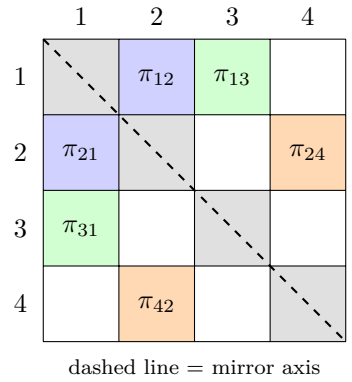
10.4 Symmetry, quasi-symmetry, and quasi-independence

Motivation. Instead of the margins one can model the joint distribution of the square table by loglinear models. Symmetry is too strict, but quasi-symmetry fits well, has a subject-specific interpretation, and yields a likelihood-ratio test of marginal homogeneity; quasi-independence describes the off-diagonal ("movers") structure.

Definition (Symmetry)

An $I \times I$ distribution satisfies **symmetry** if $\pi_{ab} = \pi_{ba}$ for all $a \neq b$.

Symmetry implies marginal homogeneity, since $\pi_{a+} = \sum_b \pi_{ab} = \sum_b \pi_{ba} = \pi_{+a}$. For $I = 2$ the two are equivalent; for $I > 2$ marginal homogeneity can hold without symmetry.



grey: diagonal, fitted exactly by S, QS, QI

mirror cells $(a,b),(b,a)$: equal under S;
under QS, $\pi_{ab}/\pi_{ba} = e^{\tau_b - \tau_a}$

Figure 10.2. The $I(I-1)/2$ mirror pairs of a square table ($I=4$).

10.4.1 Symmetry as logit and loglinear models

When all $\pi_{ab} > 0$, symmetry is the logit model $\log(\pi_{ab}/\pi_{ba}) = 0$ for $a < b$, and, for expected frequencies $\mu_{ab} = n\pi_{ab}$, the loglinear model

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.7}$$

in which both classifications share the main effects; equivalently $\log \mu_{ab} = \lambda_{ab}$ with $\lambda_{ab} = \lambda_{ba}$.

Theorem 10.16 (Fitting and testing symmetry)

For Poisson or multinomial counts:
(a) the ML fit of symmetry is $\hat{\mu}_{ab} = (n_{ab} + n_{ba})/2$ (so $\hat{\mu}_{aa} = n_{aa}$);
(b) the residual df is $I(I-1)/2$;
(c) the Pearson statistic is Bowker's statistic

$$X^2 = \sum_{a<b} \frac{(n_{ab} - n_{ba})^2}{n_{ab} + n_{ba}},$$

which for $I = 2$ is McNemar's z_0^2;
(d) the standardized Pearson residuals are $r_{ab} = (n_{ab} - n_{ba})/(n_{ab} + n_{ba})^{1/2} = -r_{ba}$, and $\sum_{a<b} r_{ab}^2 = X^2$.

Proof.

Step 1. (a) Under $\log \mu_{ab} = \lambda_{ab}$ with $\lambda_{ab} = \lambda_{ba}$, the Poisson log-likelihood $\sum_{a,b}(n_{ab} \log \mu_{ab} - \mu_{ab})$ separates into terms $(n_{ab} + n_{ba}) \log \mu - 2\mu$ for each pair $a < b$ (with common value $\mu = \mu_{ab} = \mu_{ba}$) and $n_{aa} \log \mu_{aa} - \mu_{aa}$ for each diagonal cell. Maximizing each gives $\hat{\mu} = (n_{ab} + n_{ba})/2$ and $\hat{\mu}_{aa} = n_{aa}$. The multinomial fit is the same (the fitted total equals n).

Step 2. (b) There are I^2 cells and $I + \binom{I}{2}$ free parameters $\{\lambda_{ab}, a \le b\}$, so df $= I^2 - I - I(I-1)/2 = I(I-1)/2$. Equivalently, symmetry fixes the parameter of each of the $\binom{I}{2}$ binomials $n_{ab} \mid n_{ab} + n_{ba}$ at $\frac{1}{2}$.

Step 3. (c) Diagonal cells contribute 0. With $m = (n_{ab} + n_{ba})/2$, $(n_{ab} - m)^2 = (n_{ba} - m)^2 = (n_{ab} - n_{ba})^2/4$, so the pair contributes $2 \cdot \frac{(n_{ab}-n_{ba})^2}{4m} = \frac{(n_{ab}-n_{ba})^2}{n_{ab}+n_{ba}}$.

Step 4. (d) The Pearson residual of cell (a,b) is $(n_{ab} - m)/\sqrt{m}$. Each pair is fitted by one parameter with design column $(1,1)'$ and equal weights, so the hat matrix for the pair is $\left(\begin{smallmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{smallmatrix}\right)$ and the leverage is $h = \frac{1}{2}$. Dividing by $\sqrt{1-h}$,

$$r_{ab} = \frac{(n_{ab} - n_{ba})/2}{\sqrt{m}\sqrt{1/2}} = \frac{n_{ab} - n_{ba}}{(n_{ab} + n_{ba})^{1/2}},$$

and summing r_{ab}^2 over $a < b$ reproduces (c). ■

Symmetry rarely fits (exceptions: intra-observer agreement); with substantial marginal heterogeneity it fits poorly.

10.4.2 Quasi-symmetry

Definition (Quasi-symmetry)

The **quasi-symmetry** (QS) model lets the main effects in (10.7) differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}. \tag{10.8}$$

Symmetry is the case $\lambda_a^X = \lambda_a^Y$ for all a; independence is the case $\lambda_{ab} = 0$.

Since $\lambda_a^X + \lambda_b^Y = (\lambda_a^X + \lambda_b^X) + (\lambda_b^Y - \lambda_b^X)$ and the first bracket is symmetric, QS can be rewritten as

$$\log \mu_{ab} = \lambda'_{ab} + \tau_b, \qquad \lambda'_{ab} = \lambda'_{ba}, \quad \tau_b = \lambda_b^Y - \lambda_b^X, \quad \tau_I = 0. \tag{10.9}$$

Proposition 10.17 (Likelihood equations for quasi-symmetry)

The ML fit of QS satisfies

$$\hat\mu_{a+} = n_{a+}, \qquad \hat\mu_{+b} = n_{+b}, \qquad \hat\mu_{ab} + \hat\mu_{ba} = n_{ab} + n_{ba} \quad (a \le b);$$

one of the first two sets is implied by the other two. In particular $\hat\mu_{aa} = n_{aa}$, and the residual df is $(I-1)(I-2)/2$.

Proof.

Step 1. For a Poisson loglinear model the likelihood equations equate sufficient statistics to their expectations. In (10.8) the sufficient statistics are the row totals (λ^X), column totals (λ^Y) and the symmetric sums $n_{ab} + n_{ba}$, $a \le b$ (λ_{ab}; for $a = b$ the statistic is n_{aa}). Taking $a = b$ in the third set gives $\hat\mu_{aa} = n_{aa}$.

Step 2. Redundancy: given the third set and the row equations,

$$\hat\mu_{+a} = \sum_b \hat\mu_{ba} = \sum_b (n_{ab} + n_{ba} - \hat\mu_{ab}) = n_{a+} + n_{+a} - \hat\mu_{a+} = n_{+a}.$$

Step 3. df: in form (10.9) there are $I(I+1)/2$ parameters λ'_{ab} ($a \le b$) and $I-1$ parameters τ_b, so df $= I^2 - I(I+1)/2 - (I-1) = (I-1)(I-2)/2$. ■

The equations have no closed-form solution; Newton–Raphson or IPF is used.

QS has the multiplicative form

$$\pi_{ab} = \alpha_a \beta_b \gamma_{ab}, \qquad \gamma_{ab} = \gamma_{ba}, \tag{10.10}$$

with positive parameters: the cellwise product of an independence table and a symmetric table. Symmetry is the case $\alpha_a = \beta_a$.

Theorem 10.18 (Odds-ratio characterization of quasi-symmetry)

Let all $\pi_{ab} > 0$.

(a) QS holds if and only if

$$\frac{\pi_{ab}\pi_{II}}{\pi_{aI}\pi_{Ib}} = \frac{\pi_{ba}\pi_{II}}{\pi_{bI}\pi_{Ia}} \qquad \text{for all } a, b. \tag{10.11}$$

(b) Under QS the local odds ratios satisfy $\theta_{ab} = \theta_{ba}$, where $\theta_{ab} = \pi_{ab}\pi_{a+1,b+1}/(\pi_{a,b+1}\pi_{a+1,b})$.

Proof. [Supplement]

Step 1. ($\Rightarrow$ in (a)) Substituting (10.10), the α's and β's cancel in each side, leaving $\gamma_{ab}\gamma_{II}/(\gamma_{aI}\gamma_{Ib})$ on the left and $\gamma_{ba}\gamma_{II}/(\gamma_{bI}\gamma_{Ia})$ on the right, which are equal by symmetry of γ.

Step 2. ($\Leftarrow$ in (a)) Put $\alpha_a = \pi_{aI}$, $\beta_b = \pi_{Ib}/\pi_{II}$ and $\gamma_{ab} = \pi_{ab}/(\alpha_a\beta_b) = \pi_{ab}\pi_{II}/(\pi_{aI}\pi_{Ib})$. Then $\pi_{ab} = \alpha_a\beta_b\gamma_{ab}$ by construction, and (10.11) says exactly $\gamma_{ab} = \gamma_{ba}$.

Step 3. (b) Substituting (10.10) into θ_{ab}, the α's and β's cancel, so $\theta_{ab} = \gamma_{ab}\gamma_{a+1,b+1}/(\gamma_{a,b+1}\gamma_{a+1,b})$; interchanging the roles of a and b and using $\gamma_{cd} = \gamma_{dc}$ gives the same value. ■

Goodman called QS the *symmetric association* model: odds ratios on one side of the diagonal mirror those on the other side.

The meaning of QS is clarified by subject-specific models. Generalize (10.6) to subject i:

$$\log \frac{P(Y_{it} = j)}{P(Y_{it} = I)} = \alpha_{ij} + \beta_j x_t, \qquad t = 1, 2, \quad j = 1, \ldots, I-1, \tag{10.12}$$

with $x_1 = 0$, $x_2 = 1$, $\alpha_{iI} = \beta_I = 0$, and responses independent given the subject.

Theorem 10.19 (Subject-specific logit models imply quasi-symmetry)

If (10.12) holds and subjects are sampled from any distribution F of $(\alpha_{i1}, \ldots, \alpha_{i,I-1})$, then the population-averaged table satisfies QS with $\tau_j = \lambda_j^Y - \lambda_j^X = \beta_j$ (constraint $\tau_I = 0$).

Proof. [Supplement]

Step 1. For a subject with parameter vector $\boldsymbol{\alpha}$, let $D_1(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k}$ and $D_2(\boldsymbol{\alpha}) = \sum_k e^{\alpha_k + \beta_k}$. By conditional independence,

$$P(Y_1 = a, Y_2 = b \mid \boldsymbol{\alpha}) = \frac{e^{\alpha_a}}{D_1(\boldsymbol{\alpha})} \cdot \frac{e^{\alpha_b + \beta_b}}{D_2(\boldsymbol{\alpha})}.$$

Step 2. Averaging over F,

$$\pi_{ab} = e^{\beta_b}\,\gamma_{ab}, \qquad \gamma_{ab} = E_F\Big[\frac{e^{\alpha_a + \alpha_b}}{D_1(\boldsymbol{\alpha})D_2(\boldsymbol{\alpha})}\Big],$$

and $\gamma_{ab} = \gamma_{ba}$ because $\alpha_a + \alpha_b$ is symmetric in (a, b).

Step 3. This is (10.10) with $\alpha_a \equiv 1$ and $\beta_b = e^{\beta_b}$, i.e. form (10.9) with $\tau_b = \beta_b$. ■

Conaway (1989) showed further that the conditional ML estimates of $\{\beta_j\}$ (conditioning out the α_{ij}) equal $\hat\lambda_j^Y - \hat\lambda_j^X$ from the ordinary ML fit of QS. Thus the main-effect differences of QS are subject-specific effects. QS also contains quasi-independence (Section 10.4.3) and the Bradley–Terry model (Section 10.6.3).

10.4.3 Quasi-independence

Square tables usually have more mass on the diagonal than independence predicts, but off the diagonal the structure may be simple.

Definition (Quasi-independence)

A square table satisfies **quasi-independence** (QI) if the classifications are independent given that the row and column outcomes differ:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \delta_a\, I(a = b). \tag{10.13}$$

Each diagonal cell gets its own parameter, so the likelihood equations are $\hat{\mu}_{a+} = n_{a+}$, $\hat{\mu}_{+a} = n_{+a}$, $\hat{\mu}_{aa} = n_{aa}$: the diagonal is fitted perfectly and independence holds for the remaining cells. All odds ratios formed from four off-diagonal cells in a rectangle equal 1. The model has I more parameters than independence, so df $= (I-1)^2 - I$, and it applies for $I \geq 3$.

Proposition 10.20 (Quasi-independence within quasi-symmetry)

QI is the special case of QS in which the γ_{ab}, $a \neq b$, in (10.10) are all equal. For $I = 3$ the two models are equivalent.

Proof. [Supplement]

Step 1. Under QI, $\pi_{ab} = \alpha_a \beta_b$ for $a \neq b$ and $\pi_{aa} = \alpha_a \beta_a e^{\delta_a}$; this is (10.10) with $\gamma_{ab} = 1$ off the diagonal and $\gamma_{aa} = e^{\delta_a}$. Conversely, if the off-diagonal γ_{ab} equal a constant c, absorb c into α and the diagonal γ_{aa} into δ_a.

Step 2. Let $I = 3$ and $\pi_{ab} = \alpha_a \beta_b \gamma_{ab}$ with γ symmetric. We look for positive c_1, c_2, c_3 with $c_a c_b = \gamma_{ab}$ for $a \neq b$:

$$c_1 = \left(\frac{\gamma_{12}\gamma_{13}}{\gamma_{23}}\right)^{1/2}, \quad c_2 = \left(\frac{\gamma_{12}\gamma_{23}}{\gamma_{13}}\right)^{1/2}, \quad c_3 = \left(\frac{\gamma_{13}\gamma_{23}}{\gamma_{12}}\right)^{1/2}$$

satisfy $c_1 c_2 = \gamma_{12}$, $c_1 c_3 = \gamma_{13}$, $c_2 c_3 = \gamma_{23}$. Then $\pi_{ab} = (\alpha_a c_a)(\beta_b c_b)$ for $a \neq b$, which is QI with row and column parameters $\alpha_a c_a$, $\beta_b c_b$ and free diagonal. Hence QS $\subseteq$ QI when $I = 3$; with Step 1 the models coincide (both have df $= 1$). ■

10.4.4 Migration revisited

Example 10.21 (Migration: loglinear models)

For Example 10.15, independence fits terribly ($G^2 = 125{,}923$). Symmetry is also poor ($G^2 = 243.6$, df $= 6$): e.g. 124 people moved from NE to W but only 63 the reverse way. QI ($G^2 = 69.5$, df $= 5$) is much better, mainly because it fits the diagonal exactly, but it predicts too few NE→S and too many W→S moves. QS fits well: $G^2 = 3.0$, df $= 3$. Fitted values (diagonal fitted exactly by both):

	Quasi-independence				Quasi-symmetry			
1980	NE	MW	S	W	NE	MW	S	W
NE	–	126.6	312.9	150.5	–	95.8	370.4	123.8
MW	117.4	–	531.1	255.5	91.2	–	501.7	311.1
S	133.2	243.8	–	290.0	167.6	238.3	–	261.1
W	71.4	130.6	323.0	–	63.2	166.9	294.9	–

The subject-specific effects are $\hat{\tau}_1 = -0.672$, $\hat{\tau}_2 = -0.623$, $\hat{\tau}_3 = 0.122$ (W baseline). For a given person, the estimated odds of living in the South rather than the West in 1985 were $e^{0.122} = 1.13$ times the 1980 odds. Similar analyses apply to occupational mobility tables (parent's vs child's occupation).

10.4.5 Marginal homogeneity and quasi-symmetry

Marginal homogeneity is not a loglinear model, but QS gives a convenient route to testing it.

Theorem 10.22 (Caussinus)

For a table with all $\pi_{ab} > 0$,

$$\text{quasi-symmetry} + \text{marginal homogeneity} \iff \text{symmetry}.$$

Proof.

Step 1. ($\Leftarrow$) Symmetry is QS with $\alpha_a = \beta_a$, and implies marginal homogeneity.

Step 2. ($\Rightarrow$) Write QS as $\pi_{ab} = \alpha_a \beta_b \gamma_{ab}$ and set $\kappa_a = \alpha_a/\beta_a$, $\gamma^*_{ab} = \beta_a \beta_b \gamma_{ab}$, so that

$$\pi_{ab} = \kappa_a \gamma^*_{ab}, \qquad \gamma^*_{ab} = \gamma^*_{ba} > 0.$$

Step 3. Marginal homogeneity $\pi_{j+} = \pi_{+j}$ reads $\kappa_j \sum_b \gamma^*_{jb} = \sum_a \kappa_a \gamma^*_{aj}$, i.e.

$$\kappa_j = \sum_a \kappa_a \, w_{aj}, \qquad w_{aj} = \frac{\gamma^*_{aj}}{\sum_b \gamma^*_{bj}} > 0, \quad \sum_a w_{aj} = 1,$$

using $\gamma^*_{jb} = \gamma^*_{bj}$ in the denominator. So each κ_j is a weighted average, with strictly positive weights, of all of $\kappa_1, \ldots, \kappa_I$.

Step 4. Let j be an index with $\kappa_j = \min_a \kappa_a$. If some $\kappa_a > \kappa_j$, then the positive-weight average $\sum_a \kappa_a w_{aj}$ would exceed κ_j, a contradiction. Hence all κ_a equal a common κ.

Step 5. Then $\pi_{ab} = \kappa \gamma^*_{ab} = \kappa \gamma^*_{ba} = \pi_{ba}$: symmetry holds. ■

Corollary 10.23 (Conditional test of marginal homogeneity)

Given that QS holds, marginal homogeneity is equivalent to symmetry, i.e. to $\lambda^X_a = \lambda^Y_a$ (all $\tau_a = 0$). It is tested by

$$G^2(S \mid QS) = G^2(S) - G^2(QS), \qquad \text{df} = \tfrac{I(I-1)}{2} - \tfrac{(I-1)(I-2)}{2} = I - 1.$$

Proof. The equivalence is Theorem 10.22. In form (10.9), symmetry is the nested submodel $\tau_1 = \cdots = \tau_{I-1} = 0$, so the difference of deviances is the likelihood-ratio statistic, asymptotically χ^2 with df equal to the difference of the residual df of Theorem 10.16(b) and Proposition 10.17. ■

For the migration data, $G^2(S \mid QS) = 243.6 - 3.0 = 240.6$ (df $= 3$), close to the marginal-model statistics $G^2 = 240.8$ and $W = 236.5$ of Example 10.15.

10.4.6 Ordinal quasi-symmetry model

With ordered categories and scores $u_1 \le \cdots \le u_I$ for both classifications, the **ordinal quasi-symmetry** model is

$$\log \mu_{ab} = \lambda + \lambda_a + \lambda_b + \beta u_b + \lambda_{ab}, \qquad \lambda_{ab} = \lambda_{ba}, \tag{10.14}$$

the case of QS with linear trend $\tau_b = \lambda^Y_b - \lambda^X_b = \beta u_b$. Symmetry is $\beta = 0$. Dividing μ_{ab} by μ_{ba}, the symmetric terms cancel:

$$\log \frac{\pi_{ab}}{\pi_{ba}} = \beta(u_b - u_a), \qquad a < b. \tag{10.15}$$

This is a linear logit model with no intercept for the binomial $n_{ab} \mid (n_{ab} + n_{ba})$, with predictor $x = u_b - u_a$; it is fitted that way with logit software. The larger $|\beta|$, the larger the difference between π_{ab} and π_{ba}, hence between the margins.

Proposition 10.24 (Ordinal quasi-symmetry and marginal homogeneity)

Let $u_1 < \cdots < u_I$ and all $\pi_{ab} > 0$.
(a) The likelihood equations are $\sum_a u_a \hat{\mu}_{a+} = \sum_a u_a n_{a+}$, $\sum_b u_b \hat{\mu}_{+b} = \sum_b u_b n_{+b}$ and $\hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba}$ $(a \le b)$. The fitted margins need not equal the observed ones, but the fitted row and column means equal the observed means.
(b) Under (10.14), the column mean exceeds the row mean if $\beta > 0$ and is smaller if $\beta < 0$; hence ordinal QS + marginal homogeneity = symmetry, and $H_0 : \beta = 0$ tests marginal homogeneity (df = 1).

Proof. [Supplement]

Step 1. (a) The sufficient statistics of the loglinear model $\log \mu_{ab} = \lambda'_{ab} + \beta u_b$ (symmetric λ') are the symmetric sums and $\sum_b u_b n_{+b}$, which gives the second and third sets. The first set follows from the third:

$$\sum_{a,b} (u_a + u_b)\hat{\mu}_{ab} = \sum_{a \le b} c_{ab}(u_a + u_b)(\hat{\mu}_{ab} + \hat{\mu}_{ba}) = \sum_{a,b} (u_a + u_b) n_{ab},$$

where $c_{ab} = 1$ for $a < b$ and $c_{aa} = \frac{1}{2}$; subtracting the second set leaves $\sum_a u_a \hat{\mu}_{a+} = \sum_a u_a n_{a+}$. Dividing by n, the fitted row (column) mean equals the observed row (column) mean.

Step 2. (b) The difference between the column and row means is

$$\sum_{a,b} (u_b - u_a)\pi_{ab} = \sum_{a<b} (u_b - u_a)(\pi_{ab} - \pi_{ba}) = \sum_{a<b} (u_b - u_a)\pi_{ba}(e^{\beta(u_b - u_a)} - 1),$$

using (10.15). Every term has the sign of β (and all are nonzero when $\beta \neq 0$). Marginal homogeneity forces equal means, hence $\beta = 0$, hence symmetry; the converse is immediate. ■

The likelihood-ratio test compares the deviances of symmetry and ordinal QS. Like the ordinal marginal model, it concentrates the marginal comparison on a single df.

10.4.7 Premarital and extramarital sex revisited

Example 10.25 (Premarital and extramarital sex: symmetry models)

For Example 10.13, symmetry is hopeless ($G^2 = 402.2$, df = 6), while QS fits well ($G^2 = 1.4$, df = 3), and so does ordinal QS with scores $\{1, 2, 3, 4\}$ ($G^2 = 2.1$, df = 5), with $\hat{\beta} = -2.86$. By (10.15), the probability that the premarital response is x categories more permissive than the extramarital one is $e^{2.86x}$ times the reverse probability. For instance, "premarital almost always wrong, extramarital always wrong" is $e^{2.86} = 17.4$ times as likely as the reverse combination.

10.4.8 Other ordinal models for square tables

When symmetry fails for ordered classifications, often $\pi_{ab} > \pi_{ba}$ for all $a < b$ (or the reverse). The **conditional symmetry** model (McCullagh 1978)

$$\log(\pi_{ab}/\pi_{ba}) = \tau \qquad \text{for all } a < b \tag{10.16}$$

captures this; it implies that, given $Y_1 \neq Y_2$, the probabilities above the diagonal are a mirror image (up to the factor e^{τ}) of those below. Symmetry is $\tau = 0$.

The **quasi-uniform association** model generalizes quasi-independence:

$$\log \mu_{ab} = \lambda + \lambda_a^X + \lambda_b^Y + \beta u_a u_b + \delta_a I(a = b), \tag{10.17}$$

linear-by-linear association off the diagonal. It is a special case of QS (the term $\beta u_a u_b$ is symmetric), and QI is $\beta = 0$; with equal-interval scores the local odds ratios off the diagonal all equal e^β.

For Table 10.13's data, conditional symmetry gives $\hat{\tau} = -4.130$ (SE $= 0.451$): a more negative opinion on extramarital than on premarital sex is $e^{4.13} = 62.2$ times as likely as the reverse. Quasi-uniform association gives $\hat{\beta} = 0.632$ (SE $= 0.106$), i.e. an off-diagonal local odds ratio of $e^{0.632} = 1.88$.

10.5 Measuring agreement between observers

Motivation. When two raters classify the same subjects, the square table describes their agreement. Agreement is a different property from association, so we need models (quasi-independence, quasi-symmetry) and summary indices (kappa) aimed specifically at the diagonal.

Example 10.26 (Pathologists' diagnoses)

Two pathologists A and B classified 118 slides on carcinoma of the uterine cervix: 1 = negative, 2 = atypical squamous hyperplasia, 3 = carcinoma in situ, 4 = squamous or invasive carcinoma. In parentheses: standardized Pearson residuals for independence.

	Pathologist B				
Pathologist A	1	2	3	4	Total
1	22 (8.5)	2 (−0.5)	2 (−5.9)	0 (−1.8)	26
2	5 (−0.5)	7 (3.2)	14 (−0.5)	0 (−1.8)	26
3	0 (−4.1)	2 (−1.2)	36 (5.5)	0 (−2.3)	38
4	0 (−3.3)	1 (−1.3)	17 (0.3)	10 (5.9)	28
Total	27	12	69	10	118

Independence fits poorly ($G^2 = 118.0$, df $= 9$). Large positive residuals on the diagonal show agreement beyond chance in every category, especially the first; off-diagonal residuals are mostly negative, less so for adjacent categories. The most common disagreements: B chose 3 while A chose 2 or 4.

10.5.1 Agreement: departures from independence

Let π_{ab} be the probability that A classifies a subject in a and B in b. Then $\sum_a \pi_{aa}$ is the probability of agreement, and perfect agreement is $\sum_a \pi_{aa} = 1$. Strong agreement requires strong association, but not conversely: if A always rates one category higher than B, association is strong and agreement poor. Independence is the baseline of "chance" agreement; its standardized residuals reveal patterns of agreement and disagreement (they grow with n, so their size is not a measure of agreement).

10.5.2 Using quasi-independence to analyze agreement

QI (10.13) adds agreement parameters δ_a to independence. For two subjects, suppose each observer puts one in category a and the other in b. The odds that the observers agree rather than disagree on which is which are

$$\tau_{ab} = \frac{\pi_{aa}\pi_{bb}}{\pi_{ab}\pi_{ba}} = \frac{\mu_{aa}\mu_{bb}}{\mu_{ab}\mu_{ba}}. \tag{10.18}$$

Proposition 10.27 (Agreement odds under QI and QS)

Under QI, $\tau_{ab} = \exp(\delta_a + \delta_b)$. Under QS, $\tau_{ab} = \exp(\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba})$.

Proof. [Supplement] Under QI, $\log \mu_{aa} + \log \mu_{bb} - \log \mu_{ab} - \log \mu_{ba}$ equals

$$(\lambda_a^X + \lambda_a^Y + \delta_a) + (\lambda_b^X + \lambda_b^Y + \delta_b) - (\lambda_a^X + \lambda_b^Y) - (\lambda_b^X + \lambda_a^Y) = \delta_a + \delta_b.$$

Under QS the same computation with (10.8) cancels all main effects and leaves $\lambda_{aa} + \lambda_{bb} - \lambda_{ab} - \lambda_{ba}$. ■

Larger δ_a mean stronger agreement. For Example 10.26, QI has $G^2 = 13.2$ (df = 5), much better than independence but with some lack of fit; $\hat{\delta}_2 = 0.6$, $\hat{\delta}_3 = 1.9$ and $\hat{\tau}_{23} = e^{2.5} = 12.3$.

10.5.3 Quasi-symmetry and agreement modeling

Given disagreement, some association often remains, which QI forbids. QS permits it, and for Example 10.26 it fits well: $G^2 = 1.0$, df = 2. When $n_{ab} + n_{ba} = 0$ (here categories 1 and 4), the QS likelihood equation $\hat{\mu}_{ab} + \hat{\mu}_{ba} = n_{ab} + n_{ba}$ forces both fitted values to 0; those cells should be removed from the fit to get the correct df. Under QS, $\hat{\tau}_{23} = 10.7$.

	Quasi-independence				Quasi-symmetry			
A	1	2	3	4	1	2	3	4
1	22	0.7	3.3	0.0	22	2.4	1.6	0.0
2	2.4	7	16.6	0.0	4.6	7	14.4	0.0
3	0.8	1.2	36	0.0	0.4	1.6	36	0.0
4	1.9	3.0	13.1	10	0.0	1.0	17.0	10

Fitted values for Example 10.26.

QS also informs about the margins: symmetry fits poorly ($G^2 = 39.2$, df = 5), and $G^2(S \mid QS) = 39.2 - 1.0 = 38.2$ (df = 3) shows strong marginal heterogeneity, one reason agreement is not stronger. Ordinal versions take the category order into account: given disagreement, high (low) ratings by one observer still tend to go with relatively high (low) ratings by the other.

10.5.4 Kappa measure of agreement

Definition (Cohen's kappa)

With $P_o = \sum_a \pi_{aa}$ and $P_e = \sum_a \pi_{a+}\pi_{+a}$ (agreement expected under independence),

$$\kappa = \frac{\sum_a \pi_{aa} - \sum_a \pi_{a+}\pi_{+a}}{1 - \sum_a \pi_{a+}\pi_{+a}} = \frac{P_o - P_e}{1 - P_e}.$$

The denominator is the numerator with P_o replaced by its maximum 1. So $\kappa = 0$ when agreement equals chance agreement, $\kappa = 1$ under perfect agreement, and $\kappa < 0$ (rare) when agreement is worse than chance. For given margins, stronger agreement means larger κ.

Theorem 10.28 (Asymptotic variance of sample kappa)

Under multinomial sampling, $\sqrt{n}(\hat{\kappa}-\kappa) \xrightarrow{d} N(0,\sigma_\kappa^2)$, where

$$\sigma_\kappa^2 = \frac{P_o(1-P_o)}{(1-P_e)^2} + \frac{2(1-P_o)\left[2P_oP_e - \sum_a \pi_{aa}(\pi_{a+}+\pi_{+a})\right]}{(1-P_e)^3} + \frac{(1-P_o)^2\left[\sum_a\sum_b \pi_{ab}(\pi_{b+}+\pi_{+a})^2 - 4P_e^2\right]}{(1-P_e)^4}.$$

The estimated variance of $\hat{\kappa}$ (Fleiss, Cohen and Everitt 1969) substitutes sample proportions and divides by n.

Proof. [Supplement]

Step 1. Delta method for a multinomial. If $\sqrt{n}(\mathbf{p}-\boldsymbol{\pi}) \xrightarrow{d} N(\mathbf{0}, \mathrm{diag}(\boldsymbol{\pi})-\boldsymbol{\pi}\boldsymbol{\pi}')$ and g is differentiable with gradient (g_{ab}), then $\sqrt{n}(g(\mathbf{p})-g(\boldsymbol{\pi})) \xrightarrow{d} N(0,\sigma^2)$ with

$$\sigma^2 = \sum_{a,b}\pi_{ab}g_{ab}^2 - \Big(\sum_{a,b}\pi_{ab}g_{ab}\Big)^2.$$

Step 2. Gradient. Write $\kappa = 1-(1-P_o)/(1-P_e)$. Since π_{ab} enters π_{a+} and π_{+b}, $\partial P_e/\partial\pi_{ab} = \pi_{+a}+\pi_{b+}$, and $\partial P_o/\partial\pi_{ab} = I(a=b)$. Hence

$$g_{ab} = \frac{I(a=b)(1-P_e) - (1-P_o)(\pi_{+a}+\pi_{b+})}{(1-P_e)^2}.$$

Step 3. First moment. Since $\sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+}) = \sum_a \pi_{a+}\pi_{+a} + \sum_b \pi_{+b}\pi_{b+} = 2P_e$,

$$\sum_{a,b}\pi_{ab}g_{ab} = \frac{P_o(1-P_e) - 2P_e(1-P_o)}{(1-P_e)^2}.$$

Step 4. Second moment. Let $s = \sum_a \pi_{aa}(\pi_{a+}+\pi_{+a})$ and $t = \sum_{a,b}\pi_{ab}(\pi_{+a}+\pi_{b+})^2$. Expanding the square,

$$(1-P_e)^4\sum_{a,b}\pi_{ab}g_{ab}^2 = P_o(1-P_e)^2 - 2(1-P_e)(1-P_o)s + (1-P_o)^2 t.$$

Step 5. Subtract the square of Step 3, $(1-P_e)^{-4}\left[P_o^2(1-P_e)^2 - 4P_oP_e(1-P_e)(1-P_o) + 4P_e^2(1-P_o)^2\right]$:

$$(1-P_e)^4\sigma_\kappa^2 = P_o(1-P_o)(1-P_e)^2 + 2(1-P_e)(1-P_o)(2P_oP_e - s) + (1-P_o)^2(t-4P_e^2).$$

Dividing by $(1-P_e)^4$ gives the stated formula. ■

It is rarely plausible that agreement is only at chance level, so instead of testing $\kappa=0$ one estimates κ by an interval. For Example 10.26, $P_o = 0.636$, $P_e = 0.281$, $\hat{\kappa} = (0.636-0.281)/(1-0.281) = 0.493$ and SE $= 0.057$: the excess of observed over chance agreement is about half its maximum possible value, and κ lies roughly between 0.4 and 0.6.

10.5.5 Weighted kappa: quantifying disagreement

Definition (Weighted kappa)

With weights $0 \le w_{ab} \le 1$, $w_{aa} = 1$, $w_{ab} = w_{ba}$,

$$\kappa_w = \frac{\sum_a \sum_b w_{ab}\pi_{ab} - \sum_a \sum_b w_{ab}\pi_{a+}\pi_{+b}}{1 - \sum_a \sum_b w_{ab}\pi_{a+}\pi_{+b}}.$$

Common choices are $w_{ab} = 1 - |a-b|/(I-1)$ and (Fleiss–Cohen) $w_{ab} = 1 - (a-b)^2/(I-1)^2$.

Weights let disagreements between distant categories count as more serious. Both kappa and weighted kappa depend strongly on the marginal distributions: the same rating process can give very different values for populations with different case mixes. A single index loses much information; models describing the agreement and disagreement structure are more informative.

10.5.6 Extensions to multiple observers

With R observers, ordinary loglinear models describe pairwise agreement *conditional* on the other ratings, which is usually not what is wanted. It is more natural to model the $\binom{R}{2}$ two-way marginal tables simultaneously (Becker and Agresti 1992), to use generalized kappas summarizing pairwise or multiple agreement, or to use latent-class mixture models with classes of subjects on whom observers agree or disagree (Chapter 13).

10.6 Bradley–Terry model for paired preferences

Motivation. When items (teams, wines, products) are compared two at a time, a single ranking can be built from the pairwise outcomes. The Bradley–Terry model is a logit model for these comparisons and is, in fact, quasi-symmetry in disguise.

10.6.1 Bradley–Terry model

Let Π_{ab} be the probability that a is preferred to b, with $\Pi_{ab} + \Pi_{ba} = 1$ (no ties).

Definition (Bradley–Terry model)

$$\log \frac{\Pi_{ab}}{\Pi_{ba}} = \beta_a - \beta_b, \qquad \text{equivalently} \qquad \Pi_{ab} = \frac{e^{\beta_a}}{e^{\beta_a} + e^{\beta_b}}. \tag{10.19}$$

Identifiability needs a constraint such as $\beta_I = 0$ or $\sum_a e^{\beta_a} = 1$.

So $\Pi_{ab} = \frac{1}{2}$ when $\beta_a = \beta_b$ and $\Pi_{ab} > \frac{1}{2}$ when $\beta_a > \beta_b$. The model describes $\binom{I}{2}$ probabilities by $I-1$ parameters, so residual df $= \binom{I}{2} - (I-1)$. Let N_{ab} be the number of comparisons of a and b, with a preferred n_{ab} times and b preferred $n_{ba} = N_{ab} - n_{ab}$ times; the results form a square table with empty diagonal. If comparisons are independent, $n_{ab} \sim \text{bin}(N_{ab}, \Pi_{ab})$ independently over pairs, and the model is an ordinary no-intercept logit model whose predictor row for pair (a,b) has $+1$ in position a and -1 in position b.

Proposition 10.29 (Likelihood equations for Bradley–Terry)

The ML estimates satisfy, for each item a,

$$\sum_{b \ne a} N_{ab}\hat{\Pi}_{ab} = \sum_{b \ne a} n_{ab} :$$

the fitted number of wins of each item equals its observed number of wins.

Proof. [Supplement] The log-likelihood is $L = \sum_{a<b} [n_{ab}(\beta_a - \beta_b) - N_{ab} \log(1 + e^{\beta_a - \beta_b})]$. Item a appears with sign $+1$ in pairs (a, b), $b > a$, and with sign -1 in pairs (c, a), $c < a$. Differentiating,

$$\frac{\partial L}{\partial \beta_a} = \sum_{b>a} (n_{ab} - N_{ab}\Pi_{ab}) - \sum_{c<a} (n_{ca} - N_{ca}\Pi_{ca}) = \sum_{b \neq a} (n_{ab} - N_{ab}\Pi_{ab}),$$

since $n_{ca} - N_{ca}\Pi_{ca} = -(n_{ac} - N_{ac}\Pi_{ac})$. Setting this to zero gives the result. ■

10.6.2 Home team advantage in baseball

Example 10.30 (Baseball, 1987 American League East)

Wins of the row team over the column team (Bradley–Terry fitted values in parentheses):

Winner	Milw.	Detr.	Toro.	N.Y.	Bost.	Clev.	Balt.
Milwaukee	–	7 (7.0)	9 (7.4)	7 (7.6)	7 (8.0)	9 (9.2)	11 (10.8)
Detroit	6 (6.0)	–	7 (7.0)	5 (7.1)	11 (7.6)	9 (8.8)	9 (10.5)
Toronto	4 (5.6)	6 (6.0)	–	7 (6.7)	7 (7.1)	8 (8.4)	12 (10.2)
New York	6 (5.4)	8 (5.9)	6 (6.3)	–	6 (7.0)	7 (8.3)	10 (10.1)
Boston	6 (5.0)	2 (5.4)	6 (5.9)	7 (6.0)	–	7 (7.9)	12 (9.8)
Cleveland	4 (3.8)	4 (4.2)	5 (4.6)	6 (4.7)	6 (5.1)	–	6 (8.6)
Baltimore	2 (2.2)	4 (2.5)	1 (2.8)	3 (2.9)	1 (3.2)	7 (4.4)	–

Fitting (10.19) to the $\binom{7}{2} = 21$ binomials gives $G^2 = 15.7$ (df $= 15$). Estimates ($\hat{\beta}_7 = 0$):

	Milw.	Detr.	Toro.	N.Y.	Bost.	Clev.	Balt.
Winning %	64.1	60.2	56.4	55.1	51.3	39.7	23.1
$\hat{\beta}_a$	1.58	1.44	1.29	1.25	1.11	0.68	0.00
$e^{\hat{\beta}_a}$, $\sum = 1$	0.218	0.189	0.164	0.158	0.136	0.089	0.045

For Boston vs New York, $\hat{\Pi}_{54} = e^{1.11}/(e^{1.11} + e^{1.25}) = 0.46$. Each $\hat{\beta}_a$ and $\hat{\beta}_a - \hat{\beta}_b$ has SE about 0.3, so the top five teams are not clearly distinguished.

Home advantage. Let Π^*_{ab} be the probability that a beats b when a is at home, and fit

$$\log \frac{\Pi^*_{ab}}{1 - \Pi^*_{ab}} = \alpha + (\beta_a - \beta_b) \tag{10.20}$$

to the 42 home/away binomials (e.g. Boston at home beat New York 4–2; New York at home beat Boston 4–3). This gives $G^2 = 38.6$ (df $= 35$), team estimates similar to before ($e^{\hat{\beta}_a}$: $0.220, 0.190, 0.164, 0.157, 0.137, 0.088, 0.044$) and $\hat{\alpha} = 0.302$: the home team of two evenly matched teams wins with probability $e^{0.302}/(1+e^{0.302}) = 0.575$. Boston beats New York with estimated probability 0.54 at Boston and 0.39 at New York.

Model (10.20) is useful whenever there is an order effect, e.g. the product tasted first in a taste test.

10.6.3 Bradley–Terry model and quasi-symmetry

Theorem 10.31 (Fienberg–Larntz)

Let the square table of comparisons have cell (a, b) = "a preferred to b". If the table satisfies QS (10.8), then the conditional probability of cell (a, b) given that the observation is in (a, b) or (b, a) follows the Bradley–Terry model with $\beta_a = \lambda_a^X - \lambda_a^Y$. Conversely, the Bradley–Terry model for $\{\Pi_{ab}\}$ is the logit form of QS, and the ML estimates $\hat{\lambda}_a^X - \hat{\lambda}_a^Y$ from QS give $\hat{\beta}_a$.

Proof.

Step 1. Under QS,

$$\begin{aligned}\log \frac{\mu_{ab}}{\mu_{ba}} &= (\lambda + \lambda_a^X + \lambda_b^Y + \lambda_{ab}) - (\lambda + \lambda_b^X + \lambda_a^Y + \lambda_{ba}) \\ &= (\lambda_a^X - \lambda_a^Y) - (\lambda_b^X - \lambda_b^Y) = \beta_a - \beta_b,\end{aligned}$$

since $\lambda_{ab} = \lambda_{ba}$; and $\mu_{ab}/(\mu_{ab} + \mu_{ba})$ is the conditional probability of (a, b) given the pair.

Step 2. Conversely, the QS likelihood, conditional on the pair totals $n_{ab} + n_{ba} = N_{ab}$ (which are the sufficient statistics for the symmetric terms and are fitted exactly by Proposition 10.17), is the product of binomials with logits $\beta_a - \beta_b$, i.e. the Bradley–Terry likelihood. Since the symmetric parameters are fitted by $\hat{\mu}_{ab} + \hat{\mu}_{ba} = N_{ab}$ and the remaining factor depends only on $\{\beta_a\}$, the two fits give the same $\hat{\beta}_a$. ■

10.6.4 Extensions to ties and ordinal evaluations

For an ordinal evaluation scale with I categories (e.g. much better, slightly better, same, slightly worse, much worse), let Y_{ab} be the response comparing a with b and use

$$\operatorname{logit} P(Y_{ab} \le j) = \theta_j + \beta_a - \beta_b.$$

Proposition 10.32 (Symmetry of the cutpoints)

Since comparing b with a reverses the scale, $Y_{ba} = I + 1 - Y_{ab}$, and then necessarily $\theta_j = -\theta_{I-j}$. For the scale (win, tie, lose), $\theta_1 = -\theta_2$.

Proof. $P(Y_{ab} \le j) = P(Y_{ba} \ge I + 1 - j) = 1 - P(Y_{ba} \le I - j)$, so $\operatorname{logit} P(Y_{ab} \le j) = -\operatorname{logit} P(Y_{ba} \le I - j)$. Substituting the model, $\theta_j + \beta_a - \beta_b = -(\theta_{I-j} + \beta_b - \beta_a) = -\theta_{I-j} + \beta_a - \beta_b$, so $\theta_j = -\theta_{I-j}$. ■

Chapter 11. Analyzing Repeated Categorical Response Data

Many studies observe a categorical response repeatedly for each subject: at several times (longitudinal studies), under several conditions (crossover trials), or for several units of a *cluster* (teeth in a mouth, fetuses in a litter, children in a family). Observations within a cluster tend to be more alike than observations from different clusters, and analyses ignoring this can be badly wrong. This chapter generalizes Chapter 10 from pairs to sets of T responses and adds explanatory variables. Section 11.1 compares the T margins; Section 11.2 fits marginal models by ML; Sections 11.3–11.4 develop the computationally simpler *generalized estimating equations* (GEE), a multivariate quasi-likelihood; Section 11.5 models each response given the previous ones (transitional models).

11.1 Comparing marginal distributions: multiple responses

Motivation. Usually the T marginal distributions (e.g. the success probability at each week of a treatment) matter more than the full joint dependence. We extend the marginal analyses of Chapter 10 from $T = 2$ to $T > 2$.

11.1.1 Binary marginal models and marginal homogeneity

For T binary responses $(Y_1, \ldots, Y_T)$ the marginal logit model

$$\operatorname{logit} P(Y_t = 1) = \alpha + \beta_t, \qquad t = 1, \ldots, T, \tag{11.1}$$

with a constraint such as $\beta_T = 0$, extends the matched-pairs model of Section 10.2.1. For a sequence $\mathbf{i} = (i_1, \ldots, i_T)$, $i_t \in \{0, 1\}$, let $\pi_{\mathbf{i}} = P(Y_1 = i_1, \ldots, Y_T = i_T)$; these are the cell probabilities of the 2^T table. The sample cell proportions are the ML estimates of $\{\pi_{\mathbf{i}}\}$, and the sample proportion with $y_t = 1$ is the ML estimate of $P(Y_t = 1)$. Model (11.1) is saturated; **marginal homogeneity** $P(Y_1 = 1) = \cdots = P(Y_T = 1)$ is the special case $\beta_1 = \cdots = \beta_T$. Even this one-parameter model is awkward to fit by ML, because the multinomial likelihood involves the 2^T joint probabilities rather than the T margins.

With $n_{\mathbf{i}}$ the count in cell $\mathbf{i}$, $p_{\mathbf{i}} = n_{\mathbf{i}}/n$, and $\hat{\pi}^{MH}$ the ML fit under marginal homogeneity, the likelihood-ratio statistic is

$$-2[L(\hat{\pi}^{MH}) - L(\mathbf{p})] = 2\sum_{\mathbf{i}} n_{\mathbf{i}} \log\left(p_{\mathbf{i}}/\hat{\pi}^{MH}_{\mathbf{i}}\right), \tag{11.2}$$

asymptotically χ^2_{T-1} under H_0, since (11.1) has $T-1$ more parameters than marginal homogeneity.

11.1.2 Crossover drug comparison example

Example 11.1 (Crossover study of three drugs)

Each of 46 subjects with a chronic condition used drugs A, B, C at three times; the reaction was favorable (F) or unfavorable (U). Assume no carryover effects and a stable condition.

	A favorable		A unfavorable	
	B F	B U	B F	B U
C favorable	6	2	2	6
C unfavorable	16	4	4	6

The proportions favorable are $(0.61, 0.61, 0.35)$ for (A, B, C). Statistic (11.2) equals 5.95 (df $= 2$, $P = 0.05$). For Bonferroni intervals with overall error ≤ 0.05, each of the three pairwise intervals uses confidence level $1 - 0.05/3$, so $z = 2.39$. For A vs C, $0.609 - 0.348 = 0.261$ with SE 0.108 (Theorem 10.1), giving $0.261 \pm 2.39(0.108) = (0.002, 0.520)$; the same holds for B vs C. Exact binomial tests on the discordant pairs of each 2×2 table give $P = 1.0$ (A vs B) and $P = 0.036$ (A vs C, B vs C). The sample is small, so these conclusions are tentative.

11.1.3 Modeling margins of a multicategory response

For an I-category response, the saturated baseline-category model is

$$\log \frac{P(Y_t = j)}{P(Y_t = I)} = \beta_{tj}, \qquad t = 1, \ldots, T, \quad j = 1, \ldots, I-1,$$

and marginal homogeneity is $\beta_{1j} = \cdots = \beta_{Tj}$ for all j; its likelihood-ratio test has the form (11.2) with df $= (T-1)(I-1)$. For an ordinal response, a model between these two is

$$\operatorname{logit} P(Y_t \leq j) = \alpha_j + \beta_t, \qquad t = 1, \ldots, T, \quad j = 1, \ldots, I-1, \tag{11.3}$$

with $\beta_T = 0$ and $\alpha_1 < \cdots < \alpha_{I-1}$; marginal homogeneity is $\beta_1 = \cdots = \beta_T$, a test with df $= T - 1$.

11.1.4 Wald and generalized CMH score tests of marginal homogeneity

Let $p_j(t)$ be the sample proportion in category j at occasion t, $\bar{p}_j = \sum_t p_j(t)/T$, $d_j(t) = p_j(t) - \bar{p}_j$, and $\mathbf{d}$ the vector of $d_j(t)$, $t \leq T-1$, $j \leq I-1$. With $\hat{\mathbf{V}}$ the estimated covariance matrix of $\sqrt{n}\,\mathbf{d}$, Bhapkar's Wald statistic

$$W = n\,\mathbf{d}'\hat{\mathbf{V}}^{-1}\mathbf{d} \tag{11.4}$$

is asymptotically $\chi^2_{(I-1)(T-1)}$ under marginal homogeneity, generalizing Proposition 10.14.

Other tests are generalized CMH statistics. Display the data as n strata, one per subject; stratum k is a $T \times I$ table whose row t has a 1 in the column of the outcome at occasion t. Conditional independence of occasion and response given subject means that, for each subject, the T responses are exchangeable; this implies marginal homogeneity, and the generalized CMH test of conditional independence is a test of marginal homogeneity whose null distribution is generated by this stronger exchangeability (Darroch 1981). For $I = T = 2$ it is McNemar's test (Theorem 10.5); for $I = 2$, $T > 2$ it is Cochran's Q.

Theorem 11.2 (Cochran's Q as a generalized CMH statistic)

For binary responses, let R_k be the number of successes of subject k, C_t the number of successes at occasion t and $N = \sum_t C_t = \sum_k R_k$. The generalized CMH statistic (general association) for the $T \times 2 \times n$ subject-stratified table is

$$Q = \frac{T(T-1)\sum_{t=1}^{T}(C_t - N/T)^2}{TN - \sum_k R_k^2},$$

which is asymptotically χ^2_{T-1} under marginal homogeneity. For $T = 2$, $Q = (n_{21} - n_{12})^2/(n_{21} + n_{12})$.

Proof. [Supplement]

Step 1. Null distribution. Under conditional independence with fixed stratum margins, the R_k successes of subject k are allocated to the T occasions completely at random (each of the $\binom{T}{R_k}$ allocations equally likely), independently over subjects. Since column 2 is determined by column 1,

the CMH statistic is the quadratic form in $\mathbf{C} = (C_1, \ldots, C_T)'$ about its null mean, with a generalized inverse of its null covariance.

Step 2. Moments for one subject. Let Y_{kt} indicate success at occasion t and $c_k = (R_k/T)(1-R_k/T)$. Under random allocation (sampling without replacement),

$$E(Y_{kt}) = \frac{R_k}{T}, \qquad \mathrm{Var}(Y_{kt}) = c_k, \qquad \mathrm{Cov}(Y_{kt}, Y_{ks}) = -\frac{c_k}{T-1} \ (t \neq s),$$

i.e. $\mathrm{Cov}(\mathbf{Y}_k) = \frac{T}{T-1} c_k (\mathbf{I} - \frac{1}{T}\mathbf{J})$ with $\mathbf{J}$ the matrix of ones.

Step 3. Summing over independent subjects, $E(\mathbf{C}) = (N/T)\mathbf{1}$ and

$$\mathrm{Cov}(\mathbf{C}) = \kappa(\mathbf{I} - \tfrac{1}{T}\mathbf{J}), \qquad \kappa = \frac{T}{T-1}\sum_k c_k = \frac{\sum_k R_k(T-R_k)}{T(T-1)} = \frac{TN - \sum_k R_k^2}{T(T-1)}.$$

Step 4. $\mathbf{P} = \mathbf{I} - \frac{1}{T}\mathbf{J}$ is the projection onto $\{\mathbf{v} : \mathbf{1}'\mathbf{v} = 0\}$, so $\mathbf{P}$ is its own generalized inverse and $\mathbf{C} - E\mathbf{C}$ lies in its range. The quadratic form is

$$(\mathbf{C} - E\mathbf{C})'\frac{\mathbf{P}}{\kappa}(\mathbf{C} - E\mathbf{C}) = \frac{1}{\kappa}\sum_t \left(C_t - N/T\right)^2 = Q.$$

The rank of $\mathbf{P}$ is $T-1$, which gives the χ^2_{T-1} limit (CLT for the sum of independent subject vectors).

Step 5. $T = 2$: only discordant subjects ($R_k = 1$) contribute to the denominator, which equals $2N - \sum R_k^2 = n_{12} + n_{21}$ after cancelling concordant contributions ($R_k = 0$ gives 0, $R_k = 2$ gives $2 \cdot 2 - 4 = 0$). Also $C_1 - C_2 = n_{12} - n_{21}$, so $\sum_t (C_t - N/2)^2 = (n_{12} - n_{21})^2/2$ and $Q = 2 \cdot \frac{1}{2}(n_{12} - n_{21})^2/(n_{12} + n_{21})$. ■

For an ordinal response with fixed scores, the generalized CMH statistic for variation among the T occasion means is the appropriate single-df version.

11.2 Marginal modeling: maximum likelihood approach

Motivation. Usually we want to know how the margins depend on explanatory variables (treatment, severity, time). Marginal logit models answer this, and ML fitting of them is possible by maximizing the full multinomial likelihood subject to constraints on the margins.

11.2.1 Longitudinal mental depression example

Example 11.3 (Mental depression)

Subjects with mild or severe initial depression were randomized to a standard or a new drug; after 1, 2 and 4 weeks each was classified as normal (N) or abnormal (A):

Diagnosis	Treatment	NNN	NNA	NAN	NAA	ANN	ANA	AAN	AAA
Mild	Standard	16	13	9	3	14	4	15	6
Mild	New drug	31	0	6	0	22	2	9	0
Severe	Standard	2	2	8	9	9	15	27	28
Severe	New drug	7	2	5	2	31	5	32	6

Let $s = 1$ for severe, $d = 1$ for the new drug and time $t = 0, 1, 2$ ($\log_2$ of the week number, since the logit is often linear in log time). Sample proportions normal:

Diagnosis	Treatment	Week 1	Week 2	Week 4
Mild	Standard	0.51	0.59	0.68
Mild	New drug	0.53	0.79	0.97
Severe	Standard	0.21	0.28	0.46
Severe	New drug	0.18	0.50	0.83

The proportion increases with time in every group, faster for the new drug, and is higher for mild diagnosis. Sampling is multinomial over the 8 sequences independently in the 4 groups (32 cells), while a marginal model concerns the 12 marginal probabilities. The main-effects model

$$\text{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 d + \beta_3 t$$

uses 4 parameters for 12 marginal logits (df = 8) and fits poorly, $G^2 = 34.6$: it forces a common time slope. With drug-by-time interaction,

$$\text{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 d + \beta_3 t + \beta_4 dt,$$

$G^2 = 4.2$ (df = 7). The time slope is $\hat\beta_3 = 0.48$ (SE 0.12) for the standard drug and $\hat\beta_3 + \hat\beta_4 = 1.49$ (SE 0.14) for the new drug; $\hat\beta_4 = 1.01$ (SE 0.18), and $34.6 - 4.2 = 30.4$ is the LR test of $\beta_4 = 0$. Also $\hat\beta_1 = -1.29$ (SE 0.14): severe diagnosis multiplies the odds of normal response by $e^{-1.29} = 0.27$. $\hat\beta_2 = -0.06$ (SE 0.22) shows no drug difference at week 1; at time t the new-to-standard odds ratio is $e^{-0.06+1.01t}$.

11.2.2 Modeling a repeated multinomial response

For an I-category response, observation t has $I-1$ marginal logits $\text{logit}_j(t)$ (baseline-category for nominal, cumulative logit $P(Y_t \le j)$ for ordinal responses), and a model has the form

$$\text{logit}_j(t) = \alpha_j + \boldsymbol{\beta}_j' \mathbf{x}_t, \qquad j = 1, \ldots, I-1.$$

For cumulative logits $\boldsymbol{\beta}_j$ may simplify to $\boldsymbol{\beta}$ (proportional odds). Some components of $\mathbf{x}_t$ may refer to the occasion t itself. Checking interaction is essential: are the effects of $\mathbf{x}$ the same at each t?

11.2.3 Insomnia example

Example 11.4 (Insomnia clinical trial)

In a randomized double-blind trial, patients reported time to fall asleep (minutes) before and after two weeks of an active hypnotic or a placebo:

		Follow-up			
Treatment	Initial	< 20	20–30	30–60	> 60
Active	< 20	7	4	1	0
	20–30	11	5	2	2
	30–60	13	23	3	1
	> 60	9	17	13	8
Placebo	< 20	7	4	2	1
	20–30	14	5	1	0
	30–60	6	9	18	2
	> 60	4	11	14	22

Sample marginal distributions:

Treatment	Occasion	< 20	20–30	30–60	> 60
Active	Initial	0.101	0.168	0.336	0.395
	Follow-up	0.336	0.412	0.160	0.092
Placebo	Initial	0.117	0.167	0.292	0.425
	Follow-up	0.258	0.242	0.292	0.208

Both distributions shift down, more for the active drug. With $t = 0$ (initial), 1 (follow-up) and $x = 1$ (active), 0 (placebo), the proportional odds model

$$\text{logit}\, P(Y_t \le j) = \alpha_j + \beta_1 t + \beta_2 x + \beta_3 tx \tag{11.5}$$

fitted by ML has $G^2 = 8.0$ (df $= 6$; 12 marginal logits, 6 parameters), with $\hat\beta_1 = 1.074$ (SE 0.162), $\hat\beta_2 = 0.046$ (SE 0.236), $\hat\beta_3 = 0.662$ (SE 0.244): evidence of interaction. Initially the active-vs-placebo cumulative odds ratio is $e^{0.046} = 1.04$; at follow-up it is $e^{0.046+0.662} = 2.03$. With scores $(10, 25, 45, 75)$ the initial means are 50.0 (active) and 50.3 (placebo); the mean decreases are 22.2 and 13.0, a difference of 9.2 (SE 3.0).

11.2.4 Comparisons that control for initial response

Marginal models do not always tell the whole story

If the treatment groups have identical initial marginal distributions (true up to sampling error under randomization) and identical conditional distributions of the follow-up response given the initial response, then by the law of total probability their follow-up margins are identical too. But if the initial margins differ, the changes in the margins can differ between groups even though the conditional follow-up distributions are identical. It is then more informative to compare follow-up responses controlling for the initial response y_1:

$$\text{logit}\, P(Y_2 \le j) = \alpha_j + \beta_1 x + \beta_2 y_1, \tag{11.6}$$

an ordinal analogue of analysis of covariance and a simple *transitional* model (Section 11.5).

11.3 Marginal modeling: generalized estimating equations (GEE) approach

Motivation. ML marginal fitting becomes impractical when T is large or there are many (possibly continuous) predictors, and there is no simple multivariate distribution for categorical responses. GEE specifies only the marginal mean, a variance function and a working guess at the correlation, and still estimates the regression parameters consistently.

11.3.1 Generalized estimating equation methodology: basic ideas

For a multivariate response $(Y_1, \ldots, Y_T)$ (T may vary by subject), the quasi-likelihood idea of Section 4.7 is applied to each margin: a model links $\mu_t = E(Y_t)$ to a linear predictor, and a variance function $v(\mu_t)$ describes $\text{Var}(Y_t)$. In addition one chooses a *working correlation* structure for $\{Y_t\}$. The estimates solve multivariate quasi-likelihood equations, the **generalized estimating equations** (Liang and Zeger 1986, building on econometric work of Gourieroux–Monfort–Trognon, Hansen and White).

Definition (Working correlation structures)

For cluster i with T_i observations, a working correlation matrix $\mathbf{R}(\boldsymbol{\alpha})$ specifies $\text{Corr}(Y_{it}, Y_{is})$:
(i) *independence*: $\mathbf{R} = \mathbf{I}$;
(ii) *exchangeable*: $\text{Corr}(Y_{it}, Y_{is}) = \alpha$ for all $t \neq s$;
(iii) *autoregressive* AR(1): $\text{Corr}(Y_{it}, Y_{is}) = \alpha^{|t-s|}$;
(iv) *unstructured*: a separate α_{ts} for each pair.

The key properties, proved in Section 11.4:

(a) If the mean model (link and linear predictor) is correct, the GEE estimators are consistent *even if the working correlation is wrong*: consistency depends on the first moment, not the second.

(b) The naive (model-based) standard errors are wrong when the working correlation is wrong; GEE uses the empirical dependence in the data to compute *robust* (sandwich) standard errors.

Even the naive independence structure gives good estimates, with corrected standard errors. A good working correlation improves efficiency; estimators under independence can be surprisingly efficient when correlations are weak to moderate. One can check sensitivity by comparing several structures; when they differ substantially, the correlation structure deserves study. Exchangeable is a sensible default (one extra parameter); unstructured is flexible but costs efficiency for large T.

Limitations of GEE

GEE does not specify a joint distribution, so there is no likelihood: no likelihood-ratio tests of fit or model comparison. Inference uses Wald statistics with the sandwich covariance. Unless the number of clusters is large, sandwich SEs tend to underestimate the true ones and are themselves variable; score-type tests based on the quasi-likelihood (Boos 1992; Rotnitzky and Jewell 1990) may be more trustworthy. Some critics object to the lack of likelihood; others view GEE as an estimation method rather than a model.

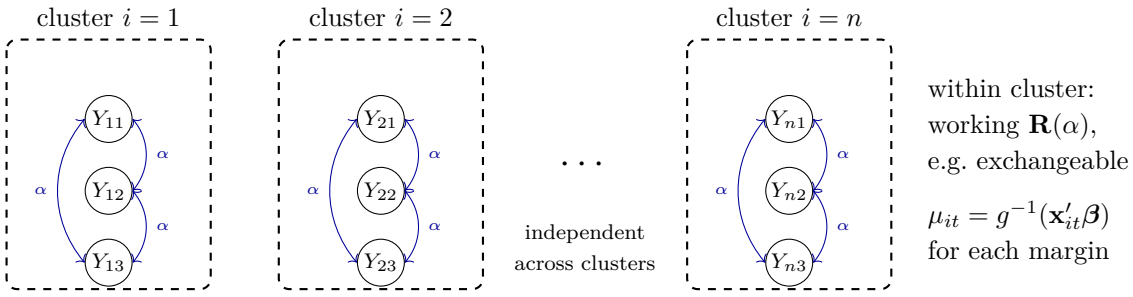


Figure 11.1. Clustered data: n independent clusters, dependent observations within a cluster. GEE models each margin and uses a working correlation within clusters; asymptotics are in the number of clusters n.

11.3.2 Longitudinal mental depression example

Example 11.5 (Depression: GEE fit)

For Example 11.3 with the interaction model, GEE gives results close to ML for every working correlation. With the exchangeable structure, $\hat{\beta}_3 = 0.48$ (SE 0.12) and $\hat{\beta}_4 = 1.02$ (SE 0.19). With the independence structure:

Parameter	Estimate	Naive SE (independence)	Robust (sandwich) SE
Intercept	−0.028	0.164	0.174
Diagnosis (severe)	−1.314	0.146	0.146
Drug (new)	−0.060	0.222	0.229
Time	0.482	0.115	0.120
Drug × time	1.017	0.189	0.188

The estimates and naive SEs are those of ordinary logistic regression on $3 \times 340 = 1020$ observations treated as independent; the robust SEs use the empirical within-subject dependence. The exchangeable estimate of the common correlation is $\hat{\alpha} = -0.003$: the three responses look pairwise uncorrelated, unusual for repeated measures, which is why all analyses agree so closely.

11.3.3 GEE approach for multinomial responses: insomnia example

GEE extends to marginal models for multinomial responses (Lipsitz, Kim and Zhao 1994): for each pair of outcome categories one chooses a working correlation for pairs of occasions, and each multinomial observation uses its $(I-1) \times (I-1)$ multinomial covariance.

Example 11.6 (Insomnia: GEE fit)

For model (11.5) with independence working correlation, GEE gives $\hat{\beta}_1 = 1.038$ (SE 0.168), $\hat{\beta}_2 = 0.034$ (SE 0.238), $\hat{\beta}_3 = 0.708$ (SE 0.244), similar to ML: time to fall asleep decreased more for the active treatment.

11.4 Quasi-likelihood and its GEE multivariate extension: details

Motivation. Why is GEE consistent when the correlation is wrong, and where does the sandwich come from? Both follow from one general fact about estimating equations with mean-zero estimating functions. We prove it once (Lemma 11.8) and apply it to quasi-likelihood and to GEE.

For subject i, $i = 1, \ldots, n$, let y_i have mean μ_i, variance function $v(\mu_i)$, and linear predictor $\eta_i = g(\mu_i) = \sum_j \beta_j x_{ij} = \mathbf{x}_i'\boldsymbol{\beta}$. The **quasi-likelihood** (QL) estimates solve the quasi-score equations

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \Big(\frac{\partial \mu_i}{\partial \boldsymbol{\beta}}\Big) \frac{y_i - \mu_i}{v(\mu_i)} = \mathbf{0}, \qquad \mu_i = g^{-1}(\mathbf{x}_i'\boldsymbol{\beta}). \tag{11.7}$$

Proposition 11.7 (Motivations for the quasi-score equations)

(a) If y_i has a natural exponential family distribution with variance function v, (11.7) are the GLM likelihood equations.
(b) With $v(\mu_i)$ replaced by known constants v_i, (11.7) are the normal equations for minimizing $\sum_i (y_i - \mu_i)^2 / v_i$.
(c) Since $\partial \mu_i / \partial \beta_j = (\partial \mu_i / \partial \eta_i) x_{ij}$, (11.7) depend only on the link and the variance function.

Proof. [Supplement] (a) For $f(y;\theta) = \exp\{y\theta - b(\theta) + c(y)\}$ we have $\mu = b'(\theta)$ and $v(\mu) = b''(\theta)$. The log-likelihood contribution $y_i\theta_i - b(\theta_i)$ has

$$\frac{\partial}{\partial \beta_j} = (y_i - \mu_i)\frac{\partial \theta_i}{\partial \beta_j} = (y_i - \mu_i)\frac{\partial \theta_i}{\partial \mu_i}\frac{\partial \mu_i}{\partial \beta_j} = \frac{y_i - \mu_i}{v(\mu_i)}\frac{\partial \mu_i}{\partial \beta_j},$$

since $\partial \mu / \partial \theta = b''(\theta) = v(\mu)$. Summing over i gives (11.7).

(b) $\partial/\partial\beta_j$ of $\sum_i (y_i-\mu_i)^2/v_i$ is $-2\sum_i (y_i-\mu_i)v_i^{-1}\partial\mu_i/\partial\beta_j$.

(c) Chain rule. ■

Wedderburn (1974) proposed using (11.7) for any link and variance function, whether or not they correspond to an exponential family member.

11.4.1 Properties of quasi-likelihood estimators

Definition (Unbiased estimating function)

A function $\mathbf{h}(\mathbf{y};\boldsymbol{\beta})$ is an **unbiased estimating function** if $E_{\boldsymbol{\beta}}[\mathbf{h}(\mathbf{Y};\boldsymbol{\beta})]=\mathbf{0}$ for all $\boldsymbol{\beta}$; the equations $\mathbf{h}(\mathbf{y};\boldsymbol{\beta})=\mathbf{0}$ that define $\hat{\boldsymbol{\beta}}$ are **estimating equations**.

The quasi-score (11.7) is unbiased as soon as $E(y_i)=\mu_i(\boldsymbol{\beta})$, whatever the true variance. It is treated as the derivative of a *quasi-log-likelihood*, which need not be a proper log-likelihood; nevertheless QL estimators behave much like ML estimators (McCullagh 1983). We now prove this rigorously, in a form that also covers GEE.

Lemma 11.8 (Asymptotics of estimating equations)

Let clusters $i=1,\dots,n$ be independent, $\boldsymbol{\beta}\in\Theta\subseteq\mathbb{R}^p$ open, with true value $\boldsymbol{\beta}_0$, and let $\boldsymbol{\gamma}$ be a nuisance parameter estimated by $\hat{\boldsymbol{\gamma}}\xrightarrow{p}\boldsymbol{\gamma}^*$. Let $\mathbf{U}_n(\boldsymbol{\beta};\boldsymbol{\gamma})=\sum_{i=1}^n \boldsymbol{\psi}_i(\boldsymbol{\beta};\boldsymbol{\gamma})$ and $\mathbf{H}_i=-\partial\boldsymbol{\psi}_i/\partial\boldsymbol{\beta}'$. Assume:
(E1) $E\,\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)=\mathbf{0}$ for all i;
(E2) $n^{-1}\sum_i \operatorname{Cov}\big(\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\big)\to\mathbf{B}$, positive definite, and $\sup_i E\|\boldsymbol{\psi}_i(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\|^{2+\delta}<\infty$ for some $\delta>0$;
(E3) $\boldsymbol{\psi}_i$ is continuously differentiable in $\boldsymbol{\beta}$, and on neighbourhoods $\mathcal{N}$ of $\boldsymbol{\beta}_0$ and $\mathcal{M}$ of $\boldsymbol{\gamma}^*$, $\sup_{\mathcal{N}\times\mathcal{M}}\|n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta};\boldsymbol{\gamma})-\mathbf{A}(\boldsymbol{\beta},\boldsymbol{\gamma})\|\xrightarrow{p}0$ for a continuous $\mathbf{A}(\cdot,\cdot)$ with $\mathbf{A}=\mathbf{A}(\boldsymbol{\beta}_0,\boldsymbol{\gamma}^*)$ non-singular;
(E4) $n^{-1/2}\big[\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})-\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\big]\xrightarrow{p}\mathbf{0}$.
Then:
(i) for every sufficiently small $r>0$, with probability tending to one, $\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})=\mathbf{0}$ has exactly one root $\hat{\boldsymbol{\beta}}$ in the ball $\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le r$, and $\hat{\boldsymbol{\beta}}\xrightarrow{p}\boldsymbol{\beta}_0$;
(ii) $\sqrt{n}(\hat{\boldsymbol{\beta}}-\boldsymbol{\beta}_0)\xrightarrow{d}N(\mathbf{0},\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1\prime})$;
(iii) if moreover $n^{-1}\sum_i \boldsymbol{\psi}_i\boldsymbol{\psi}_i'$ converges in probability, uniformly on $\mathcal{N}\times\mathcal{M}$, to a continuous limit equal to $\mathbf{B}$ at $(\boldsymbol{\beta}_0,\boldsymbol{\gamma}^*)$, then with $\hat{\mathbf{A}}=n^{-1}\sum_i \mathbf{H}_i(\hat{\boldsymbol{\beta}};\hat{\boldsymbol{\gamma}})$ and $\hat{\mathbf{B}}=n^{-1}\sum_i \boldsymbol{\psi}_i\boldsymbol{\psi}_i'(\hat{\boldsymbol{\beta}};\hat{\boldsymbol{\gamma}})$, the sandwich $\hat{\mathbf{A}}^{-1}\hat{\mathbf{B}}\hat{\mathbf{A}}^{-1\prime}\xrightarrow{p}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1\prime}$.

Proof. [Supplement]

Step 1. Set-up. Let $c=\|\mathbf{A}^{-1}\|$. By continuity of $\mathbf{A}(\cdot,\cdot)$ choose $r>0$ and a neighbourhood $\mathcal{M}'\subseteq\mathcal{M}$ of $\boldsymbol{\gamma}^*$ with $\bar{B}_r=\{\|\boldsymbol{\beta}-\boldsymbol{\beta}_0\|\le r\}\subseteq\mathcal{N}$ and $\|\mathbf{A}(\boldsymbol{\beta},\boldsymbol{\gamma})-\mathbf{A}\|\le 1/(4c)$ on $\bar{B}_r\times\mathcal{M}'$. Let ε_n be the supremum in (E3). Consider the event

$$E_n=\Big\{\hat{\boldsymbol{\gamma}}\in\mathcal{M}',\ \ \varepsilon_n\le\tfrac{1}{4c},\ \ \|n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})\|\le\tfrac{r}{2c}\Big\}.$$

By (E1)–(E2) and Chebyshev's inequality, $n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\boldsymbol{\gamma}^*)\xrightarrow{p}\mathbf{0}$; with (E4) also $n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})\xrightarrow{p}\mathbf{0}$. With $\hat{\boldsymbol{\gamma}}\xrightarrow{p}\boldsymbol{\gamma}^*$ and $\varepsilon_n\xrightarrow{p}0$, $P(E_n)\to1$.

Step 2. Linearization. For $\boldsymbol{\beta}\in\bar{B}_r$ (a convex set), the fundamental theorem of calculus along the segment from $\boldsymbol{\beta}_0$ to $\boldsymbol{\beta}$ gives

$$n^{-1}\mathbf{U}_n(\boldsymbol{\beta};\hat{\boldsymbol{\gamma}})=n^{-1}\mathbf{U}_n(\boldsymbol{\beta}_0;\hat{\boldsymbol{\gamma}})-\mathbf{A}_n(\boldsymbol{\beta})(\boldsymbol{\beta}-\boldsymbol{\beta}_0),\qquad \mathbf{A}_n(\boldsymbol{\beta})=\int_0^1 n^{-1}\sum_i \mathbf{H}_i(\boldsymbol{\beta}_0+s(\boldsymbol{\beta}-\boldsymbol{\beta}_0);\hat{\boldsymbol{\gamma}})\,ds.$$

On E_n, $\|\mathbf{A}_n(\boldsymbol{\beta}) - \mathbf{A}\| \le \varepsilon_n + \frac{1}{4c} \le \frac{1}{2c}$ for all $\boldsymbol{\beta} \in \bar{B}_r$.

Step 3. Existence. Define $F(\boldsymbol{\beta}) = \boldsymbol{\beta} + \mathbf{A}^{-1} n^{-1} \mathbf{U}_n(\boldsymbol{\beta}; \hat{\boldsymbol{\gamma}})$; its fixed points are roots. By Step 2,

$$F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0 = \mathbf{A}^{-1} n^{-1} \mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}) + \mathbf{A}^{-1}(\mathbf{A} - \mathbf{A}_n(\boldsymbol{\beta}))(\boldsymbol{\beta} - \boldsymbol{\beta}_0),$$

so on E_n, $\|F(\boldsymbol{\beta}) - \boldsymbol{\beta}_0\| \le c \cdot \frac{r}{2c} + c \cdot \frac{1}{2c} \cdot r = r$. Thus the continuous map F sends $\bar{B}_r$ into itself, and by Brouwer's fixed-point theorem it has a fixed point $\hat{\boldsymbol{\beta}} \in \bar{B}_r$.

Step 4. Uniqueness. If $\boldsymbol{\beta}_1, \boldsymbol{\beta}_2 \in \bar{B}_r$ are roots, the same integral argument gives $\mathbf{0} = \tilde{\mathbf{A}}(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_2)$ with $\|\tilde{\mathbf{A}} - \mathbf{A}\| \le \frac{1}{2c}$. Then $\|\mathbf{A}^{-1}(\tilde{\mathbf{A}} - \mathbf{A})\| \le \frac{1}{2} < 1$, so $\tilde{\mathbf{A}} = \mathbf{A}(\mathbf{I} + \mathbf{A}^{-1}(\tilde{\mathbf{A}} - \mathbf{A}))$ is invertible (Neumann series) and $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$.

Step 5. Consistency. The argument works for every smaller radius $r' \le r$, and by uniqueness the root in $\bar{B}_{r'}$ is the root in $\bar{B}_r$. Hence $P(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0\| \le r') \to 1$ for every $r' > 0$: $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$. This proves (i).

Step 6. Asymptotic normality. On E_n, Step 2 at $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ gives

$$\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) = \mathbf{A}_n(\hat{\boldsymbol{\beta}})^{-1}\, n^{-1/2} \mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}).$$

First, $\|\mathbf{A}_n(\hat{\boldsymbol{\beta}}) - \mathbf{A}\| \le \varepsilon_n + \sup_{s \in [0,1]} \|\mathbf{A}(\boldsymbol{\beta}_0 + s(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0), \hat{\boldsymbol{\gamma}}) - \mathbf{A}\| \xrightarrow{p} 0$ by (E3), continuity and consistency. Second, by (E4), $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\gamma}}) = n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) + o_p(1)$, a normalized sum of independent mean-zero vectors. For any fixed $\boldsymbol{\lambda} \neq \mathbf{0}$, Lyapunov's condition holds,

$$\frac{\sum_i E|\boldsymbol{\lambda}'\boldsymbol{\psi}_i|^{2+\delta}}{\left(\sum_i \operatorname{Var}(\boldsymbol{\lambda}'\boldsymbol{\psi}_i)\right)^{1+\delta/2}} = \frac{O(n)}{\left(n\, \boldsymbol{\lambda}'\mathbf{B}\boldsymbol{\lambda}\,(1 + o(1))\right)^{1+\delta/2}} \to 0,$$

so $n^{-1/2}\boldsymbol{\lambda}'\mathbf{U}_n \xrightarrow{d} N(0, \boldsymbol{\lambda}'\mathbf{B}\boldsymbol{\lambda})$ and, by Cramér–Wold, $n^{-1/2}\mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\gamma}^*) \xrightarrow{d} N(\mathbf{0}, \mathbf{B})$. Slutsky's theorem gives (ii).

Step 7. Sandwich. $\hat{\mathbf{A}} \xrightarrow{p} \mathbf{A}$ as in Step 6 (uniform convergence plus continuity at the consistent $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}})$), and $\hat{\mathbf{B}} \xrightarrow{p} \mathbf{B}$ by the extra assumption in (iii) in the same way. Matrix inversion and multiplication are continuous, which proves (iii). ∎

Theorem 11.9 (Consistency and asymptotic normality of QL estimators)

Suppose the mean model $g(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta}$ is correct at $\boldsymbol{\beta}_0$ (the variance function v may be wrong) and that conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i = (\partial\mu_i/\partial\boldsymbol{\beta})(y_i - \mu_i)/v(\mu_i)$ (they hold, e.g., for bounded covariates, smooth g and v, $v > 0$, and bounded fourth moments of y_i). Then the QL estimator is consistent and asymptotically normal, with

$$\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}\Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{\operatorname{Var}(Y_i)}{v(\mu_i)^2} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]\mathbf{V}, \qquad \mathbf{V} = \Big[\sum_i \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{1}{v(\mu_i)} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'}\Big]^{-1}. \tag{11.8}$$

If $\operatorname{Var}(Y_i) = v(\mu_i)$, this reduces to $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx \mathbf{V}$, the GLM formula.

Proof.

Step 1. (E1) holds because $E(y_i) = \mu_i(\boldsymbol{\beta}_0)$ and the coefficient $(\partial\mu_i/\partial\boldsymbol{\beta})/v(\mu_i)$ is nonrandom; the true variance plays no role. There is no nuisance parameter, so (E4) is vacuous.

Step 2. Differentiating,

$$\mathbf{H}_i(\boldsymbol{\beta}) = \frac{1}{v(\mu_i)} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}'} - \frac{\partial}{\partial\boldsymbol{\beta}'}\Big[\frac{1}{v(\mu_i)} \frac{\partial\mu_i}{\partial\boldsymbol{\beta}}\Big](y_i - \mu_i).$$

At $\boldsymbol{\beta}_0$ the second term has mean zero, so $\mathbf{A} = \lim n^{-1} \sum_i v(\mu_i)^{-1}(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$. Also $\operatorname{Cov}(\boldsymbol{\psi}_i) = v(\mu_i)^{-2} \operatorname{Var}(Y_i)(\partial\mu_i/\partial\boldsymbol{\beta})(\partial\mu_i/\partial\boldsymbol{\beta}')$, giving $\mathbf{B}$.

Step 3. Lemma 11.8(ii) gives $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) \approx n^{-1}\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$, which is (11.8) after writing $n\mathbf{A} \approx \mathbf{V}^{-1}$ and $n\mathbf{B} \approx \sum_i(\cdots)$. If $\operatorname{Var}(Y_i) = v(\mu_i)$, then $\mathbf{B} = \mathbf{A}$ and the sandwich collapses to $\mathbf{V}$. ■

The book's heuristic is the essence of Step 1: $\mathbf{u}(\boldsymbol{\beta})/n$ is a vector of sample means whose expectation vanishes at the true $\boldsymbol{\beta}$ whatever the variance, and $\hat{\boldsymbol{\beta}}$ is the point where it vanishes.

11.4.2 GEE methodology: technical details

For cluster i let $\mathbf{y}_i = (y_{i1}, \ldots, y_{iT_i})'$, $\boldsymbol{\mu}_i = E(\mathbf{Y}_i)$, $\mathbf{x}_{it}$ the $p \times 1$ predictor vector for y_{it} (it may vary over t), $\mathbf{X}_i$ the $T_i \times p$ matrix with rows $\mathbf{x}_{it}'$, and $\eta_{it} = g(\mu_{it}) = \mathbf{x}_{it}'\boldsymbol{\beta}$. The model refers to each margin. Suppose

$$f(y_{it}; \theta_{it}, \phi) = \exp\{[y_{it}\theta_{it} - b(\theta_{it})]/\phi + c(y_{it}, \phi)\},$$

so $\mu_{it} = b'(\theta_{it})$ and $\operatorname{Var}(Y_{it}) = \phi\, b''(\theta_{it})$.

Definition (Generalized estimating equations)

Let $\mathbf{B}_i = \operatorname{diag}\big(b''(\theta_{i1}), \ldots, b''(\theta_{iT_i})\big)$ and let $\mathbf{R}(\boldsymbol{\alpha})$ be a working correlation matrix. The **working covariance** is

$$\mathbf{V}_i = \phi\, \mathbf{B}_i^{1/2}\mathbf{R}(\boldsymbol{\alpha})\mathbf{B}_i^{1/2} \tag{11.9}$$

($= \operatorname{Cov}(\mathbf{Y}_i)$ if $\mathbf{R}$ is the true correlation). Let $\boldsymbol{\Delta}_i = \operatorname{diag}(\partial\theta_{it}/\partial\eta_{it})$ (the identity for the canonical link) and $\mathbf{D}_i = \partial\boldsymbol{\mu}_i/\partial\boldsymbol{\beta}' = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ ($T_i \times p$). The GEE estimator $\hat{\boldsymbol{\beta}}$ solves

$$\mathbf{u}(\boldsymbol{\beta}) = \sum_{i=1}^{n} \mathbf{D}_i'\mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}. \tag{11.10}$$

The formula $\mathbf{D}_i = \mathbf{B}_i\boldsymbol{\Delta}_i\mathbf{X}_i$ is the chain rule $\partial\mu_{it}/\partial\beta_j = (\partial\mu_{it}/\partial\theta_{it})(\partial\theta_{it}/\partial\eta_{it})(\partial\eta_{it}/\partial\beta_j)$. Since ϕ is a common scalar factor of $\mathbf{V}_i$, it cancels from (11.10): $\hat{\boldsymbol{\beta}}$ does not depend on ϕ (given $\boldsymbol{\alpha}$).

Proposition 11.10 (Independence working correlation gives the GLM fit)

With $\mathbf{R} = \mathbf{I}$, (11.10) becomes $\sum_i \mathbf{X}_i'\boldsymbol{\Delta}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ (for the canonical link, $\sum_i \mathbf{X}_i'(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$): the ordinary GLM likelihood equations treating all y_{it} as independent.

Proof. With $\mathbf{R} = \mathbf{I}$, $\mathbf{V}_i = \phi\mathbf{B}_i$, so $\mathbf{D}_i'\mathbf{V}_i^{-1} = \mathbf{X}_i'\boldsymbol{\Delta}_i\mathbf{B}_i(\phi\mathbf{B}_i)^{-1} = \phi^{-1}\mathbf{X}_i'\boldsymbol{\Delta}_i$ (diagonal matrices commute). Dropping the factor $1/\phi$ gives the stated equations, which are the GLM likelihood equations of Proposition 11.7(a) summed over all (i, t). ■

Definition (Moment estimators of the working parameters)

With Pearson residuals $e_{it} = (y_{it} - \hat{\mu}_{it})/b''(\hat{\theta}_{it})^{1/2}$, $N = \sum_i T_i$ (Liang and Zeger 1986):

$$\hat{\phi} = \frac{1}{N - p}\sum_i\sum_t e_{it}^2, \qquad \hat{\alpha}_{\text{exch}} = \frac{\sum_i \sum_{t<s} e_{it}e_{is}}{\hat{\phi}\big[\sum_i T_i(T_i - 1)/2 - p\big]},$$

$$\hat{\alpha}_{\text{AR1}} = \frac{\sum_i \sum_{t \le T_i - 1} e_{it}e_{i,t+1}}{\hat{\phi}\big[\sum_i (T_i - 1) - p\big]}, \qquad \hat{R}_{ts} = \frac{1}{n\hat{\phi}}\sum_i e_{it}e_{is} \quad \text{(unstructured)}.$$

The fitting algorithm alternates a modified Fisher scoring step for $\boldsymbol{\beta}$ given $(\boldsymbol{\alpha}, \phi)$,

$$\boldsymbol{\beta}^{(k+1)} = \boldsymbol{\beta}^{(k)} + \Big[\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}\mathbf{D}_i\Big]^{-1}\sum_i \mathbf{D}_i'\mathbf{V}_i^{-1}(\mathbf{y}_i - \boldsymbol{\mu}_i),$$

with moment updates of $(\boldsymbol{\alpha}, \phi)$ from the current residuals. Other algorithms solve estimating equations for $\boldsymbol{\beta}$ and $\boldsymbol{\alpha}$ jointly. GEE algorithms need not converge, but one iteration is often adequate. For time series, AR(1) treats observations farther apart in time as more weakly correlated.

Theorem 11.11 (Liang–Zeger: consistency and asymptotic normality of GEE)

Assume:
(G1) the marginal mean model is correct: $E(\mathbf{Y}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}_0)$ for all i (the working correlation may be wrong);
(G2) $\sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) = O_p(1)$ for some $\boldsymbol{\alpha}^*$ (not necessarily a "true" value);
(G3) conditions (E2), (E3) and the condition in (iii) of Lemma 11.8 hold for $\boldsymbol{\psi}_i(\boldsymbol{\beta}; \boldsymbol{\alpha}) = \mathbf{D}_i' \mathbf{V}_i^{-1}(\boldsymbol{\alpha})(\mathbf{y}_i - \boldsymbol{\mu}_i)$, and $n^{-1} \sum_i \partial \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha}) / \partial \boldsymbol{\alpha}' \xrightarrow{p} \mathbf{0}$ uniformly for $\boldsymbol{\alpha}$ near $\boldsymbol{\alpha}^*$.
Then, as the number of clusters $n \to \infty$, $\hat{\boldsymbol{\beta}} \xrightarrow{p} \boldsymbol{\beta}_0$ and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{V}_G)$, where $\mathbf{V}_G = \lim_n \mathbf{V}_{G,n}$,

$$\mathbf{V}_{G,n} = n \Big[\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \Big]^{-1} \Big[\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \operatorname{Cov}(\mathbf{Y}_i) \mathbf{V}_i^{-1} \mathbf{D}_i \Big] \Big[\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \Big]^{-1},$$

evaluated at $(\boldsymbol{\beta}_0, \boldsymbol{\alpha}^*)$. The sandwich estimator $\hat{\mathbf{V}}_{G,n}/n$, which substitutes $\hat{\boldsymbol{\beta}}$, $\hat{\boldsymbol{\alpha}}$, $\hat{\phi}$ and replaces $\operatorname{Cov}(\mathbf{Y}_i)$ by $(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)'$, is consistent for $\operatorname{Cov}(\hat{\boldsymbol{\beta}})$.

Proof. [Supplement] We verify the hypotheses of Lemma 11.8 with nuisance $\boldsymbol{\gamma} = \boldsymbol{\alpha}$ (ϕ cancels from (11.10)).

Step 1. (E1). At $\boldsymbol{\beta}_0$, $\mathbf{D}_i$ and $\mathbf{V}_i(\boldsymbol{\alpha}^*)$ are nonrandom and $E(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ by (G1). Hence $E\boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*) = \mathbf{0}$ *for any* $\boldsymbol{\alpha}^*$; this is why a wrong working correlation does not destroy consistency.

Step 2. The matrices $\mathbf{A}$ and $\mathbf{B}$. Differentiating,

$$\mathbf{H}_i = \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i - \sum_j \Big[\frac{\partial (\mathbf{D}_i' \mathbf{V}_i^{-1})}{\partial \beta_j} (\mathbf{y}_i - \boldsymbol{\mu}_i) \Big] \mathbf{e}_j',$$

and the second term has mean zero at $\boldsymbol{\beta}_0$, so $\mathbf{A} = \lim n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$. Also $\operatorname{Cov}(\boldsymbol{\psi}_i) = \mathbf{D}_i' \mathbf{V}_i^{-1} \operatorname{Cov}(\mathbf{Y}_i) \mathbf{V}_i^{-1} \mathbf{D}_i$, whose average tends to $\mathbf{B}$. (E2), (E3) hold by (G3).

Step 3. (E4). By the mean value theorem applied to each component, for some $\bar{\boldsymbol{\alpha}}$ between $\hat{\boldsymbol{\alpha}}$ and $\boldsymbol{\alpha}^*$,

$$n^{-1/2} [\mathbf{U}_n(\boldsymbol{\beta}_0; \hat{\boldsymbol{\alpha}}) - \mathbf{U}_n(\boldsymbol{\beta}_0; \boldsymbol{\alpha}^*)] = \Big[n^{-1} \sum_i \frac{\partial \boldsymbol{\psi}_i(\boldsymbol{\beta}_0; \bar{\boldsymbol{\alpha}})}{\partial \boldsymbol{\alpha}'} \Big] \sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*).$$

Here $\partial \boldsymbol{\psi}_i / \partial \alpha_k = \mathbf{D}_i' (\partial \mathbf{V}_i^{-1} / \partial \alpha_k)(\mathbf{y}_i - \boldsymbol{\mu}_i)$ has mean zero at $\boldsymbol{\beta}_0$, and by (G3) the bracket is $o_p(1)$, while $\sqrt{n}(\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha}^*) = O_p(1)$ by (G2). The product is $o_p(1)$. Estimating $\boldsymbol{\alpha}$ thus has no first-order effect on $\hat{\boldsymbol{\beta}}$.

Step 4. Lemma 11.8(i)–(ii) gives consistency and $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}_0) \xrightarrow{d} N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$, and $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} = \lim \mathbf{V}_{G,n}$. Lemma 11.8(iii) gives consistency of the sandwich, since $\hat{\mathbf{B}} = n^{-1} \sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} (\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)(\mathbf{y}_i - \hat{\boldsymbol{\mu}}_i)' \mathbf{V}_i^{-1} \mathbf{D}_i$ evaluated at $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\alpha}})$; replacing $\hat{\mathbf{A}}$ by the expected-information version $n^{-1} \sum \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ at the estimates changes it by a term that is $o_p(1)$ (Step 2). ∎

Corollary 11.12 (Model-based covariance)

If the working covariance is correct, $\operatorname{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$, then $\mathbf{V}_{G,n}/n = \big(\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i \big)^{-1}$.

Proof. Substitute $\operatorname{Cov}(\mathbf{Y}_i) = \mathbf{V}_i$ in the middle factor: it becomes $\sum_i \mathbf{D}_i' \mathbf{V}_i^{-1} \mathbf{D}_i$ and cancels one outer inverse. ∎

Theorem 11.13 (Optimality of the true covariance as weight)

Consider all estimating equations $\sum_i \mathbf{D}_i'\mathbf{W}_i(\mathbf{y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ with nonrandom positive definite weights $\mathbf{W}_i$, and let $\boldsymbol{\Sigma}_i = \mathrm{Cov}(\mathbf{Y}_i)$. Their asymptotic covariance satisfies, in the Loewner order,

$$\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \;\geq\; \mathbf{F}^{-1}, \qquad \mathbf{G} = \sum_i \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i, \quad \mathbf{M} = \sum_i \mathbf{D}_i'\mathbf{W}_i\boldsymbol{\Sigma}_i\mathbf{W}_i\mathbf{D}_i, \quad \mathbf{F} = \sum_i \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i,$$

with equality when $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$. So a correct working covariance gives the most efficient GEE estimator.

Proof. [Supplement]

Step 1. For each i, the matrix

$$\begin{pmatrix} \mathbf{D}_i'\mathbf{W}_i \\ \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1} \end{pmatrix} \boldsymbol{\Sigma}_i \begin{pmatrix} \mathbf{W}_i\mathbf{D}_i & \boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i \end{pmatrix} = \begin{pmatrix} \mathbf{D}_i'\mathbf{W}_i\boldsymbol{\Sigma}_i\mathbf{W}_i\mathbf{D}_i & \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i \\ \mathbf{D}_i'\mathbf{W}_i\mathbf{D}_i & \mathbf{D}_i'\boldsymbol{\Sigma}_i^{-1}\mathbf{D}_i \end{pmatrix}$$

is positive semidefinite (it has the form $\mathbf{K}'\boldsymbol{\Sigma}_i\mathbf{K}$). Summing over i, $\left(\begin{smallmatrix} \mathbf{M} & \mathbf{G} \\ \mathbf{G} & \mathbf{F} \end{smallmatrix}\right) \geq 0$.

Step 2. Since $\mathbf{F} > 0$, the Schur complement is positive semidefinite: $\mathbf{M} - \mathbf{G}\mathbf{F}^{-1}\mathbf{G} \geq 0$.

Step 3. Multiplying on both sides by the symmetric $\mathbf{G}^{-1}$ preserves the order: $\mathbf{G}^{-1}\mathbf{M}\mathbf{G}^{-1} \geq \mathbf{F}^{-1}$. If $\mathbf{W}_i = \boldsymbol{\Sigma}_i^{-1}$ then $\mathbf{G} = \mathbf{M} = \mathbf{F}$ and equality holds. ■

Correlation is an awkward association measure for binary data

For binary Y_t, Y_s with $p_t = P(Y_t = 1)$, $p_s = P(Y_s = 1)$, the Fréchet bounds $\max(0, p_t + p_s - 1) \leq P(Y_t = Y_s = 1) \leq \min(p_t, p_s)$ (the first because $P(A \cap B) \geq P(A) + P(B) - 1$, the second because $A \cap B \subseteq A, B$) restrict

$$\mathrm{Corr}(Y_t, Y_s) = \frac{P(Y_t = Y_s = 1) - p_t p_s}{\sqrt{p_t(1-p_t)p_s(1-p_s)}}$$

to a range depending on the means. For instance, with $p_t = 0.1$, $p_s = 0.5$ the correlation cannot exceed $(0.1 - 0.05)/\sqrt{0.09 \cdot 0.25} = 1/3$. An alternative models within-cluster association by odds ratios (e.g. exchangeable log odds ratios), which are variation-independent of the means; the *alternating logistic regressions* algorithm (Carey, Zeger and Diggle 1993) alternates a GEE step for $\boldsymbol{\beta}$ with a logistic regression step for the log odds ratios, useful when the association is itself of interest.

11.5 Markov chains: transitional modeling

Motivation. Instead of modelling margins, one can model each response given the earlier ones. The joint distribution then factors into transition probabilities, and fitting reduces to ordinary multinomial or logistic models for the transitions.

For responses Y_t at times $t = 0, 1, 2, \ldots$, the family $\{Y_t\}$ is a stochastic process; its *state space* is the set of possible values, and Y_0 is the initial state. Here state space and time are discrete.

11.5.1 Transitional models

Transitional models describe Y_t in terms of the previous responses $y_0, \ldots, y_{t-1}$ (and covariates), via the factorization

$$f(y_0, \ldots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_0, y_1) \cdots f(y_T \mid y_0, \ldots, y_{T-1}).$$

Unlike marginal models, they condition on past responses.

11.5.2 First-order Markov chains

Definition (Markov chain)

$\{Y_t\}$ is a (first-order) **Markov chain** if, for all t, Y_{t+1} given $(Y_0, \dots, Y_t)$ has the same distribution as Y_{t+1} given Y_t alone. It is a **kth-order** chain if Y_{t+1} given $(Y_0, \dots, Y_t)$ depends only on $(Y_{t-k+1}, \dots, Y_t)$. The **transition probabilities** are $\pi_{j|i}(t) = P(Y_t = j \mid Y_{t-1} = i)$, with $\sum_j \pi_{j|i}(t) = 1$; they form the one-step $I \times I$ transition matrix.

Theorem 11.14 (Markov factorization and its loglinear form)

(a) $\{Y_0, \dots, Y_T\}$ is a Markov chain if and only if

$$f(y_0, \dots, y_T) = f(y_0) f(y_1 \mid y_0) f(y_2 \mid y_1) \cdots f(y_T \mid y_{T-1}). \tag{11.11}$$

(b) A positive joint distribution is that of a Markov chain if and only if it satisfies the loglinear model $(Y_0Y_1, Y_1Y_2, \dots, Y_{T-1}Y_T)$, i.e. $\log \pi(y_0, \dots, y_T) = \sum_{t=1}^{T} h_t(y_{t-1}, y_t)$ for some functions h_t.

Proof. [Supplement]

Step 1. (a) The general factorization of Section 11.5.1 holds always; the Markov property replaces each $f(y_t \mid y_0, \dots, y_{t-1})$ by $f(y_t \mid y_{t-1})$, giving (11.11). Conversely, if (11.11) holds, summing it over $y_{t+1}, \dots, y_T$ (each later factor sums to 1, from the last backwards) gives $f(y_0, \dots, y_t) = f(y_0) \prod_{s \le t} f(y_s \mid y_{s-1})$, and dividing the expressions for $t+1$ and t gives $f(y_{t+1} \mid y_0, \dots, y_t) = f(y_{t+1} \mid y_t)$.

Step 2. (b, $\Rightarrow$) Taking logs of (11.11) and absorbing $\log f(y_0)$ into h_1 gives $\sum_t h_t(y_{t-1}, y_t)$, which contains only main effects and two-factor terms for consecutive pairs.

Step 3. (b, $\Leftarrow$) Suppose $\pi(\mathbf{y}) = \prod_{t=1}^{T} g_t(y_{t-1}, y_t)$ with $g_t = e^{h_t}$. Define $G_t(y_t) = \sum_{y_{t+1}, \dots, y_T} \prod_{s > t} g_s(y_{s-1}, y_s)$ (with $G_T \equiv 1$). Then

$$P(y_{t+1} \mid y_0, \dots, y_t) = \frac{\prod_{s \le t} g_s \cdot g_{t+1}(y_t, y_{t+1}) G_{t+1}(y_{t+1})}{\prod_{s \le t} g_s \cdot G_t(y_t)} = \frac{g_{t+1}(y_t, y_{t+1}) G_{t+1}(y_{t+1})}{G_t(y_t)},$$

which depends only on (y_t, y_{t+1}): the Markov property. ∎

Thus fitting the loglinear model $(Y_0Y_1, \dots, Y_{T-1}Y_T)$ to the I^{T+1} table of sequences tests the Markov property; adding three-factor terms $(Y_{t-2}Y_{t-1}Y_t)$ gives a second-order chain.

Theorem 11.15 (ML estimation of transition probabilities)

Let $n_{ij}(t)$ be the number of transitions from state i at time $t-1$ to state j at time t, $n_{i+}(t) = \sum_j n_{ij}(t)$, and $\{n_{i0}\}$ the initial counts, multinomial with probabilities $\{\pi_{i0}\}$. For n independent subjects following a Markov chain, the likelihood is proportional to

$$\Big[\prod_{i=1}^{I} \pi_{i0}^{n_{i0}}\Big] \prod_{t=1}^{T} \prod_{i=1}^{I} \prod_{j=1}^{I} \pi_{j|i}(t)^{n_{ij}(t)}, \tag{11.12}$$

and the ML estimates are $\hat{\pi}_{j|i}(t) = n_{ij}(t)/n_{i+}(t)$ (Anderson and Goodman 1957). For a time-homogeneous chain, $\pi_{j|i}(t) \equiv \pi_{j|i}$, the ML estimate is $\hat{\pi}_{j|i} = \sum_t n_{ij}(t) / \sum_t n_{i+}(t)$.

Proof.

Step 1. By (11.11) each subject's probability is $\pi_{y_0 0}\prod_t \pi_{y_t|y_{t-1}}(t)$; multiplying over subjects and grouping identical factors gives (11.12).

Step 2. [Supplement] The likelihood is a product of separate kernels, one for $\{\pi_{i0}\}$ and one for each (t,i), the latter being the multinomial kernel $\prod_j \pi_{j|i}(t)^{n_{ij}(t)}$ with $\sum_j \pi_{j|i}(t)=1$; the parameters of different kernels are unconstrained by each other, so each is maximized separately.

Step 3. For one kernel $\sum_j n_j \log \pi_j$ with $\sum_j \pi_j = 1$ and $m=\sum_j n_j > 0$, Gibbs' inequality gives

$$\sum_j n_j \log \pi_j - \sum_j n_j \log \frac{n_j}{m} = m \sum_j \frac{n_j}{m} \log \frac{\pi_j}{n_j/m} \le m \log \sum_j \pi_j = 0$$

(Jensen, concavity of log), with equality iff $\pi_j = n_j/m$. So $\hat\pi_{j|i}(t) = n_{ij}(t)/n_{i+}(t)$.

Step 4. Time-homogeneous case: the kernels for fixed i combine into $\prod_j \pi_{j|i}^{\sum_t n_{ij}(t)}$, and Step 3 applies with $n_j = \sum_t n_{ij}(t)$. ■

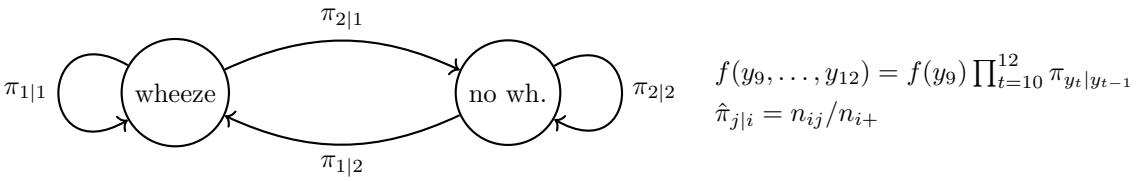


Figure 11.2. A two-state first-order Markov chain; each transition is a multinomial (here binomial) trial.

11.5.3 Respiratory illness example

Example 11.16 (Wheeze at ages 9–12)

In a Harvard study of air pollution, children were examined annually at ages 9–12 for wheeze (1 = wheeze, 2 = no wheeze):

Y_9	Y_{10}	Y_{11}	Y_{12}	Count	Y_9	Y_{10}	Y_{11}	Y_{12}	Count
1	1	1	1	94	2	1	1	1	19
1	1	1	2	30	2	1	1	2	15
1	1	2	1	15	2	1	2	1	10
1	1	2	2	28	2	1	2	2	44
1	2	1	1	14	2	2	1	1	17
1	2	1	2	9	2	2	1	2	42
1	2	2	1	12	2	2	2	1	35
1	2	2	2	63	2	2	2	2	572

The first-order chain $(Y_9Y_{10}, Y_{10}Y_{11}, Y_{11}Y_{12})$ fits poorly ($G^2 = 122.9$, df $= 8$), and so does the second-order chain $(Y_9Y_{10}Y_{11}, Y_{10}Y_{11}Y_{12})$ ($G^2 = 23.9$, df $= 4$); the poor fits may partly reflect subject heterogeneity (ignored covariates such as parental smoking). The model with all six pairwise associations fits well ($G^2 = 1.5$, df $= 5$). Its conditional log odds ratios are 1.81, 1.65, 1.85 for the pairs one year apart $(9,10),(10,11),(11,12)$ and 0.95, 1.05, 1.07 for $(9,11),(9,12),(10,12)$. The simpler model with one common value for each group fits well ($G^2 = 2.3$, df $= 9$), with estimates 1.75 (one year apart) and 1.04 (more than one year apart).

11.5.4 Transitional models with explanatory variables

With covariates $\mathbf{x}$,

$$f(y_1, \ldots, y_T; \mathbf{x}) = f(y_1; \mathbf{x}) f(y_2 \mid y_1; \mathbf{x}) \cdots f(y_T \mid y_1, \ldots, y_{T-1}; \mathbf{x}),$$

and for binary y one may specify a logistic model for each factor:

$$f(y_t \mid y_1, \ldots, y_{t-1}; \mathbf{x}_t) = \frac{\exp\left[y_t(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1} y_{t-1} + \boldsymbol{\beta}' \mathbf{x}_t)\right]}{1 + \exp(\alpha + \beta_1 y_1 + \cdots + \beta_{t-1} y_{t-1} + \boldsymbol{\beta}' \mathbf{x}_t)}, \qquad y_t = 0, 1,$$

a **regressive logistic model** (Bonney 1987): previous responses act as explanatory variables, and $\mathbf{x}_t$ may change with t. First-order Markov structure sets the coefficients of $y_1, \ldots, y_{t-2}$ to zero. Interaction between $\mathbf{x}_t$ and y_{t-1} may be needed. Note that the meaning and size of $\boldsymbol{\beta}$ depend on how many previous responses are in the model: within-cluster effects can shrink markedly once we condition on the past, unlike marginal model effects, whose interpretation does not depend on the dependence structure.

Proposition 11.17 (Fitting transitional models by ordinary logistic regression)

Conditional on the first response(s), the likelihood of a regressive logistic model is the product, over subjects and transitions, of Bernoulli likelihoods with the stated logits. Hence the conditional ML estimates are obtained by fitting an ordinary logistic regression that treats each transition as a separate independent observation, with previous responses as predictors.

Proof. [Supplement] By the factorization, subject i contributes

$$f(y_{i1}; \mathbf{x}) \prod_{t \geq 2} f(y_{it} \mid y_{i1}, \ldots, y_{i,t-1}; \mathbf{x}_{it}).$$

Ignoring (conditioning on) the first factor, each remaining factor is a Bernoulli probability $\text{expit}(\eta_{it})^{y_{it}}[1 - \text{expit}(\eta_{it})]^{1-y_{it}}$, where η_{it} is linear in the parameters with known regressors (the observed past responses and $\mathbf{x}_{it}$). Independence across subjects makes the conditional likelihood a product of such terms over all (i, t), which is exactly the likelihood of a logistic regression with one observation per transition. ■

Although transitions of one subject are dependent, the conditional likelihood has this independent-data form, so GLM software gives correct ML estimates and (likelihood-based) standard errors.

11.5.5 Child's respiratory illness and maternal smoking

Example 11.18 (Respiratory illness and maternal smoking)

Children were evaluated annually at ages 7–10 for respiratory illness; $s = 1$ if the mother smoked regularly at the start of the study:

			No maternal smoking		Maternal smoking	
Age 7	Age 8	Age 9	Age 10 No	Age 10 Yes	Age 10 No	Age 10 Yes
No	No	No	237	10	118	6
No	No	Yes	15	4	8	2
No	Yes	No	16	2	11	1
No	Yes	Yes	7	3	6	4
Yes	No	No	24	3	7	3
Yes	No	Yes	3	2	3	1
Yes	Yes	No	6	2	4	2
Yes	Yes	Yes	5	11	4	7

The regressive logistic model

$$\text{logit}\, P(Y_t = 1) = \alpha + \beta_1 s + \beta_2 t + \beta_3 y_{t-1}, \qquad t = 8, 9, 10,$$

uses three transitions per child; the data form 12 binomials for the $2 \times 3 \times 2$ combinations of (s, t, y_{t-1}) (e.g. for $(0, 8, 0)$: $y_8 = 0$ for $237 + 10 + 15 + 4 = 266$ children and $y_8 = 1$ for $16 + 2 + 7 + 3 = 28$). The ML fit is

$$\text{logit}\, \hat{P}(Y_t = 1) = -0.293 + 0.296 s - 0.243 t + 2.211 y_{t-1},$$

with SEs $(0.846, 0.156, 0.095, 0.158)$. The previous response has a strong effect; given it and age, there is slight evidence of a maternal smoking effect (LR statistic 3.55, df $= 1$, $P = 0.06$). The model fits well ($G^2 = 3.1$, df $= 8$).